

Transaction Processing Manual

BASE24-pos[®]



© 2012 by ACI Worldwide, Inc. All rights reserved.

All information contained in this documentation, as well as the software described in it, is confidential and proprietary to ACI Worldwide, Inc., or one of its subsidiaries, is subject to a license agreement, and may be used or copied only in accordance with the terms of such license. Except as permitted by such license, no part of this documentation may be reproduced, stored in a retrieval system, or transmitted in any form or by electronic, mechanical, recording, or any other means, without the prior written permission of ACI Worldwide, Inc., or one of its subsidiaries.

ACI, ACI Worldwide, and the ACI product names used in this documentation are trademarks or registered trademarks of ACI Worldwide, Inc., or one of its subsidiaries.

Other companies' trademarks, service marks, or registered trademarks and service marks are trademarks, service marks, or registered trademarks and service marks of their respective companies.

Contents

What's New	xi
Preface	xiii
Conventions Used in This Manual	xix
1: Introduction	1-1
BASE24-pos Transaction Processing	1-2
Transaction Originators	1-2
Transaction Authorizers	1-3
Issuers and Acquirers	1-6
Hosts and DPCs	1-7
Cards and Accounts	1-8
Cards	1-8
Accounts	1-11
BASE24-pos Transactions	1-15
Transaction Identifiers	1-15
Financial Transactions	1-16
Administrative Transactions	1-32
BASE24-pos Message Types	1-33
Message Functions	1-33
Message Types	1-33
Transaction Logging	1-40
Message Types Logged	1-40
Tokens Logged	1-41
Multiple Logical Networks	1-41
2: BASE24-pos Transaction Path	2-1
BASE24-pos Processes	2-2
BASE24-pos Device Handler Modules	2-3

BASE24-pos Processes <i>continued</i>	
BASE24-pos Router Module	2-6
BASE24-pos Authorization Module.....	2-6
Transaction Context Manager Process	2-7
ISO Host Interface Processes	2-7
Interchange Interface Processes	2-8
BIC ISO Interface Processes.....	2-9
Process Linkages.....	2-11
Inbound Messages from POS Terminals, Hosts, and Interchanges	2-11
Device Handler/Router/Authorization Process to Application Process Other Than the Transaction Context Manager Process	2-12
Device Handler/Router/Authorization Process to Transaction Context Manager Process	2-13
Application Process to Device Handler/Router/Authorization Process	2-13
Device Handler/Router/Authorization Process to ISO Host Interface Process ...	2-13
Device Handler/Router/Authorization Process to Interchange Interface Process .	2-14
ISO Host Interface Process to Device Handler/Router/Authorization Process ...	2-14
Interchange Interface Process to Device Handler/Router/Authorization Process .	2-14
Interchange Interface Process to Application Process	2-14
Application Process to Interchange Interface Process	2-15
Service-Based Routing	2-16
Transaction Timers	2-18
General Timers	2-18
Device Handler Module Timers	2-18
ISO Host Interface Process Timers.....	2-19
Interchange Interface Process Timers.....	2-21
3: BASE24-pos Transaction Processing Controls	3-1
Authorization Levels and Methods.....	3-3
Authorization Levels.....	3-3
Authorization Methods	3-5
Allowed Transactions	3-8
Cardholder Transactions Allowed at the Terminal.....	3-8
Transactions Allowed by the Card Issuer.....	3-12
Transactions Allowed Between the BASE24 System and an Interchange.....	3-14

BASE24-pos Transaction Processing Controls *continued*

Allowed Service Types	3-18
Service Types Allowed at the Terminal	3-18
Service Types Allowed by the Retailer	3-20
Transaction Security	3-21
PIN Verification	3-21
Card Verification	3-22
PIN Encryption	3-22
Message Authentication	3-23
Data Encryption	3-24
Derived Unique Key per Transaction (DUKPT)	3-24
Dynamic Key Management	3-25
Transaction Screening	3-26
Card Status	3-26
Expiration Date	3-27
Card Verification	3-27
PIN Check	3-28
Withdrawal Limits	3-28
Other Prescreening Checks	3-29
Draft Capture Processing Options	3-30
Authorization Only with Paper Follow-up	3-30
Authorization and Draft Capture	3-31
Authorization with Electronic Follow-up	3-31
Terminal-Defined Draft Capture	3-32
Cardholder and Transaction Limits	3-33
Card and Cardholder Limits	3-33
Transaction Limits	3-35
Usage Accumulation Totals	3-38
Establishing Usage Accumulation Periods	3-38
Clearing Usage Accumulation Totals	3-38
Store-and-Forward Processing	3-40
Interchange Store-and-Forward Processing	3-40
Host Store-and-Forward Processing	3-40
Second Card Processing	3-42
Expiration Date Checking	3-43

BASE24-pos Transaction Processing Controls *continued*

Preauthorized Holds Processing	3-44
Establishing Preauthorized Holds Processing.	3-45
Maintaining the Authorization Files	3-47
Authorizing for a Different Amount Than Requested	3-50
Lesser Amounts	3-50
Greater Amounts.	3-51
Partial Authorization Routing.	3-52
Chargeback and Representment Processing	3-54
BASE24-pos Chargeback Support	3-54
BASE24-pos Representment Support.	3-59
Controlling Chargeback and Representment Processing	3-60
Generating Completions to a Host	3-61
Token Handling.	3-62
Logging Tokens	3-62
Retrieving Tokens.	3-62
Extracting Tokens.	3-65
Sending Tokens to a Host or Co-Network	3-65
Carrying Tokens in the BASE24-pos Internal Message.	3-65
Address Verification Processing.	3-66
4: BASE24-pos Routing and Authorization	4-1
Router and Authorization Modules Overview	4-2
Router Module Processing	4-2
Authorization Module Processing	4-5
AST Routing and Authorization Selection	4-6
Information in the AST.	4-6
Information in the RTBL	4-7
Alternate Host Routing.	4-10
AST Selection Hierarchy	4-10
Selecting an AST Record	4-12
Example AST Settings	4-15
Example AST Selection Processing	4-18
RTBL Character Substitution	4-22

BASE24-pos Routing and Authorization *continued*

SPROUTE Routing	4-23
Internal Algorithms	4-24
The SPROUTE File	4-26
The SPREFIX File	4-26
The Basics of Routing (PREMAP)	4-27
Selection Hierarchy	4-29
Routing Methods	4-30
Setting Up Routing Tables	4-32
Routing Example	4-32
IDF Routing and Authorization Selection	4-37
IDF Routing Table Entries	4-38
Setting the HOLDS LVL Field for Preauthorized Holds	4-39
IDF Routing Table Examples	4-40
Deriving Parameters from the IDF Routing Table	4-44
Routing Check Authorization Transactions	4-46
Locating a CPF Record	4-47
SPROUTE Retrieval Processing	4-47
Performing AST Routing	4-48
Using the CART	4-49
Enhanced Reversal Routing	4-52
Approval Code Generation	4-53
Referral Code Generation	4-55
5: BASE24-pos Message Flows	5-1
Diagram Information	5-2
Special Diagram Notations	5-2
Diagram Limitations	5-3
Processing Assumptions	5-4
Authorization Level 1	5-5
Authorization Level 2	5-34
Authorization Level 3	5-47
Multi-Nodal Transactions	5-60
Administrative Transactions	5-63
Interchange Transactions	5-67

BASE24-pos Message Flows *continued*

APCF-Routed Transactions	5-74
Mobile Top-Up Transactions	5-80
6: BASE24-pos Transaction Impacts on the Database.....	6-1
PBF and CAF/UAF Impacts.....	6-2
PBF Totals	6-2
CAF/UAF Totals.....	6-3
Transaction Impact Table	6-5
BASE24-pos Terminal Data Files Impacts.....	6-13
A: BASE24-pos Internal Message.....	A-1
Tokens and the BASE24-pos Internal Message	A-2
POS Standard Internal Message (PSTM).....	A-3
BASE24-pos USER-DATA	A-53
B: BASE24-pos Authorization Response Codes.....	B-1
Authorization Response Codes.....	B-3
C: BASE24-pos Parametric Authorization Method	C-1
PBF/CAPF Processing Overview	C-2
Comparing Delinquency History Information	C-2
Comparing Available Credit with Overlimit Amounts.....	C-4
Verifying Single Transaction Limits	C-7
PBF/CAPF Processing Example.....	C-9
Checking the Account's Delinquency History	C-9
Checking for an Overlimit Situation.....	C-10
Calculating the Percent Overlimit.....	C-11
Calculating a Lesser Amount	C-12
Verifying the Single Transaction Limit	C-13
D: BASE24-pos Draft Capture Processing.....	D-1
Processing According to the Draft Capture Flag	D-2
Setting the Draft Capture Flag in the PSTM.....	D-4
CRT Authorization Device Handler Module	D-4
Other Device Handler Modules	D-4

E: Pregen Utility	E-1
The Pregen Utility	E-2
BUILDFILESIZE Command	E-2
BUILDSPROUTE Command	E-3
PRADD Command	E-3
PREND Command	E-7
REMOTESPROUTE Command	E-7
SPROUTEND Command	E-8
Pregen Command File	E-8
Pregen FUP Obey File	E-10
Running the Pregen Utility	E-11
Pregen Report	E-11
Using a New SPROUTE File	E-15
Error and Status Messages	E-16
Message Identification	E-16
F: Base Extended Memory Table Build Utility	F-1
Overview	F-2
Allocating a Newly Built or Rebuilt Extended Memory Table	F-4
Source Files	F-5
Acquirer Processing Code File Extended Memory Table	F-5
Exchange Rate File Extended Memory Table	F-6
Issuer Processing Code File Extended Memory Table	F-6
Response Code Description File Extended Memory Table	F-7
Split Transaction Routing File Extended Memory Table	F-7
LCONF Assigns and Params	F-8
Startup	F-9
EMTBRUN TACL Macro	F-9
Executing the Macro	F-9
Extended Memory Table Build Report	F-12
Cover Page	F-12
Hash Table Summary Page	F-13
Trailer Page	F-16
Index	Index-1

ACI Worldwide, Inc.

What's New

Here is what's new in the *BASE24-pos Transaction Processing Manual* since it was published for release 6.0, version 10.

Mar 2012

This table highlights the updates made to the *BASE24-pos Transaction Processing Manual* (release 6.0, version 10) in Mar 2012. The first column of the table lists the sections and appendixes in which the changes were made; the second column describes the changes.

Section/ Appendix	Major Changes
----------------------	---------------

B	Added 010 to the Authorization Response Code table as a valid value.
---	--

June 2011

This table highlights the updates made to the *BASE24-pos Transaction Processing Manual* (release 6.0, version 10) in June 2011. The first column of the table lists the sections and appendixes in which the changes were made; the second column describes the changes.

Section/ Appendix	Major Changes
----------------------	---------------

1	Corrects and clarifies supported card type values and descriptions.
3	Clarifies the use of the Special, Check (SC) special-use card type.
4	Corrects and clarifies the internal algorithms used by the Router module for SPROUTE routing.

ACI Worldwide, Inc.

Preface

The ***BASE24-pos Transaction Processing Manual*** describes how the BASE24-pos product processes transactions. It includes basic transaction processing concepts and definitions, as well as specifics on how transactions are identified and processed by the BASE24-pos product. It also describes how to control and configure various aspects of BASE24-pos transaction processing.

Audience

This manual is intended for the BASE24 managers and operational staff who configure the BASE24-pos database for transaction processing and must understand how transactions flow through the BASE24-pos system.

Prerequisites

Persons using this manual should be familiar with the BASE24-pos product, its basic functions, its terminology, and the BASE24 files it uses.

Additional Documentation

The BASE24 documentation set is arranged so that each BASE24 manual presents a topic or group of related topics in detail. When one BASE24 manual presents a topic that has already been covered in detail in another BASE24 manual, the topic is summarized and the reader is directed to the other manual for additional information. Information has been arranged in this manner to be more efficient for readers who do not need the additional detail and at the same time provide the source for readers who require the additional information.

This manual contains references to the following BASE24 publications:

- The ***ACI Standard POS Device Message Specifications Manual*** describes the message formats for the BASE24 Standard POS Device Handler (SPDH) module.

- The ***BASE24 Base Files Maintenance Manual*** and ***BASE24-pos Files Maintenance Manual*** provide detailed descriptions and valid values for the configuration fields discussed in this manual.
- The ***BASE24 Device Control Manual*** provides detailed information on the EMT Warmboot Control screen.
- The ***BASE24 External Message Manual*** presents detailed field descriptions and configuration information for the ISO-based external message used between the BASE24 system and a host.
- The ***BASE24 Files and Record Structures Reference Manual*** provides general information about the structure of BASE24 files and records, along with instructions for using the ACI DDLDOC utility.
- The ***BASE24 ISO Host Interface Manual***, ***BASE24 BIC ISO Interface Manual***, and ***BASE24 BIC ISO Standards Manual*** provide detailed processing information for each of the specified interfaces.
- The ***BASE24 ITS Kernel Configuration Manual*** provides information about using the Extended Memory Table Build utility to build extended memory tables used by object-oriented applications.
- The ***BASE24 Logical Network Configuration File Manual*** describes, in detail, each assign and param used in configuring BASE24-pos Device Handler/Router/Authorization and ISO Host Interface processes.
- The ***BASE24 Text Command Reference Manual*** provides detailed information on the ALTER ATTRIBUTE text command.
- The ***BASE24 Tokens Manual*** presents detailed field descriptions and configuration information for the tokens used by the BASE24-pos product.
- The ***BASE24 Transaction Security Manual*** describes BASE24 product support of PIN verification and encryption, card verification, message authentication, and dynamic key management, along with the database settings required to implement this support.
- The ***BASE24 Transaction Security Services Processing Guide*** describes how the Transaction Security Services process is integrated into a BASE24 system. It describes how BASE24 hardware security data is converted and maintained in data files, how BASE24 processes are configured to use the Transaction Security Services process, and the implementation requirements for using the process.
- The ***BASE24-atm Transaction Processing Manual*** lists other assigns and params used by the Extended Memory Table Build utility on behalf of BASE24-atm processes.

- The ***BASE24-pos Address Verification Manual*** describes the BASE24-pos add-on Address Verification module and its benefits, implementation, and requirements.
- The ***BASE24-pos CRT Authorization and CRT Administration Manual*** provides detailed information about the CRT Authorization and CRT Administration Device Handler modules.
- The ***BASE24-pos EMV Support Manual*** describes the response codes specific to the BASE24-pos EMV add-on product.
- The ***BASE24-pos Settlement and Reporting Manual*** describes, in detail, the settlement and reporting functions supported by the BASE24-pos product and how to configure them.
- The ***BASE24-pos Standard POS Device Configuration Manual*** describes the configuration requirements for the BASE24 Standard POS Device Handler (SPDH) module.
- The ***BASE24-pos Stored Value Support Manual*** describes in detail the configuration requirements for the BASE24-pos Stored Value add-on product.

This manual also refers to the following NET24-XPNET publications:

- The ***NET24-XPNET Network Control Command Reference Manual*** contains detailed information on commands and attributes that can be used to configure a BASE24 network of lines, stations, and processes.
- The ***NET24-XPNET Network Control Operations Guide***, describes the service-based routing algorithm used by the XPNET process.

Software

This manual documents standard processing as of its publication date. Software that is not current and custom software modifications (CSMs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Manual Summary

The following is a summary of the contents of this manual:

“Conventions Used in This Manual” follows this preface and describes notation and documentation conventions necessary to understand the information in the manual.

Section 1, “Introduction,” introduces the reader to the basics of BASE24-pos transaction processing. It includes definitions and discussions of transaction originators and authorizers, issuers and acquirers, hosts and DPCs, cards and accounts, supported transactions, transaction messages, and transaction logging.

Section 2, “BASE24-pos Transaction Path,” discusses the BASE24-pos processes involved in transaction processing and how they are linked together to form various transaction paths. It also describes the timers these processes use to control processing.

Section 3, “BASE24-pos Transaction Processing Controls,” describes the various transaction processing control options that enable a logical network to be tailored to the needs of its institutions and retailers. It includes discussions of such topics as authorization levels and methods, allowed transactions, transaction security options, card types, and services allowed.

Section 4, “BASE24-pos Routing and Authorization,” describes—and provides examples of—how the BASE24-pos product routes transactions and selects authorization parameters for processing transactions.

Section 5, “BASE24-pos Message Flows,” presents a series of diagrams illustrating the paths taken by BASE24-pos transactions under various types of processing situations.

Section 6, “BASE24-pos Transaction Impacts on the Database,” describes the impacts of each type of BASE24-pos transaction on the Cardholder Authorization File (CAF), Positive Balance File (PBF), BASE24-pos Terminal Data files, and Usage Accumulation File (UAF).

Appendix A, “BASE24-pos Internal Message,” describes the fields in the POS Standard Internal Message (PSTM) and describes how BASE24-pos tokens are carried in the internal message.

Appendix B, “BASE24-pos Authorization Response Codes,” describes all of the response codes that can be carried in the PSTM to identify the results of processing a transaction.

Appendix C, “BASE24-pos Draft Capture Processing,” describes BASE24-pos draft capture transaction processing and its options.

Appendix D, “BASE24-pos Parametric Authorization Method,” describes the processing that the Authorization module performs when authorizing a transaction using the Parametric Authorization method. This appendix also provides an example of transactions processed using this authorization method.

Appendix E, “Pregen Utility,” describes the Pregen utility and the procedures used to build and start using a new or revised SPROUTE File.

Appendix F, “Base Extended Memory Table Build Utility,” describes the Extended Memory Table Build utility and the procedures used to build and start using a new or revised extended memory table.

Publication Identification

Three entries appearing at the bottom of each page uniquely identify this BASE24 publication. The publication number (for example, PS-AE000-07 for the ***BASE24-pos Transaction Processing Manual***) appears on every page to assist readers in identifying the manual from which a page of information was printed. The publication date (for example, Jun-2011 for June, 2011) indicates the issue of the manual. The software release information (for example, R6.0v10 for release 6.0, version 10) specifies the software that the manual describes. This information matches the document information on the copyright page of the manual.

ACI Worldwide, Inc.

Conventions Used in This Manual

This section explains how syntax notation, required blank spaces, and BASE24-pos Terminal Data files terminology are documented in this manual.

Syntax Notation

Syntax descriptions in this manual use several symbols and conventions to denote how commands must be entered. These conventions are described in the table below.

Notation	Meaning
UPPERCASE LETTERS	Indicate key words and literals that you must enter exactly as shown.
<i>lowercase italics</i>	Identify variables that you must provide.
Brackets []	Enclose optional syntax items that you can enter in the command.
Braces { }	Enclose a group of items from which you are required to choose one item. The items are separated within the braces by a vertical bar ().
Vertical bar	Separates alternatives in a list of items.
Ellipsis ...	Indicates the immediately preceding syntactical item can occur one or more times.
Punctuation	Must be entered as shown. This includes parentheses, commas, semicolons, and other symbols not previously described.

Notation	Meaning
Spaces	Are required as delimiters wherever it would be otherwise impossible to discriminate between adjacent words or symbols. You can insert additional spaces between any adjacent words or symbols, but not within a word or multiple character symbol.
Indented lines	Indicates a continuation of the preceding line of syntax. If the syntax of a command is too long to fit on a single line, each continuation line is indented.

Required Blank Spaces

Throughout this manual when discussing the impact of required blanks or spaces in entered field data, the symbol *b* is used to denote a required blank character or space.

BASE24-pos Terminal Data File Terminology

Data entered on BASE24-pos Terminal Data files (PTD) screens is actually stored in different files, depending on the type of terminal being defined. For Hypercom and Standard POS Device Handler terminals, dynamic data is stored in the BASE24-pos Terminal Data Dynamic File—general data (PTDD1) and static data is stored in the BASE24-pos Terminal Data Static File—general data (PTDS1). For all other types of POS terminals, all data from the PTD screens is stored in the POS Terminal Data File (PTDF).

1: Introduction

This section provides an introduction to BASE24-pos transaction processing. It provides readers with some basic information on how the BASE24-pos product processes transactions.

BASE24-pos Transaction Processing

The BASE24-pos product provides the means by which a transaction originator can request and receive authorization for an action on a customer card or account from a transaction authorizer. The BASE24-pos product performs the following roles in transaction processing:

- Routing transactions from the transaction originator to the appropriate authorizer.
- Participating in or carrying out authorization on behalf of the institutions and retailers for which it processes.
- Logging all transactions for later processing and settlement.

The BASE24-pos product can accept transactions from a variety of sources. Likewise, it can involve a number of different types of authorizers in its processing. The following describes the various transaction originators and authorizers that can exist in a BASE24-pos system.

Transaction Originators

The BASE24-pos product can accept transactions for authorization from several different types of originators. These originators are described below.

Point-of-Sale Devices

Point-of-sale (POS) devices can be directly attached to the BASE24-pos system, in which case transactions are originated by customers, service personnel, or retail clerks at the point of sale. Communications between the POS device and the BASE24-pos product are controlled by BASE24-pos Device Handler modules.

Hosts

Host computers can be set up to control point-of-sale device networks and forward transactions to the BASE24-pos system. In this case, communications between these acquirer hosts and the BASE24-pos product are controlled by BASE24-pos ISO Host Interface processes.

Interchanges

Interchanges can act as switches to forward transactions to the BASE24-pos system for authorization. In this case, communications between the interchange acquiring the transaction and the BASE24-pos product are controlled by BASE24-pos Interchange Interface processes.

Co-Networks

Co-networks can be linked to a BASE24-pos system through a BASE24 Interchange (BIC) Interface process. In this case, transactions can be originated in any of the ways noted previously and forwarded through BASE24 into another BASE24 system as if through an interchange. Communications between the BASE24 networks are controlled by BIC ISO Interface processes on each BASE24 system.

CRT Entry

By purchasing the CRT Authorization or CRT Administration add-on products for the BASE24-pos product, institutions can set up CRT entry for certain types of transactions such as sales drafts, mail or telephone purchases, or chargebacks.

Transaction Authorizers

The BASE24-pos system can authorize transactions or route them to several different types of authorizers. The possible transaction authorizers are described below.

BASE24-pos System

The BASE24-pos system can be set up to authorize transactions in situations where cardholder files are maintained on the BASE24-pos system.

Hosts

Hosts can be set up to authorize transactions in situations where cardholder files are maintained on the host computer. In this case, the BASE24-pos system can screen the transactions before sending them to the host and can also be set up to stand in and authorize transactions for a host in situations where communications between the BASE24-pos system and the host system are down.

Interchanges

Interchanges can be used to forward transactions to other networks for authorization.

Co-Networks

The BASE24-pos system can be linked to co-networks through a BIC ISO Interface process, and transactions can be forwarded through the BASE24 system to another BASE24 system or co-network for authorization. The BASE24-pos system can also be set up to stand in and authorize transactions for another co-network in situations where communications between the BASE24-pos system and the co-network are down.

Application Processes

Certain BASE24-pos Device Handler modules and Interchange Interface processes can route transactions directly to another application running under the XPNET process for authorization. The authorization response returned from the application process can then be optionally logged to the POS Transaction Log File (PTLF).

The following BASE24-pos Device Handler modules and Interchange Interface processes support this functionality:

- Standard POS Device Handler (SPDH) module
- Hypercom Device Handler (HPDH) module
- Banknet Interchange Interface process
- BIC ISO Interchange Interface process
- MDS/MDSM Interchange Interface process

- PLUS ISO Interchange Interface process
- VisaNet Interchange Interface process

Issuers and Acquirers

Transaction originators and authorizers are generally defined by the EFT industry terms acquirer and issuer. An acquirer, or transaction acquirer, is the party that originates a transaction. An issuer, or card issuer, is an institution that issues cards or owns accounts that can be used in EFT transactions.

Whether an entity is defined as an issuer or an acquirer depends on the perspective from which the transaction is viewed. Intermediary networks and processors that receive transactions from a transaction originator and authorize them or forward them elsewhere for authorization can be acting on behalf of both the acquirer and the issuer.

From the acquirer's point of view, intermediary networks are acting on behalf of the issuer. For example, a cardholder initiates a transaction at a POS device belonging to institution A. In this case, institution A is the transaction acquirer. Institution A sends the transaction to an intermediary network that, in turn, sends the transaction to institution B for authorization. Institution B owns the cardholder's account and issued the cardholder the card; therefore, institution B is the true issuer. However, from institution A's point of view, the intermediary network is performing as an issuer on institution B's behalf since institution A had to send the transaction to the intermediary network before it could reach its authorizing destination.

From the issuer's point of view, intermediary networks are acting on behalf of the acquirer. As in the above example, institution B views the intermediary network as an acquirer since the intermediary network is the entity that sent it a transaction to be authorized.

The BASE24 system can process transactions on behalf of an issuer or an acquirer, depending on where a transaction originates and who is to authorize the transaction. For example, when the BASE24 system sends a transaction to a back-end host for authorization, the BASE24 system represents the acquirer in the message exchange, and the back-end host represents the issuer. On the other hand, when a host sends a transaction to the BASE24 system for authorization, the sending host represents the acquirer, and the BASE24 system represents the issuer.

Hosts and DPCs

Hosts can play a prominent role in the transaction processing performed by the BASE24 system. A host is an external mainframe computer system with which the BASE24-pos product interacts—either online or by batched tape—to authorize transactions and update cardholder records.

Although the term host generally refers to an actual computer or system of computers, the term also refers to the institution or organization responsible for the host computer system and its processing. For example, in the phrase, “the host can opt to receive advice messages,” the term host actually refers to the BASE24-defined institution in control of the host computer, rather than the computer itself.

The BASE24 system also refers to hosts as data processing centers (DPCs). Generally, the terms DPC and host are synonymous. However, the term host implies either a host computer or the organization responsible for a host computer; the term DPC always characterizes the host computer or computer network itself.

A DPC can be one mainframe computer communicating with the BASE24 system, or a DPC can represent several mainframe computers networked together at a host site. Regardless of the machine configuration in relation to the BASE24 system, a DPC represents a single host from the BASE24 system point-of-view. A host site can establish multiple DPCs; however, the BASE24 system considers each of these DPCs to be a separate host.

DPCs are assigned identification numbers and are defined to the BASE24 system using the Host Configuration File (HCF). Each DPC must have a record in the HCF to be recognized by the BASE24 system. The HCF contains options at the DPC-level that allow hosts to specify individual processing parameters for each DPC.

Cards and Accounts

Transactions in the BASE24-pos system can be defined as specific actions—usually financial—carried out on accounts of some type using a card or card number as a basis for initiating the action.

BASE24 products make a distinction between cards and accounts. Transaction limits and accumulators can be maintained at the card or account level. Balances, if maintained, are maintained at the account level. The following discusses cards and accounts as they are defined and used in BASE24 products.

Cards

Plastic cards serve as evidence of an account and as a mechanism for accessing the account using electronic funds transfer (EFT) devices such as automated teller machines (ATMs) and point-of-sale (POS) terminals. Cards are given card numbers, which may or may not match the account numbers of the accounts that they are used to access. There can be a one-to-one relationship between cards and accounts, one card can access multiple accounts, or one account can be accessed by multiple cards, depending on card issuance procedures and BASE24 parameters.

Credit Cards and Debit Cards

In the BASE24-pos system, credit cards can access credit accounts (refer to the account definitions below), but cannot access noncredit accounts.

Debit cards can access noncredit accounts (refer to the account definitions below). When defined by the card-issuer, debit cards can access credit accounts in addition to noncredit accounts. A card that accesses credit and noncredit accounts can also be known as a dual card or combination card.

Card Type Codes

In its transactions, the BASE24-pos product defines credit and debit cards with the following type identifiers. The BASE24-pos product uses these identifiers to identify transaction impacts to its various cardholder and account files.

Type	Description
0	None (used for mobile top-up with cash transactions)
1	Credit cards
2	Debit cards, dual and combination cards

To identify and maintain service totals, BASE24 products use an additional, more specific set of codes to identify the specific type of card being used to initiate transactions. These card type codes are either reserved by BASE24 products or are user-defined.

Reserved BASE24 Service Types

BASE24 products reserve the following service types across all product lines. These codes are to be used only as defined::

- AD = Administrative (BASE24-atm product only)
- AX = American Express credit
- BD = Business deposit (BASE24-atm and BASE24-teller products only)
- C* = Private label credit (includes C, C0–C9, CA, and CC–CZ)
- CB = Carte Blanche credit
- D~~b~~ = Demonstration (BASE24-atm product only)
- DC = Diners Club credit
- DS = Discover (Sears) credit
- JB = Japan Credit Bureau (JCB) credit
- M~~b~~ = MasterCard credit
- MD = MasterCard debit (See BASE24-pos product note below)
- MM = MasterCard dual (See BASE24-pos product note below)
- P* = Proprietary debit (includes P, P0–P9, and PA–PZ)
- S1 = Secure Internet Validation Virtual PAN (BASE24-pos Secure Internet Validation add-on product only)
- SC = Special, Check (BASE24-pos product only)
- SD = Bulk data maintenance (BDM) seed (BASE24-pos Stored Value add-on product and BASE24-card product only)
- SN = Stored value no reload (BASE24-pos Stored Value add-on product only)

- SP = Special purpose (BASE24-atm self-service banking (SSB) Enhanced Check Application only)
- SR = Stored value reload (BASE24-pos Stored Value add-on product only)
- ST = Super teller (BASE24-atm self-service banking (SSB) Base Application only)
- UP = China UnionPay (CUP) credit
- V~~b~~ = Visa credit
- VD = Visa debit (See BASE24-pos product note below)
- VV = Visa dual (See BASE24-pos product note below)

Codes with a first character of C, except code CB, can be user-defined but must be used to identify private label credit cards.

Codes with a first character of P can be user-defined but must be used to identify proprietary debit cards. BASE24 products treat cards with proprietary debit codes and codes MD and VD as debit cards and treats cards with all other codes as credit cards.

Administrative (AD), business deposit (BD), demonstration (D~~b~~), special purpose (SP), and super teller (ST) are special-use card types used by the BASE24-atm product.

Business deposit (BD) is also a special-use card type used by the BASE24-teller product to identify cards that can be used to initiate deposit transactions only. The BASE24-teller product does not perform any other processing based on card type; however, BASE24 guidelines should still be used when establishing card types for the BASE24-teller product.

MasterCard dual (MM) and Visa dual (VV) can be processed as debit or credit card types, based on the default combo card type specified in the Card Prefix File (CPF).

Special, Check (SC) is a special-use card type used to initiate BASE24-pos check guarantee and check verification transactions only. Secure Internet Validation Virtual PAN (S1) is a special-use card type used to identify a virtual PAN in a SIV transaction. Stored Value No Reload (SN) and Stored Value Reload (SR) are special-use card types used by the BASE24-pos Stored Value add-on product only. BDM Seed (SD) is a special-use card type used in Bulk Data Maintenance (BDM) in the BASE24-pos Stored Value add-on product and BASE24-card product only.

BASE24-pos Note: The BASE24-pos product does not allow MasterCard debit (MD), MasterCard dual (MM), Visa debit (VD), or Visa dual (VV) card types in the PRDF and BASE24-pos Terminal Data files. The BASE24-pos product

automatically includes the MD and MM card types with the MasterCard credit (M~~b~~) card type, and automatically includes the VD and VV card types with the Visa credit (V~~b~~) card type.

User-Defined Card Types

Users can define and use any one- or two-character code, not reserved by BASE24 products, according to the following guidelines:

- The first character must be alphabetic: A, B, D through O, and Q through Z.
- The second character can be A through Z, 0 through 9, or a blank.

A valid COBNAMES table entry is recommended for each user-defined code.

Note: Codes starting with C and P can also be user-defined, but these are subject to restrictions imposed by BASE24 products (refer to the reserved BASE24 card types list above). The standard COBNAMES table already contains entries for these codes; however, the COBNAMES entries can be modified as needed by the user.

Accounts

BASE24 products classify all accounts as one of two types: credit and noncredit. This classification determines the account-level limits and accumulators that apply to a particular transaction.

Note: BASE24 products also support the acquiring, routing, and logging of transactions for a third type of account—electronic benefit transfer. These accounts, however, are not maintained on the BASE24 database. Electronic benefit transfer transactions are supported in a pass-through mode only.

Credit and Noncredit Accounts

Credit accounts involve funds advanced to an accountholder, by a financial institution or retailer, based on a credit agreement with the accountholder.

Noncredit accounts involve accountholder funds on deposit with a financial institution (e.g., DDA/checking or savings).

The BASE24 system allows processors to set up various transaction limits and accumulators. Throughout the BASE24 system, the account classification determines which limits and accumulators apply to a particular transaction.

Account Type Codes

The BASE24-pos product identifies accounts by account type as described in the table below. Institutions can set up other account types in the account type ranges, but the BASE24-pos product treats these account types as if they are the first number in the range. For example, an institution may want to distinguish between different types of checking accounts for their own record keeping and reporting purposes. They can assign type 02 to checking account type A, 03 to checking account type B, and 04 to checking account type C. However, during processing, the BASE24-pos product treats types 01, 02, 03, and 04 the same.

Type	Account Description	Credit	Noncredit
01	Checking (DDA)		✓
02–09	Checking (DDA) as set up by the institution. These types are processed the same as type 01.		✓
11	Savings		✓
12–19	Savings as set up by the institution. These types are processed the same as type 11.		✓
31	Credit	✓	
32–39	Credit as set up by the institution. These types are processed the same as type 31.	✓	
96 ¹	Cash benefit		
98 ¹	Food stamp benefit		

¹ Cash benefit and food stamp benefit accounts for electronic benefit transfer transactions are not maintained in the BASE24-pos database. These transactions are supported in a pass-through mode only.

Account Type Names

Screens for certain BASE24 files use account type names defined in the Account Type Table File (ATT) instead of the account type codes listed above. Account type names are used on Acquirer Processing Code File (APCF) and Issuer Processing Code File (IPCF) screens, which the BASE24-pos product uses for defining transaction profiles. Transaction profiles define a set of transactions allowed and the accounts on which the transactions can be performed.

ACI provides a default ATT that contains account type names for a number of account type codes. The ATT uses account type codes based on the International Organization for Standardization (ISO), rather than the account type codes used internally by the BASE24-pos product that are listed above. The following table shows you how ISO and BASE24 account type codes relate to account type names provided in the default ATT. The first two columns of the table list the ISO and BASE24-pos account type codes, respectively. The third column lists the corresponding account type name in the ATT, and the last column describes the account.

Account Type		Account Type Name	Description
ISO	BASE24		
00	Not applicable	NONE	
10	11	SAV	Savings accounts
1A	14	SAV1	
1B	15	SAV2	
1C	16	SAV3	
1D	17	SAV4	
1E	18	SAV5	
1F	19	SAV6	

Account Type		Account Type Name	Description
ISO	BASE24		
20	01	DDA	Demand deposit accounts
2A	02	DDA1	
2B	03	DDA2	
2C	04	DDA3	
2D	05	DDA4	
2E	06	DDA5	
2F	07	DDA6	
2G	08	DDA7	
2H	09	DDA8	
30	31	CR	Credit accounts
3A	33	CR1	
3B	34	CR2	
3C	35	CR3	
3D	36	CR4	
3E	37	CR5	
3F	38	CR6	
3G	39	CR7	
96	96	CSHBNF	Cash Benefit
98	98	FDSTMP	Food Stamp Benefit

BASE24-pos Transactions

The BASE24-pos product supports the transactions described on the following pages. Unless noted to the contrary, the BASE24-pos product supports these transactions in all respects. However, not all POS devices and interchanges support all of the listed transactions. Refer to the appropriate POS device support manual for a complete list of transactions supported by a specific POS device. Refer to the appropriate Interchange Interface documentation for a complete list of transactions supported by a specific Interchange Interface process.

Transaction Identifiers

Transactions are identified in the BASE24-pos product by a series of four values, which are carried in the TRAN.TRAN-CDE field of the BASE24-pos Standard Internal Message (PSTM). These four values, labeled together as TCTAAC, are defined as follows:

PSTM TRAN.TRAN-CDE FIELD (TCTAAC)		
Value	Position	Description
TC	1–2	Transaction Code — A transaction code identifying the type of transaction.
T	3	Card Type — A code identifying the type of card used in the transaction. Card types include credit cards, debit and dual cards, and none (cash). These types are described on the previous pages.
AA	4–5	Account Type — A code identifying the type of account associated with the transaction. Account types include checking accounts, savings accounts, credit card accounts, cash benefit accounts, and food stamp benefit accounts. These types are described on the previous pages.
C	6	Category Code — A category code used as further identification of the type of transaction.

For more information about the four values in the TRAN.TRAN-CDE field of the PSTM, refer to appendix A.

Financial Transactions

The BASE24-pos product supports the following financial transactions. The entry information in the table for each transaction identifies how the transaction can enter the BASE24-pos system. The possibilities include all of the transaction originators defined earlier in this section.

Note: Not all POS devices and interchanges support all transactions listed in the table below. The BASE24 system and the BASE24-pos product are capable of handling these transactions when they are received from a POS device or an interchange.

TC	T	AA	C	Transaction
10	-	—	0	Normal Purchase — A purchase of goods and services at a point-of-sale using a credit or debit card. Entry: directly-connected POS terminals, hosts, interchanges, co-networks, CRT Authorization terminal, batch process
10	1	31	0	Normal purchase from a credit account using a credit card.
10	2	01	0	Normal purchase from a checking account using a debit card.
10	2	11	0	Normal purchase from a savings account using a debit card.
10	2	31	0	Normal purchase from a credit account using a debit card.
10	2	96	0	Normal purchase from a cash benefit account using an electronic benefits transfer card.
10	2	98	0	Normal purchase from a food stamp benefit account using an electronic benefits transfer card.
10	-	—	1	Normal Sales Draft Purchase — A normal purchase transaction entered by the retailer without the customer being present. Entry: directly-connected POS terminals, hosts, interchanges, co-networks, CRT Administration terminal

TC	T	AA	C	Transaction
10	1	31	1	Normal sales draft purchase from a credit account using a credit card.
10	2	01	1	Normal sales draft purchase from a checking account using a debit card.
10	2	11	1	Normal sales draft purchase from a savings account using a debit card.
10	2	31	1	Normal sales draft purchase from a credit account using a debit card.
10	2	96	1	Normal sales draft purchase from a cash benefit account using an electronic benefits transfer card.
10	2	98	1	Normal sales draft purchase from a food stamp benefit account using an electronic benefits transfer card.
10	-	—	2	Representation of a Normal Purchase — An administrative transaction that reverses a chargeback when the acquiring retailer determines that the cardholder incorrectly disputed an earlier transaction. For example, the retailer determines a cardholder did actually make the purchase. A representation re-presents the charge to the cardholder's account and removes responsibility from the retailer. Entry: directly-connected POS terminals, hosts, co-networks, CRT Administration terminal
10	1	31	2	Representation of a normal purchase from a credit account using a credit card.
10	2	01	2	Representation of a normal purchase from a checking account using a debit card.
10	2	11	2	Representation of a normal purchase from a savings account using a debit card.
10	2	31	2	Representation of a normal purchase from a credit account using a debit card.
10	2	96	2	Representation of a normal purchase from a cash benefit account using an electronic benefits transfer card.

TC	T	AA	C	Transaction
10	2	98	2	Representation of a normal purchase from a food stamp benefit account using an electronic benefits transfer card.
10	-	—	3	Chargeback of a Normal Purchase — An administrative transaction that charges an earlier transaction back to the acquiring retailer when that earlier transaction is disputed by the cardholder. For example, the cardholder receives a statement with a purchase the cardholder did not make. A chargeback presents the charge to the retailer's account and removes responsibility from the cardholder. Entry: directly-connected POS terminals, hosts, co-networks, CRT Administration terminal
10	1	31	3	Chargeback of a normal purchase from a credit account using a credit card.
10	2	01	3	Chargeback of a normal purchase from a checking account using a debit card.
10	2	11	3	Chargeback of a normal purchase from a savings account using a debit card.
10	2	31	3	Chargeback of a normal purchase from a credit account using a debit card.
10	2	96	3	Chargeback of a normal purchase from a cash benefit account using an electronic benefits transfer card.
10	2	98	3	Chargeback of a normal purchase from a food stamp benefit account using an electronic benefits transfer card.
11	-	—	0	Preauthorization of Purchase — An online check of a customer's account to verify that funds are available for an intended purchase. The preauthorized amount is placed on hold in the customer's account until the purchase is completed, the hold time limit is exceeded, or the hold is removed manually. If the purchase is completed, this transaction must be followed by a completion transaction. Entry: directly-connected POS terminals, hosts, interchanges, co-networks, batch process

TC	T	AA	C	Transaction
11	1	31	0	Preauthorization of purchase from a credit account using a credit card.
11	2	01	0	Preauthorization of purchase from a checking account using a debit card.
11	2	11	0	Preauthorization of purchase from a savings account using a debit card.
11	2	31	0	Preauthorization of purchase from a credit account using a debit card.
11	2	96	0	Preauthorization of purchase from a cash benefit account using an electronic benefits transfer card.
12	-	—	0	Completion of a Preauthorized Purchase — A completion of a purchase that was preauthorized. This transaction follows and completes a preauthorization of purchase transaction. The actual amount of the purchase is posted from this transaction and the hold for the amount specified in the preauthorization of purchase transaction is released. Entry: directly-connected POS terminals, hosts, interchanges, co-networks, batch process
12	1	31	0	Completion of a preauthorized purchase from a credit account using a credit card.
12	2	01	0	Completion of a preauthorized purchase from a checking account using a debit card.
12	2	11	0	Completion of a preauthorized purchase from a savings account using a debit card.
12	2	31	0	Completion of a preauthorized purchase from a credit account using a debit card.
12	2	96	0	Completion of a preauthorized purchase from a cash benefit account using an electronic benefits transfer card.

TC	T	AA	C	Transaction
12	-	—	1	Completion of a Preauthorized Sales Draft Purchase — A completion of a preauthorized sales draft purchase. Entry: directly-connected POS terminals, hosts, interchanges, co-networks
12	1	31	1	Completion of a preauthorized sales draft purchase from a credit account using a credit card.
12	2	01	1	Completion of a preauthorized sales draft purchase from a checking account using a debit card.
12	2	11	1	Completion of a preauthorized sales draft purchase from a savings account using a debit card.
12	2	31	1	Completion of a preauthorized sales draft purchase from a credit account using a debit card.
13	-	—	0	Mail or Telephone Purchase — A purchase of goods or services where the customer is not present at the point of sale (e.g., through the mail or by telephone). Entry: directly-connected POS terminals, hosts, interchanges, co-networks, CRT Authorization terminal, batch process
13	1	31	0	Mail or telephone purchase from a credit account using a credit card.
13	2	01	0	Mail or telephone purchase from a checking account using a debit card.
13	2	11	0	Mail or telephone purchase from a savings account using a debit card.
13	2	31	0	Mail or telephone purchase from a credit account using a debit card.

TC	T	AA	C	Transaction
13	-	—	1	Mail or Telephone Sales Draft Purchase — A draft purchase of goods or services where the customer is not present at the point of sale (e.g., through the mail or by telephone). Entry: directly-connected POS terminals, hosts, interchanges, co-networks, CRT Administration terminal, batch process
13	1	31	1	Mail or telephone sales draft purchase from a credit account using a credit card.
13	2	01	1	Mail or telephone sales draft purchase from a checking account using a debit card.
13	2	11	1	Mail or telephone sales draft purchase from a savings account using a debit card.
13	2	31	1	Mail or telephone sales draft purchase from a credit account using a debit card.
14	-	—	0	Merchandise Return — A return of merchandise to a retailer for a refund. The refund is electronically credited to the customer's account. Entry: directly-connected POS terminals, hosts, interchanges, co-networks, CRT Authorization terminal, batch process
14	1	31	0	Merchandise return to a credit account using a credit card.
14	2	01	0	Merchandise return to a checking account using a debit card.
14	2	11	0	Merchandise return to a savings account using a debit card.
14	2	31	0	Merchandise return to a credit account using a debit card.
14	2	96	0	Merchandise return to a cash benefit account using an electronic benefits transfer card.
14	2	98	0	Merchandise return to a food stamp benefit account using an electronic benefits transfer card.

TC	T	AA	C	Transaction
14	-	—	1	Sales Draft Merchandise Return — A merchandise return transaction entered by the retailer without the customer being present. Entry: directly-connected POS terminals, hosts, interchanges, co-networks, CRT Administration terminal
14	1	31	1	Sales draft merchandise return to a credit account using a credit card.
14	2	01	1	Sales draft merchandise return to a checking account using a debit card.
14	2	11	1	Sales draft merchandise return to a savings account using a debit card.
14	2	31	1	Sales draft merchandise return to a credit account using a debit card.
14	2	96	1	Sales draft merchandise return to a cash benefit account using an electronic benefits transfer card.
14	2	98	1	Sales draft merchandise return to a food stamp benefit account using an electronic benefits transfer card.
15	-	—	0	Cash Advance — A disbursement of cash from a customer's account at the point-of-sale. Entry: directly-connected POS terminals, hosts, interchanges, co-networks, CRT Authorization terminal, batch process
15	1	31	0	Cash advance from a credit account using a credit card.
15	2	01	0	Cash advance from a checking account using a debit card.
15	2	11	0	Cash advance from a savings account using a debit card.
15	2	31	0	Cash advance from a credit account using a debit card.
15	2	96	0	Cash advance from a cash benefit account using an electronic benefits transfer card.

TC	T	AA	C	Transaction
15	-	—	1	Sales Draft Cash Advance — A cash advance transaction entered by the retailer without the customer being present. Entry: directly-connected POS terminals, hosts, interchanges, co-networks
15	1	31	1	Sales draft cash advance from a credit account using a credit card.
15	2	01	1	Sales draft cash advance from a checking account using a debit card.
15	2	11	1	Sales draft cash advance from a savings account using a debit card.
15	2	31	1	Sales draft cash advance from a credit account using a debit card.
15	-	96	-	Cash Benefit — A disbursement of cash from a customer's cash benefit account at the point-of-sale. Entry: directly-connected POS terminals, hosts, interchanges, co-networks
16	-	—	0	Card Verification — An online check to verify that a customer's card is valid. Entry: directly-connected POS terminals, hosts, interchanges, co-networks, CRT Authorization terminal
16	1	00	0	Verification of a credit card.
16	2	00	0	Verification of a debit card.
17	-	—	0	Balance Inquiry — An online inquiry into the balance of a customer account. Entry: directly-connected POS terminals, hosts, interchanges, co-networks
17	1	31	0	Credit account balance inquiry using a credit card.
17	2	01	0	Checking account balance inquiry using a debit card.
17	2	11	0	Savings account balance inquiry using a debit card.

TC	T	AA	C	Transaction
17	2	31	0	Credit account balance inquiry using a debit card.
17	2	96	0	Cash benefit account balance inquiry using an electronic benefits transfer card.
17	2	98	0	Food stamp benefit account balance inquiry using an electronic benefits transfer card.
18	-	—	0	Purchase with Cash Back — A combination of a purchase and a cash withdrawal at the point-of-sale. The requested transaction amount is greater than the price of the actual purchase, and the customer receives the difference in cash. This transaction is only valid with debit cards. Entry: directly-connected POS terminals, hosts, interchanges, co-networks, batch process
18	2	01	0	Purchase with cash back from a checking account using a debit card.
18	2	11	0	Purchase with cash back from a savings account using a debit card.
18	2	96	0	Purchase with cash back from a cash benefit account using an electronic benefits transfer card.
18	-	—	1	Sales Draft Purchase with Cash Back — A purchase with cash back transaction entered by the retailer without the customer being present. This transaction is only valid with debit cards. Entry: directly-connected POS terminals, hosts, interchanges, co-networks
18	2	01	1	Sales draft purchase with cash back from a checking account using a debit card.
18	2	11	1	Sales draft purchase with cash back from a savings account using a debit card.

TC	T	AA	C	Transaction
19	0	00	0	Check Verification — An online verification that a customer's check is tied to a valid checking account. The BASE24-pos standard product supports this transaction in a pass-through mode only. It does not perform any type of verification. Instead, it routes this transaction to an appropriate authorizer and logs the response. Note: When the BASE24-pos eCheck add-on product is installed, this transaction can also be authorized using the Positive Balance authorization method with any authorization level. Entry: directly-connected POS terminals, hosts, co-networks
19	0	00	4	An online verification that a customer's personal check is valid for cash.
19	0	00	5	An online verification that a customer's personal check is valid for a purchase.
19	0	00	6	An online verification that a customer's personal check is valid for a purchase with cash back.
19	0	00	7	An online verification that a customer's government check is valid.
19	0	00	8	An online verification that a customer's payroll check is valid.
19	0	00	9	An online verification that a customer's electronic check is valid.
20	0	00	0	Check Guarantee — An electronic reservation of funds in a customer's checking account to cover a check written by the customer. The BASE24-pos product supports this transaction in a pass-through mode only. It does not perform any type of guarantee. Nor does it place any funds on reserve. Instead, it routes this transaction to an appropriate authorizer and logs the response. <i>Continued on next page</i>

TC	T	AA	C	Transaction
				<i>Continued</i> Note: When the BASE24-pos eCheck add-on product is installed, this transaction can also be authorized using the Positive Balance authorization method with any authorization level. Positive Balance File (PBF) balances can also be impacted. When the transaction is authorized offline on the BASE24-pos system, PBF balances can be impacted immediately upon authorization. When the transaction is authorized online by the host or an interchange, PBF balances on the BASE24 system can be impacted when the response is returned from the host or interchange, or the authorizer can choose not to impact the PBF. Entry: directly-connected POS terminals, hosts, co-networks
20	0	00	4	An electronic reservation of funds in a customer's checking account to cover a personal check for cash.
20	0	00	5	An electronic reservation of funds in a customer's checking account to cover a personal check for a purchase.
20	0	00	6	An electronic reservation of funds in a customer's checking account to cover a personal check for a purchase with cash back.
20	0	00	7	An electronic reservation of funds in a customer's checking account to cover a government check.
20	0	00	8	An electronic reservation of funds in a customer's checking account to cover a payroll check.
20	0	00	9	An electronic reservation of funds in a customer's checking account to cover an electronic check.
21	-	—	0	Purchase Adjustment — A correction of an error in a previously completed purchase transaction. Entry: directly-connected POS terminals, hosts, interchanges, co-networks, CRT Authorization terminal, batch process

TC	T	AA	C	Transaction
21	1	31	0	Purchase adjustment to a credit account using a credit card.
21	2	01	0	Purchase adjustment to a checking account using a debit card.
21	2	11	0	Purchase adjustment to a savings account using a debit card.
21	2	31	0	Purchase adjustment to a credit account using a debit card.
21	-	—	1	Sales Draft Purchase Adjustment — A correction of an error in a previously completed sales draft purchase transaction. Entry: directly-connected POS terminals, hosts, interchanges, co-networks
21	1	31	1	Sales draft purchase adjustment to a credit account using a credit card.
21	2	01	1	Sales draft purchase adjustment to a checking account using a debit card.
21	2	11	1	Sales draft purchase adjustment to a savings account using a debit card.
21	2	31	1	Sales draft purchase adjustment to a credit account using a debit card.
22	-	—	0	Merchandise Return Adjustment — A correction of an error in a previously completed merchandise return transaction. Entry: directly-connected POS terminals, hosts, interchanges, co-networks, CRT Authorization terminal, batch process
22	1	31	0	Merchandise return adjustment from a credit account using a credit card.
22	2	01	0	Merchandise return adjustment from a checking account using a debit card.

TC	T	AA	C	Transaction
22	2	11	0	Merchandise return adjustment from a savings account using a debit card.
22	2	31	0	Merchandise return adjustment from a credit account using a debit card.
22	-	—	1	Sales Draft Merchandise Return Adjustment — A correction of an error in a previously completed sales draft merchandise return transaction. Entry: directly-connected POS terminals, hosts, interchanges, co-networks
22	1	31	1	Sales draft merchandise return adjustment from a credit account using a credit card.
22	2	01	1	Sales draft merchandise return adjustment from a checking account using a debit card.
22	2	11	1	Sales draft merchandise return adjustment from a savings account using a debit card.
22	2	31	1	Sales draft merchandise return adjustment from a credit account using a debit card.
23	-	—	0	Cash Advance Adjustment — A correction of an error in a previously completed cash advance transaction. Entry: directly-connected POS terminals, hosts, interchanges, co-networks, CRT Authorization terminal, batch process
23	1	31	0	Cash advance adjustment to a credit account using a credit card.
23	2	01	0	Cash advance adjustment to a checking account using a debit card.
23	2	11	0	Cash advance adjustment to a savings account using a debit card.
23	2	31	0	Cash advance adjustment to a credit account using a debit card.

TC	T	AA	C	Transaction
23	-	—	1	Sales Draft Cash Advance Adjustment — A correction of an error in a previously completed cash advance transaction. Entry: directly-connected POS terminals, hosts, interchanges, co-networks
23	1	31	1	Sales draft cash advance adjustment to a credit account using a credit card.
23	2	01	1	Sales draft cash advance adjustment to a checking account using a debit card.
23	2	11	1	Sales draft cash advance adjustment to a savings account using a debit card.
23	2	31	1	Sales draft cash advance adjustment to a credit account using a debit card.
24	-	—	0	Cash Back Amount Adjustment — A correction of an error in a previously completed purchase with cash back transaction. Entry: directly-connected POS terminals, hosts, co-networks, batch process
24	2	01	0	Cash back amount adjustment to a checking account using a debit card.
24	2	11	0	Cash back amount adjustment to a savings account using a debit card.
24	-	—	1	Sales Draft Cash Back Amount Adjustment — A correction of an error in a previously completed sales draft purchase with cash back transaction. Entry: directly-connected POS terminals, hosts, co-networks
24	2	01	1	Sales draft cash back amount adjustment to a checking account using a debit card.
24	2	11	1	Sales draft cash back amount adjustment to a savings account using a debit card.

TC	T	AA	C	Transaction
25	-	—	1	Card Activation — A card activation transaction is initiated when a customer swipes the card at the POS device. Entry: directly-connected POS terminals
25	2	01	0	Card activation transaction from a checking account for a stored value card.
26	-	—	0	Additional Card Activation — An additional card activation transaction allows the customer to obtain additional cards that access the same account as their original card. Entry: directly-connected POS terminals
26	2	01	0	Additional card activation transaction from a checking account for a stored value card.
27	-	—	0	Replenishment — A replenishment transaction is initiated when the customer requests additional funds to be loaded to an existing stored value card. Entry: directly-connected POS terminals
27	2	01	0	Replenishment transaction from a checking account for a stored value card.
28	-	—	0	Full Redemption — A full redemption transaction subtracts the remaining funds from an account. Entry: directly-connected POS terminals
28	2	01	0	Full redemption transaction from a checking account for a stored value card.
29	-	—	0	Mobile Top-Up With Cash — A mobile top-up with cash transaction requests a secondary provider (e.g., telephone company) to load additional funds to a mobile telephone account. This transaction uses cash received at the terminal instead of funds from a debit or credit account. <i>Continued on next page</i>

TC	T	AA	C	Transaction
29	-	—	0	<i>Continued</i> The Authorization module does not handle this transaction code. Entry: directly-connected POS terminals
29	0	00	0	Mobile top-up with cash transaction performed at a POS terminal.
30	-	—	0	Mobile Top-Up With Funds — A mobile top-up with funds transaction requests a secondary provider (e.g., telephone company) to load additional funds to a mobile telephone account. This transaction uses funds from a debit or credit account. The Authorization module handles this transaction as a purchase (transaction code 10). Entry: directly-connected POS terminals
30	1	31	0	Mobile top-up with funds from a credit account using a credit card.
30	2	01	0	Mobile top-up with funds from a checking account using a debit card.
30	2	11	0	Mobile top-up with funds from a savings account using a debit card.
30	2	31	0	Mobile top-up with funds from a credit account using a debit card.
30	-	—	1	Mobile Top-Up With Funds by Sales Draft — A mobile top-up with funds transaction entered by the retailer without the customer being present. The Authorization module handles this transaction as a purchase (transaction code 10). Entry: directly-connected POS terminals
30	1	31	1	Mobile top-up with funds by sales draft from a credit account using a credit card.

TC	T	AA	C	Transaction
30	2	01	1	Mobile top-up with funds by sales draft from a checking account using a debit card.
30	2	11	1	Mobile top-up with funds by sales draft from a savings account using a debit card.
30	2	31	1	Mobile top-up with funds by sales draft from a credit account using a debit card.

Administrative Transactions

The BASE24-pos product supports the following administrative transactions.

TC	T	AA	C	Transaction
50	0	00	0	Batch Totals — Ends the batch to which data is currently being logged and accumulated. Data begins accumulating in the next batch at the time this transaction is authorized and approved.
51	0	00	0	Shift Totals — Ends the shift to which data is currently being logged and accumulated. Data begins accumulating in the next shift at the time this transaction is authorized and approved.
52	0	00	0	Day Totals — Ends the day to which data is currently being logged and accumulated. Data begins accumulating in the next day at the time this transaction is authorized and approved.

BASE24-pos Message Types

All transactions are carried out using messages exchanged between the originator and authorizer of the transactions. These messages are assigned numbers to identify their message type and function.

Message Functions

Messages can be grouped into the following functional categories.

Request — A request from a transaction originator to a transaction authorizer asking that a transaction be approved, denied, or referred.

Response — A response from a transaction authorizer to a transaction originator approving, denying, or referring a transaction.

Advice/Force Post — A message from a transaction originator to a transaction authorizer so that the transaction authorizer can update its files. Advices are also known as force post transactions because the authorizer must accept the transactions.

Reversal — A message from a transaction originator to a transaction authorizer indicating that any database impacts from an earlier transaction should be reversed because the transaction either could not be completed or was completed incorrectly. (The BASE24-pos product supports only whole reversals; it does not support partial reversals.)

Adjustment — A message from a transaction originator to a transaction authorizer requesting that the amount of a previously approved transaction be adjusted or corrected.

Message Types

Message types are used inside and outside of the BASE24-pos system. The following pages identify the message types used inside the BASE24-pos product along with the message types as they would be sent to, or received from, a host or BIC co-network. A table is provided for each of the following message groups used by the BASE24-pos product:

- Authorization Messages
- Financial Transaction Messages

- Balance Inquiry Messages
- Batch Upload Messages
- Reversal Messages
- Reconciliation Control Messages
- Network Management Messages

In each of the tables below, internal message types are defined in the table column identified with INT. External message types are provided for ISO communications with hosts and BIC co-networks. The ISO external message types are defined in the table column identified with ISO.

Note: Message types used between the BASE24-pos product and POS devices, and between the BASE24 system and interchanges, are proprietary to the devices and interchanges. Therefore, they are not included in the table.

Authorization Messages

The following table describes the message types for authorization messages.

INT	ISO	Description
Not used	0100	Authorization Request — A request for preauthorization. The BASE24-pos product does not apply a preauthorization to a cardholder's account for the purpose of issuing a bill or statement. The BASE24-pos product handles authorization requests as 0200 messages internally—identified with a value of 11 (preauthorization purchase) in the TRAN-CDE.TC field of the PSTM.
Not used	0110	Authorization Request Response — An approval or denial of a request for preauthorization. An authorization request response is returned in response to an authorization request to approve or deny the request. The BASE24-pos product handles authorization request responses as 0210 messages internally—identified with a value of 11 (preauthorization purchase) in the TRAN-CDE.TC field of the PSTM.

INT	ISO	Description
Not used	0120	Authorization Advice — A notification that a transaction was preauthorized on behalf of the card issuer. The BASE24-pos product does not apply the advice to the cardholder's account for the purpose of issuing a bill or statement. The BASE24-pos product handles authorization advices as 0220 messages internally—identified with a value of 11 (preauthorization purchase) in the TRAN-CDE.TC field of the PSTM.
Not used	0121	Authorization Advice Repeat — A reissue of an authorization advice. An authorization advice repeat is identical to the original authorization advice it duplicates, except that it denotes to the receiver that it is a possible duplicate message. An authorization advice repeat is used when some type of acknowledgment is expected to an authorization advice but not received.
Not used	0130	Authorization Advice Response — An acknowledgment of receipt sent in response to an authorization advice or an authorization advice repeat.

Financial Transaction Messages

The following table identifies the message types used for financial transaction messages.

INT	ISO	Description
0200	0200	Financial Transaction Request — A request to authorize a transaction, which if approved can be immediately applied to the cardholder's account for billing or statement purposes.
0210	0210	Financial Transaction Response — An approval or denial response to a financial transaction request.

INT	ISO	Description
0220	0220	Financial Transaction Advice — A notification that a transaction was previously completed on behalf of a card issuer. When the BASE24-pos product receives a financial transaction advice, it considers the advice to be a force post transaction. The BASE24-pos product generates a financial transaction advice to notify a host that the BASE24-pos product has approved a transaction or received a force post transaction.
Not used	0221	Financial Transaction Advice Repeat — A reissue of a financial transaction advice. A financial transaction advice repeat is identical to the original financial transaction advice it duplicates, except that it denotes to the receiver that it is a possible duplicate message. A financial transaction advice repeat is used when some type of acknowledgment is expected to a financial transaction advice but not received.
Not used	0230	Financial Transaction Advice Response — An acknowledgment of receipt sent in response to a financial transaction advice or a financial transaction advice repeat.

Balance Inquiry Messages

The following table identifies the message types used for balance inquiry messages.

INT	ISO	Description
0200	Not used	Balance Inquiry Request — A request for account balance information.
0210	Not used	Balance Inquiry Response — A response to a balance inquiry request. This message type contains account balance information.

Batch Upload Messages

The following table identifies the message types used for batch upload messages. These messages are used only between a Hypercom device and the BASE24-pos system.

INT	ISO	Description
0320	Not used	Batch Upload Message — A message containing information to be uploaded from the device and logged to the PTLF. This message is used only between the Hypercom device and the BASE24-pos system. It is not sent to a host or a co-network.
0330	Not used	Batch Upload Message Response — An acknowledgment sent in response to a batch upload message.

Reversal Messages

The following table identifies the message types used for reversal messages.

INT	ISO	Description
0402	0402	Card Issuer Reversal Request — A reversal request initiated by the card issuer or the acquirer due to a dispute. This is known as a chargeback. The BASE24-pos product can generate a response to a 0402, but cannot generate a 0402 request without the addition of the CRT Administration process.
0412	0412	Card Issuer Reversal Request Response — A response to a chargeback. The BASE24-pos product can generate this message type, but cannot generate the chargeback without the addition of the CRT Administration process.
0420	0420	Acquirer Reversal Advice — A whole reversal of a financial transaction that did not complete as authorized.
Not used	0421	Acquirer Reversal Advice Repeat — A reissue of an acquirer reversal advice. An acquirer reversal advice repeat is identical to the original acquirer reversal advice, except that it denotes to the receiver that it is a possible duplicate message. The BASE24-pos product uses an acquirer reversal advice repeat when some type of acknowledgment to an acquirer reversal advice is expected but never received.

INT	ISO	Description
Not used	0430	Reversal Response — An acknowledgment of receipt generated by a host in response to an acquirer reversal advice.
Not used	Not used	Reversal Response Repeat — A reissue of an acknowledgment of receipt generated by a host in response to an acquirer reversal advice. The BASE24-pos product uses a reversal response repeat when some type of acknowledgment to an acquirer reversal advice is expected but never received.

Reconciliation Control Messages

The following table identifies the message types used for reconciliation control messages. Only the BIC ISO Interface process is capable of maintaining acquirer totals and exchanging 05XX messages with a co-network. The ISO Host Interface, however, can forward 05XX messages that it receives from the Authorization module.

INT	ISO	Description
0500	0500	Acquirer Reconciliation Request — A request for a confirmation of acquirer totals in order to affect settlement between parties. An acquirer reconciliation request response is expected in return for an acquirer reconciliation request.
Not used	0510	Acquirer Reconciliation Request Response — A response to an acquirer reconciliation request.
Not used	0520	Acquirer Reconciliation Advice — An advice of totals sent in order to affect settlement between the parties.
Not used	0521	Acquirer Reconciliation Advice Repeat — A reissue of an acquirer reconciliation advice. An acquirer reconciliation advice repeat is identical to the original acquirer reconciliation advice it duplicates, except that it denotes to the receiver that it is a possible duplicate message. An acquirer reconciliation advice repeat is used when some type of acknowledgment is expected to an acquirer reconciliation advice but not received.

INT	ISO	Description
Not used	0530	Acquirer Reconciliation Advice Response — An acknowledgment of receipt generated by a host in response to an acquirer reconciliation advice.

Network Management Messages

The following table identifies the message types used for network management messages.

INT	ISO	Description
Not used	0800	Network Management Message Request — A request to log on, log off, verify a connection, or exchange keys with a communications entity.
Not used	0810	Network Management Message Response — A response to a network management message request.

Transaction Logging

The BASE24-pos Authorization module of the Device Handler/Router/Authorization process logs all of the transactions it processes to the POS Transaction Log File (PTLF). The Interchange Interface processes and BIC ISO Interface processes log all the transactions they process to the appropriate Interchange Log Files (ILFs) if configured to do so. The Transaction Context Manager process logs the secondary provider (e.g., telephone company) portion of mobile top-up transactions to the PTLF.

Message Types Logged

The following BASE24-pos financial message types are logged to the PTLF or the ILF:

Type	Description
0210	Authorization response
0220	Force-post advice—incoming only
0412	Chargeback response
0420	Authorization reversal

The following BASE24-pos administrative message types are logged to the PTLF or the ILF:

Type	Description
0500	Batch close request
0500	Shift close request
0500	Day close request

Note: Only the BIC ISO Interface process logs the above messages to the ILF.

The following BASE24-pos message types are not logged to the PTLF or ILF:

Type	Description
0200	Authorization request
0220	Completion advice—created locally (i.e., completion)
0402	Chargeback request
0800	Logon/logoff requests
----	Clerk totals requests
----	Mail requests

Tokens Logged

The Token File (TKN) controls which tokens in the internal message are logged to the PTLF or ILF. All tokens that can be logged to the PTLF or ILF are displayed on TKN screen 2. As each message is logged to the PTLF or ILF, tokens identified to be logged are arranged in the PTLF or ILF record in the order in which they were added to the internal message.

At terminal cutover, the Authorization module optionally logs terminal totals to the PTLF. Institutions can configure which terminals totals they want to have logged to the PTLF using the LOG TOTALS field on screen 3 of the BASE24-pos Terminal Data files (PTD).

Multiple Logical Networks

When a transaction involves multiple logical networks in the same BASE24 system, the BASE24-pos product logs transactions to two POS Transaction Log Files (PTLFs)—one for the transaction acquirer and one for the card issuer. The BASE24-pos product adds a special token to transactions that are SPROUTE routed using a CPF to a different logical network for authorization so that products such as the ACI Settlement Manager and ACI Payments Manager can consistently detect transactions having entries in two log files. The ACQ-LN field in the Multiple Logical Network token contains the acquirer logical network ID.

ACI Worldwide, Inc.

2: BASE24-pos Transaction Path

A number of processes and processing entities form the BASE24-pos transaction path—that is, the series of BASE24-pos processes and processing entities through which a specific transaction passes in order to be authorized.

The actual path each transaction takes varies based on a number of factors, including the type of transaction being processed, the transaction originator, the transaction authorizer, and the processing set up by the BASE24 processor.

This section discusses the following aspects of the BASE24-pos transaction path:

- BASE24-pos processes that can be part of the transaction path.
- How the various BASE24-pos processes and processing entities are linked together for the purpose of forwarding and receiving transaction messages.
- Service-based routing within the BASE24-pos system.
- The various timers used by BASE24-pos processes to ensure the timely completion of transactions.

BASE24-pos Processes

Along the transaction path, processing is carried out in stages by several BASE24-pos processes, each of which has its own responsibilities. The BASE24-pos processes that can be involved in transaction processing are listed below and described on the following pages.

- Device Handler/Router/Authorization processes. These processes are made up of three individual modules bound together: a Device Handler module, a Router module, and an Authorization module (the individual modules are described separately below because each has its own responsibilities)
- Transaction Context Manager process.
- ISO Host Interface processes.
- Interchange Interface processes.
- BIC ISO Interface processes.
- BASE24-pos processes all communicate with one another through the XPNET process.

Note: The Device Handler, Router, and Authorization modules are bound together to form one process; therefore, messages passed between these modules do not pass through the XPNET process.

BASE24-pos Device Handler Modules

Device Handler modules are responsible for communications between the point-of-sale (POS) devices, or terminals, and the BASE24-pos product. The basic functions of the BASE24-pos Device Handler modules are described below, followed by a list of modules supported by the BASE24-pos product.

BASE24-pos Device Handler Module Functions

BASE24-pos Device Handler modules typically perform the following functions as appropriate for the type of terminals they support:

- Recognize native-mode messages sent from the POS terminal and translate these messages into BASE24-pos internal message format.

Each type of POS terminal sends and receives messages in a device-specific format called native mode. The BASE24-pos product is not capable of using the various native-mode message formats internally. Therefore, the Device Handler module translates the messages between the terminal and the BASE24-pos product.

- Determine whether the requested transaction is allowed, and if so, whether it needs to be authorized by the BASE24-pos system or sent to a different authorization destination.

Note: The above alternate routing functionality is supported for Standard POS and Hypercom Device Handler modules only.

- Assign transaction sequence numbers to transactions if the originating POS terminals are not capable of assigning them.
- Pass messages to the Router module.
- Set and stop transaction response timers as transactions pass in and out of the Device Handler module.

These response timers control the maximum length of time the Device Handler module waits for a response from the Authorization module before cancelling the transaction. For example, when the Device Handler module receives a transaction from a terminal, it sets a response timer and then passes the transaction on for processing. When processing completes, the Device Handler module deletes the timer. If processing does not complete before the timer expires, the transaction is declined.

- Update fields in the BASE24-pos Terminal Data files records for each POS device.

When a transaction passes through the Device Handler/Router/Authorization process, the Device Handler module updates BASE24-pos Terminal Data files fields that track transaction totals for the terminal and provide operators with information about the terminal's current state.

- Cut over terminals for daily settlement, either using a cutover transaction or a Device Handler force cutover option.

If the Device Handler module does not cut over the terminal, a Secondary Settlement process cuts over the terminal. Refer to the ***BASE24-pos Settlement and Reporting Manual*** for more information about the Secondary Settlement process and terminal cutover.

- Facilitate PIN and message authentication support between the terminal and the BASE24-pos system. For information on the PIN and message authentication support provided by BASE24-pos Device Handler modules, refer to the ***BASE24 Transaction Security Manual***.

- Downline load terminal configuration information for some types of terminals.

The information is downline loaded from load files. For example, the Visa II Device Handler module downline loads information from the VVCNFG file to Visa II terminals.

Supported BASE24-pos Device Handler Modules

ACI supports several Device Handler modules to communicate between industry standard terminals and the BASE24-pos product. The terminals themselves can be classified as first-generation, second-generation, or third generation terminals. First-generation terminals do not retain totals independently of the BASE24-pos product and do not support electronic draft capture. Second-generation terminals allow retailers to compare totals retained by the terminal with totals from paper receipts. These terminals support draft capture, but require two transactions to complete the process. Third-generation terminals retain totals independently of the BASE24-pos product and support draft capture with a one-transaction process.

The Device Handler modules that ACI provides to support POS terminals are described below.

Visa I Device Handler module	Supports first-generation terminals.
Visa II Device Handler module	Supports first- and second-generation terminals.
BASE24 Standard POS Device Handler (SPDH) module	Supports third-generation terminals. For more information about the BASE24 Standard POS Device Handler (SPDH) module, refer to the <i>BASE24-pos Standard POS Device Configuration Manual</i> and the <i>ACI Standard POS Device Message Specifications Manual</i> .
Hypercom Device Handler (HPDH) module	Supports Hypercom T4, T5, T6, and T7 terminals. The HPDH module can communicate with any POS terminal having the capability of sending and receiving messages using the message format supported by the HPDH module.
Merchant Host Interface (MHI) Device Handler module	Supports terminals connected to a host, enabling the BASE24 system to accept transactions originating from terminals controlled by a host. The BASE24-pos product then authorizes these transactions and accumulates transaction totals either by terminal or by groups of terminals.
CRT Authorization Device Handler module	Supports CRT Authorization terminals. The CRT Authorization Device Handler module enables credit transactions and referrals to be entered at CRT terminals for authorization and also accepts transactions referred from the Authorization module. Referred transactions are those that cannot be authorized based on the cardholder information in the BASE24-pos database.

CRT Administration Device Handler module	Supports CRT Administration terminals. The CRT Administration Device Handler module enables CRT operators to enter sales drafts, chargebacks, and representations on formatted BASE24 screens.
--	--

Note: The CRT Authorization and CRT Administration Device Handler modules can be bound with the Router and Authorization modules, either together or separately. For more information on these Device Handler modules, refer to the ***BASE24-pos CRT Authorization and CRT Administration Manual***.

BASE24-pos Router Module

The Router module routes transactions through the Device Handler/Router/Authorization process and the BASE24-pos system. To do this, the Router module verifies the retailer and verifies that the transaction is allowed for the card type initiating the transaction. The Router module then classifies the transaction according to whether it is an on-us or not-on-us transaction and according to the card type. This classification determines how the transaction is routed. The Router module also selects the authorization destination according to the card prefix.

Any transaction sent to the Device Handler/Router/Authorization process is first received by the Router module with which the Device Handler and Authorization modules are bound. The Router module then passes the message to the Device Handler or Authorization module, or sends it to an Interchange Interface process, ISO Host Interface process, or Transaction Context Manager process depending on the requirements of the transaction. The Router module may also perform several checks on the transaction before passing it on.

BASE24-pos Authorization Module

The Authorization module is responsible for authorizing each transaction it receives or passing each transaction to the appropriate interface process. The interface process then sends the transaction to the entity that is to authorize the transaction. The Authorization module logs all transactions it receives to the transaction log file and can generate completion messages to notify hosts regarding transactions approved or declined by the BASE24-pos system.

Transaction Context Manager Process

The BASE24-pos Transaction Context Manager process is responsible for receiving a mobile top-up transaction from the Device Handler/Router/Authorization process and splitting the transaction into funds and service provider (e.g., telephone company) portions. The Transaction Context Manager process routes the required information to the funds authorizer for approval, then routes the required information to the service provider for approval and logs the service provider's response to the POS Transaction Log File (PTLF). Finally, the Transaction Context Manager process formats the responses into a single message that can be returned to the Device Handler/Router/Authorization process.

ISO Host Interface Processes

BASE24-pos ISO Host Interface processes are responsible for communicating with host systems. They become involved in processing when the host is the appropriate authorizer or is to receive completion messages for transactions authorized by the BASE24-pos product. They also become involved in processing when the BASE24 system is the authorizer, and the host is the transaction acquirer. For information on the ISO Host Interface process, refer to the ***BASE24 ISO Host Interface Manual*** and ***BASE24 External Message Manual***.

The ISO Host Interface process performs the following functions:

- Translates messages between the BASE24-pos internal message format and the ISO external format recognizable to a host.
- Provides a store-and-forward function, whereby if a host is unavailable, certain types of messages bound for the host can be stored and sent to the authorizing entity at a later time.
- Sets and stops transaction timers that monitor the time it takes for messages to return to the ISO Host Interface process from another entity.
- Provides for PIN, MAC, and key management between the BASE24-pos system and the host. For information on the transaction security functions performed by the ISO Host Interface process, refer to the ***BASE24 Transaction Security Manual***.

Note: An ISO Host Interface process can process host messages for multiple BASE24 products. If an institution purchases multiple BASE24 products, the appropriate product modules are bound into one ISO Host Interface process. This allows the ISO Host Interface process to recognize and process transaction messages for different products.

Interchange Interface Processes

BASE24-pos Interchange Interface processes are responsible for communicating with interchanges. They become involved in processing when the interchange is the appropriate authorizer or when the BASE24 system is the authorizer, and the interchange is the transaction acquirer.

Each interchange has its own Interchange Interface process because each interchange has its own message format and message handling requirements. Interchange Interface processes typically perform for the following:

- Translate messages between the BASE24-pos internal message format and the external message format recognizable to the interchange.
- Determine whether outbound transactions are allowed to be sent from the BASE24 system to the interchange.
- Determine whether inbound transactions are allowed from the interchange, and if so, whether they need to be authorized by the BASE24-pos system or sent to a different authorization destination.

Note: The above inbound transaction allowed checking and alternate routing functionality is supported for the following Interchange Interface processes only:

- ❑ Banknet Interchange Interface process
- ❑ BIC ISO Interchange Interface process
- ❑ MDS/MDSM Interchange Interface process
- ❑ PLUS ISO Interchange Interface process
- ❑ VisaNet Interchange Interface process
- Log transactions to the Interchange Log File (ILF).
- Provide a store-and-forward function, whereby, if the interchange is unavailable, certain types of messages bound for the interchange can be stored and sent to the interchange at a later time.
- Set and stop transaction timers that monitor the time it takes for messages to return to the Interchange Interface process from another entity.
- Provide for PIN, MAC, and key management between the BASE24-pos system and the interchange. For information on the transaction security functions performed by Interchange Interface processes, refer to the ***BASE24 Transaction Security Manual***.

Transactions entering the BASE24-pos system from an interchange must first pass through the Interchange Interface process. The Interchange Interface process translates the message from the interchange's external message to the POS Standard Internal Message (PSTM) before routing the message to the other BASE24 processes.

Transactions bound for an interchange that originated at a host or at a POS device connected to the BASE24-pos system must first pass through the appropriate Interchange Interface process. The Interchange Interface process translates the message into the interchange's external message format before sending it to the interchange.

BIC ISO Interface Processes

BASE24 Interchange (BIC) ISO Interface processes are responsible for communicating with other BASE24 systems and co-networks. They become involved in processing when a transaction must be forwarded to another BASE24 system or co-network for authorization or a transaction is being sent from another BASE24 system or co-network for authorization. For information on the BIC ISO Interface process, refer to the ***BASE24 BIC ISO Interface Manual*** and ***BASE24 BIC ISO Standards Manual***.

The BIC ISO Interface process performs the following:

- Translates messages between the BASE24-pos internal message format and the external message format recognizable to the BIC co-network.
- Determines whether outbound transactions are allowed to be sent from the BASE24 system to the interchange.
- Determines whether inbound transactions are allowed from the interchange, and if so, whether they need to be authorized by the BASE24-pos product or sent to a different authorization destination.

Note: The above inbound transaction allowed checking and alternate routing functionality is supported by the BIC ISO Interchange Interface process only if it uses the Enhanced Interchange Configuration File (ICFE).

- Logs transactions to the Interchange Log File (ILF).
- Provides a store-and-forward function, whereby, if the co-network is unavailable, certain types of messages bound for the co-network can be stored and sent to the co-network at a later time.

- Sets and stops transaction timers that monitor the time it takes for messages to return to the BIC ISO Interface process from another entity.
- Provides for PIN, MAC, and key management between the BASE24-pos system and a co-network. For information on the transaction security functions performed by BIC ISO Interface processes, refer to the ***BASE24 Transaction Security Manual***.

Transactions entering the BASE24-pos system from a BIC co-network must first pass through the BIC ISO Interface process. The BIC ISO Interface process translates the message from the co-network's external message to the BASE24-pos Standard Internal Message (PSTM) before routing the message to the other BASE24 processes.

Transactions bound for a BIC co-network that originated at a host or at a POS device connected to the BASE24-pos system must first pass through the BIC ISO Interface process. The BIC ISO Interface process translates the message into the co-network's external message format before sending it to the co-network.

Process Linkages

Since there are multiple processes involved in transaction processing, transactions must be sent between them for processing. The links between these processes are configurable. Each link in the transaction path is defined below.

Inbound Messages from POS Terminals, Hosts, and Interchanges

All transactions inbound to the BASE24-pos system are sent by stations. All POS terminals, host links, and interchange links are defined as stations. A station is a physical point in the BASE24 system that is designated to send or receive messages.

The XPNET process controls the BASE24 system's interaction with its stations. The XPNET process receives inbound transactions and routes them to the appropriate processes. The XPNET process determines the process or service to which to send a message using the destination table.

Each station definition contains a DESTINATION or DESTSTRING attribute, which identifies the process or service to which all messages from the station are to be sent.

- In the case of POS terminals, the DESTINATION or DESTSTRING attribute specifies a Device Handler/Router/Authorization process or service.
- In the case of host stations, the DESTINATION or DESTSTRING attribute specifies an ISO Host Interface process.
- In the case of interchange stations, the DESTINATION or DESTSTRING attribute specifies an Interchange Interface process.

Using the DESTINATION or DESTSTRING attribute, the XPNET process forwards messages to the designated process or service.

Once the designated process receives the message from the station, that process verifies that it should be receiving messages from the station.

Inbound Messages to an ISO Host Interface Process

When an ISO Host Interface process receives a message from a host station, it verifies that it is supposed to receive messages from that particular station. It does this based on an internal table of stations that it builds at initialization from the STATION fields on screen 2 of its records in the Host Configuration File (HCF). If a message is received from a station that is not in the table, the ISO Host Interface logs a message indicating the situation and writes the incoming message to the log.

Inbound Messages to an Interchange Interface Process

When an Interchange Interface process receives a message from an interchange station, it verifies that it is supposed to receive messages from that particular station. It does this based on an internal table of stations that it builds at initialization from the STATION fields on screen 1 of its record in the Interchange Configuration File (ICF) or the Enhanced Interchange Configuration File (ICFE). If a message is received from a station that is not in the table, the Interchange Interface process logs a message indicating the situation and writes the incoming message to the log (this action can vary by Interchange Interface).

Device Handler/Router/Authorization Process to Application Process Other Than the Transaction Context Manager Process

When the Device Handler module receives a transaction from a POS terminal it determines whether the transaction is allowed. Device Handler/Router/Authorization processes with Standard POS and Hypercom Device Handler modules can also route a transaction directly to an application process for authorization, thereby bypassing the normal routing and authorization procedures described in section 4. If the transaction is allowed and the AUTHORIZATION DESTINATION field on Acquirer Processing Code File (APCF) screen 2 contains a value, the Standard POS or Hypercom Device Handler module calls the Router module to send the transaction to the specified application process. In this case, no further routing or authorization processing is performed by the Device Handler/Router/Authorization process.

Device Handler/Router/Authorization Process to Transaction Context Manager Process

When the Standard POS Device Handler module receives a mobile top-up transaction from a POS terminal, it determines whether the transaction is allowed. Device Handler/Router/Authorization processes with the Standard POS Device Handler modules route mobile top-up transactions directly to the Transaction Context Manager process. The Standard POS Device Handler module calls the Router module to send the transaction to the Transaction Context Manager process specified in the POS-SPLIT-TXN-RTG-DEST param in the Logical Network Configuration File (LCONF). In this case, the Transaction Context Manager process routes the transaction to the Device Handler/Router/Authorization process and to the service provider.

Application Process to Device Handler/Router/Authorization Process

When an application process returns a transaction response to the Device Handler/Router/Authorization process, the Router module determines whether the response is logged to the POS Transaction Log File (PTLF). If the LOG AUTH DEST RESPONSE field on APCF screen 2 is set to a value of Y, the module logs the transaction response to the PTLF and sends the transaction response back to the Device Handler module. This processing is supported only for Device Handler/Router/Authorization processes with a Standard POS or Hypercom Device Handler module.

Device Handler/Router/Authorization Process to ISO Host Interface Process

The Device Handler/Router/Authorization process determines whether a transaction must be sent to a host by using the routing methods described in section 4. If a transaction is to be sent to a host, the routing steps determine the ISO Host Interface process to which the transaction must be sent for forwarding to the host.

Device Handler/Router/Authorization Process to Interchange Interface Process

The Device Handler/Router/Authorization process determines whether a transaction must be sent to an interchange by using the routing methods described in section 4. If a transaction is to be sent to an interchange, the routing steps determine the Interchange Interface process to which the transaction must be sent for forwarding to the interchange.

ISO Host Interface Process to Device Handler/Router/Authorization Process

The ISO Host Interface process, handling requests from acquirer networks, determines the Device Handler/Router/Authorization process to which it is to send inbound transaction requests from the host using the value in the AUTH PROCESS field on screen 7 of the Host Configuration File (HCF).

Interchange Interface Process to Device Handler/Router/Authorization Process

The Interchange Interface process sends incoming transactions to the process named in the AUTH PROCESS field on screen 10 of the Interchange Configuration File (ICF) or Enhanced Interchange Configuration File (ICFE).

Interchange Interface Process to Application Process

When one of the following Interchange Interface processes receives a transaction from a POS terminal it determines whether the transaction is allowed and whether the transaction is to be authorized by the Device Handler/Router/Authorization process or by an application process:

- Banknet Interchange Interface process
- BIC ISO Interchange Interface process
- MDS/MDSM Interchange Interface process
- PLUS ISO Interchange Interface process
- VisaNet Interchange Interface process

If the transaction is allowed and a value is present in the AUTHORIZATION DESTINATION field on Acquirer Processing Code File (APCF) screen 2, the transaction is sent to the specified application process. This destination overrides the process specified in the AUTH PROCESS field on ICFE screen 10.

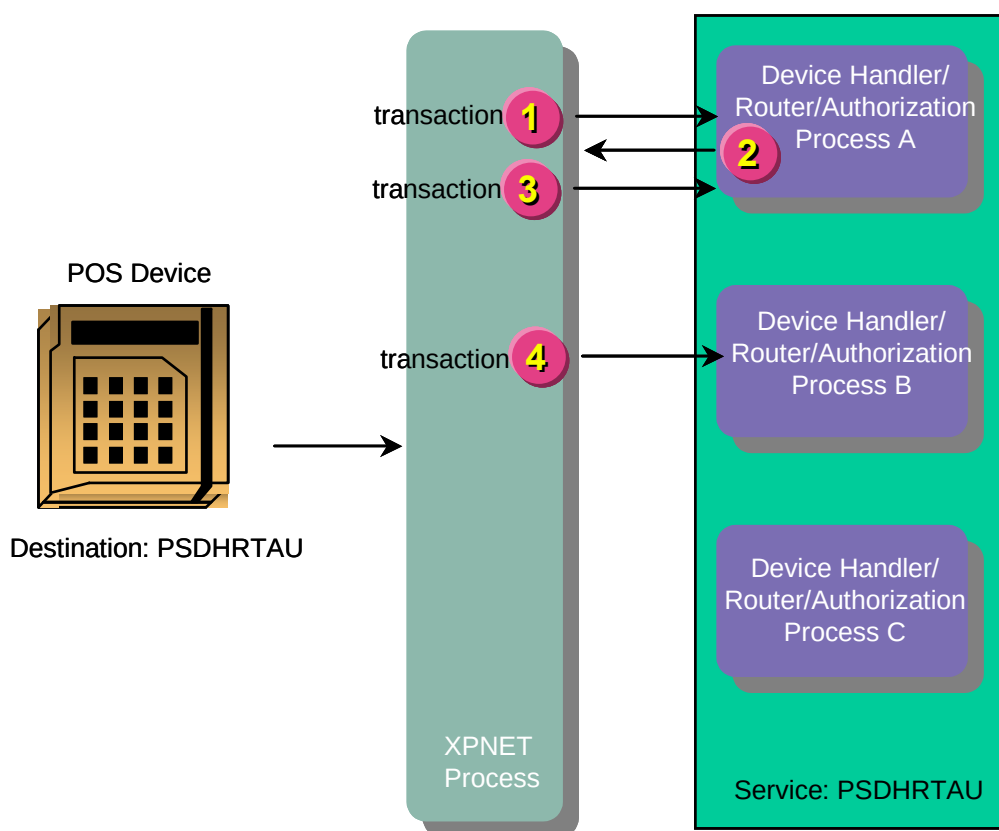
Application Process to Interchange Interface Process

When an application process returns a transaction response to an Interchange Interface process, the process determines whether the response is logged to the POS Transaction Log File (PTLF). If the LOG AUTH DEST RESPONSE field on APCF screen 2 is set to Y for the transaction, the module sends the transaction response as a log-only transaction to the Device Handler/Router/Authorization process (i.e., the process specified in the AUTH PROCESS field on ICFE screen 10) for logging to the PTLF.

Service-Based Routing

When multiple Device Handler/Router/Authorization processes are in use, the BASE24-pos system allows devices to send transaction requests to a service instead of directly to a Router/Authorization process (or Device Handler/Router/Authorization process). This option is designed to avoid the processing inefficiencies that could develop when a specific process is assigned to receive transaction requests. Without service-based routing, the requesting processes assigned to one server process must wait for that server process to complete its transactions even if other server processes are available. With service-based routing, transactions can be distributed efficiently among multiple server processes as they become available.

The following illustration shows how service-based routing functions with a POS device sending transactions to multiple Device Handler/Router/Authorization processes.



In this example, the XPNET process can send transactions to three Device Handler/Router/Authorization processes that have been associated with the service PSDHRTAU. The POS device station sends all BASE24-pos transactions it receives to the PSDHRTAU service. The XPNET process sends each transaction it receives from the station to the first available Device Handler/Router/Authorization process for the PSDHRTAU service as follows:

1. The XPNET process sends the first transaction to the Device Handler/Router/Authorization process A, the first provider available for the PSDHRTAU service.
2. Device Handler/Router/Authorization process A completes its processing of the first transaction and passes the transaction through the XPNET process to another process for further processing.
3. The XPNET process sends a second transaction to Device Handler/Router/Authorization process A. Because the first transaction processing is complete, process A is still the first available provider for the PSDHRTAU service.
4. The XPNET process sends a third transaction to Device Handler/Router/Authorization process B because Device Handler/Router/Authorization process A is still processing the second transaction when the XPNET process receives the third transaction for the PSDHRTAU service.

The XPNET process uses a service-based routing algorithm to determine which provider receives each message sent to a service. Messages sent to a service are distributed on a rotary basis. A provider is considered to be available if it is started and is not currently busy. Providers configured with a startup logic of DEMAND that are stopped are started only if they are needed to handle the processing load. Once started, they remain in rotation. For a detailed description of the service-based algorithm used by the XPNET process, refer to the ***NET24-XPNET Network Control Operations Guide***.

Processes are configured to a service using the SERVICE attribute. For detailed information on configuring the SERVICE attribute, refer to the ***NET24-XPNET Network Control Command Reference Manual***.

Transaction Timers

Timers enable processes to take action in situations where responses have not been received as expected. The BASE24-pos product uses timers for logical network processing, device handler processing, host processing, and interchange processing. These various timers are configured in the Logical Network Configuration File (LCONF), the Host Configuration File (HCF), and the Interchange Configuration File (ICF) or Enhanced Interchange Configuration File (ICFE).

General Timers

The following general timers are used by multiple BASE24 processes.

Security Device Timer Limit — The SECURE-DEV-TIM-LMT param in the LCONF specifies the maximum length of time the Device Handler/Router/Authorization process waits for a response from the security module before the message is considered timed out. All processes that initialize a hardware security module, such as the ISO Host Interface process and Interchange Interface processes, use this param.

Device Handler Module Timers

The following timers are used in transaction processing by BASE24-pos Device Handler modules.

Device Handler Timer Limit — The POS-DH-AUTH-RESP-TIMEOUT param in the LCONF specifies the maximum length of time that the Device Handler module waits for an authorization response to a transaction request message before sending an unable to respond message to the device.

Note: The Hypercom Device Handler (HPDH) module and the BASE24 Standard POS Device Handler (SPDH) module do not use the POS-DH-AUTH-RESP-TIMEOUT param. The HPDH uses the Hypercom Device Handler Timer Limit. The SPDH determines this timer by using a setting in the configuration file for BASE24 Standard POS Devices (ACNF).

Hypercom Device Handler Timer Limit — The POS-HPDH-AUTH-RESP-TIMEOUT param in the LCONF specifies the maximum length of time that the Hypercom Device Handler module waits for an authorization response to a transaction request message before sending an unable to respond message to the Hypercom device.

ISO Host Interface Process Timers

The following timers are used in transaction processing by ISO Host Interface processes. Not all are directly related to transaction processing; some are related to network management.

Network Management Timer Limit — This timer limit is set in the NETWORK MANAGEMENT field on screen 1 of the HCF. It specifies the length of time a BASE24 ISO Host Interface process waits for a response after sending a network management message to a data processing center (DPC) station. If a response message is not received within the time specified, the ISO Host Interface process marks the station down.

Extended Network Timer Limit — This timer limit is set in the EXTENDED NETWORK field on screen 1 of the HCF. It specifies the length of time an ISO Host Interface process waits, once a station has been marked down, before sending an echo test message.

Store and Forward Timer Limit — This timer limit is set in the STORE AND FORWARD field on screen 1 of the HCF. It specifies the length of time an ISO Host Interface process waits for an acknowledgment after sending a store-and-forward message to the DPC.

Wait for Traffic Timer Limit — This timer limit is set in the WAIT FOR TRAFFIC field on screen 1 of the HCF. It specifies the length of time an ISO Host Interface process waits for traffic on the line to a station after sending a network management message. Once this length of time has passed and no traffic has occurred on the line, the ISO Host Interface process sends another network management message.

Outbound Timer Limit — This timer limit is set in the OUTBOUND LIMIT field on screen 7 of the HCF. It specifies the length of time an ISO Host Interface process waits for a response message after sending a BASE24-pos financial transaction message to the host. The EXTRN-RQST-TIMEOUT param in the LCONF determines the actions taken when a transaction request times out.

If the EXTRN-RQST-TIMEOUT param is set to a value of 0 and a request times out, the ISO Host Interface process fails the transaction and returns it to the Device Handler/Router/Authorization process for alternate or stand-in routing. The ISO Host Interface process generates a reversal and sends it to the host if it receives a late response.

If the EXTRN-RQST-TIMEOUT param is set to a value of 1 and a request times out, the ISO Host Interface process fails the transaction and returns it to the Device Handler/Router/Authorization process for alternate or stand-in routing and also generates a reversal that it sends to the host. If the ISO Host Interface process then receives a late response from the host, it drops the message.

Inbound Timer Limit — This timer limit is set in the INBOUND LIMIT field on screen 7 of the HCF. It specifies the length of time an ISO Host Interface process waits for a response message after sending a financial transaction message to a BASE24-pos Device Handler/Router/Authorization process.

Completion Acknowledgment Timer Limit — This timer limit is set in the COMPLETION ACK field on screen 7 of the HCF. It specifies the length of time an ISO Host Interface process waits for an acknowledgment message after sending an advice or reversal message to the DPC.

Queue Subtract Timer Limit — This timer limit is set in the QUEUE SUBTRACT field on screen 7 of the HCF. It specifies the length of time the ISO Host Interface process is to subtract from the value in the OUTBOUND LIMIT field to compute the maximum time the XPNET process can wait before sending the BASE24-pos message to the host. If the XPNET process cannot send the message to the host within this time, it returns the message to the ISO Host Interface process as stale. This option circumvents processing when it becomes apparent that sending a message will not give a host time to respond before the message times out.

Interchange Interface Process Timers

The following timers are used in transaction processing by Interchange Interface processes. Not all are directly related to transaction processing; some are related to network management. Not all Interchange Interface processes support all of these timer types. The timers supported depend on the specific Interchange Interface process.

Note: Although the secondary provider (e.g., telephone company) for mobile top-up transactions functions like a host, its timers are considered to be interchange interface timers and are configured in the ICF or ICFE.

Network Management Timer Limit — This timer limit is set in the NETWORK MANAGEMENT field on screen 3 of the ICF or ICFE. It specifies the length of time an Interchange Interface process waits before attempting to retry a network management message (sign on, sign off, etc.) once a message has failed.

Extended Network Timer Limit — This timer limit is set in the EXTENDED NETWORK field on screen 3 of the ICF or ICFE. It specifies the length of time an Interchange Interface process waits before attempting to retry a network management message during periods when communications to the interchange are down.

Wait for Traffic Timer Limit — This timer limit is set in the WAIT FOR TRAFFIC field on screen 3 of the ICF or ICFE. It specifies the length of time an Interchange Interface process waits for traffic on the line from the interchange after sending a network management message. Once this length of time has passed and no traffic has occurred on the line, the Interchange Interface sends another network management message.

Performance Period Timer Limit — This timer is set in the PERFORMANCE PERIOD field on screen 3 of the ICF or ICFE. It specifies the length of the time that the interchange's performance is monitored.

Store and Forward Timer Limit — This timer limit is set in the STORE AND FORWARD field on screen 10 of the ICF or ICFE. It specifies the length of time an Interchange Interface process waits for an acknowledgment after sending a BASE24-pos store-and-forward message to the interchange.

Outbound Timer Limit — This timer limit is set in the OUTBOUND field on screen 10 of the ICF or ICFE. It specifies the length of time an Interchange Interface process waits for a response message after sending a BASE24-pos financial transaction message to the interchange.

Inbound Timer Limit — This timer limit is set in the INBOUND field on screen 10 of the ICF or ICFE. It specifies the length of time an Interchange Interface process waits for a response message after sending a financial transaction message from the interchange to a BASE24-pos Device Handler/Router/Authorization process.

Completion Timer Limit — This timer is set in the COMPLETION field on screen 10 of the ICF or ICFE. It specifies the length of time an Interchange Interface process waits for a completion message from either the interchange or the BASE24 system after passing a response message from one entity to another.

Completion Acknowledgment Timer Limit — This timer limit is set in the COMPLETION ACK field on screen 10 of the ICF or ICFE. It specifies the length of time an Interchange Interface process waits for an acknowledgment message after sending an advice, reversal, or adjustment message to the interchange.

3: BASE24-pos Transaction Processing Controls

This section describes the basic BASE24-pos transaction processing controls that are available at the terminal, retailer, institution, host, and interchange levels. It provides information on the following topics and features:

- Authorization levels and methods
- Allowed transactions
- Allowed service types
- Transaction security
- Transaction screening
- Draft capture processing options
- Cardholder and transaction limits
- Usage accumulation totals
- Store-and-forward processing
- Second card processing
- Expiration date checking
- Preauthorized holds processing
- Chargeback and representment processing
- Generating completions to a host
- Token handling
- Address verification processing

This section should be used in conjunction with section 4, which provides additional information on how the BASE24-pos product performs the various types of processing. For information on additional configuration issues, along with screen layouts and valid values for file settings presented in this section, refer to the following manuals:

- ***BASE24 Base Files Maintenance Manual***
- ***BASE24-pos Files Maintenance Manual***
- ***BASE24 ISO Host Interface Manual***
- ***BASE24 Logical Network Configuration File Manual***

Authorization Levels and Methods

The Authorization module authorizes transactions based on the authorization levels and methods specified by each institution. These levels and methods are described below, along with other authorization functions that the Authorization module performs.

Authorization Levels

The authorization level defines the level of participation that the BASE24-pos product has in authorizing transactions. The BASE24-pos product can perform all of an institution's authorizations, or it can forward transactions to a host for full or final approval. The BASE24-pos product can also authorize transactions in place of the host in cases where communications to the host are down.

The authorization level is specified in the AUTH LEVEL field on screen 16 of the Institution Definition File (IDF). The possible authorization levels are online authorization (level 1), offline authorization (level 2), and online/offline authorization (level 3).

A single institution can process using multiple authorization levels, depending on the number of hosts it uses and the types of transactions it processes. An institution can control processing with multiple authorization levels using the IDF routing table on screen 16 of the IDF. For more information on the IDF routing table, refer to section 4.

Note: When the BASE24-pos eCheck add-on product is installed, the authorization level specified in the AUTH LEVEL field on screens 1, 2, and 3 of the Check Authorization Routing Table File (CART) is used for check verification and check guarantee transactions.

Online Authorization

Online authorization is also referred to as level 1 authorization. At this level, the host performs all transaction authorizations. If the host is offline and, therefore, unable to authorize transactions, the BASE24-pos product declines all transaction requests. Typically, the BASE24-pos product does not maintain customer account files at this level of authorization.

Note: Although the BASE24-pos product cannot authorize transactions when an institution is using the online authorization level, it can optionally screen transactions before sending them to the host. If a transaction does not pass the prescreening checks, the BASE24-pos product can deny the transaction. Prescreening is described later in this section.

Offline Authorization

Offline authorization is also referred to as level 2 authorization. At this level, the Authorization module of the Device Handler/Router/Authorization process performs all transaction authorizations; no authorization requests are forwarded to the host. The institution can, however, choose to have completion messages sent to the host to notify the host of transactions authorized by the BASE24 system.

With the offline authorization level, the BASE24-pos product must maintain cardholder accounts in its database. These account files, as well as transaction log files, are then periodically extracted to tape or disk and used to update host files. These accounts are also refreshed periodically using information from host files.

Online/Offline Authorization

Online/offline authorization is also referred to as level 3 authorization. At this level, the Authorization module of the Device Handler/Router/Authorization process optionally screens transactions based on prescreening settings in the IDF and forwards those transactions that pass the screening checks to the host for final approval. The Authorization module denies transactions that do not pass the screening checks.

The Authorization module also stands in for the host and authorizes transactions when communications to the host are down. The transactions the BASE24-pos product authorizes are then stored and forwarded to the host when communications with the host are restored.

With the online/offline authorization level, the BASE24 system must maintain cardholder account files in its database. These account files, as well as transaction log files, are periodically extracted to tape or disk and used to update host files. These accounts are also refreshed periodically using information from host files.

Authorization Methods

The authorization method an institution uses establishes the file configuration and authorization checks that the BASE24-pos product performs while authorizing transactions.

Institutions specify the authorization method or methods in the AUTH TYPE field on screen 16 of the IDF. Possible authorization methods are online host processing, Negative Authorization with Usage Accumulation, Positive Authorization, Positive Balance Authorization, Negative Authorization without Usage Accumulation, and Parametric Authorization.

Note: When the BASE24-pos eCheck add-on product is installed, the authorization method specified in the AUTH TYPE field on screens 1, 2, and 3 of the Check Authorization Routing Table File (CART) is used for check verification and check guarantee transactions. Only the online host processing or Positive Balance Authorization methods can be configured in this field.

Online Host Processing

Online host processing must be used in conjunction with the online authorization level (level 1). Online authorization does not support the use of BASE24-pos authorization files; therefore, the host must perform all authorizations online. The BASE24-pos product can, however, screen transactions for an institution using the online host processing authorization method. This screening involves the Institution Definition File (IDF) and the Card Prefix File (CPF).

Negative Authorization with Usage Accumulation Method

The Negative Authorization with Usage Accumulation method uses the Negative Card File (NEG) and the Usage Accumulation File (UAF). The Authorization module authorizes transactions using negative identification. If a record is found in the NEG, the Authorization module takes the action specified in the record. Otherwise, the Authorization module assumes the account is good and checks the usage accumulation (tracked in the UAF) against the institution's transaction limits in the CPF.

Positive Authorization Method

The Positive Authorization method uses the Cardholder Authorization File (CAF). The Authorization module authorizes transactions using positive identification. For the transaction to be valid, a CAF record must exist for the card that initiated the transaction, the status of the account in the CAF must be good, and the amount previously withdrawn plus any holds and any new transaction amounts cannot exceed the cardholder's transaction limits.

Positive Balance Authorization Method

The Positive Balance Authorization method uses the Cardholder Authorization File (CAF) and the Positive Balance File (PBF). The Authorization module authorizes transactions using positive identification. For the transaction to be valid, a CAF record must exist for the card that initiated the transaction, the status of the account in the CAF must be good, and the amount previously withdrawn plus any holds and any new transaction amounts cannot exceed the cardholder's transaction limits. A PBF record must also exist for the account, and the amount of the transaction cannot exceed the balance in the account.

Note: When the BASE24-pos eCheck add-on product is installed, check guarantee transactions can be authorized using this method, thereby impacting PBF balances immediately for approved checks.

Negative Authorization without Usage Accumulation Method

The Negative Authorization without Usage Accumulation method uses the Negative Card File (NEG). The BASE24-pos product does not maintain card usage totals for this method. The Authorization module authorizes transactions using negative identification. If a record is found in the NEG, the Authorization module takes the action specified in the NEG record. If no record is found in the NEG, the Authorization module assumes the account is good and checks the amount of the transaction against the institution's transaction limits in the CPF.

Parametric Authorization Method

The Parametric Authorization method uses the Cardholder Authorization File (CAF), the Positive Balance File (PBF), and the Card Authorization Parameters File (CAPF). The Authorization module first verifies transactions by using the card status and account activity during the current usage accumulation period as

maintained in the CAF. If the card status and account activity are valid according to the CAF, the Authorization module then compares the delinquency history as maintained in the PBF with the delinquency table in the CAPF. Next, the Authorization module computes the amount of available credit and uses the CAPF overlimit table in an attempt to authorize the transaction if sufficient credit is not available. Finally, the Authorization module checks the single transaction limits in the CAPF before approving the transaction. For an example of Parametric Authorization method processing, refer to appendix D.

Allowed Transactions

Institutions can define which transactions they will allow at their terminals; for credit, checking, and savings accounts; and at the interchange. The ability to restrict the transactions allowed at the terminal, retailer, cardholder account, and interchange levels enables the institution to better control its transaction processing.

Cardholder Transactions Allowed at the Terminal

Institutions that own POS terminals can define which cardholder transactions a certain terminal can accept. One terminal can accept a different transaction set than another terminal belonging to the same institution. Institutions can base the allowed transactions for a certain terminal on the type of terminal, the location of the terminal, the most frequent types of transactions made at the terminal, or other business criteria.

For terminals supported by Standard POS Device Handler or Hypercom Device Handler modules, the transactions allowed can be specified at the terminal level, at the retailer level, or at the institution level using acquirer transaction profiles. For all other terminal types, the transactions allowed are specified at the terminal level only. Institutions can define acquirer transaction profiles to group terminals for the purpose of defining allowed transactions. For example, an institution might define an acquirer transaction profile for all of its POS devices at a particular location and define a different acquirer transaction profile for all of its POS devices at another location. Acquirer transaction profiles defined at the terminal level override acquirer transaction profiles defined at the retailer level. Likewise, acquirer transaction profiles defined at the retailer level override acquirer transaction profiles defined at the institution level.

Institutions can also define the transactions for which the retailer is required to enter an administrative card using retailer transaction profiles. Administrative card transaction profiles define the administrative transactions allowed for each administrative card. As with acquirer transaction profiles, institutions can define these transaction profiles to group retailers or administrative cards for the purpose of defining allowed transactions. For example, an institution might define a different retailer transaction profile for each retailer in the system or use the same retailer transaction profile for all retailers in the system. Retailer transaction profiles defined at the retailer level override default retailer transaction profiles defined at the institution level. Administrative card profiles at the administrative card level override default administrative card profiles defined at the institution level.

Note: The transactions allowed at the terminal can be performed only if the terminal is capable of performing them. For example, some terminals have limited capabilities and, although the institution may define a transaction as allowed and the BASE24-pos product can process that transaction, the terminal itself may not have the capability of handling that particular transaction.

Terminal Level

At the terminal level, institutions define the allowed transactions for a specific terminal on screen 6 of the BASE24-pos Terminal Data files (PTD). The format of screen 6 varies according to the type of terminal defined in the BASE24-pos Terminal Data files record. For Standard POS Device Handler and Hypercom terminals, this screen displays the ACQUIRER TXN PROFILE field. The acquirer transaction profile value is used to read the Acquirer Processing Code File (APCF) extended memory table, which defines all the transactions allowed for the profile and the accounts on which the transactions can be performed. If an acquirer transaction profile is not defined in the BASE24-pos Terminal Data files for a terminal, the value from the acquirer transaction profile defined in the PRDF is used to read the APCF extended memory table to determine the transactions allowed. For other types of POS terminals, screen 6 displays a field for each possible transaction allowed. Refer to the ***BASE24-pos Files Maintenance Manual*** for descriptions of both screen 6 formats and a list of possible allowed transactions.

Retailer Level

At the retailer level, institutions can define the acquirer transaction profile for a specific retailer in the ACQUIRER TXN PROFILE field on screen 4 of the POS Retailer Definition File (PRDF). If an acquirer transaction profile is not defined in the PRDF for the retailer, the value from the acquirer transaction profile defined in the IDF is used to read the APCF extended memory table to determine the transactions allowed.

The retailer transaction profile for a specific retailer is defined in the RETAILER TXN PROFILE field on PRDF screen 4. If a retailer transaction profile is not defined in the PRDF for the retailer, the value from the default retailer transaction profile defined in the IDF is used to read the APCF extended memory table to determine whether the retailer is required to swipe an administrative card for the transaction.

The administrative card transaction profile for a specific administrative card is defined in the ADMIN TXN PROFILE field on Administrative Card File (ADMN) screen 1. If the administrative card transaction profile is not defined in the ADMN for the retailer, the value from the default administrative card transaction profile defined in the IDF is used to read the APCF extended memory table to determine whether an administrative transaction is allowed for a particular administrative card.

Institution Level

A default acquirer transaction profile can be defined for the institution in the ACQUIRER TXN PROFILE field on Institution Definition File (IDF) screen 19. If an acquirer transaction profile is not defined in the BASE24-pos Terminal Data files for a terminal or in the PRDF for the retailer, the value from the acquirer transaction profile defined in the IDF is used to read the APCF extended memory table. Finally, if an acquirer transaction profile is not defined in the BASE24-pos Terminal Data files, PRDF, or IDF, a value of all asterisks (wildcard characters) is used to read the APCF extended memory table.

Default retailer and administrative card transaction profiles are defined for the institution in the RETAILER TXN PROFILE and ADMIN TXN PROFILE fields on IDF screen 19. If these profiles are not defined in the PRDF and ADMN, respectively, the profile values from the IDF are used to read the APCF extended memory table.

Acquirer Transaction Profile Processing

Once the acquirer transaction profile value used to read the APCF extended memory table has been determined, the Device Handler module attempts to read the APCF extended memory table using the acquirer transaction profile value, the message category code from the transaction, and the ISO processing code for the transaction. The APCF supports the use of wildcard characters for the acquirer transaction profile and message category, so that one APCF record can cover many combinations. The Device Handler module calls a utility to map the internal BASE24 transaction code for the transaction to the corresponding ISO processing code before reading the APCF extended memory table.

Using the acquirer transaction profile, message category, and ISO processing code, the Device Handler module steps through the records in the APCF extended memory table looking for a match. If an exact match is not found, certain other alternatives are allowed. Below is the hierarchy used for selection.

Preference	Acquirer Transaction Profile	Message Category	ISO Processing Code
1	Exact match	Exact match	Exact match
2	Exact match	*	Exact match
3	*****	Exact match	Exact match
4	*****	*	Exact match

Once a match is found, the record is read to determine whether the transaction is allowed.

The APCF also allows you to specify an authorization destination for a transaction other than the Router/Authorization module bound in with the Device Handler module. This mechanism allows you to route transactions received from a POS terminal to an application process in the XPNET system. When a transaction is routed in this way, you can also specify whether the transaction response returned from the authorization destination is logged to the POS Transaction Log File (PTLF).

Retailer and Administrative Card Transaction Profile Processing

Once the retailer transaction profile value used to read the APCF extended memory table has been determined, the Router/Authorization module attempts to read the APCF extended memory table using the retailer transaction profile value, the message category code from the transaction, and the ISO processing code for the transaction. The APCF supports the use of wildcard characters for the retailer transaction profile and message category, so that one APCF record can cover many combinations. The Router/Authorization module calls a utility to map the internal BASE24 transaction code for the transaction to the corresponding ISO processing code before reading the APCF extended memory table.

Using the retailer transaction profile, message category, and ISO processing code, the Router/Authorization module steps through the records in the APCF extended memory table looking for a match. If an exact match is not found, certain other alternatives are allowed. Below is the hierarchy used for selection.

Preference	Retailer Transaction Profile	Message Category	ISO Processing Code
1	Exact match	Exact match	Exact match
2	Exact match	*	Exact match
3	*****	Exact match	Exact match
4	*****	*	Exact match

Once a match is found, the record is read to determine whether the transaction requires an administrative card. If it does and the track data is present for an administrative card, the Router/Authorization module searches the APCF extended memory table again using the same search hierarchy, but this time using the administrative card transaction profile to determine whether the transaction is allowed for the administrative card. This search is also performed if an administrative card is not required for the transaction, but track data for an administrative card is present in the transaction. The APCF supports the use of wildcard characters for the administrative card transaction profile and message category, so that one APCF record can cover many combinations.

Note: Administrative transactions cannot be routed to an authorization destination specified in the APCF extended memory table. They must always be authorized by the Router/Authorization module bound in with the Device Handler module.

Transactions Allowed by the Card Issuer

Institutions can define which transactions can be made in association with each account type using issuer transaction profiles. Account types include credit, checking, and savings accounts. For example, the institution may allow cardholders to perform some transactions using credit accounts but not using savings or checking accounts.

As with acquirer transaction profiles, issuer transaction profiles can be established at different levels in the BASE24 database. Issuer transaction profiles can be set at the account level in the Cardholder Account File (CAF), at the card prefix level in the Card Prefix File (CPF), or at the card issuer level in the Institution Definition

File (IDF). The Router/Authorization module uses the issuer transaction profile to read the Issuer Processing Code File (IPCF) extended memory table, which defines all the transactions allowed for the profile and the accounts on which the transactions can be performed.

Account Level

At the account level, institutions can define the issuer transaction profile for a specific account in the ISSUER TXN PROFILE field on screen 10 of the Cardholder Authorization File (CAF). If an issuer transaction profile is not defined in the CAF for an account, the value from the issuer transaction profile defined in the CPF is used to read the IPCF extended memory table to determine the transactions allowed.

Card Prefix Level

At the card prefix level, institutions can define the issuer transaction profile for a specific card prefix in the ISSUER TXN PROFILE field on screen 6 of the Cardholder Prefix File (CPF). If an issuer transaction profile is not defined in the CPF for the card prefix, the value from the default issuer transaction profile defined in the IDF is used to read the IPCF extended memory table to determine the transactions allowed.

Institution Level

A default issuer transaction profile can be defined for the institution in the ISSUER TXN PROFILE field on Institution Definition File (IDF) screen 19. If an issuer transaction profile is not defined in the CAF for the account or in the CPF for the card prefix, the value from the issuer transaction profile defined in the IDF is used to read the IPCF extended memory table. Finally, if an issuer transaction profile is not defined in the CAF, CPF, or IDF, a value of all asterisks (wildcard characters) is used to read the IPCF extended memory table.

Issuer Transaction Profile Processing

Once the issuer transaction profile value used to read the IPCF extended memory table has been determined, the Router/Authorization module attempts to read the IPCF extended memory table using the issuer transaction profile value, the message category code from the transaction, and the ISO processing code for the transaction. The IPCF supports the use of wildcard characters for the acquirer

transaction profile and message category, so that one IPCF record can cover many combinations. The Router/Authorization module calls a utility to map the internal BASE24 transaction code for the transaction to the corresponding ISO processing code before reading the IPCF extended memory table.

Using the issuer transaction profile, message category, and ISO processing code, the Router/Authorization module steps through the records in the IPCF extended memory table looking for a match. If an exact match is not found, certain other alternatives are allowed. Below is the hierarchy used for selection.

Preference	Issuer Transaction Profile	Message Category	ISO Processing Code
1	Exact match	Exact match	Exact match
2	Exact match	*	Exact match
3	*****	Exact match	Exact match
4	*****	*	Exact match

Once a match is found, the record is read to determine whether the transaction is allowed. If the ON-US TRANSACTION ALLOWED field on IPCF screen 2 is set to a non-zero value, the transaction is allowed. If the transaction is allowed and the COMPLETION REQUIRED TO HOST field on IPCF screen 2 is set to a value of Y, the module will generate a completion message indicating whether the transaction is approved or declined and send it to the host after the transaction has been authorized.

Transactions Allowed Between the BASE24 System and an Interchange

Institutions can define which transactions it will send to a particular interchange for authorization and—for some interchanges—which transactions it will receive from an interchange for authorization.

Outbound Transactions

Institutions define the transactions that can be sent to a particular interchange on screen 9 of the Interchange Configuration File (ICF) or by specifying an issuer transaction profile value in the ISSUER TXN PROFILE field on screen 10 of the Enhanced Interchange Configuration File (ICFE). The value of the issuer

transaction profile is used to read the Issuer Processing Code File (IPCF) extended memory table, which defines all the transactions allowed for the profile and the accounts on which the transactions can be performed. If an issuer transaction profile is not specified for the interchange, the Interchange Interface process uses a value of all asterisks (wildcard characters) to read the IPCF extended memory table.

If the BASE24 system attempts to send a transaction to the interchange, and the transaction is not allowed, the Interchange Interface process denies the transaction and the transaction is never sent to the interchange.

The following interchanges must be configured in the ICFE if they are installed in release 6.0 or newer:

- Banknet Interchange Interface process
- BIC ISO Interchange Interface process
- MDS/MDSM Interchange Interface process
- PLUS ISO Interchange Interface process
- VisaNet Interchange Interface process

Issuer Transaction Profile Processing. Once the issuer transaction profile value used to read the IPCF extended memory table has been determined, the Interchange Interface process attempts to read the IPCF extended memory table using the issuer transaction profile value, the message category code from the transaction, and the ISO processing code for the transaction. The IPCF supports the use of wildcard characters for the issuer transaction profile and message category, so that one IPCF record can cover many combinations. The Interchange Interface process calls a utility to map the internal BASE24 transaction code for the transaction to the corresponding ISO processing code before reading the IPCF extended memory table.

Using the issuer transaction profile, message category, and ISO processing code, the Interchange Interface process steps through the records in the IPCF extended memory table looking for a match. If an exact match is not found, certain other alternatives are allowed. Below is the hierarchy used for selection.

Preference	Issuer Transaction Profile	Message Category	ISO Processing Code
1	Exact match	Exact match	Exact match
2	Exact match	*	Exact match
3	*****	Exact match	Exact match
4	*****	*	Exact match

Once a match is found, the record is read to determine whether the transaction is allowed. If the ON-US TRANSACTION ALLOWED field on IPCF screen 2 is set to a non-zero value, the transaction is allowed. If the transaction is allowed and the COMPLETION REQUIRED TO HOST field on IPCF screen 2 is set to a value of Y, the Router/Authorization module will generate a completion message indicating whether the transaction is approved or declined and send it to the host after the transaction has been authorized.

Inbound Transactions

For interchanges configured in the ICFE, the ACQUIRER TXN PROFILE field on ICFE screen 10 defines the transactions allowed from the interchange. The value of the acquirer transaction profile is used to read the Acquirer Processing Code File (APCF) extended memory table, which defines all the transactions allowed for the profile and the accounts on which the transactions can be performed. If an acquirer transaction profile is not specified for the interchange, the Interchange Interface process uses a value of all asterisks (wildcard characters) to read the APCF extended memory table. If the BASE24 system receives a transaction from an interchange that has not been defined as allowed, the Interchange Interface process denies the transaction.

Note: The Interchange Interface process does not determine whether inbound transactions are allowed when the interchange is configured in the ICF.

Acquirer Transaction Profile Processing. Once the acquirer transaction profile value used to read the APCF extended memory table has been determined, the Interchange Interface process attempts to read the APCF extended memory table using the acquirer transaction profile value, the message category code from the transaction, and the ISO processing code for the transaction. The APCF supports the use of wildcard characters for the acquirer transaction profile and message category, so that one APCF record can cover many combinations. The Interchange Interface process calls a utility to map the internal BASE24 transaction code for the transaction to the corresponding ISO processing code before reading the APCF extended memory table.

Using the acquirer transaction profile, message category, and ISO processing code, the Interchange Interface process steps through the records in the APCF extended memory table looking for a match. If an exact match is not found, certain other alternatives are allowed. Below is the hierarchy used for selection.

Preference	Acquirer Transaction Profile	Message Category	ISO Processing Code
1	Exact match	Exact match	Exact match
2	Exact match	*	Exact match
3	*****	Exact match	Exact match
4	*****	*	Exact match

Once a match is found, the record is read to determine whether the transaction is allowed.

The APCF also allows you to specify an authorization destination for a transaction other than the Device Handler/Router/Authorization process specified in the AUTH PROCESS field on ICFE screen 10. This mechanism allows you to route transactions received from an interchange to an application process. When a transaction is routed in this way, you can also specify whether the transaction response returned from the authorization destination is logged to the POS Transaction Log File (PTLF).

Allowed Service Types

The BASE24-pos product uses a set of one- and two-character codes to identify the service type requested in a particular transaction. The BASE24-pos product makes the distinction between service type and card type. Card types identify whether the transaction was initiated with a credit card, debit card, or dual card. Service types indicate the type of service (e.g., MasterCard, American Express) related to the card type used to initiate the transaction. For mobile top-up transactions, the service types indicate the issuer identification number (IIN) supported at the terminal.

Service type codes are either reserved by BASE24 products or are defined by the institution. Institutions must use the reserved codes as they are defined. Refer to section 1 for the reserved service types and for the requirements of user-defined service types.

Service types an institution allows for each terminal and retailer are defined in the BASE24-pos Terminal Data files, the POS Retailer Definition File (PRDF), the Host Configuration File (HCF), and the Interchange Configuration File (ICF) or Enhanced Interchange Configuration File (ICFE) as described below. For mobile top-up transactions, the service type for a secondary provider (e.g., telephone company) is defined in the Mobile Operator File (MOF) and used to update totals contained in the BASE24-pos Terminal Data (PTD) files.

Service Types Allowed at the Terminal

To define the service types allowed at a particular terminal, an institution must configure fields in the BASE24-pos Terminal Data files, HCF, ICF or ICFE, and MOF.

BASE24-pos Terminal Data Files

On screens 8, 9, and 10 of the BASE24-pos Terminal Data files (PTD), the institution establishes card floor limits for purchases, cash advances, and mail orders. These card floor limits are established by service type in the CARD TYPE field. Limits for mobile phone card service types must also be established in the PTD.

Note: Although the card floor limits are established by service type, the field on PTD screens 8, 9, and 10 for the service type is called CARD TYPE. With the exception of this field, card type when used in the BASE24-pos product actually refers to whether the card is a credit, debit, or dual card.

Refer to the **BASE24-pos Files Maintenance Manual** for more information about establishing card floor limits using the CARD TYPE field on PTD screens 8, 9, and 10.

Host Configuration File

On screen 8 of the Host Configuration File (HCF), the institution establishes the types of services that can be requested at the terminal and accepted by a particular host for a retailer. These service types are established in the ALLOWED SERVICES fields.

Refer to the **BASE24 Base Files Maintenance Manual** for more information about establishing the service types allowed by a particular host using the ALLOWED SERVICES fields in the HCF.

Interchange Configuration Files

On screen 11 of the Interchange Configuration File (ICF) or Enhanced Interchange Configuration File (ICFE), the institution establishes the types of services that can be requested at the terminal and sent to a particular interchange. These service types are established in the ALLOWED SERVICES fields.

Refer to the **BASE24 Base Files Maintenance Manual** for more information about establishing the service types allowed by a particular interchange using the ALLOWED SERVICES fields in the ICF or ICFE.

Service Types Allowed by the Retailer

To define the service types allowed by a particular retailer, an institution must configure fields in the PRDF.

POS Retailer Definition File

On screens 2 and 3 of the POS Retailer Definition File (PRDF), the institution establishes authorization parameters for transactions processed by a particular retailer. These authorization parameters are established by service type in the CARD TYPE field.

Note: Although the card floor limits are established by service type, the field on PRDF screens 2 and 3 for the service type is called CARD TYPE. With the exception of this field, card type actually refers to whether the card is a debit card, credit card, or dual card when used in the BASE24-pos product.

Refer to the ***BASE24-pos Files Maintenance Manual*** for more information about establishing authorization parameters using the CARD TYPE field in the PRDF.

Transaction Security

The BASE24-pos product offers several forms of transaction security, including PIN and card verification, PIN encryption, message authentication, data encryption, and dynamic key management. The information below summarizes these transaction security features. Refer to the ***BASE24 Transaction Security Manual*** for additional information. Transaction security parameters are specified for the institution on Institution Definition File (IDF) screen 2, for the card prefix on Card Prefix File (CPF) screen 2, for the POS terminal on BASE24-pos Terminal Data files (PTD) screen 7, and, if applicable, on Derivation Key File (KEYD) screen 1. Transaction security parameters for each financial institution, DPC, ISO Host Interface process, and Interchange Interface process are also specified in the Key File (KEYF) or Key 6 File (KEY6), the Key Authorization File (KEYA), and the External Message File (EMF).

Depending on the security features selected, the BASE24-pos product uses the Transaction Security Services process to convert BASE24 hardware security data. Refer to the ***BASE24 Transaction Security Services Processing Guide*** for additional information about this process.

PIN Verification

The Authorization module can perform PIN verification for transactions authorized by the BASE24-pos product or as a prescreening step for transactions authorized by a host. The value in the VALIDATE PIN field on screen 7 of the PTD indicates whether the POS terminal can accept PINs. The value in the VALIDATE PIN field does not indicate whether the PIN is required or not. The value in the PIN PROCESSING FLAG field on CPF screen 7 identifies whether PINs are required with transactions from POS terminals.

If the terminal can accept PINs, the BASE24 system can use software or security devices to perform PIN verification. The BASE24 system supports four methods of PIN verification:

- DES (IBM 3624) PIN verification
- Diebold PIN verification
- Identikey PIN verification
- Visa PVV PIN verification

PIN verification information is maintained on screens 2 through 5 of the Key Authorization File (KEYA) in addition to the IDF, CPF, and PTD screens mentioned earlier.

Note: The BASE24-pos product verifies any PIN it receives, even if a PIN is not required to complete a transaction.

Card Verification

Card verification confirms the validity of the card presented by a customer. While the BASE24 system can perform PIN verification based on parameters established at the institution or card prefix level, card verification parameters must be established at the card prefix level.

Card verification can be performed electronically or manually. For electronic card verification, the card verification data is captured from the track data on the card's magnetic strip when it is swiped through a card reader on the POS terminal. For manual card verification, the merchant manually enters the PAN, expiration date, and manual CVD value printed on the card into the POS terminal. Manual card verification allows merchants to verify a card even if it cannot be read electronically.

The BASE24 system uses security devices or software to perform card verification. Card verification information is maintained on KEYA screen 6 in addition to the CPF and the PTD screens mentioned earlier.

PIN Encryption

PINs can be sent between the BASE24 system and a host or interchange encrypted or in the clear. PINs are encrypted and decrypted within the BASE24 system by the BASE24 processes that send transactions to and receive transactions from processing entities outside the system. These processes include ISO Host Interface processes, Interchange Interface processes, and Device Handler modules.

When the BASE24 system receives a transaction, the PIN may need to be encrypted, decrypted, or translated from one encrypted form to another, depending on how the BASE24 system and the external processor are set up to handle their PINs. The same is true for transactions that the BASE24 system is sending to another entity. PIN encryption data for messages sent between the POS device and a Device Handler module is configured on screen 7 of the PTD.

PINs can also be sent from the ISO Host Interface process or the Interchange Interface process encrypted or in the clear. PIN encryption data for messages sent between the ISO Host Interface process and a host or between the Interchange Interface process and an interchange is configured on screens 1 through 3 of the KEYF or on screens 1 through 4 of the KEY6.

The ISO Host Interface process and Interchange Interface processes use the KEY6 instead of the KEYF when the LCONF param DES-TRIPLE-SINGLE param is set to a value of Y, indicating that PINs need to be translated from Triple Data Encryption (3DES) encryption to DES encryption and vice versa. Otherwise, these processes use the KEYF.

The Key 6 File (KEY6) allows for double-length encryption keys, which are required when translating PINs from encryption under a double-length key using the Triple Data Encryption algorithm (3DEA) to encryption under a single-length key using the Data Encryption algorithm (DEA) and vice versa.

Message Authentication

Message authentication is a method used to monitor the integrity of data flowing between EFT devices, such as POS devices and ATMs, and the transaction authorizer. A message can change as the result of tampering or because of communications difficulties. Message authentication identifies whether or not the message has changed.

Message authentication is supported between the POS device and the BASE24-pos product using the Device Handler module, between the host and the BASE24-pos product using the BASE24-pos ISO Host Interface process, between the interchange and the BASE24-pos product using the Interchange Interface process, and between a BIC co-network and the BASE24-pos product using the BIC ISO Interface process. Message authentication information is maintained on ACI Standard Device Configuration File (ACNF) processing screen 2, PTD screen 7, KEYF screens 1 through 3 or KEY6 screens 1 through 4 (as described above for PIN encryption), and External Message File (EMF) screen 2.

Note: Not all interchanges and their respective Interchange Interface processes support message authentication. Message authentication support between devices and Device Handler modules also depends on the specific device and Device Handler module.

Data Encryption

Data encryption is an additional licensable module. With this module, the BASE24 Standard POS Device Handler (SPDH) module can return the available balance to a POS device encrypted or in the clear. To encrypt the available balance, one of the following valid values must be entered in the DATA ENCRYPTION TYPE field on PTD screen 7.

- 01 = ASCII encryption
- 02 = Binary encryption

The SPDH module requires the Transaction Security Services process to perform data encryption. Refer to the ***BASE24 Transaction Security Services Processing Guide*** for additional information about this process. Software data encryption is not available.

Derived Unique Key per Transaction (DUKPT)

The BASE24 Standard POS Device Handler (SPDH) module supports derived unique key per transaction (DUKPT) processing for PIN and MAC encryption.

The SPDH module can perform DUKPT processing for single-length key PIN encryption with or without the Transaction Security Services process. However, the SPDH module requires the Transaction Security Services process to perform DUKPT PIN encryption and verification with triple DES encryption and the longer keys used with triple DES encryption. The Transaction Security Services process is required to perform any DUKPT processing for MAC encryption. The terminal and the appropriate BASE24 process must be configured for the same key length. Refer to the ***BASE24 Transaction Security Services Processing Guide*** for information about the Transaction Security Services process.

If DUKPT PIN or MAC encryption is selected, the terminal must support the manual entry or injection of a base derivation key. When the Transaction Security Services process is not used, these keys are also entered on screen 1 of the Derivation Key File (KEYD). The SPDH module uses double-length (32-byte) derivation keys stored in the KEYD to translate incoming DUKPT-encrypted PIN blocks and MACs received from the terminal into a single-length Master/Session key PIN block or MAC, which the Router/Authorization module can verify normally. The Transaction Security Services process uses double-length derivation keys stored in its own database to translate incoming DUKPT-encrypted PIN blocks and MACs. When the Transaction Security Services process is used, the KEYD contains a key locator instead of the base derivation key.

Dynamic Key Management

Several processes use working keys to protect the PINs, message authentication codes (MACs), and available balance information moving through the BASE24 network. Working keys should be changed regularly to maintain proper security. Monitoring and changing these keys is known as key management, and can be performed manually or automatically. Automated key management is known as dynamic key management. Dynamic key management is an additional licensable module. This module requires the use of security modules to perform PIN encryption, PIN verification, message authentication, and data encryption. Dynamic key management is not supported when these functions are performed in software.

The following processes provide dynamic key management support.

- The BASE24 Standard POS Device Handler (SPDH) module provides support for POS devices.
- The ISO Host Interface process provides support for host transactions.
- The BASE24 Interchange Interface process provides support for interchange transactions.
- The BIC ISO Interface process provides support for a BIC co-network.

Key management information is maintained on KEYF screens 1 through 3 or KEY6 screens 1 through 4 (as described above for PIN encryption). Key management information for POS devices is maintained on ACI Standard Device Configuration File (ACNF) processing screen 3. Refer to the ***BASE24-pos Standard POS Device Configuration Manual*** for additional information on maintaining POS device key management thresholds.

Note: Not all interchanges and their respective Interchange Interface processes support dynamic key management.

Transaction Screening

Transaction screening can be performed in situations where the BASE24 system shares authorization responsibilities with the host. In such situations, the Authorization module of the Device Handler/Router/Authorization process can make preliminary standard checks on the transactions and decline those that do not meet the preliminary requirements. This prevents unnecessary processing by the host or co-network.

Transaction screening is also referred to as prescreening. Prescreening is not to be confused with a preauthorized purchase transaction or a preauthorized purchase completion transaction. The preauthorized purchase transaction allows retailers to put a hold on a customer's funds before a purchase is completed. The preauthorized purchase completion transaction completes the transaction for which funds have previously been placed on hold.

The BASE24-pos product can perform prescreening only when configured to do so by the institution and when the institution is processing transactions using the online or online/offline authorization level. The BASE24-pos product automatically performs these checks as part of its authorization routine when transactions are authorized offline by the BASE24 system. Transaction screening checks include those described below.

Card Status

The card status indicates whether the card that initiated the transaction is valid, has some type of restriction on it, or is a special use card. Depending on the status, the BASE24-pos product either denies the transaction or sends it on to the host for authorization. Institutions indicate whether they want the BASE24-pos product to prescreen the card status using the CARD STATUS field on screen 2 of the IDF.

The Device Handler/Router/Authorization process uses card status information set in the Cardholder Authorization File (CAF) when the institution is using the Positive Authorization method or the Positive Balance Authorization method. A CAF record can have one or two cards associated with it. Refer to "Second Card Processing" later in this section for more information about configuring card status checks.

When the institution is using the Negative Authorization with Usage Accumulation method or the Negative Authorization without Usage Accumulation method, the Device Handler/Router/Authorization process uses the value set in the REASON ON FILE field on screen 1 of the Negative Card File (NEG) to determine the card status.

Note: The card status is not checked when the institution is processing transactions using online host processing. The CAF and NEG do not exist for the institution when using the online authorization level.

Expiration Date

A series of checks can be performed to determine whether the card has expired. In the EXP DATE field on IDF screen 2, institutions indicate whether the BASE24-pos product should prescreen the card's expiration date. If the EXP DATE field indicates that the expiration date should be checked during prescreening, the Device Handler/Router/Authorization process then uses values set in the CPF to determine if the expiration date should be checked for this card prefix. Several configuration options are available for expiration date checking. Refer to "Second Card Processing" and "Expiration Date Checking" later in this section for a description of these options.

Card Verification

Card verification ensures that the card presented by the customer for a transaction is valid.

Card verification uses a value the BASE24 system calls the card verification digits (CVD) that is located in the track data on the card. The CVD can also be printed on the card for manual card verification when a card reader is unavailable. The card issuer generates the CVD using card verification keys, the primary account number, the card expiration date, and the card's service code. To verify a card, the BASE24 system uses the same card verification keys and track information to compute another CVD. If the two CVDs match, the BASE24 system considers the card to be valid.

For electronic card verification, all of the card verification data is captured electronically by swiping the card through a card reader. For manual card verification, the CVD, primary account number, and expiration date printed on the card are manually entered by the merchant. The BASE24-pos product automatically uses a service code value of 000. Card verification processing by the

BASE24 system for electronic and manual verification is the same, with the exception that manual verification can be performed once or twice using different date formats for the CVD.

Institutions indicate whether the BASE24 system should perform card verification as a prescreening check using the CHECK IF HOST ONLINE CV field on screen 2 of the Card Prefix File (CPF).

Refer to the **BASE24 Transaction Security Manual** for more information about card verification.

PIN Check

Institutions can configure the BASE24-pos product to determine whether cardholders entered the correct PINs when initiating transactions. This is configured in the PIN field on IDF screen 2 and the CHECK IF HOST ONLINE PIN field on CPF screen 2.

The PIN field in the IDF establishes PIN prescreening at the institution level, meaning that all transactions initiated with cards issued by that institution should or should not be prescreened for valid PINs. However, an institution may want to prescreen all PINs except for those belonging to cards with specific prefixes or prescreen only those cards with certain prefixes. In that case, the CHECK IF HOST ONLINE PIN field in the CPF overrides the PIN field in the IDF.

Withdrawal Limits

Institutions can configure the BASE24-pos product to determine during prescreening whether the requested transaction will cause the cardholder to exceed the withdrawal limits the institution established for either the card prefix or the cardholder. Institutions can configure this prescreening option using the LIMITS field on screen 2 of the IDF.

If the LIMITS field indicates that the BASE24-pos product should prescreen the withdrawal limits, the Device Handler/Router/Authorization process checks the ACTIVITY LIMITS fields on CPF screens 1 and 6 or CAF screens 1 and 10. The values in the ACTIVITY LIMITS fields on CAF screens 1 and 10 override those on CPF screens 1 and 6. This enables institutions to establish limits for entire card prefixes while allowing the flexibility to establish different limits for specific cardholders. For example, if a cardholder is consistently delinquent, the institution may want to establish lower limits than those established for the prefix of the card the cardholder carries.

Other Prescreening Checks

In addition to the prescreening checks described above, the Device Handler/Router/Authorization process also performs the following:

- Checks the transaction message for valid data.
- Checks the currency code in the transaction against the value in the CURRENCY CODE field on Institution Definition File (IDF) screen 3.
- Determines whether the transaction is allowed for the institution. Transactions allowed for the BASE24-pos product are set on IDF screens 18 and 19.
- Checks the transaction amount against ceiling limits in the POS Retailer Definition File (PRDF) or transaction limits in the BASE24-pos Terminal Data files and Authorization Selection Table (AST).

Draft Capture Processing Options

Using the BASE24-pos product, institutions can process transactions as draft capture or non-draft capture transactions. Non-draft capture transactions are transactions for which the paper receipt represents the actual settlement instrument. Paper receipts must be physically presented for deposit in order for the transaction to be settled. While the BASE24-pos product may have authorized the transaction, the BASE24-pos record cannot be used to settle the item.

Draft capture transactions are electronically captured at the time of the transaction and are sufficient in themselves to enact a funds transfer for the transaction amount. The BASE24-pos product identifies draft capture transactions in its files and includes draft capture totals on its retailer reports so that the retailer sponsor can settle with the retailer.

Draft capture is a settlement tool only. It has no impact on the authorization of transactions. Cardholders are still checked against the same authorization criteria without regard for whether a transaction is draft capture or non-draft capture. Thus, the retailer is not incurring additional risk of cardholder fraud by using or not using draft capture.

For a description of draft capture processing, refer to appendix D.

The BASE24-pos product provides the draft capture options described below. Institutions set these options for each service type using the TP (transaction profile) field on screens 8, 9, and 10 of the BASE24-pos Terminal Data files (PTD). If the institution sets the TP field to indicate a draft capture method that requires an electronic follow-up transaction and the institution is performing referral processing, the institution also must set the EF (electronic follow-up) field on screens 8, 9, and 10 of the PTD.

Authorization Only with Paper Follow-up

With the authorization only with paper follow-up option, the BASE24-pos product obtains an online authorization for the amount of the customer purchase. In an authorization-only environment, the card issuer waits for paper follow-up before posting the transaction and settling with the retailer. If an institution sets up a service type in the BASE24-pos Terminal Data files with this option, none of the transactions for that service type from the specified terminal are draft capture transactions.

This option should be used with Visa I terminals because the Visa I terminal is a first-generation terminal and is not capable of draft capture processing. This option can also be used with Visa II terminals, BASE24 Standard POS Device terminals, and Hypercom devices. This option is always used for CRT Administration devices.

Authorization and Draft Capture

With the authorization and draft capture option, the BASE24-pos product performs an authorization on the transaction and, at the same time, automatically captures the draft electronically. Using electronic draft capture, the card issuer does not require a paper follow-up before posting the transaction and settling with the retailer.

This option can be used with BASE24 Standard POS Device terminals and Hypercom devices.

Authorization with Electronic Follow-up

The authorization with electronic follow-up option provides for a two transaction process that authorizes and settles transactions when using referral processing.

The first transaction obtains an online authorization. This is similar to processing under the authorization-only option. If the transaction is authorized, a second transaction is generated to collect the ticket electronically and transmit the information to the card issuer for settlement. If the first transaction is not authorized, the second transaction is not generated.

With this option, the institution must also set the EF field on screens 8, 9, and 10 of the PTD to indicate how the BASE24-pos product should process electronic follow-up transactions if using referral processing.

Note: If not using referral processing, the BASE24-pos product updates the cardholder files with information from the first transaction. The follow-up transaction indicates that the transaction can be settled.

This option can be used with Visa II terminals.

Terminal-Defined Draft Capture

The terminal-defined draft capture option allows the terminal to determine the draft capture mode for each transaction based on service type. If a service type is set up with this option, the terminal selects one of the draft capture options described previously (authorization only, authorization and draft capture, or authorization with electronic follow-up) for each transaction involving this service type.

This option can be used only with BASE24 Standard POS Device terminals.

Cardholder and Transaction Limits

Institutions can establish amount and usage limits for cards, transactions, and individual cardholders. These limits are configured using the Card Prefix File (CPF), Cardholder Authorization File (CAF), Authorization Selection Table (AST), Card Authorization Parameters File (CAPF), POS Retailer Definition File (PRDF), and BASE24-pos Terminal Data files. BASE24 and BASE24-pos products use these limits when authorizing transaction requests. Transactions that exceed the limits as set by the institution are declined.

In addition, accumulator fields in the CAF and the Usage Accumulation File (UAF) are used to track transaction activity performed using the BASE24-pos product during a single usage accumulation period. A usage accumulation period is a set amount of time during which transaction amounts and counts are accumulated. At the end of the usage accumulation period, the amounts and counts are cleared. The accumulator fields in the CAF and UAF are compared to corresponding limits established in the CAF or CPF to determine if a transaction should be approved or declined. The Negative Authorization without Usage Accumulation method and the host-only method do not use accumulator fields. CPF limits are checked on a per transaction basis and must be set accordingly.

Refer to the ***BASE24 Base Files Maintenance Manual*** for more information about the accumulator fields in the CAF and UAF.

Card and Cardholder Limits

Institutions can establish card and cardholder limits as described in the following paragraphs.

Card Prefix File

The Card Prefix File (CPF) defines each card prefix that can be processed within the BASE24 logical network. One record exists in the CPF for each card prefix to be processed. Each CPF record contains processing parameters and authorization limits for that specific card prefix.

Screen 1 — Using the CARD PROCESSING INFORMATION fields on Card Prefix File (CPF) screen 1, institutions can limit the transaction activity allowed by the BASE24 system for cards with a specific prefix during a single usage accumulation period. These card usage limits apply to both BASE24-atm and BASE24-pos transactions combined.

When the institution is using the Negative Authorization without Usage Accumulation method, the authorization level is online, and the LIMITS field on IDF screen 2 indicates to do so, the values in these fields are checked with every transaction. When the institution is using the host-only authorization method, the LIMITS field on IDF screen 2 indicates to do so, and the BASE24 system must prescreen transactions for these limits, the values in these fields are checked with every transaction.

When the institution is using the Positive Authorization method or the Positive Balance Authorization method, the limits established on CPF screen 1 are used if the value in the TOTAL AGGR field on Cardholder Authorization File (CAF) screen 1 is set to zero; otherwise, the limits set in the CAF are used.

The transactions controlled by these limits include cash disbursements against credit and noncredit accounts and purchases made against noncredit accounts. Credit card purchases are not governed by these limits.

Screen 6 — Using the POS AUTHORIZATION INFORMATION and ACTIVITY LIMITS fields on Card Prefix File (CPF) screen 6, institutions can limit the transaction activity allowed by the BASE24-pos product for cards with a specific prefix during a single usage accumulation period. These card usage limits apply only to BASE24-pos transactions.

If the institution is using the Negative Authorization without Usage Accumulation method, the authorization level is online, and the LIMITS field on IDF screen 2 indicates to do so, the values in these fields are checked with every transaction. If the institution is using the host-only authorization method, the LIMITS field on IDF screen 2 indicates to do so, and the BASE24 system must prescreen transactions for these limits, the values in these fields are checked with every transaction.

The limits established on CPF screen 6 are used if the value in the TOTAL AGGR field on CAF screen 1 is set to zero; otherwise, the limits set in the CAF are used.

Using the TIMES USED PER PERIOD LIMIT field on CPF screen 6, institutions can specify the maximum number of times a card can be used to perform BASE24-pos transactions during a single usage accumulation period. This field is used only if the TOTAL AGGR field on CAF screen 1 contains a value of zero.

The transactions controlled by these limits include cash disbursements, purchases, replenishments and refunds against credit and noncredit accounts.

Refer to the ***BASE24 Base Files Maintenance Manual*** for more information about the fields on screens 1 and 6 of the CPF.

Cardholder Authorization File

The Cardholder Authorization File (CAF) contains one record for each cardholder whose card-issuing institution uses the Positive Authorization method, Positive Balance Authorization method, or Parametric Authorization method. Each CAF record contains authorization parameters and usage accumulation information for a specific cardholder.

Screen 1 — Using the ACTIVITY LIMITS fields on CAF screen 1, institutions can establish usage accumulation limits for the cardholder. The limits defined in these fields are for both ATM and POS transactions combined. The transactions affected by these usage limits are cash disbursements made against credit and noncredit accounts and purchases made against noncredit accounts. Credit purchases are not governed by these limits.

Screen 10 — Using the ACTIVITY LIMITS fields on CAF screen 10, institutions can establish usage accumulation limits for the cardholder. The limits defined in these field are for POS transactions only. Using the TIMES USED PER PERIOD LIMIT field on CAF screen 10, institutions can specify the maximum number of times a card can be used to perform BASE24-pos transactions during a single usage accumulation period. This field is used only if the TOTAL AGGR field on CAF screen 1 contains a value other than zero.

Refer to the ***BASE24 Base Files Maintenance Manual*** for more information about the fields on screens 1 and 10 of the CAF.

Transaction Limits

Institutions can establish transaction limits as described in the following paragraphs.

Authorization Selection Table

The Authorization Selection Table (AST) defines authorization routing for BASE24-pos transactions based on floor limits. The floor limits are also defined in the AST.

Screens 1, 3, and 5 — Using the ONLINE fields on screens 1, 3, and 5 of the AST, institutions can establish floor limits, by transaction type, for a particular service type and issuer code. The issuer codes are defined in the ISSUER field on screen 7

of the Card Prefix File (CPF). Whether or not the BASE24-pos product uses the floor limits established in the AST depends on parameters set in the BASE24-pos Terminal Data files.

Refer to the ***BASE24-pos Files Maintenance Manual*** for more information about the fields on screens 1, 3, and 5 of the AST.

Card Authorization Parameters File

The Card Authorization Parameters File (CAPF) contains one record for each credit card type supported by each card-issuing institution that uses the Parametric Authorization method. Each record contains transaction limits, delinquency limits, and overlimits for those transactions authorized by the Parametric Authorization method.

Screen 1 — Using the SINGLE PURCHASE LIMIT, SINGLE PURCHASE PERCENT, SINGLE ADVANCE LIMIT, and SINGLE ADVANCE PERCENT fields on screen 1 of the CAPF, institutions can establish purchase and advance limits for each credit card type.

Using the DELINQUENT TABLE fields, institutions can establish the number of days and billing cycles an account can be delinquent and still let transactions be authorized.

Using the OVERLIMIT TABLE fields, institutions can establish the percentage an account can be overlimit and the number of cycles it can be delinquent and still let transactions be authorized.

Refer to the ***BASE24-pos Files Maintenance Manual*** for more information about the fields on screen 1 of the CAPF.

POS Retailer Definition File

The POS Retailer Definition File (PRDF) contains one record for each retailer in the logical network. Each record contains transaction limits as allowed by the retailer.

Screens 2 and 3 — Using the CEILING LIMIT fields on screens 2 and 3 of the PRDF, institutions can establish the maximum amount the retailer allows for a single transaction associated with a particular service type.

Refer to the ***BASE24-pos Files Maintenance Manual*** for more information about the fields on screens 2 and 3 of the PRDF.

BASE24-pos Terminal Data Files

The BASE24-pos Terminal Data files contains one record for each terminal in the logical network. Each record contains transaction limits as allowed by the terminal.

Screen 3 — Using the TERM. ADJUSTMENT LIMIT, ADJUSTMENT COUNT, RETURN LIMIT, and RETURN COUNT fields on screen 3 of the BASE24-pos Terminal Data files (PTD), institutions can establish transaction limits as allowed by this terminal for adjustments and merchandise return transactions.

Screens 8, 9, and 10 — Using the CARD FLOOR LIMITS fields on screens 8, 9, and 10 of the PTD, institutions can establish card floor limits for the terminal. These card floor limits are established according to service type.

If both the CARD FLOOR LIMITS fields on PTD screens 8, 9, and 10 and the ONLINE fields in the AST contain values greater than zero, the lesser floor limit for the service type is used. If the CARD FLOOR LIMITS field on the PTD screens contain a zero for the transaction type, the ONLINE field entry for the transaction type in the AST is used. If the ONLINE field entry in the AST is zero for the transaction type, any amount is considered to be over the floor limit.

Refer to the ***BASE24-pos Files Maintenance Manual*** for more information about the fields on screens 3, 8, 9, and 10 of the PTD.

Usage Accumulation Totals

BASE24 products keep card and account usage totals in the Cardholder Authorization File (CAF) if the institution is using the Positive Balance Authorization method and in the Usage Accumulation File (UAF) if the institution is using the Negative Authorization with Usage Accumulation method. Institutions can specify whether they clear usage accumulation totals in the Cardholder Authorization File (CAF) and the Usage Accumulation File (UAF) every day of the week, every day but weekends, every day but weekends and holidays, or after a specified number of days.

Institutions establish their usage accumulation periods and how they clear their usage accumulation totals on screens 3 and 4 of the Institution Definition File (IDF). Refer to the ***BASE24 Base Files Maintenance Manual*** for a detailed description of how to set these fields.

Establishing Usage Accumulation Periods

A usage accumulation period is a set amount of time during which each cardholder's transaction statistics, such as the number of times a card is used, are tracked and compared to established transaction limits.

For example, an institution can set a usage accumulation period of one week and restrict the number of times a card can be used during that period to 10. Each time a cardholder uses his or her card, a counter is increased by one. If the counter reaches 10 within the usage accumulation period, subsequent transactions may be denied. At the end of the usage accumulation period, the BASE24 system clears the counter and all other usage accumulation totals.

An institution defines the length of its usage accumulation period in the PERIOD LENGTH field and the first day of the usage accumulation period in the BEGINNING DATE field on screen 4 of the Institution Definition File (IDF). The setting in the PERIOD LENGTH field is impacted by the setting in the WORK DAY CODE field on the same screen.

Clearing Usage Accumulation Totals

Institutions can choose to clear their usage accumulation totals every day of the week, every day but weekends and holidays, every day but Saturdays and holidays, or every day but Sundays and holidays. An institution defines how often usage accumulation totals are cleared by setting the WORK DAY CODE field on

screen 4 of the IDF. The setting in this field is impacted by the setting in the PERIOD LENGTH field on the same screen. If totals are not cleared on holidays, the institution must also specify the holiday dates in the HOLIDAYS field on screen 4 of the IDF.

The BASE24-pos Authorization module clears usage accumulation totals and removes expired preauthorized holds contained in the CAF for an account as part of processing the first transaction of a new usage accumulation period.

Usage accumulation totals and expired preauthorized holds in the UAF can be cleaned up or purged, depending upon the value in the PERSISTENT UAF field on IDF screen 3, as follows:

- Without a persistent UAF, all records in the UAF (including records with unexpired preauthorized holds) are purged at the end of each usage accumulation period at the time specified by the code in the FIELD CUTOVER field on IDF screen 3. If multiple BASE24 products are enabled and using the UAF, they all must cut over at the same time.
- With a persistent UAF, records are cleaned up instead of being purged. Each UAF record is read and examined. If any amounts in the base segment, ATM segment, or POS segment have expired, the values are reset. The expiration date and time are calculated using the BEGINNING DATE field on IDF screen 4, LAST RESET DATE field on UAF screen 1, LAST USED DATE field for the BASE24-atm product on UAF screen 4, and LAST USED DATE field for the BASE24-pos product on UAF screen 6. A current timestamp is taken from the system. Standard and enhanced preauthorization hold data is examined to see if it has expired and a hold data expired flag is set appropriately. If all usage amounts have been reset and there are no valid preauthorization hold flags, a delete record sequence is performed. If any usage amount or preauthorization hold flag is updated, the record is updated; otherwise, the record is skipped. The following initiation options are available.
 - With Settlement support, the UAF cleanup function can be initiated automatically by the Settlement process at Logical Network cutover or at Midnight Cutover, initiated automatically based on a timer set in the Extract Configuration File (ECF), or initiated manually by an operator.
 - Without Settlement support, the UAF cleanup function can be initiated either manually by an operator or automatically based on a timer set in the ECF.

Store-and-Forward Processing

When communications to a host system or an interchange are down for maintenance or other processing needs, the BASE24-pos system can continue to process transactions, storing them in a data file to be transmitted when communications are reestablished. This function is known as store-and-forward processing. The data file used in this processing is the Store-and-Forward File (SAF).

Interchange Store-and-Forward Processing

When communications are reestablished with an interchange, the BASE24-pos system can forward the stored transactions to the interchange, interspersing the stored transactions with live transactions. Some interchanges also provide the option of forwarding the stored transactions to the interchange consecutively. This option, however, is specific to particular interchanges.

The maximum number of times the Interchange Interface process should attempt to transmit a transaction from an interchange SAF to the interchange before the transaction is logged to the error log and deleted from the SAF is specified in the MAX SAF RETRY field on screen 3 of the Interchange Configuration File (ICF) or Enhanced Interchange Configuration File (ICFE).

For more information on store-and-forward processing to the interchange, refer to the appropriate interchange documentation or the ***BASE24 Base Files Maintenance Manual***.

Host Store-and-Forward Processing

When communications are reestablished with a host, the BASE24-pos system can forward the stored transactions to the host by interspersing the stored transactions with live transactions or by sending the stored transactions consecutively. The method used to send stored transactions to the host is specified in the STORE AND FORWARD PROCESSING FLAGS field on screen 1 of the Host Configuration File (HCF).

HCF screen 1 also contains other fields used for store-and-forward processing with the host. The MAXIMUM OUTSTANDING SAFS field is used to specify the maximum number of store-and-forward messages that can be outstanding to a multithreaded host at one time. The STORE AND FORWARD field is used to specify the number of seconds the ISO Host Interface process waits for an

acknowledgment after transmitting a stored message to the DPC. The MAX SAF RETRY field is used to specify the number of times the BASE24-pos ISO Host Interface process should attempt to send the same store-and-forward message before logging it to the error log and deleting it from the Store-and-Forward File (SAF).

For detailed information on store-and-forward processing to the host system, refer to the ***BASE24 ISO Host Interface Manual***.

Second Card Processing

Institutions can issue two cards with the same primary account number (PAN) and member number. The only way to differentiate between the two cards is by using the expiration date information. This is not the same as issuing multiple cards with the same card status and expiration date information. An example is when an institution reissues a card shortly before the existing card expires.

The BASE24-pos product maintains card status and expiration date information in the Cardholder Authorization File (CAF), which means the institution must be using the Positive Authorization method or the Positive Balance Authorization method to perform second card processing. The PAN and the member number form the primary key to the CAF, so a single CAF record must be able to have one or two cards associated with it.

A single CAF record can hold two card expiration dates. Similarly, as the card status can be different for each card (e.g., one is stolen and the other is open), a single CAF record can hold two card status values.

When the CAF record has one card, the Device Handler/Router/Authorization process uses the card status set in the CARD STATUS field on CAF screen 1 and the expiration date set in the EXPIRATION DATE field on CAF screen 2. When the CAF record has two cards, the Device Handler/Router/Authorization process uses the information on CAF screens 1 and 2 for the first card and information set in the CARD STATUS and EXPIRATION DATE fields on CAF screen 7 for the second card.

If a CAF record has two cards associated with it, the Device Handler/Router/Authorization process first checks the expiration date received with the transaction. If the expiration date matches the value in the EXPIRATION DATE field on CAF screen 2, the Device Handler/Router/Authorization process uses that expiration date and the value set in the CARD STATUS field on CAF screen 1. If the expiration date matches the value in the EXPIRATION DATE field on CAF screen 7, the Device Handler/Router/Authorization process uses the card status set in the CARD STATUS field on CAF screen 7.

If no expiration date is received with the transaction, the Device Handler/Router/Authorization process uses the date in the EXPIRATION DATE field on CAF screen 2 and the value set in the CARD STATUS field on CAF screen 1.

Expiration Date Checking

If the expiration date is checked, either as a prescreening check or during transaction authorization, the following fields control the processing performed:

- The EXP CHECK DISP field on IDF screen 2 controls the response code (return card or retain card) the Device Handler/Router/Authorization process returns to the merchant when a card fails the expiration date check.
- The EXP CHECK TYPE field on CPF screen 1 controls whether to check the expiration date and tells where the expiration date is located. Options are do not check, use Track 1 or Track 2, and use the CAF.

The Device Handler/Router/Authorization process can use the CAF option only if the institution is using the Positive Authorization method or the Positive Balance Authorization method.

- The TRACK 1 SETTINGS EXP DATE and TRACK 2 SETTINGS EXP DATE fields on CPF screen 2 provide the location of the expiration date on card tracks.
- The EXPIRATION DATE PROCESSING FLAG field on CPF screen 3 indicates whether an expiration date is required in transaction requests.
- The EXPIRATION DATE COMPARISON field on CPF screen 3 indicates whether the expiration date on Track 1 or Track 2 of the card is compared against the expiration date in the CAF.
- The EXP CHECK TYPE field on CPF screen 1 must be set to the value 1 (use Track 1 or Track 2) if expiration date comparison processing is performed.

The CAF can contain card expiration dates in the EXPIRATION DATE fields on screens 2 and 7. Expiration date comparison processing must be performed for the Device Handler/Router/Authorization process to consider the second card expiration date.

- The CARD VALIDITY PERIOD field on CPF screen 3 indicates the maximum number of years into the future a card expiration date is valid.
- The RECURRING PAYMENT EXP CHECK TYPE field on CPF screen 8 indicates the type of card expiration date check to use for recurring payment transactions. Options are do not check, use Track 1 or Track 2, use the CAF, and use the setting in the EXP CHECK TYPE field on CPF screen 1.

Preauthorized Holds Processing

A preauthorized hold transaction requests that funds be placed on hold against a future purchase or payment. These funds are not available to the cardholder until the preauthorized hold request expires or the BASE24 system receives a preauthorized completion transaction indicating that the purchase or payment for which funds were being held was completed as authorized. The BASE24-pos product offers standard and enhanced preauthorized holds processing.

A preauthorized completion transaction can be initiated as a request (0200 message) or a force post (0220 message). The BASE24-pos product can reject a preauthorized completion request that does not meet the approval requirements, but cannot reject a preauthorized completion force post transaction. However, the BASE24-pos product can be configured to generate a chargeback (0402 message) following a force post preauthorization completion transaction that does not meet the approval requirements.

- In standard preauthorized holds processing, the BASE24-pos Router/Authorization module processes a preauthorization completion request as a purchase transaction. If the hold can be found and has not expired, the BASE24-pos Router/Authorization module deletes the hold from the database. If the hold is not found or has expired, the BASE24-pos Router/Authorization module can be configured to deny the request.
- In standard preauthorized holds processing, the BASE24-pos Router/Authorization module processes a preauthorization completion force post as a purchase transaction. If the hold can be found and has not expired, the BASE24-pos Router/Authorization module deletes the hold from the database. If the hold is not found or has expired, the BASE24-pos Router/Authorization module can be configured to generate a chargeback.
- In enhanced preauthorized holds processing, the BASE24-pos Router/Authorization module can decline a preauthorization completion request if any of the following processing requirements are not met.
 - ❑ The preauthorization completion amount must not exceed the preauthorization hold amount.
 - ❑ The preauthorization hold must not have expired.
 - ❑ The preauthorization hold completion timer must not have expired.

- In enhanced preauthorized holds processing, the BASE24-pos Router/Authorization module can generate a chargeback following the preauthorization completion force post if any of the following processing requirements are not met.
 - ❑ The preauthorization completion amount must not exceed the preauthorization hold amount.
 - ❑ The preauthorization hold must not have expired.
 - ❑ The preauthorization hold completion timer must not have expired.

Establishing Preauthorized Holds Processing

The authorization files that contain information about preauthorized holds depend on the authorization method or methods that the institution is using. The files that can contain preauthorized holds information are the Positive Balance File (PBF), the Cardholder Authorization File (CAF), and the Usage Accumulation File (UAF). The preauthorized holds information in each of these files, if it exists, is contained in its own file segment. The PBF contains standard preauthorized holds segments, while the CAF and UAF contain standard and enhanced preauthorized holds segments. To indicate that these files may have preauthorized holds segments, the institution must set file segment indicators on screen 5 of the Institution Definition File (IDF). The PRE-AUTH field is the indicator for the standard preauthorized holds segment and the ENHNC-PRE-AUTH field is the indicator for the enhanced preauthorized holds segment. If no preauthorized holds exist, these file segments do not exist.

Also in the IDF, the institution can indicate which authorization file (i.e., CAF, PBF, or UAF) contains preauthorized holds information that the BASE24 system must update when it receives a preauthorized purchase transaction or a reversal of a preauthorized purchase completion transaction. The institution indicates in the HOLDS LVL (DESCR) field on IDF screen 16 which authorization files contain the preauthorized holds information that must be updated. The institution indicates in the AUTH TYPE (DESCR) field on IDF screen 16 which authorization file contains the preauthorized holds information to be checked when authorizing transactions.

The standard and enhanced preauthorized holds segments in the authorization file records each can contain information for up to 10 preauthorized holds. However, an institution may want to limit the number of preauthorized holds that the Authorization module can add to these segments to less than 10 for some card

prefixes. Institutions can impose this limit using the MAXIMUM NUMBER OF PRE-AUTH-HOLDS field on screen 7 of the Card Prefix File (CPF). The same limit applies to both segments.

Note: The MAXIMUM NUMBER OF PRE-AUTH-HOLDS field limits only the number of preauthorized holds that can be added to each of the preauthorized holds segments by the Authorization module. The number of preauthorized holds added to each the preauthorized holds segments by the Refresh process and the BASE24-from host maintenance product are not limited by this field.

The ENHANCED PREAUTH field on PTD screen 6 controls which preauthorization holds segment is used to capture transaction information for transactions performed at a specific terminal. Options include the following:

- Standard segment for all transactions (default value)
- Enhanced segment for noncredit account transactions and standard segment for credit account transactions
- Enhanced segment for credit account transactions and standard segment for noncredit account transactions
- Enhanced segment for debit and credit account transactions

The PREAUTH CHARGEBACKS field on PTD screen 6 controls the processing performed when a preauthorization hold completion transaction does not meet all of the processing requirements. If this flag is set, the BASE24-pos Router/Authorization module denies a request (0200) message or issues a chargeback (0402) message for a force post (0220) message. These requirements are discussed at the start of the “Preauthorized Holds Processing” topic. This setting applies to standard and enhanced preauthorization holds.

If the institution does not indicate which authorization file contains preauthorized holds information, the BASE24-pos product still can perform preauthorization transactions without putting any funds on hold. For example, a preauthorized transaction can be used to verify that the account holds sufficient funds but those funds cannot be placed on hold.

Maintaining the Authorization Files

The standard and enhanced preauthorized holds information in the CAF, PBF, and UAF is organized as shown in the following table. Operators can access the appropriate screen to obtain information about a preauthorized hold or to cancel a preauthorized hold.

File	Screen	Preauth Segment	Available Fields
CAF	5	Standard	HOLD STATUS, TRAN NUM, ACCT TYP, AMOUNT, ACCOUNT NUMBER
CAF	6	Enhanced	HOLD STATUS, APPRV CODE, ACCT TYP, ACCT NUMBER, SEQUENCE NUMBER, HOLD AMOUNT
PBF	3	Standard	HOLD STATUS, TRANSACTION NUMBER, AMOUNT
UAF	2	Standard	HOLD STATUS, TRANSACTION NUMBER, AMOUNT
UAF	3	Enhanced	HOLD STATUS, APPROVAL CODE, ACCOUNT TYPE, SEQUENCE NUMBER, HOLD AMOUNT

For standard preauthorized holds, the BASE24 system uses the transaction number, contained in the TRAN NUM and TRANSACTION NUMBER fields, to match the preauthorized completion to the preauthorized hold request.

For enhanced preauthorized holds, the BASE24 system uses the approval code, contained in the APPROVAL CODE field, to match the preauthorized completion to the preauthorized hold request.

The HOLD STATUS fields on these screens indicates whether a preauthorized hold is currently in effect or has expired, and the AMOUNT and HOLD AMOUNT fields contain the amount of a preauthorized hold.

Controlling the Length of a Preauthorized Hold

For standard and enhanced preauthorized hold processing, the card issuer controls the length of a preauthorized hold. A preauthorized hold is no longer considered when this time limit is reached, regardless of the completion timer that is part of enhanced preauthorized hold processing.

- When the preauthorized hold request is received from a terminal, either the terminal or the BASE24 system determines how long the preauthorized hold will remain in effect. For standard preauthorized holds, the BASE24 system uses the hold time length specified in the PRE-AUTH HOLD TIME field on screen 3 of the BASE24-pos Terminal Data files (PTD). For enhanced preauthorized holds, the BASE24 system uses the hold time specified in the ENHNC-PRE-AUTH-HLD-STORE-TIME param in the LCONF.
- When the preauthorized hold request is received from an ISO host, the hold time length is included in the external message.
- When the request is received from a BIC ISO co-network, the hold time length is also included in the external message. If the hold time length is not included in the external message from the co-network, the BASE24 system uses the hold time length specified in the PRE-AUTH HOLD TIME field on screen 11 of the Interchange Configuration File (ICF) or Enhanced Interchange Configuration File (ICFE).

For enhanced preauthorized hold processing, the card issuer can control how much time is allowed for a completion message to get from the terminal to the transaction acquirer and how long the completion message is stored in the UAF or CAF.

Determining the Amount of a Preauthorized Hold

For standard and enhanced preauthorized hold processing, how the amount of a preauthorized hold is determined depends on the transaction originator and whether a transaction amount is available.

- When a preauthorized hold request contains a hold amount, that amount is used for the preauthorized hold.
- When the preauthorized hold request originates at a terminal and an amount is not entered, the BASE24 system uses the amount specified in the DEFAULT PRE-AUTH AMOUNT field on screen 3 of the BASE24-pos Terminal Data files (PTD).

- When the preauthorized hold request originates at an ISO host and an amount is not entered, the BASE24 system assumes an amount of zero; no default amount is available.

Authorizing for a Different Amount Than Requested

Lesser Amounts

The BASE24-pos product provides authorizers with the option of approving the following requests that would normally be denied because they exceed the funds available by enabling the requests to be authorized for a lesser amount.

- Normal purchase (transaction code 10)
- Preauthorization of purchase (transaction code 11)
- Mail or telephone purchase (transaction code 13)
- Cash advance (transaction code 15)
- Purchase with cash back (transaction code 18)

When a request is processed and the funds are not available, the authorizer can approve the request by using a lesser amount than was originally requested. The BASE24-pos system determines this lesser amount by calculating the maximum amount that can be authorized without exceeding any transaction, card, or usage limits.

The AUTH PARTIAL AMT field on PTD screen 6 controls whether transactions performed at a device attached to a BASE24-pos Standard POS Device Handler module can be approved for a lesser amount. This function is also controlled by the PARTIAL AMOUNT SUPPORT field on CPF Screen 6. Additionally, this function can also be controlled by the AUTH PARTIAL AMOUNT field on PRDF Screen 4, if the transaction originates from a Host, and the Host message doesn't specify whether partial authorizations are allowed.

When a request comes in to the BASE24-pos system from a device or interchange that supports authorizing transactions for a lesser amount, the Interchange Interface process or the Device Handler module sets the PRE-AUTH-OPT field in the PSTM to indicate that the transaction originator supports authorizing requests for a lesser amount. This is also applicable with transactions originating from a Host and received through the HISO Interface. If the transaction originator does not support requests for a lesser amount, the value in this field is not changed.

The transaction is then authorized by the appropriate authorizer according to the configured authorization method. Requests can be authorized for a lesser amount using any authorization method.

Greater Amounts

The BASE24-pos product provides authorizers with the option of approving the following requests by authorizing a greater amount than requested.

- Normal purchase (transaction code 10)
- Preauthorization of purchase (transaction code 11)
- Mail or telephone purchase (transaction code 13)

When a request is processed, the authorizer can approve the request by using a greater amount than requested. The BASE24-pos system sets the transaction amount to a maximum value (from the PARTIAL AUTH LIMIT field on CPF Screen 8). The transaction is then approved for this amount, unless a limit or balance would be exceeded. If this happens, the BASE24-pos system reduces the transaction amount by calculating the maximum amount that can be authorized without exceeding any transaction, card, or usage limits, or the available credit in the account."

The AUTH PARTIAL AMT field on PTD screen 6 controls whether transactions performed at a device attached to a BASE24-pos Standard POS Device Handler module can be approved for a greater amount. This function is also controlled by the PARTIAL AMOUNT SUPPORT field on CPF Screen 6. Additionally, this function can also be controlled by the AUTH PARTIAL AMOUNT field on PRDF Screen 4, if the transaction originates from a Host, and the Host message doesn't specify whether partial authorizations are allowed.

When a request comes in to the BASE24-pos system from a device or interchange that supports authorizing transactions for a greater amount, the Interchange Interface process or the Device Handler module sets the PRE-AUTH-OPT field in the PSTM to indicate that the transaction originator supports authorizing requests for a greater amount. This is also applicable with transactions originating from a Host and received through the HISO Interface. If the transaction originator does not support requests for a greater amount, the value in this field is not changed.

The transaction is then authorized by the appropriate authorizer according to the configured authorization method. Requests can be authorized for a greater amount using any authorization method.

Partial Authorization Routing

The PARTIAL AUTH ROUTING field on Screen 8 of the CPF is used when the transaction may be authorized for a greater amount, and where a Host is involved in the transaction (Level 1 or Level 3 authorization).

The Host may have any one of the following capabilities:

- Support authorization for both a lesser amount and a greater amount.
- Support authorization for a lesser amount, but not for a greater amount.
- Support authorization for neither a lesser amount nor a greater amount.

The PARTIAL AUTH ROUTING field on Screen 8 of the CPF allows each of these capabilities to be supported.

When the Host supports authorization for both a lesser amount and a greater amount, the PARTIAL AUTH ROUTING field can be set to '0' (standard), for which standard routing is performed. There is no need to modify any logic because the Host can perform authorization for a greater amount.

Note: The AST limit checking is performed using the PARTIAL AUTH LIMIT field on Screen 8 of the CPF to provide a better estimate of the transaction value than the requested (nominal) amount.

When the Host supports authorization for a lesser amount, but not for a greater amount, the PARTIAL AUTH ROUTING field can be set to '1' (auth limit), for which the transaction amount is set to the PARTIAL AUTH LIMIT field on Screen 8 of the CPF. The Host needs only perform authorization for a lesser amount because BASE24 has notified the Host of the maximum authorization amount.

Note: The AST limit checking is performed using the PARTIAL AUTH LIMIT field on Screen 8 of the CPF to provides a better estimate of the transaction value than the requested (nominal) amount.

When the Host supports authorization for neither a lesser amount nor a greater amount, the PARTIAL AUTH ROUTING field can be set to '2' (under floor), for which AST limit checking is performed using a transaction amount of zero. Thus, "under floor" routing is performed, and the "under floor" authorizer is likely to be BASE24-pos. In this case, BASE24 can authorize the transaction for a greater amount, whereas the Host would not be able to do so.

Limit Checking

BASE24 can be configured to perform limit checking as a pre-screen function. If the issuer wants BASE24 to perform this function as a meaningful check on transactions that can be authorized for a greater amount, then the PARTIAL AUTH ROUTING field should be set to '1', even if the Host supports authorization for a greater amount. When the PARTIAL AUTH ROUTING field is set to '1', BASE24 will perform all limit checking against the PARTIAL AUTH LIMIT field on Screen 8 of the CPF.

If the PARTIAL AUTH ROUTING field is set to '0', then BASE24 will perform all limit checking against the requested amount, which is likely to be a nominal amount, such as \$1.

If the PARTIAL AUTH ROUTING field is set to '2', no limit checking is performed as a pre-screen function since all authorization processing is performed by BASE24-pos.

Chargeback and Representment Processing

A chargeback is a completed transaction that a cardholder has disputed by contacting the institution that issued the cardholder's debit or credit card and the account statement on which the transaction appears. The acquiring institution or retailer must research the transaction to determine if, in fact, the cardholder performed the original transaction and is responsible for paying the transaction amount.

For example, if a monthly account statement indicates that the cardholder has been charged twice for the same transaction, the cardholder can call the card-issuing institution to dispute the statement. The card-issuing institution then charges the transaction back to the retailer or the acquiring institution, thus placing the responsibility of payment back on the retailer.

A representment acts as a reversal of a chargeback. When the retailer or acquiring institution researches a chargeback, they may determine that the cardholder is responsible for paying the amount of the original transaction under dispute. Thus, the retailer or acquiring institution issues a representment, which again charges the cardholder for the original transaction.

BASE24-pos Chargeback Support

Institutions must handle the research and transfer of funds for chargebacks outside of the BASE24-pos product. However, the BASE24-pos product does allow BASE24-pos files to reflect the disputed transaction processing that institutions are handling outside of the BASE24-pos product. Because chargebacks are reflected in BASE24-pos files, the chargeback information appears on BASE24-pos reports. Institutions can then track disputed transactions using these BASE24-pos reports.

The BASE24-pos product supports the cardholder side of a chargeback by routing the chargeback transaction using the cardholder account number and optionally updating the cardholder file information. The BASE24-pos product does not notify retailers of chargebacks. Therefore, the issuing institution to which the cardholder reported the disputed transactions must notify the retailer or acquiring institution by transmitting appropriate documentation to the retailer or acquiring institution.

However, the BASE24-pos product does log chargebacks to the POS Transaction Log File (PTLF). The PTLF records then appear on reports that the card issuer can use when settling disputes between the retailer and the cardholder.

Although notifying retailers of chargebacks is not a function of the BASE24-pos product, in situations where the cardholder and retailer are within the same logical network, the retailer information would also be part of the chargeback transaction and, as such, would be part of the PTLF records. In that case, the retailer can also use the reports from the PTLF records to research disputed transactions.

The movement of funds as the result of a chargeback, like all other aspects of settlement, is the institution's responsibility. The BASE24-pos product reports the activity and provides extractable records for processing electronically. Chargeback activity is reflected on the Daily Disputed Transactions Report (B24POS05). Refer to the ***BASE24-pos Settlement and Reporting Manual*** for more information about this report.

A chargeback request enters the BASE24-pos system as a 0402 message, and the chargeback response is processed as a 0412. The 0412 message is always logged to the PTLF and contains one of the following response codes indicating the action the BASE24-pos product has taken.

- 001 = The acquirer was found and notified (approval without balances).
- 955 = The issuer was found and the customer files were updated.
- 956 = The acquirer was not found and, therefore, was not notified.
- 957 = The transaction originated on the BASE24-pos system but the card was not-on-us; therefore the issuer was not notified.
- 958 = The BASE24-pos system could not route the chargeback or could not find the issuer because of a bad response or bad prefix.
- 959 = Chargebacks are not supported. This response code is not currently used.
- 960 = The issuer was found, but the cardholder files were not updated.
- 961 = The BASE24-pos system tried to route the chargeback and failed. Files were not updated and no acquirer was found.
- 962 = The BASE24-pos system was unable to route the not-on-us transaction.

The POS-CHRGBCK-NO-RESP-ORIG param in the LCONF indicates whether the BASE24-pos Router module sends the 0412 message to the transaction originator when it receives a chargeback request (0402) message. If the value set in this param is the same as the value of the TRAN.ORIG field of the PSTM, the Router module does not send response to the transaction originator.

Chargebacks can originate from one of the following sources:

- Host
- CRT Administration terminal
- BIC Co-network

- Interchange
- BASE24-pos Router/Authorization module

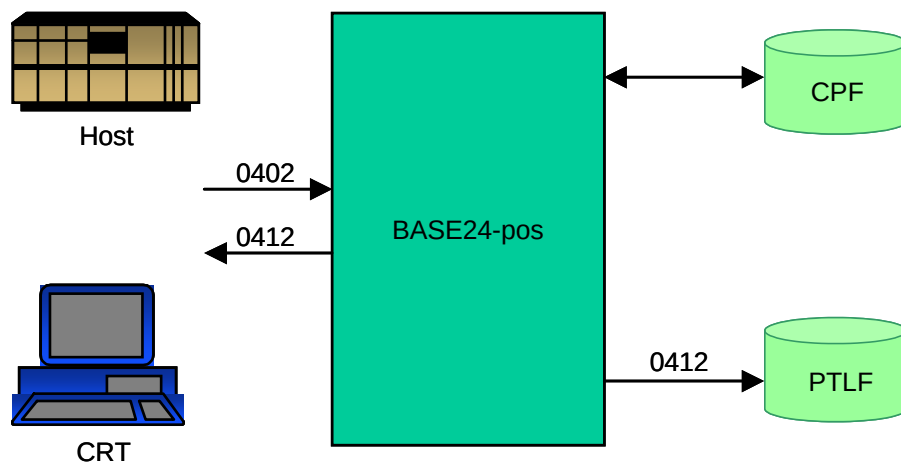
Chargebacks Originating from the Host or a CRT Administration Terminal

When a chargeback originates at a host or a CRT Administration terminal, the BASE24-pos product acts on the chargeback only if it recognizes the card prefix of the card involved.

If the card prefix is in the CPF, the Authorization module assigns the transaction a response code, creates a 0412 response, and logs the transaction to the PTLF. If both the issuer and acquirer are found, two transactions are logged to the PTLF. If only the issuer is found, the transaction is logged to the PTLF and the BASE24-pos product routes the transaction using SPROUTE to find the acquirer. (SPROUTE routing is explained in section 4.)

If a CRT Administration terminal originated the chargeback, the terminal also displays a message indicating the response code as assigned by the Authorization module. The host can then take the appropriate action.

The following illustration is an overview of the processing that occurs when a chargeback originates at a host or a CRT Administration terminal.



Chargebacks Originating from a BIC Co-network

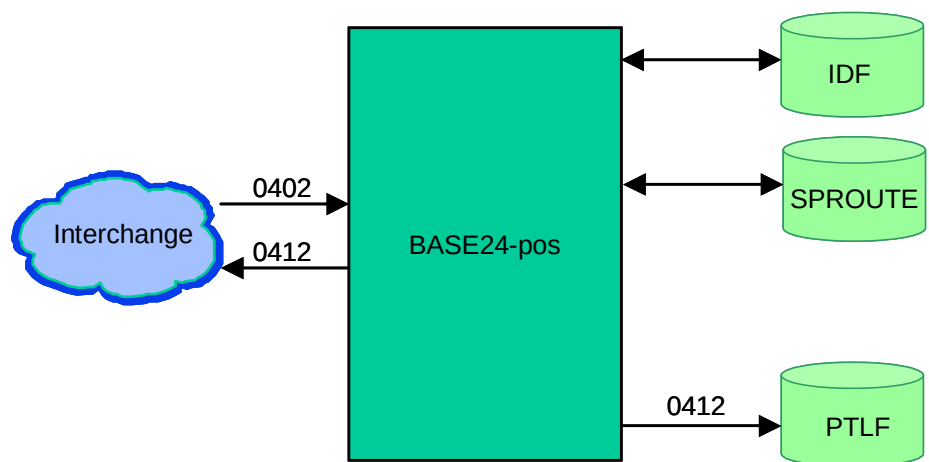
ISO BIC co-networks send chargebacks to the BASE24-pos system in an attempt to locate the acquirer. The BASE24-pos product accepts chargebacks from a BIC co-network if the chargeback has a response code of 955 or 000. These response codes indicate that the co-network has updated the issuing institution's cardholder record but is attempting to locate the acquirer in the BASE24-pos system.

For chargebacks from a co-network, the Device Handler/Router/Authorization process does not check the CPF, but instead uses the routing transit number in the transaction to check IDF records. If a match exists, the acquiring institution is in the IDF and the Device Handler/Router/Authorization process logs the transaction to the PTLF. The PTLF records then appear on BASE24-pos reports that the retailer can use to track disputed transactions.

If a match does not exist, the acquiring institution is not in the IDF. If the response code is 955, the Device Handler/Router/Authorization process then uses SPROUTE to route the transaction using the routing transit number. (SPROUTE routing is explained in section 4.)

The following illustration is an overview of the processing that occurs when a chargeback originates at a BIC co-network.

Note: The BASE24-pos system may process chargebacks from other interchanges in a similar fashion as those chargebacks it receives from a BIC co-network. Refer to the documentation for the specific interchange for more information.



Chargebacks Originating from an Interchange

The BASE24-pos system accepts chargebacks from an interchange if the chargeback has a response code of 955 or 000. These response codes indicate that the interchange has updated the issuing institution's cardholder record. However, unlike when the BASE24-pos system receives a chargeback from a BIC co-network, the BASE24-pos product simply logs the chargeback to the Interchange Log File (ILF) and drops the transaction. Chargebacks must then be researched and processed manually.

Note: The BASE24-pos system may process chargebacks from other interchanges in a similar fashion as those chargebacks it receives from a BIC co-network. Refer to the documentation for the specific interchange for more information.

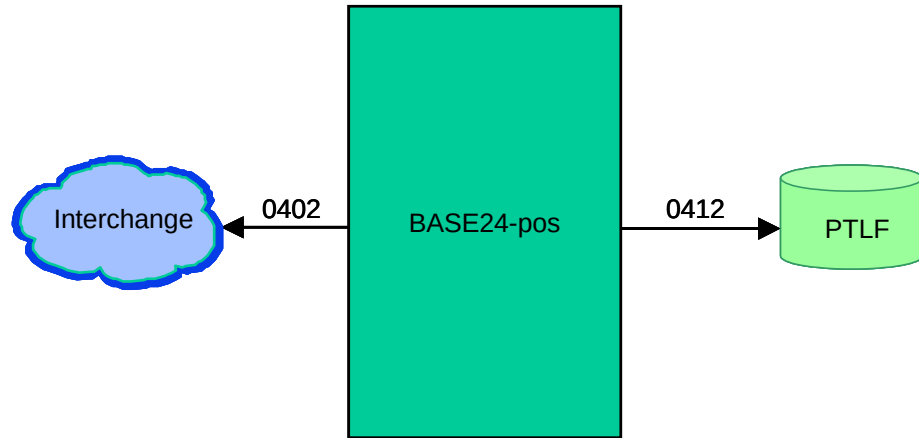
Chargebacks Originating from BASE24-pos Device Handler/Router/Authorization Module

When the BASE24-pos Device Handler/Router/Authorization module serves as an issuer, it can originate a chargeback when it receives a force post (0220 message) preauthorization completion transaction that does not meet the approval requirements.

- In standard preauthorized holds processing, a chargeback can be generated if the hold has expired.
- In enhanced preauthorized holds processing, a chargeback can be generated for any of the following conditions:
 - ❑ The preauthorization hold has expired
 - ❑ The preauthorization hold completion timer has expired
 - ❑ The preauthorization completion amount exceeds the preauthorization hold amount

The PREAUTH CHARGEBACKS field on PTD screen 6 controls whether a chargeback is issued for a force post (0220 message) preauthorization hold completion transaction that does not meet all of the processing requirements. The BASE24-pos Standard POS Device Handler (SPDH) module places the value from this PTD field in the PRE-AUTH-CHRGBK field of the POS Merchant token.

The Router/Authorization module sends a 0402 request to the interchange and logs a 0412 response to the PTLF.



BASE24-pos Representment Support

A representment acts as a reversal of a chargeback. When the retailer or acquiring institution researches a chargeback, they may determine that the cardholder is responsible for the original transaction under dispute. Thus the retailer or acquiring institution issues a representment, which again places the responsibility for the original transaction on the cardholder.

Representments enter the BASE24-pos system as 0200 messages and are processed as requests. This means that a representment can be declined if the transaction does not meet the processing requirements of the BASE24-pos system or the host.

Representments appear with chargebacks on the Daily Disputed Transactions Report (B24POS05). On this report, representments and chargebacks are distinguished from one another only by the point of view of the institution (e.g., issuing institutions generate chargebacks while acquiring institutions generate representments). Refer to the ***BASE24-pos Settlement and Reporting Manual*** for more information about this report.

Controlling Chargeback and Representment Processing

BASE24-pos processes chargebacks and representments according to the card prefix level. Institutions configure how the BASE24-pos product handles chargebacks and representments using the CHARGEBACK UPDATE and REPRESENTMENT UPDATE fields on screen 7 of the Card Prefix File (CPF).

The Authorization module can log the transaction to the PTLF and update the BASE24 database, or the Authorization module can log the transaction to the PTLF without updating the BASE24 database.

Refer to the ***BASE24 Base Files Maintenance Manual*** for more information on these fields.

Generating Completions to a Host

Completions (0220 messages) notify the host of authorizations made by the BASE24-pos system. The institution controls completion generation at the institution level. For terminal totals transactions, the Device Handler/Router/Authorization process generates completions based on the POS COMPLETIONS REQUIRED fields on screen 19 of the Institution Definition File (IDF). For all other transactions, the Device Handler/Router/Authorization process generates completions based on the COMPLETION REQUIRED TO HOST field on screen 2 of the Issuer Processing Code File (IPCF).

If the POS COMPLETIONS REQUIRED or COMPLETION REQUIRED TO HOST field for a certain transaction indicates that completions are required, the Device Handler/Router/Authorization process generates a completion for each approved 0210 message it processes.

If the POS COMPLETIONS REQUIRED or COMPLETION REQUIRED TO HOST field for a certain transaction indicates that completions are not required, the Device Handler/Router/Authorization process generates completions only when the host is unavailable for authorizing a 0200 message and the Authorization module stands in to approve the transaction. In this circumstance, the Authorization module generates a completion to notify the host of the approval, regardless of the setting in the POS COMPLETIONS REQUIRED or COMPLETION REQUIRED TO HOST field.

Since completions are controlled at the institution level, all DPCs processing for an institution either receive completions or do not receive completions. Completions cannot be turned on or off by specific DPC.

Token Handling

Tokens provide the BASE24-pos product with flexibility in the information carried in its internal messages; logged to its log files; extracted by the Super Extract process; and sent to a host, interchange, or co-network using the ISO Host Interface or ISO BASE24 Interchange (BIC) Interface processes.

A token is a collection of related data which may or may not be needed to process a particular transaction. A token can be made up of one field or a series of fields. If the information is needed to process a transaction, it can be added to the internal message, logged, and extracted. If the information is not needed, the token need not be involved in the transaction at all—carried in the message, logged, or extracted.

The tokens used by the BASE24-pos product are described in detail in the ***BASE24 Tokens Manual***.

BASE24-pos processes use the Token File (TKN) to determine which tokens are logged, extracted, or sent using the ISO Host Interface and ISO BASE24 Interchange (BIC) Interface processes to other entities.

Logging Tokens

TKN screen 2 is used to specify which tokens contained in the internal message are logged to the POS Transaction Log File (PTLF) or any Interchange Log File (ILF). These tokens are arranged in the log record in the order they were added to the internal message by BASE24-pos processes. The option of logging or not logging exists for all BASE24-pos tokens; however, to process reversals correctly, tokens always need to be logged to the ILF.

Retrieving Tokens

The options available for retrieving tokens differ according to BASE24-pos Device Handler module.

Hypercom and POS Standard Device Handler Modules

The TOKEN RETRIEVAL OPTION field on Institution Definition File (IDF) screen 19 indicates whether BASE24-pos Hypercom and POS Standard Device Handler modules include tokens in reversal messages, and if so, whether the token

data is retrieved from the second BASE24-pos Terminal Data Dynamic File—scratch pad (PTDD2) or from the POS Transaction Log File (PTLF). This option can be overridden at the terminal level using the TOKEN RETRIEVAL OPTION field on BASE24-pos Terminal Data files (PTD) screen 2.

Retrieving tokens from the PTLF and appending them to the PSTM is the default setting. In addition, this option impacts transaction processing as follows:

- The request rate to the PTLF is doubled, even if the transaction is not a reversal.
- If the PTLF is unavailable for any reason, such as the primary PTLF being unavailable during backup processing, the Device Handler module forwards the reversal transaction to the Authorization module without any token data. The Authorization module logs the reversal as an exception item that must be reconciled manually.

When one of these Device Handler modules receives a request from a POS device, it creates the TLF token with a field indicating the token reversal processing option. If the option is set to a value of 2 in the TLF token, the Router/Authorization module updates the token with POS Transaction Log File (PTLF) key data when processing the transaction response. The Device Handler module updates device-dependent PTD extended memory with the TLF Token data and writes this data to the BASE24-pos Terminal Data Dynamic File—general data (PTDD1). It can then use this information to retrieve the record from the PTLF during reversal processing. If the token retrieval option is set to a value of 0 or 1, the Router/Authorization module does not update the TLF token.

When one of the above Device Handler modules receives a response from the Router/Authorization module, it checks the token retrieval option in the TLF token. If the option is set to a value of 1, the Device Handler stores token data from the response message in the PTDD2 file. If no record exists in the PTDD2 file for the terminal from which the transaction originated, a new record is created using the terminal ID as the key. If a record already exists for the terminal, the record is updated with the token data from the current message. The Device Handler module can then retrieve the record from the PTDD2 file during reversal processing.

CRT Authorization Device Handler Module

The BASE24-pos CRT Authorization Device Handler module can handle tokens four ways for reversal processing.

- Retrieve tokens from the token buffer space in the Transaction Queue File (TQF). This buffer space is less than 280 bytes per transaction, which may or may not be sufficient for all of the tokens used in a transaction.
- Retrieve tokens from the Transaction Queue File Token File (TQFT).
- Retrieve tokens from the POS Transaction Log File (PTLF). Only tokens that have been logged to the PTLF are available.
- Not retrieve any token information. Token data is not written to the TQF or TQFT.

One assign and two params in the LCONF control how the BASE24-pos CRT Authorization Device Handler module handles tokens. The CRT-AUTH-TQFT assign identifies the TQFT, the CRT-AUTH-MAX-TQFTS param indicates the maximum number of TQFT records the BASE24-pos CRT Authorization Device Handler module can store in memory at one time, and the CRT-AUTH-TKN-RETRV-OPT param indicates where tokens from approved responses are stored to be appended to reversals.

When the params are not defined, tokens are stored in the TQF if the TQFT assign is not defined. The CRT-AUTH-MAX-TQFTS param is an optional param with a default value of 75. The CRT-AUTH-TKN-RETRV-OPT param has the following valid values to indicate where the tokens are stored.

- 0 = No token data is stored in or retrieved from the TQF or TQFT.
- 1 = Token data is stored in and retrieved from the TQFT if the CRT-AUTH-TQFT assign exists or the TQF if the CRT-AUTH-TQFT assign does not exist. This is the default value if this param is missing or is invalid.
- 2 = Token data is retrieved from the PTLF record associated with the transaction.

Extracting Tokens

TKN screen 3 is used to specify which tokens in the PTLF or ILF records are extracted by the Super Extract process. This screen is also used to specify the order in which these tokens are arranged within each PTLF or ILF extract record. The option of extracting or not extracting exists for all BASE24-pos tokens; however, a token must be logged to the PTLF or ILF in order to be extracted.

Sending Tokens to a Host or Co-Network

TKN screen 4 is used to specify which tokens contained in the internal message are sent to a host using the ISO Host Interface process or to a co-network using the BIC ISO Interface process. This screen is also used to specify the order in which the tokens are carried in element P-63 (BASE24-pos Additional Data) of the BASE24 external message. If tokens are to be sent in the external message, data element P-63 must be set to conditional or mandatory. Refer to the ***BASE24 External Message Manual*** for detailed descriptions of the BASE24 external message.

Carrying Tokens in the BASE24-pos Internal Message

Information in the TKN does not affect the tokens carried in the internal message. Tokens are added to the internal message if the token data is present and the token is required for processing. Tokens are then used in the internal message throughout the transaction path.

Address Verification Processing

Optionally, transaction processing can include address verification. Address verification assists in preventing the fraudulent use of credit and debit cards when other means of identification are not readily available (i.e., during mail or phone order purchases).

Cardholder addresses are stored in a host database, an interchange database, or a BASE24-pos database. The BASE24-pos database is located in an Address Verification segment of the Cardholder Authorization File (CAF). Addresses can be stored in the Address Verification segment of the CAF if BASE24 Address Verification, an add-on BASE24-pos product, is being used. When this segment of the CAF is to be used, the appropriate file segment indicator for address verification must be set on screen 5 of the Institution Definition File (IDF).

When a cardholder makes a purchase, the retailer enters the cardholder's address at the terminal. If the address entered does not match the cardholder address in the database, the retailer has the option to decline the transaction.

When address verification is performed using an interchange database or a host database, either the interchange or the host compares the addresses, respectively. When address verification is performed on the BASE24-pos system, the Authorization module compares the addresses using an algorithm. The Authorization module can also perform address verification as a part of transaction screening before sending the transaction elsewhere to be authorized.

Refer to the ***BASE24-pos Address Verification Manual*** for more information.

4: BASE24-pos Routing and Authorization

This section summarizes the processing the BASE24-pos product performs during transaction routing and authorization. The information should be used with section 3, which describes the controls available for the transaction processing performed by the BASE24-pos product. This section includes information on the following topics:

- Router and Authorization modules
- AST routing and authorization selection using the Authorization Selection Table (AST) and Routing Table File (RTBL)
- SPROUTE routing using the SPROUTE File
- IDF routing and authorization using the Institution Definition File (IDF)
- Routing for check authorization transactions
- Enhanced reversal routing
- Approval code generation
- Referral code generation

Note: This section does not address the routing of a transaction by a Device Handler module or Interchange Interface process to an authorization destination specified in the Acquirer Processing Code File (APCF) extended memory table. APCF routing is discussed in sections 2, 3, and 5 of this manual.

Router and Authorization Modules Overview

The following is an overview of the routing and authorization functions performed by the BASE24-pos Router and Authorization modules. These modules are bound together with a Device Handler module and are primarily responsible for the routing and authorization performed by the BASE24-pos system.

Router Module Processing

The Router module receives messages and routes them to hosts, interchanges, a Transaction Context Manager process, Router modules in other logical networks, or its bound Authorization module. The Router module can receive messages from its bound Device Handler module, an ISO Host Interface process, an Interchange Interface process, a Transaction Context Manager process, or another logical network.

Tasks of the Router Module

When the Router module receives a transaction message, it performs the following tasks:

- Determines whether the message can be routed to an authorizing entity. The Router module determines how to route the transaction according to information from files and the POS Standard Internal Message (PSTM). If the Router module cannot obtain the information it needs to route a transaction, it denies the transaction request.
- Determines the routing parameters to use for the transaction and sets fields in the PSTM based on those parameters. Routing parameters include the primary authorizer, alternate authorizers, override destination, and offline limits. At this point, the Router module can determine an authorizer.
- Sends the message to the appropriate authorizer. If the authorizer is the Authorization module, the Router module passes the message to that module. If the authorizer is an interchange, host, or Router/Authorization process in another logical network, the Router module sends the message to the XPNET process to be routed to the appropriate Interchange Interface, ISO Host Interface, or Router/Authorization process.
- Sends the mobile top-up transactions to the Transaction Context Manager process for split routing.

Determining Authorizers

The Router module ensures a high availability of transaction authorizers through its flexible transaction routing capabilities. The Router module can determine the primary authorizer for the transaction, up to two alternate authorizers, and a default authorization procedure. The alternate authorizers are used only if the primary authorizer is unavailable. When all specified authorizers are unavailable, the Router module's bound Authorization module implements default authorization procedures.

An authorizer can be the BASE24 system for local authorization against the BASE24-pos database, a host, an interchange, or another BASE24 network. In addition, national credit card transactions (Visa, MasterCard, and American Express) can be checked against National Negative Files, which are maintained as part of the BASE24-pos database, before they proceed to another authorizing entity. The National Negative Files can also be used as the sole authorizing check for not-on-us transactions.

The primary and alternate authorizers can differ for each transaction according to service type and issuer code, depending on the preferences of the institution and the retailer involved in the particular transaction. Retailers and their corresponding institutions set floor limits for each card type. For example, a Visa credit card transaction that is over the floor limit can be sent to a host for authorization, whereas transactions that are under the floor limit can be processed by the Router/Authorization process.

Transactions with Recognized Prefixes

To process a transaction, the Router module must locate the prefix of the card used to initiate the transaction in the Card Prefix File (CPF). If the Router module locates a CPF record, the Router module attempts to perform AST routing to determine a primary authorizer and up to two alternate authorizers. The Router module can perform AST routing only if it is able to locate an Authorization Selection Table (AST) record for the card prefix. If the Router module cannot locate an AST record, the Router module passes the transaction to the Authorization module to perform IDF routing. Both AST routing and IDF routing are described later in this section.

The primary authorizer determined by AST routing can be overridden by the destination specified in the PRDF record for the retailer associated with the transaction. If the PRDF record contains an override destination, the transaction is sent there instead of to the primary authorizer determined in the AST routing process.

The Router module can also check the National Negative Card File (NNF) for the card number of the card initiating the transaction. If the card number is in the NNF, the Router module denies the transaction. If the card number is not in the NNF, the Router module attempts to send the transaction to the primary authorizer.

Transactions with Unrecognized Prefixes

If the Router module cannot locate a card prefix in the Card Prefix File (CPF), the transaction must be sent elsewhere for processing.

An example of this might be an interchange cardholder using this institution's terminal. In this case the Device Handler/Router/Authorization process does not have access to the cardholder record and the transaction must be routed elsewhere—to the interchange—for processing. Another example would be a cardholder whose institution belongs to another BASE24 logical network. Here, again, the Device Handler/Router/Authorization process does not have access to the cardholder record (because it is in the database for the other logical network) and, therefore, cannot find the prefix in the CPF.

The Router module routes transactions for which it does not recognize the prefix using the SPROUTE File (SPROUTE), which is a BASE24 configuration file built when the BASE24 system is configured. From SPROUTE, the Router module can derive primary and alternate destinations for transactions for which it does not recognize the prefix.

Mobile Top-Up Transactions

For mobile top-up with cash or mobile top-up with funds transactions, the Router module routes the transaction to the Transaction Context Manager process identified by the POS-SPLIT-TXN-RTG-DEST param in the Logical Network Configuration File (LCONF). The Transaction Context Manager process then identifies the funds portion of the mobile top-up transaction, if applicable, and sends it back to the Router module as a purchase transaction.

Authorization Module Processing

The Authorization module receives transactions from the Router module and authorizes them according to an institution's authorization parameters and controls. These controls are described in section 3. The Authorization module is responsible for selecting appropriate routing and authorization parameters from the Institution Definition File (IDF) Routing Table, which is discussed later in this section. The Authorization module also logs transactions to a POS Transaction Log File (PTLF), which can then be extracted for affecting settlement immediately without waiting for transaction receipts.

AST Routing and Authorization Selection

Using the Authorization Selection Table (AST) and the Routing Table File (RTBL), the Router module determines the appropriate authorizer for each transaction.

To determine the appropriate authorizer, the Router module first accesses the AST. The AST defines authorization routing for each transaction. Depending on card type, issuer code, and retailer group, the AST designates a primary authorizer, up to two alternate authorizers, and a default action in the event that all authorizers specified are unavailable.

The authorizers in the AST are identified by code, not name. For example, the primary authorizer might be identified by the value 05. Institutions actually name authorizers and assign authorizer codes in the RTBL. The RTBL contains the names of authorizers along with their corresponding codes. For example, the value 05 found in the AST might correspond to the authorizer VisaNet.

After accessing the AST to determine routing parameters and authorizer codes, the Router module accesses the RTBL to determine the corresponding authorizer for each code.

Information in the AST

AST information can be broken down into three sets of information: key fields, selection fields, and action fields. The Router module uses the first set, the key fields, to select an AST record. It uses the second set, the selection fields, to select an AST entry within the record. And it uses the third set, the action fields, to determine a course of action for the transaction.

AST Key Fields

The two key values that the Router module uses to select an AST record are the FIID and the retailer ID. These values are contained in the FIID and RETAILER ID fields, respectively, on screens 1, 3, and 5 of the AST. Values can be entered in these fields only on screen 1. The values are then carried over to screens 3 and 5.

AST Selection Fields

The selection values that the Router module uses to locate the correct entry within the AST record include the service and issuer code. These values are contained in the SERVICE and ISSUER fields, respectively, on screens 1, 3, and 5 of the AST.

In addition to the service and issuer codes, several values enable the Router module to define a floor limit for the transaction. These values include online limits for purchases, cash advances, and mail orders and offline limits. These values are contained in the ONLINE PUR (online purchase), ONLINE ADV (online cash advance), ONLINE MO (online mail or telephone order), and OFFLINE LIMIT fields.

AST Action Fields

Several values allow the Router module to define an authorizer or default action for the transaction. Authorizers and default actions can be defined for transactions that are either under the floor limit or over the floor limit. The AST action fields include OVER P (over the floor primary authorizer), OVER A1 (over the floor first alternate authorizer), OVER A2 (over the floor second alternate authorizer), OVER DA (over the floor default action), UNDER P (under the floor primary authorizer), UNDER A1 (under the floor first alternate authorizer), UNDER A2 (under the floor second alternate authorizer), and UNDER DA (under the floor default action). These fields are contained on AST screens 2, 4, and 6.

Information in the RTBL

An institution can designate authorizers and assign authorizer codes in the RTBL according to the institution's needs. However, some authorizers are hard-coded authorizers, meaning the institution cannot designate them. Hard-coded authorizers include AUTHF, AUTHH, NEGN, NEGF, and DFLT. These authorizers must be assigned the following codes:

00 = AUTHF
01 = AUTHH
02 = NEGN
03 = NEGF
14 = DFLT

Each hard-coded authorizer is explained in more detail in the following paragraphs.

AUTHF

When the Router module determines AUTHF to be the authorizer, transactions are authorized using the authorization methods and levels specified in the IDF Routing Table, except when the authorization level specifies that the transaction should be handled online/offline. (The IDF Routing Table is explained in detail later in this section.) AUTHF overrides the online/offline setting and forces the transaction to be processed by the BASE24-pos system without being sent to the host, even if the host is online and available. The authorization method used is the authorization method specified in the IDF Routing Table for the transaction.

When AUTHF is used and the authorization level is online/offline, the BASE24-pos system sends a completion to the host for each financial transaction that it approves if the POS COMPLETIONS REQUIRED field on Institution Data File (IDF) screen 19 or the COMPLETION REQUIRED TO HOST field on Issuer Processing Code File (IPCF) screen 2 for that transaction indicates that the host requires completions. However, if the transaction is a force-post (0220), the BASE24-pos system does not send a completion to the host. This difference in sending completions is intended to offload processing from the host.

Using the AST and the RTBL, AUTHF can be designated as a primary authorizer for under the floor destinations. AUTHF can also be used as an alternate authorizer for both over and under the floor destinations. However, AUTHF cannot be used as an alternate authorizer in the AST if AUTHH is configured as the primary authorizer.

AUTHH

When the Router module determines AUTHH to be an authorizer, transactions are authorized using the authorization methods and levels specified in the IDF Routing Table. (The IDF Routing Table is explained in detail later in this section.) However, when the authorization level is online/offline, the BASE24-pos system authorizes transactions only when the host is unavailable. If the host is available, the BASE24-pos system sends the transactions to the host for authorization. The BASE24-pos system sends completions to the host for the financial transactions that it approves.

Using the AST and the RTBL, AUTHH can be a primary or alternate authorizer and can be used for both over and under the floor destinations. If more than one ISO Host Interface process is configured as an authorizer in the AST, then AUTHH must be specified for the first of these entries (i.e., where the actual destination is

configured in the IDF Routing Table) because the Router/Authorization module must retrieve the DPC number from the IDF Routing Table before routing the transaction to an ISO Host Interface process.

NEGN

When the Router module determines NEGN to be an authorizer, transactions are authorized using the National Negative Card File (NNF). If the card number is found in the NNF, the transaction is approved VIP, denied, or referred based on the card status value in the NNF. If the card number is not found in the NNF, the transaction is approved.

Using the AST and the RTBL, NEGN can be used only as a primary authorizer. It can be used for either over or under the floor destinations.

NEGF

When the Router module determines NEGF to be an authorizer, transactions are authorized using the National Negative Card File (NNF). If the card number is found in the NNF, the transaction can be approved VIP, denied, or referred based on the card status value in the NNF record. If the card number is not found in the NNF, the transaction amount is compared to the offline limit amount in the AST. If the transaction amount is less than or equal to the offline limit amount, the BASE24-pos product takes the default action specified in the AST. If the transaction amount is greater than the offline limit amount, the transaction is denied.

Using the AST and the RTBL, NEGF can be an alternate authorizer for both over and under the floor destinations.

DFLT

When the Router module determines DFLT to be an authorizer, transactions are authorized using the default action (authorize blind, issue a call, or deny) specified in the AST. Under this option, transactions are not routed to an authorizing entity.

This option can be used as a primary authorizer and can be used for either under or over the floor limit. If DFLT is used as a primary authorizer in the AST, alternate authorizers are not allowed.

Alternate Host Routing

It is possible to configure more than one ISO Host Interface process in the AST as a destination for transactions that are under the floor, over the floor, or both.

If more than one ISO Host Interface process is configured in the AST as an under or over the floor destination, then the first of these entries must be specified as AUTHH (i.e., where the actual destination is configured in the IDF Routing Table). If more than one ISO Host Interface process is configured in the AST as an under or over the floor destination, then all of these processes must use an HCF record with the same DPC number configured in the IDF Routing Table.

It is also possible to configure one or more ISO Host Interface processes in conjunction with one or more interchange interface processes in the AST as destinations for transactions that are under the floor, over the floor, or both. However, the Router/Authorization module performs stand-in authorization (as opposed to default authorization) only if an ISO Host Interface process is configured as the last of the routing destinations.

These restrictions are required because the IDF Routing Table is the only location from which the DPC number can be retrieved. Also, the Router/Authorization module performs stand-in authorization only if various PSTM fields are set appropriately, and these values are set only by the ISO Host Interface process.

AST Selection Hierarchy

There are two levels involved in AST selection. First, the Router module must select the correct AST record to use to route the transaction. Then, the Router module must select an entry within the AST record. The entry provides the floor limits that determine the authorizers to which the transaction can be sent.

Record Selection Hierarchy

The Router module selects an AST record by attempting to match the FIID and the retailer ID associated with the transaction with an FIID and retailer ID in an AST record. The Router module looks for a match according to the hierarchy presented

in the table below. An asterisk (*) for the retailer ID applies to all retailers that do not have an AST record and do not participate in a retailer program that has its own AST record.

Preference	FIID	Retailer ID
1	Exact Match	Exact Match
2	Exact Match	Retailer Program
3	Exact Match	*

Note: If the transaction is a check authorization transaction, the routing group from the Check Authorization token in the PSTM is used instead of the retailer ID when using this hierarchy. If the transaction is a check authorization transaction and the routing group in the Check Authorization token is blank, the retailer ID is used in this hierarchy. A match of a retailer program or an asterisk is never used with check authorization transactions.

Record Entry Selection Hierarchy

Once the Router module has located the correct AST record, it must locate the appropriate entry within that record. The Router module determines which entry to use when routing the transaction by attempting to match the service and issuer code associated with the transaction with the service and issuer code entries in the AST record. The Router module looks for a match according to the hierarchy presented in the table below, where an asterisk (*) represents a wildcard character.

Preference	Service	Issuer Code
1	Exact Match	Exact Match
2	Exact Match	*
3	**	Exact Match
4	**	*

When the Router module searches for a match on the service and issuer code, it starts with the first entry in the AST record. The first entry that meets the hierarchical requirements in the table above is the entry that the Router module uses when routing the transaction.

Selecting an AST Record

The Router module selects the AST record to use in routing based on the FIID and the retailer ID values it associates with the transaction. The actual FIID and retailer ID values used depend on several factors:

- The value set in the POS-RTAU-AST-ROUTE param in the Logical Network Configuration File (LCONF). This param could be set to 0 (normal routing) or 1 (enhanced routing).
- The originator of the transaction: an interchange, host, or device connected to the BASE24-pos system.
- Whether or not a Card Prefix File (CPF) record can be found for the card prefix involved in the transaction.

FIID Possibilities

Depending on the situation, the Router module obtains the FIID to use for AST record selection from one of the following sources:

- FIID field on screen 1 of the CPF record for the card prefix involved in the transaction
- TERM-FIID field of the PSTM
- FIID field on screen 1 of the retailer sponsor's PRDF record

Retailer ID Possibilities

Depending on the situation, the Router module uses one of the following as a retailer ID value for AST record selection:

- A specific retailer ID taken from the RETL-ID field in the PSTM.
- The retailer program code taken from the RETAILER PROG field on screen 1 of the retailer sponsor's PRDF record.
- A check routing group—for check guarantee and check verification transactions only—taken from the CHK-RTG-GRP field in the Check Guarantee/Verification token (token 05) in the internal message.
- An asterisk, if an AST record cannot be found with a more specific value used as a retailer ID.

Determining the FIID/Retailer ID Combination to Use for AST Record Selection

The following tables describe how the Router module determines the FIID and Retailer ID combination to use for selecting an AST record in situations where the POS-RTAU-AST-ROUTE param in the LCONF is set to 0 (normal routing) and 1 (enhanced routing), respectively. The first column of each table identifies the originator of the transaction; the second column indicates whether or not a CPF record can be found for the card prefix involved in the transaction; the third column indicates the Router module's preference for using each FIID/Retailer ID key combination. An asterisk (*) for the retailer ID applies to all retailers that do not have an AST record and do not participate in a retailer program that has its own AST record.

Table 1: POS-RTAU-AST-ROUTE Param Set to 0 (Normal Routing)				
Originator	CPF Record Found	Preference	Key Combination	
			FIID	Retailer ID
Interchange	Yes	1	FIID from the CPF	Retailer ID in the PSTM
		2	Same as above	*
	No	1	FIID from the TERM-FIID field in the PSTM	Retailer ID in the PSTM
		2	Same as above	*
Device or Host	Yes	1	FIID from the CPF	Retailer ID in the PSTM
		2	Same as above	Retailer program specified in the PRDF
		3	Same as above	*

Table 1: POS-RTAU-AST-ROUTE Param Set to 0 (Normal Routing)

Originator	CPF Record Found	Preference	Key Combination	
			FIID	Retailer ID
Device or Host <i>continued</i>	No	1	FIID from the TERM-FIID field in the PSTM, if it is not zeros or blanks Otherwise, the FIID from the PRDF is used	Retailer ID in the PSTM
		2	Same as above	Retailer program specified in the PRDF
		3	Same as above	*

Table 2: POS-RTAU-AST-ROUTE Param Set to 1 (Enhanced Routing)

Originator	CPF Record Found	Preference	Key Combination	
			FIID	Retailer ID
Interchange	Yes	1	FIID from the CPF	Retailer ID in the PSTM
		2	Same as above	*
		3	FIID from the TERM-FIID field in the PSTM	Retailer ID in the PSTM
		4	Same as above	*
	No	1	FIID from the TERM-FIID field in the PSTM	Retailer ID in the PSTM
		2	Same as above	*

Table 2: POS-RTAU-AST-ROUTE Param Set to 1 (Enhanced Routing)				
Originator	CPF Record Found	Preference	Key Combination	
			FIID	Retailer ID
Device or Host	Yes	1	FIID from the CPF	Retailer ID in the PSTM
		2	Same as above	Retailer program specified in the PRDF
		3	Same as above	*
		4	FIID from the TERM-FIID field in the PSTM, if it is not zeros or blanks Otherwise, the FIID from the PRDF is used	Retailer ID in the PSTM
		5	Same as above	Retailer program specified in the PRDF
		6	Same as above	*
	No	1	FIID from the TERM-FIID field in the PSTM, if it is not zeros or blanks Otherwise, the FIID from the PRDF is used	Retailer ID in the PSTM
		2	Same as above	Retailer Program
		3	Same as above	*

Example AST Settings

Institutions can set up their AST to meet their specific needs. The first step in setting up an AST is determining how many AST records are needed to route the transactions that the BASE24-pos system will receive from the various transaction acquirers. Some example AST settings are shown on the following pages. These are only examples and are not recommended settings.

Example AST Records

Example 1: In the following example, the AST contains three records. Each record is for a specific institution and all the retailers associated with that specific institution.

Record	FIID	Retailer ID
1	BNK1	*
2	BNK2	*
3	BNK3	*

Example 2: In the following example, the AST contains eight records. In this case, retailer 01 and retailer 02 belong to bank 4 and have special routing requirements. With the AST set up as shown below, transactions from retailer 01 would be routed according to AST record 1, transactions from retailer 02 would be routed according to AST record 2, and transactions from all other retailers belonging to bank 4 would be routed according to AST record 3.

Similarly, retailer A and retailer C, belonging to bank 5, have special routing requirements, as do retailers belonging to bank 5 who participate in the Visa and MasterCard retailer programs. With the AST set up as shown below, transactions from retailer A would be routed according to AST record 4, while transactions from retailer C would be routed according to AST record 5. Transactions from other retailers belonging to bank 5 who participate in the Visa or MasterCard retailer programs would be routed according to AST records 6 and 7, respectively. Finally, transactions from other retailers belonging to bank 5 who do not participate in the Visa or MasterCard retailer programs would be routed according to AST record 8.

Record	FIID	Retailer ID
1	BNK4	RETAILER01
2	BNK4	RETAILER02
3	BNK4	*
4	BNK5	RETAILERA
5	BNK5	RETAILERC

Record	FIID	Retailer ID
6	BNK5	T (Visa retailer program)
7	BNK5	A (MasterCard retailer program)
8	BNK5	*

Example AST Entries

The example below shows how the entries in an AST record might look. This is an example only. These are not recommended entries. The actual entries in each AST record will depend on the needs of each institution.

AST Screen 1 Data					
SERVICE	ISSUER CODE	ONLINE			OFFLINE LIMIT
		PUR	ADV	MO	
VV	00	1000	1000	1000	500
MD	99	1000	1000	1000	500
AX	00	1000	1000	1000	500
DS	00	1000	1000	1000	500
P	00	1000	1000	1000	500

AST Screen 2 Data									
SERVICE	ISSUER CODE	OVER				UNDER			
		P	A1	A2	DA	P	A1	A2	DA
VV	00	01			C	00			A
MD	99	02			D	01			D
AX	00	14			C	00			A

AST Screen 2 Data									
SERVICE	ISSUER CODE	OVER				UNDER			
		P	A1	A2	DA	P	A1	A2	DA
DS	00	01			D	00			A
P	00	03			C	14			C

The order of the entries in the AST depends on the service hierarchy defined by the BASE24 system. Some services are reserved, meaning that they have already been assigned values. Institutions can also establish their own services, called proprietary services. The BASE24-pos product sorts reserved services according to a predefined hierarchy. Proprietary services appear in the table after the reserved services in the order in which they were entered in the table. If a wildcard service (**) has been defined in the AST, that appears as the last entry in the AST.

Example AST Selection Processing

The following example describes the basic steps involved in routing a transaction to the appropriate authorizer. The transaction involved is a purchase for \$300 using a MasterCard card at Retailer D. The transaction was initiated from Terminal 001, which has a record in the BASE24-pos Terminal Data files. Retailer D belongs to BNK5 and does not participate in any retailer programs. The Router module could not find a Card Prefix File record for this card, so the card type determined by an algorithm is M and the issuer code is 99 (not-on-us).

Determining Which AST Record to Use

The Router module first must determine which AST record to use when routing this transaction. The Router module searches the AST for an appropriate record using the FIID for BNK5 and the retailer ID for Retailer D.

Record	FIID	Retailer ID
1	BNK1	*
2	BNK2	*
3	BNK3	*

Record	FIID	Retailer ID
4	BNK4	RETAILER01
5	BNK4	T (Visa retailer program)
6	BNK4	*
7	BNK5	RETAILERA
8	BNK5	RETAILERC
9	BNK5	A (MasterCard retailer program)
10	BNK5	*

Records 1 through 6 are not a part of BNK5, so those records are not considered. Records 7 through 10 all apply to BNK5. Retailers A and C have separate AST records (records 7 and 8), but Retailer D does not. Record 9 applies to retailers participating in the MasterCard retailer program. Record 10 applies to all other retailers belonging to BNK5. Retailer D falls into this category. The Router module selects record 10 because that is the first applicable record in the AST that it finds for this transaction.

Selecting Which AST Entry to Use

Once the Router module has selected an AST record, it must select an appropriate entry from that record. This entry determines the authorizer to which the transaction is routed. To determine the appropriate entry, the Router module uses the service and the issuer code of the transaction.

The following table shows the entries that are in record 10, which the Router module selected in the previous step, where *b* represents a blank space and an asterisk (*) is a wildcard character.

AST Record 10 Data						
ENTRY	SERVICE	ISSUER CODE	ONLINE			OFFLINE LIMIT
			PUR	ADV	MO	
1	V <i>b</i>	99	1000	1000	1000	200
2	M <i>b</i>	00	1000	1000	1000	200

AST Record 10 Data						
ENTRY	SERVICE	ISSUER CODE	ONLINE			OFFLINE LIMIT
			PUR	ADV	MO	
3	AX	99	1000	1000	1000	200
4	DS	00	1000	1000	1000	200
5	DS	99	1000	1000	1000	200
6	P	00	1000	1000	1000	200
7	**	99	1000	1000	1000	200
8	**	*	1000	1000	1000	200

The Router module searches through the entries beginning with entry 1. Although entry 2 is the correct service (M), the issuer code does not match. Nor is the issuer code an asterisk. The Router module's next preference is entry 7, with a service of ** and an issuer code of 99.

Identifying the Transaction's Floor Limit

The Router module now must identify the floor limit of the transaction by comparing the transaction amount to the floor limits established in either the BASE24-pos Terminal Data files or the AST as described earlier in this section. For this example, the AST floor limits are used.

The AST floor limits are established in the ONLINE PUR, ONLINE ADV, and ONLINE MO fields of the AST. (The OFFLINE LIMIT field is used differently and is explained later in this example.)

AST Record 10 Data						
ENTRY	SERVICE	ISSUER CODE	ONLINE			OFFLINE LIMIT
			PUR	ADV	MO	
7	**	99	1000	1000	1000	200

The transaction is a purchase for \$300. Therefore, the Router module uses the ONLINE PUR field value of \$1000 when determining whether the transaction is under or over the floor. In this case, the transaction is under the floor.

Identifying the Transaction's Authorizers

Now that the Router module has determined whether the transaction is over or under the floor limits, the Router module can determine the primary, first alternate, and second alternate authorizers and a default action for this transaction.

Note: If the service is MasterCard or Visa and the retailer participates in one or both of the retailer merchant programs, only the over the floor limit authorization destinations are valid. The Router module does not route such transactions to the under the floor authorization destinations even if the transaction is actually under the floor limit.

Because the transaction is under the floor, the Router module locates the appropriate service and issuer code (**, 99) on screen 2 of the AST and attempts to route the transaction to the primary authorizer specified in the UNDER P field, shown below—the Router module determines the actual authorizer to which a value of 08 equates using information in the Routing Table File (RTBL).

SERVICE	ISSUER CODE	OVER				UNDER			
		P	A1	A2	DA	P	A1	A2	DA
**	99					08	04		A

For this example, the primary authorizer is not available, and the Router module attempts to route the transaction to the first alternate authorizer specified in the UNDER A1 field—the authorizer associated with a value of 04 in the RTBL. For this example, the first alternate authorizer is not available either. The Router module cannot attempt to route the transaction to a second alternate authorizer because none is specified in the UNDER A2 field.

Instead, the Router module passes the transaction to the Authorization module, which attempts to authorize the transaction. To authorize the transaction, the Authorization module compares the transaction amount of \$300 to the limit specified in the OFFLINE LIMIT field of the AST. The value in the OFFLINE LIMIT field is \$200. Because this limit is less than the transaction amount, the Authorization module declines the transaction as over-the-floor. If the transaction had been less than \$200, the Authorization module would have performed the default action specified in the UNDER DA field: A (authorize blind).

RTBL Character Substitution

When multiple XPNET nodes or logical networks have the same processes configured, you can define destination process names in the RTBL that include an asterisk (*) in the second or third positions of the process name. Using asterisks allows you to define one symbolic process name that identifies multiple Interchange Interface processes as destinations. This feature, which is also known as *star routing*, requires that you have the same processes configured in each XPNET node or logical network, that the processes use standard naming conventions, and that all of your Router modules access the same AST and RTBL.

The AST uses values defined in the RTBL for its routing. When you define destination process codes in the RTBL, you can use an asterisk (*) as a wildcard character for the second or third character or both characters of a process name (for example, P1*^BNET or P**^BNET). When the Router module builds its routing tables, it substitutes the character from the same position of its own symbolic name for each asterisk that it finds.

For example, assume you have Device Handler/Router/Authorization processes named P1A^RTAU and P2B^RTAU, and your RTBL includes an entry for P**^BNET. When P1A^RTAU reads the RTBL, it will substitute the second and third character of its own name for the asterisks, and add an entry to its internal table of P1A^BNET. Likewise, P2B^RTAU adds an entry of P2B^BNET to its internal table.

SPROUTE Routing

When determining routing destinations, the Router module uses routing information from the Card Prefix File (CPF) maintained in the current logical network whenever it is available. However, when this CPF does not contain a record for the prefix of the card used to initiate a transaction, the Router module can use internal algorithms and the SPROUTE File (SPROUTE) to determine where a transaction is to be authorized as follows:

- The Router module uses SPROUTE to determine whether a record for the card prefix exists in the CPF of another logical network.
- The Router module uses SPROUTE to determine whether a record for the card prefix exists in the Interchange Prefix File (IPF) of any logical network.
- The Router module uses algorithms specified in SPROUTE to determine if the card is an American Express credit card, MasterCard credit card, Visa credit card, or Discover credit card.
- The Router module uses internal algorithms to determine if the card is an American Express credit, Carte Blanche credit, China UnionPay credit, Diners Club credit, Discover credit, Japan Credit Bureau credit, MasterCard credit, or Visa credit card.

If the Router module finds the card prefix in SPROUTE or identifies the card type using an algorithm, it continues to process the transaction. Otherwise, the Router module declines the transaction with a referral response.

The Router module can also use SPROUTE to select the appropriate destination for the transaction. This occurs when an Authorization Selection Table (AST) record is not available or the AST record does not contain an entry for the card type being used. If an AST record is found and it contains an entry for the card type being used, the Router module uses the destinations specified in the AST record to route the transaction. Otherwise, the Router module routes the transaction based on the destinations in SPROUTE.

Internal Algorithms

The Router module can use eight internal algorithms in conjunction with SPROUTE to identify the card type (for the purpose of selecting an AST record entry for routing). When the card type is identified with the internal algorithms, an AST record is still necessary to provide routing information. If an AST record still cannot be found, SPROUTE provides routing destinations.

The POS-RTAU-ALGO-CHECK param in the Logical Network Configuration File (LCONF) specifies whether the internal algorithms are performed before using SPROUTE, after using SPROUTE, or are not performed at all. Refer to the ***BASE24 Logical Network Configuration File Manual*** for more information about this param.

The internal algorithms use the first digits of the card number in the TRAN.TRACK2 field in the BASE24-pos Standard Internal Message (PSTM) to identify the card type. Some of the algorithms also use the length of the card number to identify the card type. The Diners Club and MasterCard credit card algorithms also use the TERM-CNTRY-CDE field in the PSTM to identify the card type because the card type depends on the country in which the terminal accepting the card is located. The internal algorithms check for the leading digit and card number length combinations shown in the following table.

Card Type	Leading Digits	Card Number Length
American Express credit	34	Not checked
	37	Not checked
Carte Blanche credit	39	14
China UnionPay credit	622126–622925	16, 19
	624–626	16
	6282–6288	16

Card Type	Leading Digits	Card Number Length
Diners Club credit	30	14
	36 and terminal located outside US and Canada	14
	38	14
	300–305	16
	3095	16
	38	16
	39	16
Discover credit	60110	16
	60112–60114	16
	601174	16
	601177–601179	16
	601186–601199	16
	644–659	16
Japan Credit Bureau credit	3528–3589	16
MasterCard credit	36 and terminal located inside US or Canada	14
	51	Not checked
	52	Not checked
	53	Not checked
	54	Not checked
	55	Not checked
Visa credit	4	13
	4	16

The SPROUTE File

The SPROUTE File (SPROUTE) is built during BASE24 configuration by the Pregon utility. It contains information from the PREMAP and SPREFIX Files, two other BASE24 configuration files built by the Pregon utility. The PREMAP File (PREMAP) contains the actual routing logic of the BASE24 system and the SPREFIX File (SPREFIX) contains all of the prefixes from the prefix files in the system (i.e., all the CPFs in the BASE24 system, if you have more than one logical network, plus any interchange prefix files in the system). If you have only one logical network in the BASE24 system, the CPF is not included in SPREFIX. SPREFIX is a temporary file that the Pregon utility creates to build SPROUTE. Once SPROUTE is built, the utility purges the file. The Router module uses SPROUTE to look up prefixes which are recognized as valid by the BASE24 system.

SPROUTE is a CPU memory-resident file that contains all of the information needed and available to determine proper destinations for transactions whose prefixes are not recognized by a BASE24-pos Router module. It is maintained in a common CPU memory area and is available to all processes running in a particular CPU. SPROUTE is paged in and out of memory for each CPU as needed. If a process running on another CPU needs to read SPROUTE, SPROUTE is paged into memory for that CPU. The Router module uses SKEL library procedures to access SPROUTE and perform prefix routing. SKEL is an XPNET library that is used by BASE24 application processes to route transactions.

When a BASE24-pos Router module cannot find a CPF record for the card originating the transaction, the process calls the SPROUTE^LOOKUP routing procedure in the SKEL library. The SKEL library procedures access SPROUTE to determine a primary or alternate authorizer for the transaction. This information is returned to the application process, and the transaction is routed accordingly.

The SPROUTE created by the Pregon utility resides on the XPNET system data subvolume. The Pregon utility also copies SPROUTE to any remote HP NonStop systems in the XPNET system, if required.

The SPREFIX File

The SPREFIX File (SPREFIX) serves as a temporary point of reference for all prefixes used in prefix routing. The Pregon utility builds SPREFIX on the XPNET system data subvolume from all of the Card Prefix Files (CPFs) and Interchange Prefix Files (IPFs) specified in the Pregon Command File (PRECMD). Once the file is built, the utility uses it to build the SPREFIX table in SPROUTE. After SPROUTE is built, the utility purges SPREFIX.

The Basics of Routing (PREMAP)

PREMAP contains the routing logic for the BASE24 system as a whole. It is a multiple entry table, built by the Pregon utility from commands entered in the Pregon Command File (PRECMD). Refer to appendix E for detailed procedures on setting up the PRECMD and using the Pregon utility. Once PREMAP is created, the utility reads it to build the PREMAP table in SPROUTE. Each entry in the PREMAP table contains the following fields:

```
source-ln, sym-source, routing-group, routing-order,
method, card-type, dest, alt-dest, file-name, type, sprefix-key
method, card-type, dest, alt-dest, file-name, type, sprefix-key
method, card-type, dest, alt-dest, file-name, type, sprefix-key
...
```

PREMAP entries can be divided into two parts: key fields and action fields. The key fields allow processes to select the appropriate entry from the PREMAP data. The action fields prescribe the actions to take once a PREMAP entry has been selected.

PREMAP contains one record for each transaction source, symbolic source, routing group, and destination combination in the system. A transaction source is any logical network or interchange from which transactions can be routed.

A description of each of the fields in a PREMAP entry is included below. These descriptions summarize the fields for the purpose of understanding the routing performed by the BASE24-pos Router module.

Key Fields

The *source-ln*, *sym-source*, *routing-group*, and *routing-order* fields are the key fields of the table entry. The Router module compares transactions to these four fields to determine whether the subsequent routing actions should be taken.

source-ln — The logical network or interchange from which the transaction originated (e.g., PRO1, PRO2, TES1, VISA, BNET). This field must always contain the name of a logical network or interchange.

sym-source — The symbolic name of the Device Handler/Router/Authorization process that received the transaction. This field can contain an actual process name or an asterisk (*) if the source process need not be specific.

routing-group — The routing group associated with the transaction originator. This field can contain an actual routing group identifier, or, if the routing group need not be specific, this field can contain an asterisk (*).

An asterisk can be entered in this field if routing based on the routing group is not required. The asterisk is used with the last two entries in the selection hierarchy when the Router module searches SPROUTE for a PREMAP entry.

You can enter zeros in this field if routing based on the routing group and source process is not required. When the routing group is zeros and the source process contains an asterisk, the Router module matches on the source logical network or interchange only as the second entry in the selection hierarchy. The selection hierarchy is described later in this section.

routing-order — A field generated by the Pregen utility that specifies the order in which routing methods are to be applied for this record. This order is determined by the order in which you enter routing methods in the PRECMD.

Action Fields

The *method*, *card-type*, *dest*, *file-name*, *type*, and *alt-dest* fields specify the routing actions to take if a transaction matches on the key fields of the table entry. Multiple actions can be set to allow for several routing alternatives. The Router module checks multiple actions in the order they appear in the PREMAP table in SPROUTE.

method — The method to use for taking action on this table entry. The methods available are LGNT, AMEX, BASE or VISA, DSCV, INAS or BNET, and DFLT. Each is discussed later in this section.

card-type — The service type associated with the card that initiated the transaction.

dest*, *alt-dest — The symbolic names of the primary and alternate destinations to use for routing the transaction if the account number meets the criteria set forth for the selected method.

Note: Alternate destinations are intended for use by BASE24 Interchange Interface processes to use if their respective interchange links are down. They are not intended to allow for alternate routing by the Router modules. While a Router

module will place an alternate destination in the internal message if one is found in SPROUTE, it does not route to an alternate destination if a message is found to be undeliverable to the primary destination.

file-name, type — The fully qualified name of the prefix file, and its type, for use with the LGNT method. Valid values for *type* are as follows:

CP = Card Prefix File
IP = Interchange Prefix File

sprefix-key — A field generated by the Pregen utility that contains the key used to access SPREFIX for this routing method.

Selection Hierarchy

To select a PREMAP table entry, the Router module provides the following information:

- The logical network or interchange from which the transaction originated.
- Its own symbolic Device Handler/Router/Authorization process name.
- The routing group associated with the terminal or interchange originating the transaction.

Using the logical network or interchange, symbolic process name, and routing group, the Router module steps through the PREMAP entries in SPROUTE looking for a match. If an exact match is not found, certain other alternatives are allowed. Below is the hierarchy used for selection.

Preference	Logical Network	Process Name	Routing Group
1	match	match	match
2	match	*	zeros
3	match	*	match
4	match	match	*
5	match	*	*

When the Router module finds an entry that meets the above selection criteria, it takes the routing actions specified in the entry to derive the primary and alternate destinations.

For the BASE24-pos product, if the Router module cannot find an entry based on the above hierarchy, the Router module may not be able to process the transaction, depending on whether it has tried its internal routing algorithms. The POS-RTAU-ALGO-CHECK param in the Logical Network Configuration File (LCONF) indicates whether to perform SPROUTE routing or attempt to use an algorithm to route the transaction first. If the param indicates to perform an algorithm first, the algorithm did not identify a card type, and the Router module cannot find a SPROUTE entry based on the above hierarchy, the transaction cannot be processed.

Routing Methods

Once the PREMAP entry is selected, the Router module uses the action fields for the entry to determine the primary and alternate destinations.

In order to evaluate a transaction against the PREMAP action fields, the Router module supplies the account number used in the transaction and the account number length. The Router module obtains this information from the track data carried in the PSTM. The methods of checking the account number are described below.

If the account number matches one of the algorithms, the Router module attempts to determine the card type using SPROUTE. With that card type, the Router module attempts AST routing. If a destination cannot be determined using AST routing, the Router module uses the destinations determined during SPROUTE routing.

Note: Once the Router module selects a PREMAP entry, there is no hierarchy for the methods the entry specifies. The Router module applies the methods in the order in which they appear within the PREMAP entry. The PREMAP entry can be written to contain any desired combination of the five methods. The methods can be arranged in whatever order is the most effective for routing transactions. Methods can also be omitted if they are not needed.

LGNT (Logical Network)

In a LGNT, or logical network search, the Router module searches the designated SPREFIX data—specified by the file name (and type)—contained in SPROUTE for a card prefix that matches the account number. If the account number's prefix is located in the SPREFIX data, the primary and alternate destinations are taken from that set of action fields.

If an AST entry is available for the account number's card type, routing destinations are taken from the AST entry instead of the SPREFIX data.

AMEX (American Express Algorithm)

The Router module checks the account number against an AMEX algorithm. In this case, if the account number starts with 34 or 37, the primary and alternate destinations are taken from that set of action fields.

BASE or VISA (Visa Algorithm)

The Router module checks the account number against a Visa algorithm. In this case, if the account number is 13 or 16 positions in length and the first position is a 4, the primary and alternate destinations are taken from that set of action fields.

DSCV (Discover Algorithm)

The Router module checks the account number against a Discover algorithm. In this case, the primary and alternate destinations are taken from that set of action fields only when the leading digits fall into one of the following ranges.

60110000–60119999
62212600–62292599
64400000–64499999
650000–659999

INAS or BNET (MasterCard Algorithm)

The Router module checks the account number against a MasterCard algorithm. In this case, if the leading digits are 51, 52, 53, 54, or 55 for any card number length, the primary and alternate destinations are taken from that set of action fields.

DFLT (Default Routing)

No checking is done. The Router module takes the primary and alternate destinations from that set of action fields.

Setting Up Routing Tables

The order in which multiple entries are made in the PREMAP routing table should be based on business and performance considerations. Some of the factors to take into account include:

- Does the destination accept invalid prefixes? DFLT routing has the lowest overhead because no prefix checks are preformed.
- Can an algorithm check be performed? AMEX, BASE, DSCV, and INAS routing methods require less overhead than the prefix table lookup performed with LGNT.
- Which set of action fields will handle the most traffic? The earlier a routing method can be selected, the less time and overhead are required.

Routing Example

The following pages provide an example of how the Router module derives primary and alternate destination from SPROUTE.

PREMAP Data

The example uses the PREMAP information shown below. This PREMAP information is for illustrative purposes only. It is not intended to represent a complete or recommended version of PREMAP. The internal algorithms the Router module can use to identify the card type could be used in conjunction with or in place of the algorithms defined in SPROUTE. For information on how to use the Pregen utility to build the PREMAP data for an actual system, refer to appendix E of this manual.

PREMAP Table

```

PRO1, P1A^AUTH1, 00000000000,
    INAS,          ,dest,          , ,alt-dest,
    BASE,          ,dest,          , ,alt-dest,
    LGNT,          ,dest, $DATA.PRO2DATA.CPF, CP,alt-dest,
    DFLT,          ,dest,          , ,alt-dest;
PRO1, P1A^AUTH1, 00000000001,
    INAS,          ,dest,          , ,alt-dest,
    BASE,          ,dest,          , ,alt-dest,
    AMEX,          ,dest,          , ,alt-dest,
    LGNT,          ,dest, $DATA.PRO2DATA.CPF, CP,alt-dest,
    DFLT,          ,dest,          , ,alt-dest;
PRO1, P1A^AUTH1,      *      ,
    INAS,          ,dest,          , ,alt-dest,
    BASE,          ,dest,          , ,alt-dest,
    LGNT,          ,dest, $DATA.PRO2DATA.CPF, CP,alt-dest,
    DFLT,          ,dest,          , ,alt-dest;
PRO1, P1A^SPDH1, 00000000000,
    INAS,card-type,dest,          , ,alt-dest,
    BASE,card-type,dest,          , ,alt-dest,
    LGNT,card-type,dest, $DATA.PRO2DATA.CPF, CP,alt-dest,
    LGNT,card-type,dest, $DATA.PRO1DATA.IPF, IP,alt-dest,
    DFLT,card-type,dest,          , ,alt-dest;
PRO1,      *      , 00000000000,
    BNET,card-type,dest,          , ,alt-dest,
    DSCV,card-type,dest,          , ,alt-dest,
    AMEX,card-type,dest,          , ,alt-dest,
    LGNT,card-type,dest, $DATA.PRO2DATA.CPF, CP,alt-dest,
    DFLT,card-type,dest,          , ,alt-dest;
PRO1,      *      ,      *      ,
    INAS,card-type,dest,          , ,alt-dest,
    BASE,card-type,dest,          , ,alt-dest,
    AMEX,card-type,dest,          , ,alt-dest,
    LGNT,card-type,dest, $DATA.PRO2DATA.CPF, CP,alt-dest,
    DFLT,card-type,dest,          , ,alt-dest;

```

Processing Steps

The following steps describe the processing that the Router module goes through in order to derive primary and alternate destinations from SPROUTE.

In this example, a Device Handler/Router/Authorization process, P1C^SPDH1, receives a transaction request. The card issuer is part of logical network PRO2. The POS-RTAU-ALGO-CHECK param in the LCONF is set to 1 (use internal algorithms first).

1. After not finding the cardholder's prefix in the CPF for logical network PRO1, the Router module performs the internal algorithm checks and the internal algorithm checks fail. The Router module then looks for an entry in SPROUTE. It first looks for an exact match on the following combination of values:

PRO1, P1C^SPDH1, 00000000000

The Router module steps through the key fields in the PREMAP table sequentially as follows:

Step 1	PRO1, P1A^AUTH1, 00000000000	No match
Step 2	PRO1, P1A^AUTH1, 00000000001	No match
Step 3	PRO1, P1A^AUTH1, *	No match
Step 4	PRO1, P1A^SPDH1, 00000000000	No match
Step 5	PRO1, *, 00000000000	No match
Step 6	PRO1, *, *	No match

2. Since no match was found, the Router module looks for its second preference:

PRO1, *, 00000000000

The Router module steps through the key fields in the PREMAP table sequentially as follows:

Step 1	PRO1, P1A^AUTH1, 00000000000	No match
Step 2	PRO1, P1A^AUTH1, 00000000001	No match
Step 3	PRO1, P1A^AUTH1, *	No match
Step 4	PRO1, P1A^SPDH1, 00000000000	No match

Step 5

PRO1,	*	,	00000000000
PRO1,	*	,	*

Match found

3. Once an appropriate PREMAP entry is located, the Router module checks the card number involved in the transaction, shown below, against the actions specified in the selected entry.

CARD NUMBER: 8100643254987611

The first method tried is the BNET algorithm. To meet this criteria, the card number must start with 51, 52, 53, 54, or 55:

PRO1,	*	,	*	,
BNET,	<i>card-type,</i>	<i>dest,</i>		<i>, alt-dest</i>
DSCV,	<i>card-type,</i>	<i>dest,</i>		<i>, alt-dest</i>
AMEX,	<i>card-type,</i>	<i>dest,</i>		<i>, alt-dest</i>
LGNT,	<i>card-type,</i>	<i>dest,</i>	\$DATA.PRO2DATA.CPF,	CP, <i>alt-dest</i>
DFLT,	<i>card-type,</i>	<i>dest,</i>		<i>, alt-dest</i>

4. The card number does not meet the BNET criteria, so the Router module tries the DSCV algorithm. To meet this criteria, the card number must be 16 positions in length and the leading digits fall into one of the following ranges.

60110000–60119999

650000–659999

PRO1,	*	,	*	,
BNET,	<i>card-type,</i>	<i>dest,</i>		<i>, alt-dest</i>
DSCV,	<i>card-type,</i>	<i>dest,</i>		<i>, alt-dest</i>
AMEX,	<i>card-type,</i>	<i>dest,</i>		<i>, alt-dest</i>
LGNT,	<i>card-type,</i>	<i>dest,</i>	\$DATA.PRO2DATA.CPF,	CP, <i>alt-dest</i>
DFLT,	<i>card-type,</i>	<i>dest,</i>		<i>, alt-dest</i>

5. The card number does not meet the DSCV criteria, so the Router module tries the AMEX algorithm. To meet this criteria, the card number must start with 34 or 37:

PRO1,	*	,	*	,	
BNET,	card-type,	dest,		,	,alt-dest
DSCV,	card-type,	dest,		,	,alt-dest
AMEX,	card-type,	dest,		,	,alt-dest
LGNT,	card-type,	dest,	\$DATA.PRO2DATA.CPF,	CP,	alt-dest
DFLT,	card-type,	dest,		,	,alt-dest

6. The card number does not meet the AMEX criteria, so the Router module tries a logical network search using the card number. In this case, the Router module checks the \$DATA.PRO2DATA.CPF portion of the SPREFIX information in SPROUTE:

PRO1,	*	,	*	,	
BNET,	card-type,	dest,		,	,alt-dest
DSCV,	card-type,	dest,		,	,alt-dest
AMEX,	card-type,	dest,		,	,alt-dest
LGNT,card-type,dest,\$DATA.PRO2DATA.CPF, CP,alt-dest					
DFLT,	card-type,	dest,		,	,alt-dest

7. Because the cardholder is a part of the PRO2 logical network, the Router module locates the cardholder prefix in the designated SPREFIX information. Since a prefix can be located, the Router module uses the primary and alternate destinations from that entry if an AST record cannot be found. When routing destinations are available in the AST record and in SPROUTE, the Router module uses the AST destinations.

```
LGNT, card-type, dest, $DATA.PRO2DATA.CPF, CP, alt-dest
```

IDF Routing and Authorization Selection

The IDF Routing Table specifies authorization parameters according to account type and prefix routing group. The Authorization module checks the BASE24-pos routing table on screen 16 of the IDF and authorizes the transaction according to that table in the following cases:

- The Router module finds an appropriate record in the CPF but cannot find an appropriate record in the AST.
- The Router module finds appropriate records in the CPF and in the AST and the AST and RTBL indicate AUTHH or AUTHF as authorizers.

The routing table enables institutions to specify multiple DPCs for processing authorizations. In this table, institutions can segregate transactions by account type and prefix routing group to be processed by separate DPCs and under separate authorization parameters.

Institutions can set up a table entry for each DPC involved in authorizing transactions and then establish the authorization parameters individually for each DPC. For example, one DPC could authorize transactions for a specific group of card prefixes, while another handles authorizations on credit accounts.

Using the routing table, an institution can distribute processing according to its database. An institution whose account database is distributed among several DPCs can also tailor the Authorization module to involve all of the DPCs according to the types of transactions and card prefixes for which the institution has files.

IDF Routing Table Entries

The routing table in the IDF can contain up to nine entries. Each entry contains the fields listed and briefly described below. For more information about these fields and the valid values that can be entered in these fields, refer to the ***BASE24 Base Files Maintenance Manual***.

PRIMARY DPC — The DPC number of the host computer to which transactions included under this entry's processing parameters are to be routed.

SYMBOLIC NAME — The symbolic name of the ISO Host Interface process or the Interchange Interface process used for sending this group of transactions to a back-end authorizer.

ACCT TYPE — The transaction account type to be processed according to this table entry's parameters.

When the routing table contains multiple entries and includes an entry with a value of AL (all account types) in this field and a value of A (all card prefixes) in the PRFX RTG field, the entry with a value of AL in this field and a value of A in the PRFX RTG field must follow all of the other entries.

PRFX RTG — The prefix routing group to be processed according to this table entry's parameters. Card prefixes can be grouped in the CPF, using the PREFIX ROUTING field on screen 1, for different types of authorization processing. The PREFIX ROUTING field in the CPF then ties the card prefix groups to this table entry.

When the routing table contains multiple entries and includes an entry with a value of A (all card prefixes) in this field and a value of AL (all account types) in the ACCT TYPE field, the entry with a value of A in this field and a value of AL in the ACCT TYPE field must follow all of the other entries.

AUTH TYPE — The authorization method for processing transactions that use this table entry. This value also controls which authorization file is checked for existing preauthorized holds information.

AUTH LVL — The authorization level for processing transactions that use this table entry.

HOLDS LVL — A code indicating which authorization file should receive new preauthorized holds information resulting from preauthorized hold request transactions or reversals of preauthorized hold completion transactions that use this table entry. This value does not control which authorization file is checked for existing preauthorized holds.

Setting the HOLDS LVL Field for Preauthorized Holds

Preauthorized hold information can be stored in three different BASE24 files: the CAF, the PBF, or the UAF. Preauthorized hold information can also be maintained in both the CAF and PBF. ACI offers the following guidelines for determining the authorization file in which preauthorized hold information is stored.

- When the BASE24-teller product is using the preauthorized hold information, the information should be stored in the PBF only or in the CAF and PBF because the BASE24-teller product only tracks hold information at the account level and cannot access hold information stored in the UAF or CAF.
- When preauthorized hold information is needed for tracking account-level usage but not card-level usage, preauthorized hold information should be stored in the PBF only. Usage information takes less space to store at the account level than at the card level.
- When preauthorized hold information is needed for tracking card-level usage but not account-level usage and each PBF account number appears on only one CAF record, preauthorized hold information should be stored in the CAF. In this situation, preauthorized hold information could be stored in the CAF and PBF; however, the CAF and PBF contain duplicate hold information and the PBF information is only necessary for tracking account-level usage.
- When preauthorized hold information is needed for tracking card-level usage and each PBF account can appear on more than one CAF record, preauthorized hold information should be stored in the CAF and PBF. CAF-only storage of hold information is not recommended when a PBF account can appear on multiple CAF records because all outstanding holds for a specific PBF account may not be known at the time of a transaction. For example, if PBF account 1234 can be accessed by cards 22222 and 33333, a hold placed on PBF account 1234 with card 22222 would not be known to the BASE24 system if PBF account 1234 is then accessed with card 33333 because the hold information is maintained for the individual cards, not the PBF account.

- When preauthorized hold information is stored in the CAF or the UAF, each hold is considered a use of the card when the BASE24-pos Authorization module checks the card usage limit. For example, when the CAF contains three holds, the first transaction of the usage accumulation period is considered the fourth use of the card for the period when the BASE24-pos Authorization module compares the card usage to the limit in the TIMES USED PER PERIOD LIMIT field on CPF screen 6 or CAF screen 10. These limits should be set to allow for preauthorized holds.
- Preauthorized hold information does not need to be stored in a BASE24 file if the host is responsible for tracking holds or the institution plans to authorize transactions without reserving funds.

IDF Routing Table Examples

The BASE24-pos routing table in the IDF was designed for one entry per DPC, whereby authorization parameters can be controlled individually for each DPC. Institutions can, however, set up multiple entries for the same DPC in order to modify the authorization level or method based on the account type or card prefix being processed.

For example, a DPC might maintain all of the files for a certain card prefix and be set up as authorization level 1 (online) for that prefix. At the same time, it might share authorization responsibilities with the BASE24-pos system in other cases, such as authorization level 3 (online/offline). In that case, one DPC would have two table entries.

An institution could also set up credit accounts for Negative Authorization with or without Usage Accumulation, while they set up checking and savings accounts for Positive Balance Authorization, using the same DPC as a back-end host. Again, one DPC would have two table entries.

The routing table allows for virtually any combination of authorization levels and methods, with the exception that when the AUTH TYPE field contains a value of 0 (online host processing), the AUTH LVL field requires a value of 1 (online) and vice versa.

The following examples illustrate how transactions are processed under each type of routing table.

Example 1: Single-Entry Routing Table (One DPC)

If only one host DPC is being used, the IDF routing table can contain a single entry. An example is shown below:

#	PRIMARY DPC	SYMBOLIC NAME	ACCT TYPE	PREFIX RTG	AUTH TYPE	AUTH LVL	HOLDS LVL
1	1	P1A^HISF1	AL (all)	A (all)	3 (CAF/PBF)	3 (on/off)	0 (none)

In this example, the Authorization module processes all transactions using authorization level 3 (online/offline). The Authorization module sends messages to P1A^HISF1 (the ISO Host Interface process) to be sent to DPC1, regardless of card and account type. If DPC1 is unavailable, the Authorization module authorizes or declines the transaction using the Positive Balance Authorization method.

Example 2: Single-Entry Routing Table (No Host)

If a host is not involved in authorization, the IDF routing table can contain a single entry. An example is shown below:

#	PRIMARY DPC	SYMBOLIC NAME	ACCT TYPE	PRFX RTG	AUTH TYPE	AUTH LVL	HOLDS LVL
1	0	NO HOST ROUTING	AL (all)	A (all)	3 (CAF/PBF)	2 (off)	0 (none)

In this example, the Authorization module processes all transactions using the Positive Balance Authorization method. No messages are sent to a host for authorization.

Example 3: Multiple-Entry Routing Table

To process credit accounts differently from other account types using the same DPC, a routing table must be set up with multiple entries. An example is shown below.

Note: One entry in this table contains default values in the ACCT TYPE and PRFX RTG fields. To avoid the possibility of misrouting transactions, the entry containing two default values (i.e., the ACCT TYPE field is set to AL and the PRFX RTG field is set to A) must be the last entry in the table.

#	PRIMARY DPC	SYMBOLIC NAME	ACCT TYPE	PRFX RTG	AUTH TYPE	AUTH LVL	HOLDS LVL
1	1	P1A^HISF1	31 (ccd)	A (all)	1 (NEG/UAF)	2 (off)	0 (none)
2	1	P1A^HISF1	AL (all)	A (all)	0 (no files)	1 (on)	0 (none)

In this example, transactions on credit accounts (entry 1) are authorized offline using the Negative Authorization with Usage Accumulation method. Completions can be sent to the host. Transactions on all non-credit accounts (entry 2) are sent to DPC1 for online authorization.

Example 4: Multiple-Entry Routing Table

This example contains a number of DPCs with varying parameters.

Note: One entry in this table contains default values in the ACCT TYPE and PRFX RTG fields. To avoid the possibility of misrouted transactions, the entry containing two default values (i.e., the ACCT TYPE field is set to AL and the PRFX RTG field is set to A) must be the last entry in the table. The order of other table entries does not matter.

#	PRIMARY DPC	SYMBOLIC NAME	ACCT TYPE	PRFX RTG	AUTH TYPE	AUTH LVL	HOLDS LVL
1	1	P1A^HISF1	01 (chk)	1	3 (CAF/PBF)	3 (on/off)	0 (none)
2	1	P1A^HISF1	11 (sav)	1	3 (CAF/PBF)	2 (off)	0 (none)
3	2	P1A^HISF1	31 (ccd)	1	0 (no files)	1 (on)	0 (none)
4	5	P1A^HISF1	31 (ccd)	4	6 (CAF/PBF) (CAPF)	2 (off)	0 (none)
5	3	P1A^HISF1	AL (all)	2	0 (no files)	1 (on)	0 (none)
6	4	P1A^HISF1	AL (all)	3	0 (no files)	1 (on)	0 (none)
7	6	P1A^HISF1	AL (all)	A	3 (CAF/PBF)	2 (off)	0 (none)

In this example, transactions on checking accounts (entry 1) initiated by cards in prefix group 1 are sent to DPC1 if it is available. If DPC1 is not available, transactions are authorized offline using the Positive Balance Authorization method.

Transactions on savings accounts (entry 2) initiated by cards in prefix group 1 are authorized offline using the Positive Balance Authorization method. Transactions on credit accounts (entry 3) initiated by cards in prefix group 1 are sent to DPC2 for online authorization.

Transactions on credit accounts (entry 4) initiated by cards in prefix group 4 are authorized offline using the Parametric Authorization method. All transactions initiated by cards in prefix group 2 (entry 5) are sent to DPC3 for online authorization. All transactions initiated by cards in prefix group 3 (entry 6) are sent to DPC4 for online authorization.

All transactions initiated by cards that do not fit into any of the other defined prefix groups (entry 7) are authorized offline using the Positive Balance Authorization method.

Deriving Parameters from the IDF Routing Table

To derive the routing and authorization parameters required for processing a transaction, the Authorization module compares the account type and prefix routing group for the transaction to the routing table on screen 16 of the IDF. The Authorization module checks all routing table entries in the routing table and selects the entry with account type and prefix routing group values that best satisfy the requirements of the transaction based on the following hierarchy:

Preference	Account Type	Prefix Group
1	match	match
2	match	default
3	default	match
4	default	default

If the Authorization module cannot find an entry based on account type and prefix routing group, the transaction cannot be processed.

Example Transaction

The following example illustrates how the BASE24-pos product uses routing table entries to process a transaction. For a normal purchase initiated using a credit card in prefix group 1, the BASE24-pos Authorization module searches its routing table for the following combinations according to hierarchy preference.

Preference	ACCT TYPE	PRFX RTG
1	31	1
2	31	A
3	AL	1
4	AL	A

The IDF routing table that the Authorization module uses in this example is shown below. The entries in this table are arranged as required so that default account type and prefix routing group entries follow the more specific entries.

The Authorization module uses entry 3 because it is the first preference in the hierarchy and the most specific combination of account type and prefix routing group that appears in the routing table for this transaction.

#	PRIMARY DPC	SYMBOLIC NAME	ACCT TYPE	PRFX RTG	AUTH TYPE	AUTH LVL	HOLDS LVL
1	1	P1A^HISF1	01	1	3 (CAF/PBF)	3 (on/off)	0 (none)
2	1	P1A^HISF1	11 (sav)	1	3 (CAF/PBF)	2 (off)	0 (none)
3	2	P1A^HISF1	31 (ccd)	1	0 (no files)	1 (on)	0 (none)
4	5	P1A^HISF1	31 (ccd)	4	6 (CAF/PBF/ CAPF)	2 (off)	0 (none)
5	3	P1A^HISF1	AL (all)	2	0 (no files)	1 (on)	0 (none)
6	4	P1A^HISF1	AL (all)	3	0 (no files)	1 (on)	0 (none)
7	6	P1A^HISF1	AL (all)	A	3 (CAF/PBF)	2 (off)	0 (none)

Routing Check Authorization Transactions

Check guarantee and check verification transaction requests are routed somewhat differently than other transaction requests. The Router module uses the Authorization Selection Table (AST), Routing Table File (RTBL), and SPROUTE File to route these transactions correctly. If the customer uses a credit or debit card as a piece of identification when enacting the transaction, the Router module also uses the Card Prefix File (CPF).

An additional file, the Check Authorization Routing Table File (CART), is also used when routing check authorization transactions. For more information about the CART, refer to the ***BASE24-pos Files Maintenance Manual***.

The BASE24-pos standard product provides check authorization processing as a pass-through function only, unless the eCheck add-on product is installed. The BASE24-pos product can receive check authorization transactions from a device, an ISO host, or an interchange. The BASE24-pos product then uses the CART to route the transactions to another destination where check authorization transactions are authorized. The authorizing entity then sends the transaction response back to the BASE24-pos system, which returns the transaction response to the entity that originated the transaction request. The BASE24-pos product cannot authorize check authorization transactions, unless the BASE24-pos eCheck add-on product is installed.

When the BASE24-pos eCheck add-on product is installed, check guarantee and check verification transactions also can be authorized using the Positive Balance Authorization method with any authorization level. The BASE24-pos eCheck add-on product enables the authorizer to impact Positive Balance File (PBF) account balances upon authorization of check guarantee transactions. When check guarantee transactions are authorized offline by the BASE24-pos product, PBF balances on the BASE24 system are impacted immediately upon authorization. When check guarantee transactions are authorized online by the host or an interchange, PBF balances on the BASE24 system can be impacted when the response is returned from the host or interchange, or the authorizer can choose not to impact the PBF.

To determine an appropriate transaction authorizer, the Router module performs the following steps:

1. Searches for a Card Prefix File (CPF) record to obtain the card type (i.e., the service type for the BASE24-pos product) if a debit or credit card was used as a form of identification or Track 2 data was created as a service by the acquiring interchange. The Router module also retrieves the FIID from the CPF if a record is found.

2. If a CPF record could not be located in the CPF, the Router module attempts to obtain the service type from an internal algorithm or by searching the SPROUTE File.
3. Performs AST routing using the service type retrieved from steps 1 or 2 to locate an AST entry.
4. Performs CART routing if the authorizer is determined to be AUTHH during AST routing.

Each of the above steps are explained in detail below.

Locating a CPF Record

The Router module attempts to locate a Card Prefix File (CPF) record when the person issuing the check uses a debit or credit card as identification or Track 2 data was created as a service by the acquiring interchange. The Router module searches the CPF only to obtain data needed for AST routing—the card type and card issuer FIID. If the person issuing the check uses another form of identification, such as a driver's license, the Router module cannot search for a CPF record and instead uses a default card type of SC (Special, Check) and the acquiring terminal or interchange FIID to perform AST routing. Special, Check (SC) is a special-use card type used to initiate BASE24-pos check guarantee and check verification transactions only.

If the Router module can search the CPF and a CPF record is found, the Router module retrieves the card type and FIID from the record to perform AST routing. If the Router module can search the CPF and a CPF record is not found, the Router module attempts to retrieve the service type from an internal algorithm or by searching the SPROUTE File. See the “SPROUTE Retrieval Processing” topic below.

SPROUTE Retrieval Processing

The Router module performs SPROUTE retrieval processing when it could not locate a CPF record and the person issuing the check used a debit or credit card as identification or Track 2 data was provided by the acquiring interchange. The Router module performs SPROUTE retrieval processing only to determine the service type for use in AST routing; it does not route the check authorization transaction according to the destination identified in the PREMAP statement.

The Router module attempts to retrieve the service type from an internal algorithm or by searching the SPROUTE File. If a service type cannot be found, the Router module uses a default service type of SC (Special, Check).

Performing AST Routing

The Router module locates an AST record using different key values, depending on whether a CPF record could be located in the previous steps. The Router module assigns primary key values to search the AST as follows:

- If a CPF record was found, the Router module uses the card issuer FIID from the CPF and either the check routing group from the check authorization token or the retailer ID in the PSTM if the check routing group is blank in the token. The FIID value is used to search against the FIID field on AST screens 1, 3, and 5, while the check routing group or retailer ID value is used to search against the RETAILER ID field on AST screens 1, 3, and 5.
- If a CPF record was not found, the Router module uses the acquiring terminal or interchange FIID from the PSTM and either the check routing group or retailer ID as described above. The FIID value and check routing group or retailer ID value are used to search the AST as described above.

Once the Router module locates an AST record, it searches for an entry based on the following qualifications:

- If a CPF record was found or a service type was determined using an internal algorithm or SPROUTE retrieval processing, the Router module uses the actual service type and issuer code to locate an AST entry.
- If the CPF could not be searched or no CPF record was found and a service type could not be determined using an internal algorithm or SPROUTE routing, the Router module uses a service type of SC and an issuer code of **. The SC (Special, Check) is a service specifically for check authorization transactions.

The Router module then determines whether the transaction is over or under the floor limit and routes the transaction accordingly.

If the authorizer indicated in the AST is AUTHF, NEGN, or NEGF, the Router module declines the request (force post transactions are not allowed for check authorization transactions). If the authorizer indicated in the AST is not AUTHF, NEGN, NEGF, or AUTHH, the Router module sends the transaction to the destination in the AST.

If the Router module cannot locate an AST record, it does not attempt to find a record using either the retailer program or * as it does when routing other transactions. Instead, the Router module automatically uses AUTHH as the primary authorizer.

Using the CART

If the Router module determines AUTHH to be the authorizer, the Router module attempts to read the CART according to the information available in the transaction message. The key values used to read the CART vary according to whether the transaction was acquired by an on-us terminal and the data present in the transaction message. The following two tables list the key values used to read the CART for on-us (terminal acquired) and not-on-us (interchange acquired) transactions.

On-Us Terminals		
CART Keys Used	PAN Present in Track 2	RTTN Present in MICR Data
1. Partial primary key lookup using the check routing group from the PRDF only.		
2. Alternate key lookup using the routing and transit number. If no record found, use key 1 above.		✓
3. Primary key lookup using the check routing group from PRDF and the FIID from the CPF. If no record found, use key 1 above.	✓	
4. Primary key lookup using the check routing group from PRDF and the FIID from CPF. If no record found, use alternate key. If still no record found, use key 1 above.	✓	✓

Not-On-Us Terminals		
CART Keys Used	PAN Present in Track 2	RTTN Present in MICR Data
1. Primary key lookup using the check routing group from the CHK-RTG-GRP param in the LCONF and the FIID from ICF or ICFE		
2. Alternate key lookup using the routing and transit number. If no record found, use key 1 above.		✓
3. Primary key lookup using the check routing group from the CHK-RTG-GRP param in the LCONF and the FIID from the CPF. If no record found, use key 1 above.	✓	
4. Primary key lookup using the check routing group from the CHK-RTG-GRP param in the LCONF and the FIID from the CPF. If no record found, use alternate key. If still no record found, use key 1 above.	✓	✓

Check authorization transactions are routed according to transaction code, check verification/guarantee service provider code from the PRDF, and check ID type. When the eCheck add-on product is installed, check authorization transactions are also routed according to authorization method and authorization level.

The CART is a table similar in function to the AST. The Router module uses a combination of transaction code, check verification/guarantee service provider code, and check ID type to select an entry in the CART. From this entry, the Router module determines a routing destination for the check authorization transaction.

The Router module searches the CART for an entry with the appropriate transaction code, check verification/guarantee service provider code, and check ID type according to the hierarchy presented in the table below.

Preference	Transaction Code	Service Provider Code	Check ID Type
1	Match	Match	Match
2	Match	Match	*
3	Match	*	Match
4	Match	*	*
5	*	Match	Match
6	*	Match	*
7	*	*	Match
8	*	*	*

When the eCheck add-on product is installed, the Router module updates the Check Guarantee/Verification 2 token in the PSTM with information obtained from the CART. If the token does not already exist, the Router module creates it.

The Router module then sends the transaction request to the process specified in the SYMBOLIC NAME field of the CART entry.

Refer to the ***BASE24-pos Files Maintenance Manual*** for more information about the CART.

Enhanced Reversal Routing

The POS-RVSL-ROUTE param in the LCONF indicates whether the Router module uses normal or enhanced routing logic when sending reversal transactions to Interchange Interface processes acting as transaction authorizers (issuers). When this param is set to a value of 0, the Router module uses normal reversal routing logic (i.e., using the AST, SPROUTE, or IDF routing methods, followed by additional star routing). When this param is set to a value of 1, the Router module saves the symbolic name of the authorizer when the original transaction response is sent to an Interchange Interface process. The module then sends a subsequent reversal transaction to the symbolic name saved from the original transaction response.

Enhanced reversal routing is required when multiple Interchange Interface processes are used to communicate with a single interchange, because each process writes to its own ILF. When an Interchange Interface process receives a reversal, it attempts to retrieve the original transaction data from its ILF. If the original transaction was processed by another Interchange Interface process, the process cannot retrieve the original transaction data from its ILF and, therefore, drops the reversal.

Enhanced reversal routing ensures that the reversal transaction is sent to the same Interchange Interface process that received the original transaction response. For more information on the POS-RVSL-ROUTE param, refer to the ***BASE24 Logical Network Configuration File Manual***.

Note: When enhanced reversal routing is enabled, the Router/Authorization module does not reverse the transaction from the BASE24 database. For this reason, enhanced reversal routing must be used only with authorization type 0 (host) and authorization level 1 (online).

Approval Code Generation

The TRAN.APPRV-CDE field in the POS Standard Internal Message (PSTM) contains the approval code for a transaction. If the transaction is sent to an interchange that returns approval codes to be printed on the receipt, the returned approval code is placed in this field. If the transaction is approved by the Authorization module or a host, the value in this field is generated by the Authorization module or the host.

When the Authorization module generates the approval code, it first calculates a temporary eight-digit approval code based upon the transaction time, the primary account number (PAN), and the transaction amount. The following factors determine which digits from this temporary approval code are then used for the approval code that is placed in the TRAN.APPRV-CDE field in the PSTM.

- The TRAN.APPRV-CDE-LGTH field in the PSTM contains the approval code length set by the Device Handler module, ISO Host Interface process, or Interchange Interface process.

The Device Handler module retrieves this length from the APPRV-CDE-LGTH field in the PTDF or PTDS1. The ISO Host Interface process retrieves this length from the POS message. The Interchange Interface process retrieves this length from the message received from the interchange or from the APPRV-CDE-LGTH field in the BASE24-pos segment of the ICF or ICFE.

- The APPROVAL CODE LENGTH field on CPF screen 8 contains the approval code length specified by a card issuer.

The Authorization module checks for this value in extended memory. If this value exists and is less than the value currently in the TRAN.APPRV-CDE-LGTH field in the PSTM, the Authorization module moves this value to the TRAN.APPRV-CDE-LGTH field in the PSTM.

- The approval code placed in the TRAN.APPRV-CDE field in the PSTM cannot be all zeros. It is not possible for all eight digits of the temporary approval code to be zeros, although it is possible for a number of contiguous digits to be zeros.

The Authorization module extracts the right-most portion of the temporary approval code. The adjusted value in the TRAN.APPRV-CDE-LGTH field in the PSTM controls the number of digits extracted. If all of the extracted digits are zero, the Authorization module drops the right-most digit and extracts the next available digit from the temporary approval code. The Authorization module repeats this process until the extracted portion no longer contains all zeros.

As an example, the following table shows the processing performed if the Authorization module requires four attempts to find a four-digit approval code that does not contain all zeros. The digits used from the eight-digit temporary approval code are bolded.

Attempt	Temporary Approval Code	Approval Code	Comment
1	2100 0000	0000	The four right-most digits give an approval code of all zeros.
2	210 00000	0000	Moving one position to the left still gives an approval code of all zeros.
3	21 000000	0000	Moving another position to the left still gives an approval code of all zeros.
4	2 1000000	1000	Moving another position to the left gives an approval code that is not all zeros. No further processing is necessary.

Referral Code Generation

The POS Referral File (PRF) is used with CRT Authorization. The key to the PRF is a combination of the retailer ID and the referral code (2 to 6 digits). The retailer ID usually identifies a specific retailer, making the chance of duplicate key errors when writing PRF records is relatively small. However, certain interchanges use the retailer ID to identify another member of the interchange that acquires the transaction. This means that all retailers associated with a particular acquirer have the same retailer ID, increasing the likelihood of duplicate key errors when writing PRF records.

When writing a PRF record, the BASE24-pos Router/Authorization process checks for a system error that indicates an attempt to add a record with a duplicate key. The Router/Authorization process does not drop the PRF record after a single unsuccessful attempt to write it because of a duplicate key error. Instead, the Router/Authorization process increases the value of the referral code by one and attempts to write the PRF record with the new key. If a duplicate key error is returned again, the procedure of increasing the referral code by one and attempting to write the PRF record with the new key is repeated. The PRF record is dropped only after the procedure is performed unsuccessfully 100 times.

If adding one to the referral code increases the length of the referral code (e.g., 9999 plus 1 equals 10000), the Router/Authorization process subtracts 100 from the referral code (e.g., 10000 minus 100 equals 9900) before attempting to write the PRF record.

As an example, the following table shows the processing performed if the Router/Authorization process requires four attempts to write a PRF record successfully.

Attempt	Referral Code	Comment
1	9998	A duplicate key error is returned.
2	9999	A duplicate key error is returned.
3	9900	The referral code is 9999 plus 1 minus 100. A duplicate key error is returned.
4	9901	The PRF record is written successfully with this referral code. The procedure does not need to be performed again.

ACI Worldwide, Inc.

5: BASE24-pos Message Flows

Transactions that route to or through the BASE24-pos system for authorization do so in a series of transaction messages. This section provides examples of transaction message flows through the BASE24-pos system under various types of processing situations. The transaction message flow diagrams are organized as follows:

- Authorization level 1
- Authorization level 2
- Authorization level 3
- Multi-nodal transactions
- Administrative transactions
- Interchange transactions
- APCF-routed transactions
- Mobile top-up transactions

Note: This section does not include all possible transaction message flows, only a representative sampling. The examples included are intended to familiarize you with typical message flows and give you a sense of how messages move through the BASE24 system under different types of processing situations.

Diagram Information

The following provides some general information on the transaction message flow diagrams in this section.

Special Diagram Notations

In addition to message types, the following special notations are used in the transaction message flow diagrams in this section.

Notation	Description
PREQ	Authorization request initiated by a POS device connected to the BASE24-pos system. This message is in the native-mode language of the POS device. The Device Handler module translates this native-mode message into a PSTM request message (0200).
PRES	Authorization response sent to a POS device connected to the BASE24-pos system. The Device Handler module translates PSTM response messages (0210s) into the native-mode language of the POS device.
PFAI	Authorization response returned to the Device Handler/Router/Authorization process from the XPNET process if the message could not be delivered to the originator. The message is the 0210 message in native format that was sent to the originator. However, the XPNET process has changed one field in the header to indicate to the Device Handler/Router/Authorization process that the message is being returned as undeliverable.
FAI	A message failure, the result of a no-response situation with an authorizing issuer host.
ADM	An administrative transaction.
CUST	The customer portion of an administration transaction.
IREQ	Authorization request sent to, or received from, an interchange. The Interchange Interface process translates PSTM request messages (0200s) into requests understandable to the interchange and external requests from the interchange into 0200 messages.

Notation	Description
IRES	Authorization response sent to, or received from, an interchange. The Interchange Interface process translates incoming responses into responses understandable to the BASE24-pos system and outgoing responses into responses understandable to the interchange.
IADV	Transaction advice sent to an interchange. The Interchange Interface process generates transaction advices to interchanges.
IRVSL	Transaction reversal sent to an interchange. The Interchange Interface process translates PSTM messages (0420s) into messages understandable to the interchange.
TREQ	Authorization request sent to a secondary provider (e.g., telephone company (telco)) host. The Telco Interface process translates PSTM request messages (0200s) into requests understandable to the telco host.
TRES	Authorization response received from a secondary provider (e.g., telephone company (telco)) host. The Telco Interface process translates incoming responses into responses understandable to the BASE24-pos system.

Diagram Limitations

Although the diagrams in this section describe the general transaction flow through a BASE24-pos system, some of the finer points of transaction flows cannot adequately be shown in diagram format. Some of these points include differences between device types and steps taken before actual processing begins.

For example, all diagrams except mobile top-up depict the normal transaction flow for transactions originating at a Visa II device. Although the transaction flows will be similar for transactions originating at a Visa I or a BASE24 Standard POS device, specific differences are not noted. Only BASE24 Standard POS devices support mobile top-up transactions.

Also, when a transaction enters a bound Device Handler/Router/Authorization process, the Router module is the first module to receive the transaction. Without actually processing the transaction, the Router module passes the transaction to the Device Handler module, which converts the transaction message from the device

native format into PSTM format. The Device Handler module then passes the transaction to the Router module. The Router module continues processing, often times by passing the transaction on to the Authorization module. The diagrams on the following pages do not show the first time the transaction enters the Router module as described above, but instead begin the transaction flow with the Device Handler module where actual processing begins.

Processing Assumptions

Several processing assumptions have been made for the transaction message flow diagrams in this section. These assumptions are described below.

Transaction Allowed Processing

Transaction allowed processing is not depicted in these diagrams. Refer to section 3 of this manual for detailed descriptions of transaction allowed processing.

Host Multithreading and Text-Level Acknowledgments

Unless specifically stated to the contrary, all of the transaction message flow diagrams in this section assume that hosts are multithreaded and are not configured to provide or require text-level acknowledgments for the messages they receive and send.

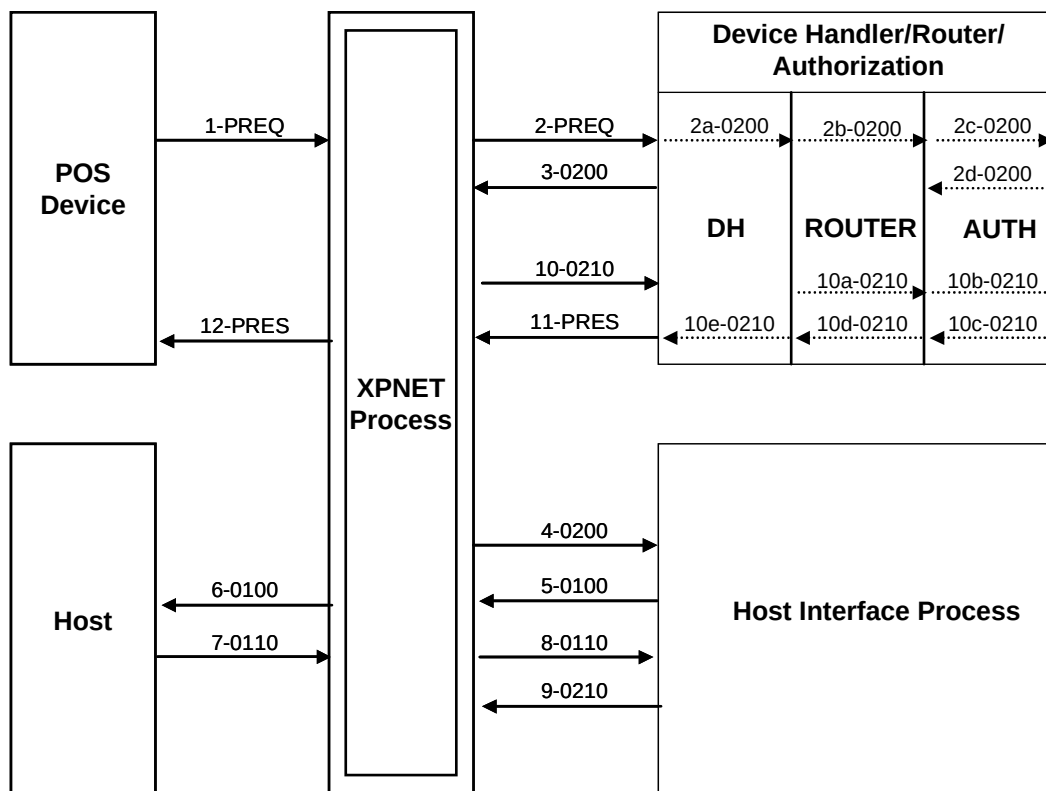
Note: Single-threaded hosts and multithreaded hosts can be configured in the Host Configuration File (HCF) to respond with text-level acknowledgments or responses to each message they receive (e.g., 0230 to a 0220, or 0430 to a 0420).

Authorization Level 1

The following pages contain message flows for transactions being processed by the BASE24-pos system using level 1 (online) authorization processing. Authorization level 1 requires that a host authorize all transactions. The BASE24-pos system can perform several preauthorization checks, but only a host can authorize transactions. If a host is unavailable, the BASE24 system tries to route the transaction to an alternate authorizer according to the Authorization Selection Table (AST). If the alternate routing suggested in the AST fails or alternate routing is not defined, the BASE24 system declines the transaction.

Approved Preauthorization

In the diagram below, the transaction request is for a preauthorization. AUTHH is the primary authorizer and the host approves the transaction. A completion is not required for this transaction.



Steps

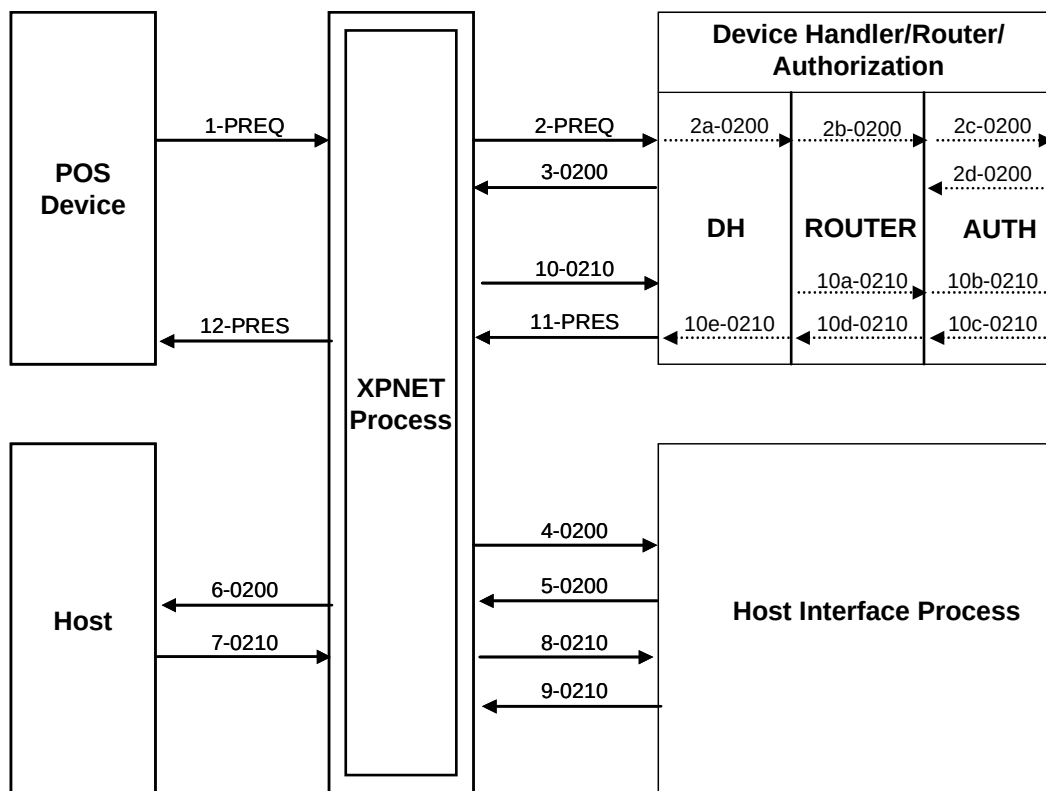
Processing

- 1-2** The POS device sends the transaction request to the BASE24-pos system as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Device Handler/Router/Authorization process.
- 2a-2d** The Device Handler module reads the BASE24-pos Terminal Data files record for the POS device to determine how to process the request. The Device Handler module then reformats the transaction request into a POS Standard Internal Message (PSTM), sets a response timer, and passes the transaction request as a 0200 message to the Router module.

Steps	Processing
	The Router module finds the retailer's record in the POS Retailer Definition File (PRDF), locates a Card Prefix File (CPF) record for the card involved in the transaction, and determines AST routing. The Router module then passes the message to the Authorization module. From the CPF, the Authorization module locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 1. The request passes all prescreening checks.
3–4	The Device Handler/Router/Authorization process sends the 0200 message to the ISO Host Interface process.
5–6	The ISO Host Interface process reformats the request into an external message format, sets an outbound request timer, and sends the request as a 0100 message to the host.
7–8	The host authorizes the transaction and returns a 0110 response message to the ISO Host Interface process.
9–10	The ISO Host Interface process deletes its outbound request timer, reformats the external message into a PSTM, and sends the response as a 0210 message to the Device Handler/Router/Authorization process.
10a–10e	The Authorization module updates its authorization files if they are being maintained by the BASE24-pos system, logs the response to the POS Transaction Log File (PTLF), and passes the response to the Device Handler module.
11–12	The Device Handler module translates the 0210 response message into a native-mode response message, deletes the response timer, and sends the message back to the POS device by way of the XPNET process.

Approved Financial Transaction

In the diagram below, the transaction is a financial transaction request. AUTHH is the primary authorizer and the host approves the transaction. A completion is not required for this transaction.



Steps

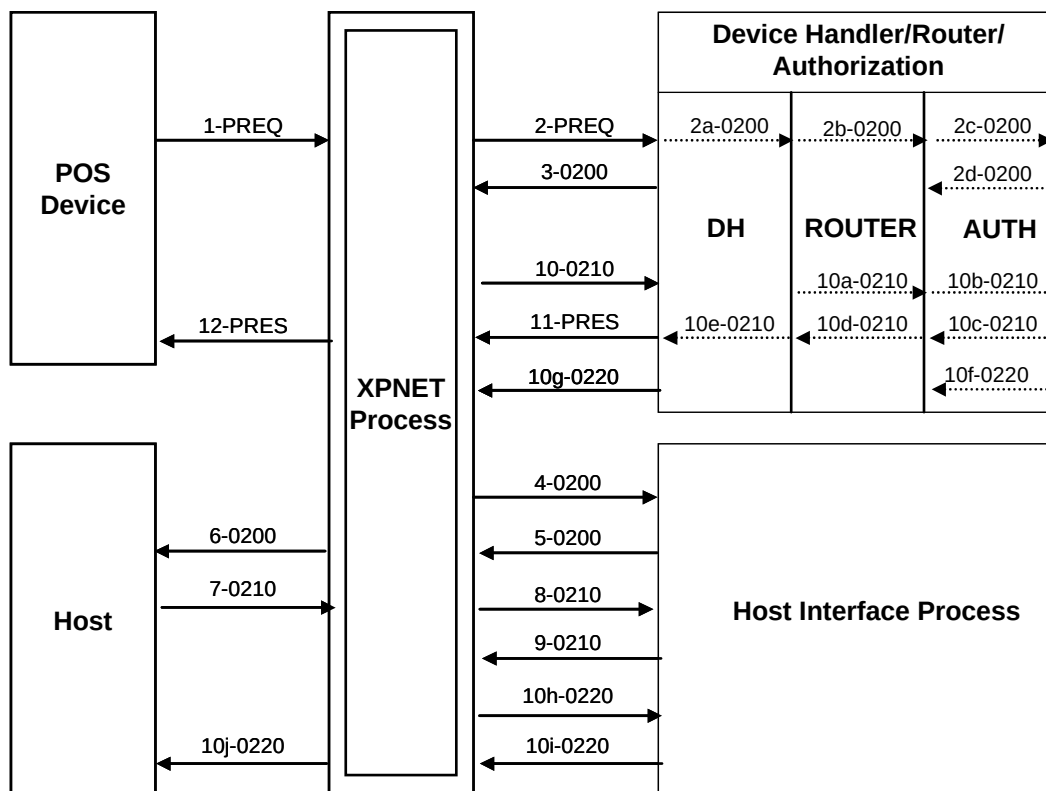
Processing

- 1–2 The POS device sends the transaction request to the BASE24-pos system as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Device Handler/Router/Authorization process.
- 2a–2d The Device Handler module reads the BASE24-pos Terminal Data files record for the POS device to determine how to process the request. The Device Handler module then reformats the transaction request into a POS Standard Internal Message (PSTM), sets a response timer, and passes the transaction request as a 0200 message to the Router module.

Steps	Processing
	The Router module finds the retailer's record in the POS Retailer Definition File (PRDF), locates a Card Prefix File (CPF) record for the card involved in the transaction, and determines AST routing. The Router module then passes the message to the Authorization module. From the CPF, the Authorization module locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 1. The request passes all prescreening checks.
3–4	The Device Handler/Router/Authorization process sends the 0200 message to the ISO Host Interface process.
5–6	The ISO Host Interface process reformats the request into an external message format, sets an outbound request timer, and sends the request as a 0200 message to the host.
7–8	The host authorizes the transaction and returns a 0210 response message to the ISO Host Interface process.
9–10	The ISO Host Interface process deletes its outbound request timer, reformats the external message into a PSTM, and sends the response as a 0210 message to the Device Handler/Router/Authorization process.
10a–10e	The Authorization module updates its authorization files if they are being maintained by the BASE24-pos system, logs the response to the POS Transaction Log File (PTLF), and passes the response to the Device Handler module.
11–12	The Device Handler module translates the 0210 response message into a native-mode response message, deletes the response timer, and sends the message back to the POS device by way of the XPNET process.

Approved Financial Transaction with Completions

In the diagram below, the transaction is a financial transaction request. AUTHH is the primary authorizer and the host approves the transaction. A completion is required for this transaction.



Steps

Processing

- 1–2 The POS device sends the transaction request to the BASE24-pos system as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Device Handler/Router/Authorization process.
- 2a–2d The Device Handler module reads the BASE24-pos Terminal Data files record for the POS device to determine how to process the request. The Device Handler module then reformats the transaction request into a POS Standard Internal Message (PSTM), sets a response timer, and passes the transaction request as a 0200 message to the Router module.

Steps	Processing
	The Router module finds the retailer's record in the POS Retailer Definition File (PRDF), locates a Card Prefix File (CPF) record for the card involved in the transaction, and determines AST routing. The Router module then passes the message to the Authorization module. From the CPF, the Authorization module locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 1. The request passes all prescreening checks.
3–4	The Device Handler/Router/Authorization process sends the 0200 message to the ISO Host Interface process.
5–6	The ISO Host Interface process reformats the request into an external message format, sets an outbound request timer, and sends the request as a 0200 message to the host.
7–8	The host authorizes the transaction and returns a 0210 response message to the ISO Host Interface process.
9–10	The ISO Host Interface process deletes its outbound request timer, reformats the external message into a PSTM, and sends the response as a 0210 message to the Device Handler/Router/Authorization process.
10a–10e	The Authorization module updates its authorization files if they are being maintained by the BASE24-pos system, logs the response to the POS Transaction Log File (PTLF), and passes the response to the Device Handler module.
10f	The Authorization module formats a 0220 completion message intended for the host.
10g–10h	The Device Handler/Router/Authorization process sends the 0220 completion message to the ISO Host Interface process.
10i–10j	The ISO Host Interface process reformats the 0220 completion message into an external message format and sends the completion message to the host. The completion message notifies the host that the transaction completed as authorized.

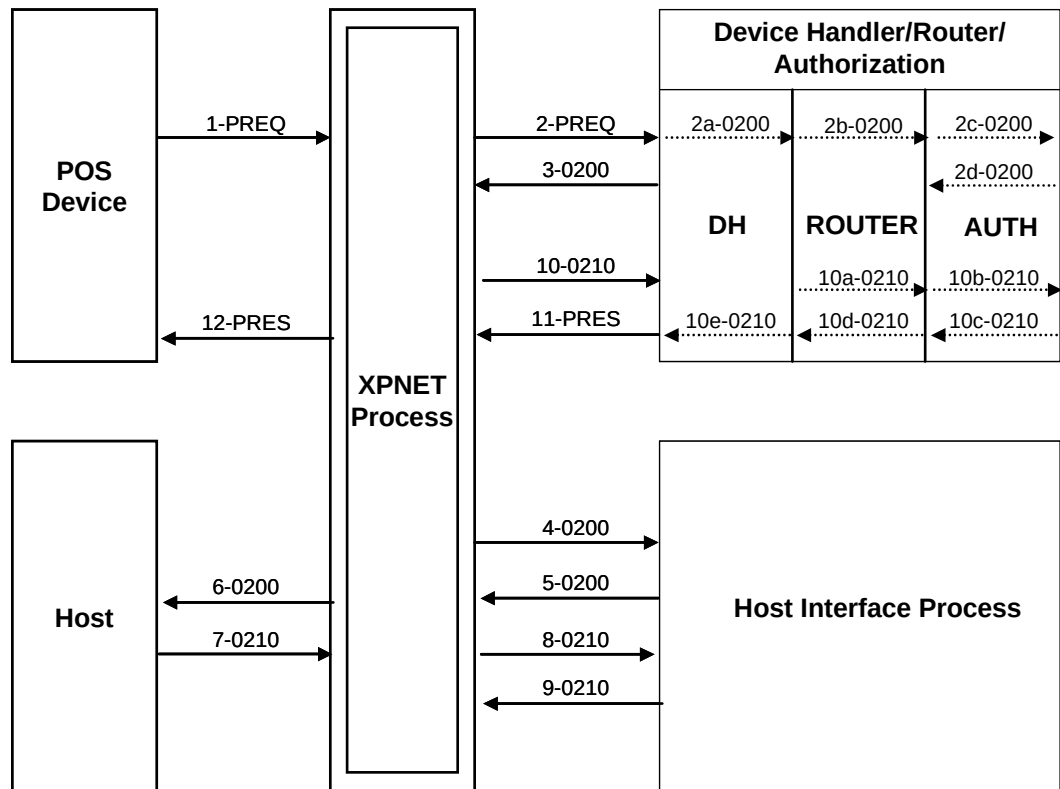
Steps

Processing

- 11–12 The Device Handler module translates the 0210 response message into a native-mode response message, deletes the response timer, and sends the message back to the POS device by way of the XPNET process.

Denied Financial Transaction

In the diagram below, the transaction is a financial transaction request. AUTHH is the primary authorizer and the host denies the transaction. Although a completion is required for this transaction type, a completion is not generated because the transaction was denied.



Steps

Processing

- 1–2 The POS device sends the transaction request to the BASE24-pos system as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Device Handler/Router/Authorization process.
- 2a–2d The Device Handler module reads the BASE24-pos Terminal Data files record for the POS device to determine how to process the request. The Device Handler module then reformats the transaction request into a POS Standard Internal Message (PSTM), sets a response timer, and passes the transaction request as a 0200 message to the Router module.

Steps**Processing**

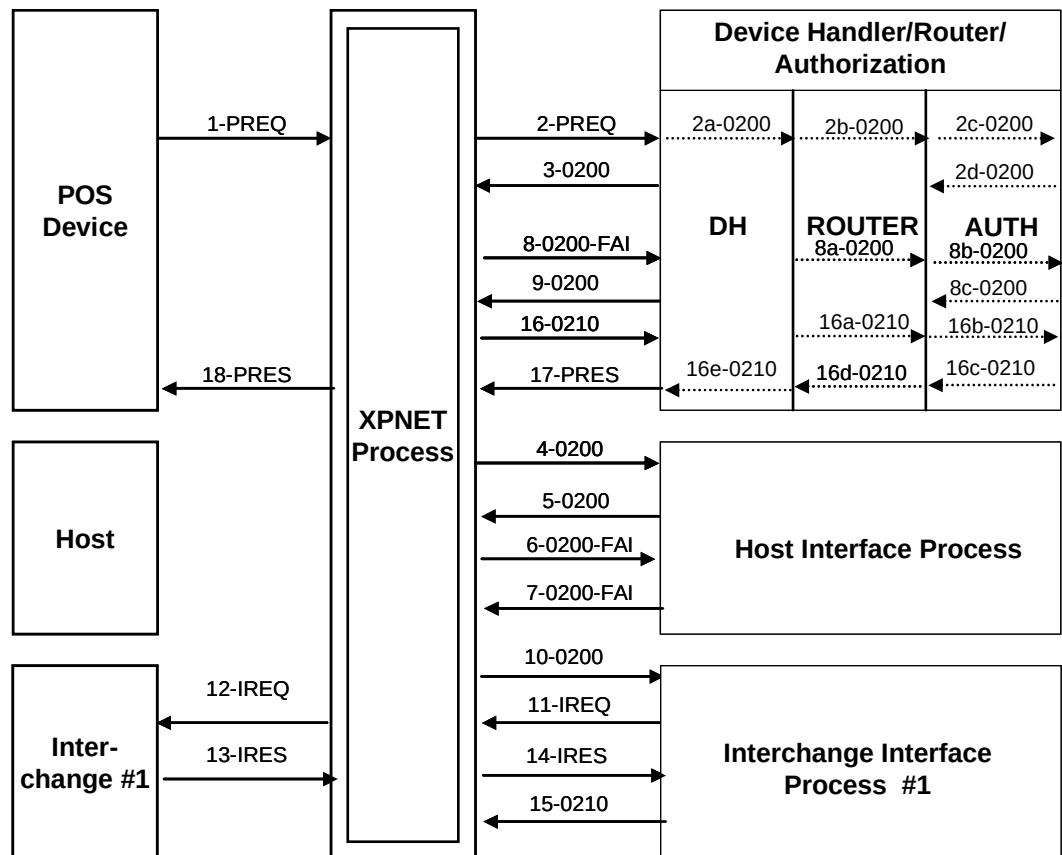
The Router module finds the retailer's record in the POS Retailer Definition File (PRDF), locates a Card Prefix File (CPF) record for the card involved in the transaction, and determines AST routing. The Router module then passes the message to the Authorization module.

From the CPF, the Authorization module locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 1. The request passes all prescreening checks.

- | | |
|---------|---|
| 3–4 | The Device Handler/Router/Authorization process sends the 0200 message to the ISO Host Interface process. |
| 5–6 | The ISO Host Interface process reformats the request into an external message format, sets an outbound request timer, and sends the request as a 0200 message to the host. |
| 7–8 | The host denies the transaction request and returns a 0210 response message to the ISO Host Interface process. |
| 9–10 | The ISO Host Interface process deletes its outbound request timer, reformats the external message into a PSTM, and sends the response as a 0210 message to the Device Handler/Router/Authorization process. |
| 10a–10e | The Authorization module logs the response to the POS Transaction Log File (PTLF) and passes the response to the Device Handler module. The Authorization module does not update its authorization files for denied transactions. The Device Handler module then deletes the authorization timer. |
| 11–12 | The Device Handler module translates the 0210 response message into a native-mode response message, deletes the response timer, and sends the message back to the POS device by way of the XPNET process. |

Interchange as Alternate Authorizer (Denied)

In the diagram below, the transaction is a financial transaction request initiated with an on-us debit or credit card. AUTHH is the primary authorizer, but communications to the host are down and, therefore, the host is not available to authorize the transaction. The ISO Host Interface process is not yet aware that the host is unavailable. Interchange #1 is the first alternate authorizer and, therefore, the transaction is sent there for authorization but authorization is denied.



Steps

1-2

The POS device sends the transaction request to the BASE24-pos system as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Device Handler/Router/Authorization process.

Processing

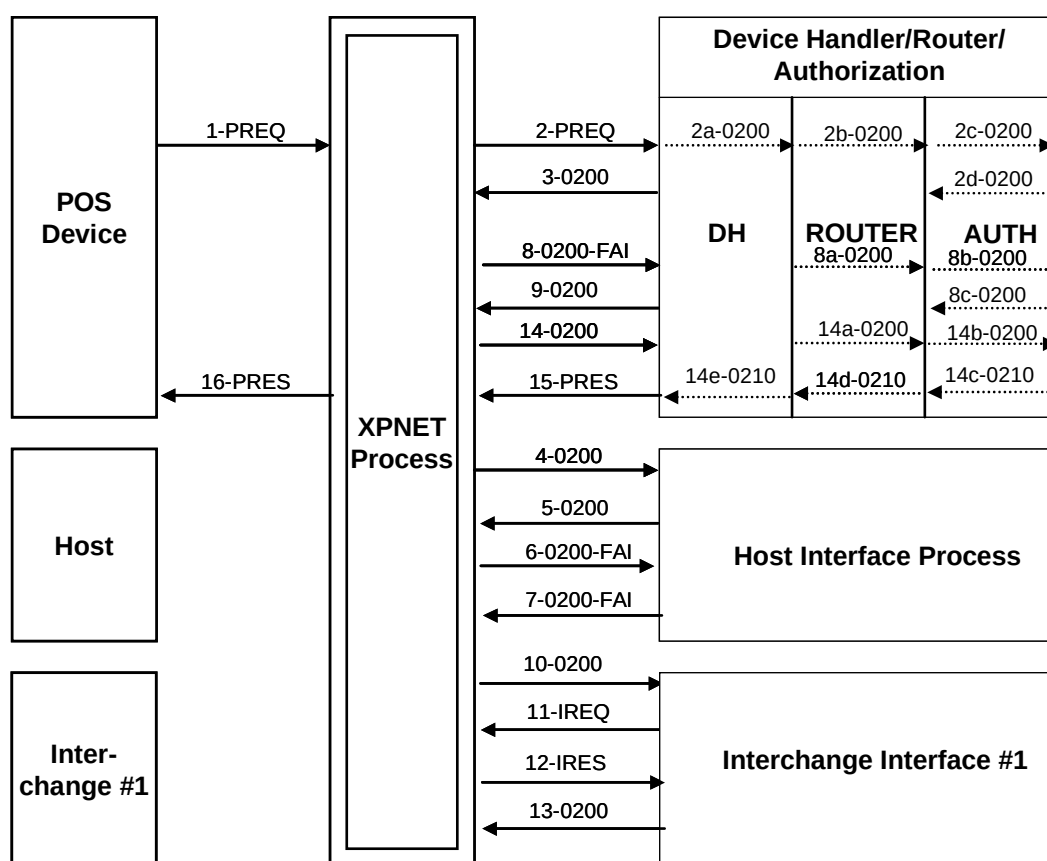
Steps	Processing
2a–2d	The Device Handler module reads the BASE24-pos Terminal Data files record for the POS device to determine how to process the request. The Device Handler module then reformats the transaction request into a POS Standard Internal Message (PSTM), sets a response timer, and passes the transaction request as a 0200 message to the Router module. The Router module finds the retailer’s record in the POS Retailer Definition File (PRDF), locates a Card Prefix File (CPF) record for the card involved in the transaction, and determines AST routing. The Router module then passes the message to the Authorization module. From the CPF, the Authorization module locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 1. The request passes all prescreening checks.
3–4	The Device Handler/Router/Authorization process sends the 0200 message to the ISO Host Interface process.
5–6	The ISO Host Interface process reformats the request into an external message format, sets an outbound request timer, and attempts to send the request as a 0200 message to the host. Communications to the host are down and, therefore, the 0200 message returns to the ISO Host Interface process from the XPNET process.
7–8	The ISO Host Interface process returns the 0200 message to the Device Handler/Router/Authorization process.
8a–8c	The Router and Authorization modules determine the first alternate authorizer to be Interchange #1.
9–10	The Device Handler/Router/Authorization process sends the 0200 message to the Interchange Interface process.
11–12	The Interchange Interface process reformats the request into an external message format, sets an outbound request timer, and sends the request to the interchange.
13–14	The interchange denies the transaction request and sends the response message to the Interchange Interface process.

Steps	Processing
15–16	The Interchange Interface process deletes the outbound request timer, translates the response from external message format to a 0210 internal response message, and sends the 0210 response message to the Device Handler/Router/Authorization process.
16a–16e	The Authorization module logs the response to the POS Transaction Log File (PTLF) and passes the response to the Device Handler module. The Authorization module does not update its authorization files because the transaction is denied.
17–18	The Device Handler module translates the 0210 response message into a native-mode response message, deletes the response timer, and sends the message back to the POS device by way of the XPNET process.

Alternate Authorization Not Possible

In the diagram below, the transaction is a financial transaction request. AUTHH is the primary authorizer according to AST routing, but communications to the host are down and, therefore, the host is not available to authorize the transaction. The ISO Host Interface process is not yet aware that the host is unavailable.

Interchange #1 is the first alternate authorizer but communications to it are down. No second alternate authorizer is named in the AST, so the BASE24 system takes the AST default action, which is to deny the transaction. Although a completion is required for this transaction type, a completion is not sent because the transaction was denied.



Steps

- 1–2 The POS device sends the transaction request to the BASE24-pos system as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Device Handler/Router/Authorization process.

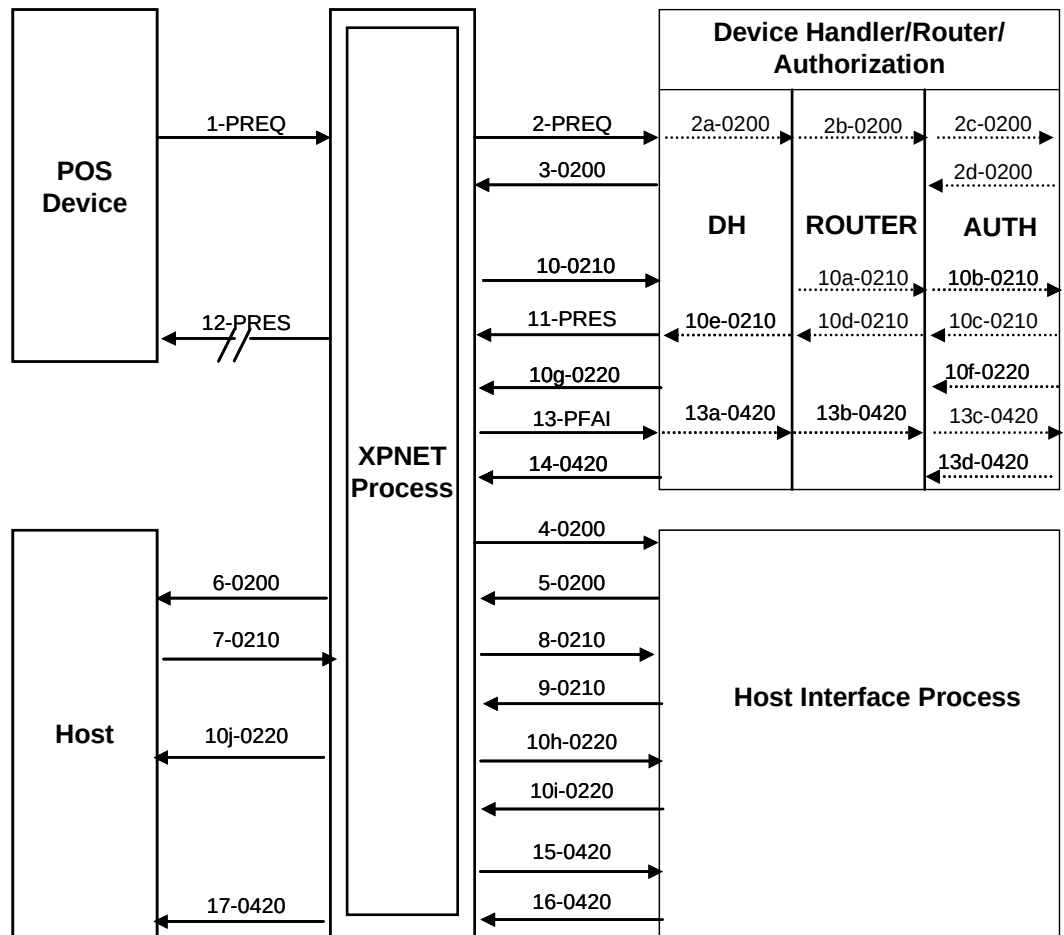
Processing

Steps	Processing
2a–2d	The Device Handler module reads the BASE24-pos Terminal Data files record for the POS device to determine how to process the request. The Device Handler module then reformats the transaction request into a POS Standard Internal Message (PSTM), sets a response timer, and passes the transaction request as a 0200 message to the Router module. The Router module finds the retailer’s record in the POS Retailer Definition File (PRDF), locates a Card Prefix File (CPF) record for the card involved in the transaction, and determines AST routing. The Router module then passes the message to the Authorization module. From the CPF, the Authorization module locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 1. The request passes all prescreening checks.
3–4	The Device Handler/Router/Authorization process sends the 0200 message to the ISO Host Interface process.
5–6	The ISO Host Interface process reformats the request into an external message format, sets an outbound request timer, and attempts to send the request as a 0200 message to the host. Communications to the host are down and, therefore, the 0200 message returns to the ISO Host Interface process from the XPNET process.
7–8	The ISO Host Interface process returns the 0200 message to the Device Handler/Router/Authorization process.
8a–8c	The Router and Authorization modules determine the first alternate authorizer to be Interchange #1.
9–10	The Device Handler/Router/Authorization process sends the 0200 message to the Interchange Interface process.
11–12	The Interchange Interface process reformats the request into an external message format, sets an outbound request timer, and attempts to send the request to the interchange. Communications to the interchange are down and, therefore, the request message returns to the Interchange Interface process from the XPNET process.

Steps	Processing
13–14	The Interchange Interface process returns the 0200 message to the Device Handler/Router/Authorization process.
14a–14e	The Router and Authorization modules determine that no other alternate authorizer has been named. Therefore, the Authorization module uses default authorization, which is to deny the transaction request.
15–16	The Device Handler module translates the 0210 response message into a native-mode response message, deletes the response timer, and sends the message back to the POS device by way of the XPNET process.

Reversal Without Host Acknowledgments

In the diagram below, the transaction is a financial transaction request that must be reversed because the POS device went offline after the transaction was authorized but before the transaction response reached the device. AUTHH is the primary authorizer and completions are required for this transaction type.



Steps

Processing

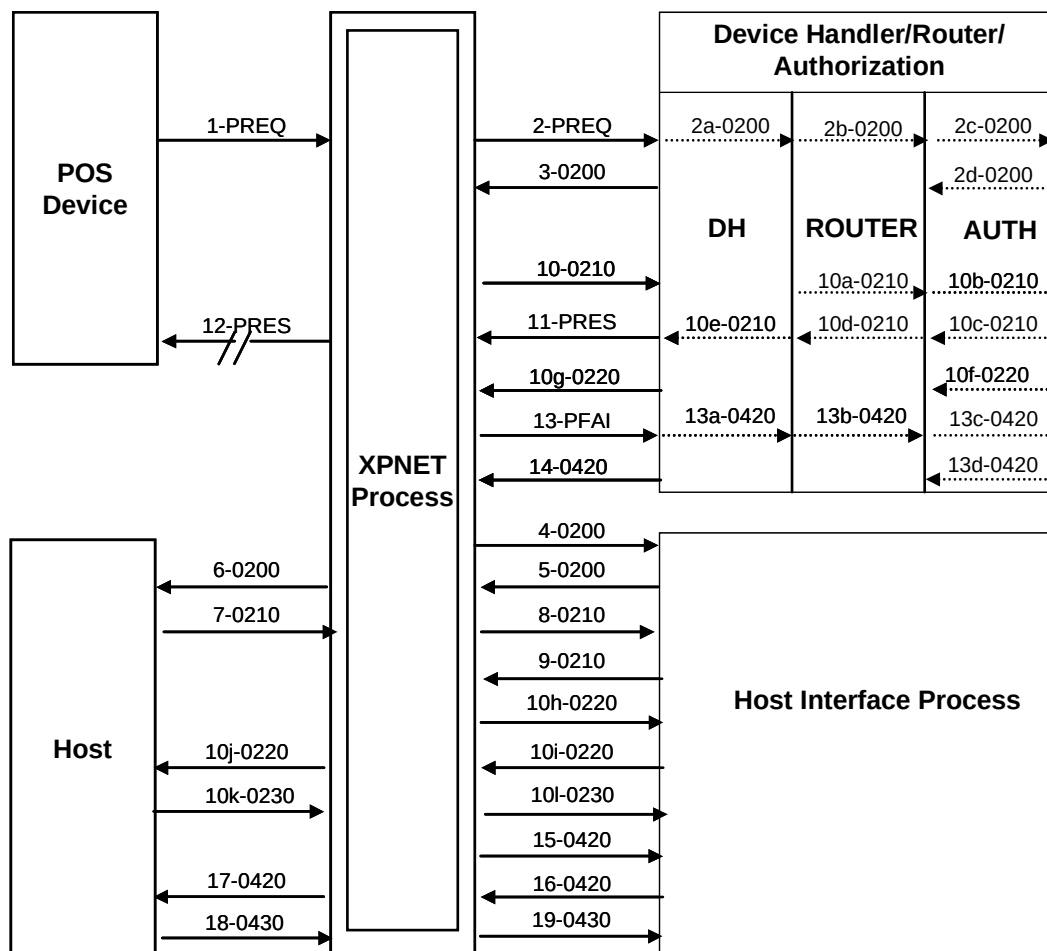
- 1-2 The POS device sends the transaction request to the BASE24-pos system as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Device Handler/Router/Authorization process.

Steps	Processing
2a–2d	The Device Handler module reads the BASE24-pos Terminal Data files record for the POS device to determine how to process the request. The Device Handler module then reformats the transaction request into a POS Standard Internal Message (PSTM), sets a response timer, and passes the transaction request as a 0200 message to the Router module. The Router module finds the retailer’s record in the POS Retailer Definition File (PRDF), locates a Card Prefix File (CPF) record for the card involved in the transaction, and determines AST routing. The Router module then passes the message to the Authorization module. From the CPF, the Authorization module locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 1. The request passes all prescreening checks.
3–4	The Device Handler/Router/Authorization process sends the 0200 message to the ISO Host Interface process.
5–6	The ISO Host Interface process reformats the request into an external message format, sets an outbound request timer, and sends the request as a 0200 message to the host.
7–8	The host authorizes the transaction request and returns a 0210 response message to the ISO Host Interface process.
9–10	The ISO Host Interface process deletes its outbound request timer, reformats the external message into a PSTM, and sends the response as a 0210 message to the Device Handler/Router/Authorization process.
10a–10e	The Authorization module updates its authorization files if they are being maintained by the BASE24-pos system, logs the response to the POS Transaction Log File (PTLF), and passes the response to the Device Handler module.
10f	The Authorization module formats a 0220 completion message intended for the host.

Steps	Processing
10g–10h	The Device Handler/Router/Authorization process sends a 0220 completion message to the ISO Host Interface process.
10i–10j	The ISO Host Interface process reformats the 0220 completion message into an external message format and sends the completion message to the host. The completion message notifies the host that the transaction completed as authorized.
11–12	The Device Handler module translates the 0210 response message into a native-mode response message, deletes the response timer, and attempts to send the message back to the POS device by way of the XPNET process. Communications with the POS device, however, are now down and the response message cannot reach the POS device to complete the transaction.
13	The XPNET process marks the response message as undeliverable and returns the message to the Device Handler/Router/Authorization process.
13a–13d	The Authorization module updates its authorization files if they are being maintained by the BASE24-pos system and logs the reversal to the POS Transaction Log File (PTLF). The Device Handler/Router/Authorization process determines where the original transaction was authorized and formats a 0420 reversal message to notify the authorizer that the transaction did not complete as authorized.
14–15	The Device Handler/Router/Authorization process sends the 0420 reversal message to the ISO Host Interface process.
16–17	The ISO Host Interface process reformats the 0420 reversal message into external message format and sends it to the host. The host can then use the reversal message to update its database appropriately.

Reversal With Host Acknowledgments

In the diagram below, the transaction is a financial transaction request that must be reversed because the POS device went offline after the transaction was authorized but before the transaction response reached the device. AUTHH is the primary authorizer and completions are required for this transaction type. In response to completion messages, including both advices and reversals, the host generates acknowledgment messages.



Steps

Processing

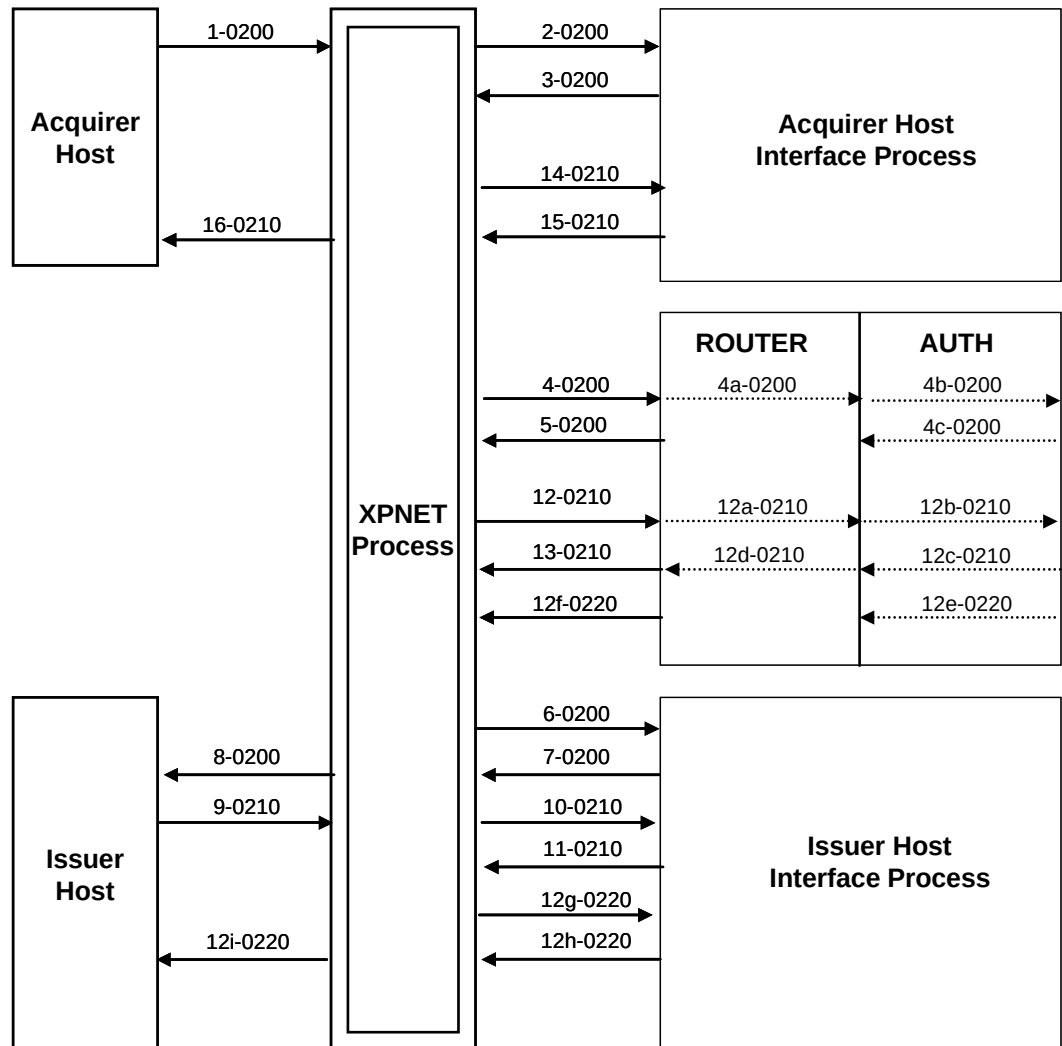
- 1–2 The POS device sends the transaction request to the BASE24-pos system as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Device Handler/Router/Authorization process.

Steps	Processing
2a–2d	The Device Handler module reads the BASE24-pos Terminal Data files record for the POS device to determine how to process the request. The Device Handler module then reformats the transaction request into a POS Standard Internal Message (PSTM), sets a response timer, and passes the transaction request as a 0200 message to the Router module. The Router module finds the retailer’s record in the POS Retailer Definition File (PRDF), locates a Card Prefix File (CPF) record for the card involved in the transaction, and determines AST routing. The Router module then passes the message to the Authorization module. From the CPF, the Authorization module locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 1. The request passes all prescreening checks.
3–4	The Device Handler/Router/Authorization process sends the 0200 message to the ISO Host Interface process.
5–6	The ISO Host Interface process reformats the request into an external message format, sets an outbound request timer, and sends the request as a 0200 message to the host.
7–8	The host approves the transaction request and returns a 0210 response message to the ISO Host Interface process.
9–10	The ISO Host Interface process deletes its outbound request timer, reformats the external message into a PSTM, and sends the response as a 0210 message to the Device Handler/Router/Authorization process.
10a–10e	The Authorization module updates its authorization files if they are being maintained by the BASE24-pos system, logs the response to the POS Transaction Log File (PTLF), and passes the response to the Device Handler module.
10f–10h	The Device Handler/Router/Authorization process formats and sends a 0220 completion message to the ISO Host Interface process.

Steps	Processing
10i–10j	The ISO Host Interface process reformats the 0220 completion message into an external message format, sets a response timer, and sends the completion message to the host. The completion message notifies the host that the transaction completed as authorized.
10k–10l	The host formats a 0230 advice response message and sends it to the ISO Host Interface process. The 0230 message notifies the ISO Host Interface process that the host received the 0220 advice. When the ISO Host Interface process receives the 0230 message, it deletes the response timer.
11–12	The Device Handler module translates the 0210 response message into a native-mode response message, deletes the response timer, and attempts to send the message back to the POS device by way of the XPNET process. Communications with the POS device, however, are down and the response message cannot reach the POS device to complete the transaction.
13	The XPNET process marks the response message as undeliverable and returns the message to the Device Handler/Router/Authorization process.
13a–13d	The Authorization module updates its authorization files if they are being maintained by the BASE24-pos system. The Device Handler/Router/Authorization process determines where the original transaction was authorized and formats a 0420 reversal message to notify the authorizer that the transaction did not complete as authorized.
14–15	The Device Handler/Router/Authorization process sends the 0420 reversal message to the ISO Host Interface process.
16–17	The ISO Host Interface process reformats the 0420 reversal message into external message format, sets a response timer, and sends it to the host. The host can then use the reversal message to update its database appropriately.
18–19	The host formats a 0430 acknowledgment message and sends it to the ISO Host Interface process. The 0430 message notifies the ISO Host Interface process that the host received the reversal message. When the ISO Host Interface process receives the 0430 message, it deletes the response timer.

Approved With Completions

In the diagram below, the transaction is a financial transaction request. AUTHH is the primary authorizer and the host approves the transaction. Completions are required for this transaction type.



Steps

1–2

A host acquires the transaction request. This acquirer host sends the transaction request to the BASE24-pos system in external message format, where it is received by the acquirer ISO Host Interface process.

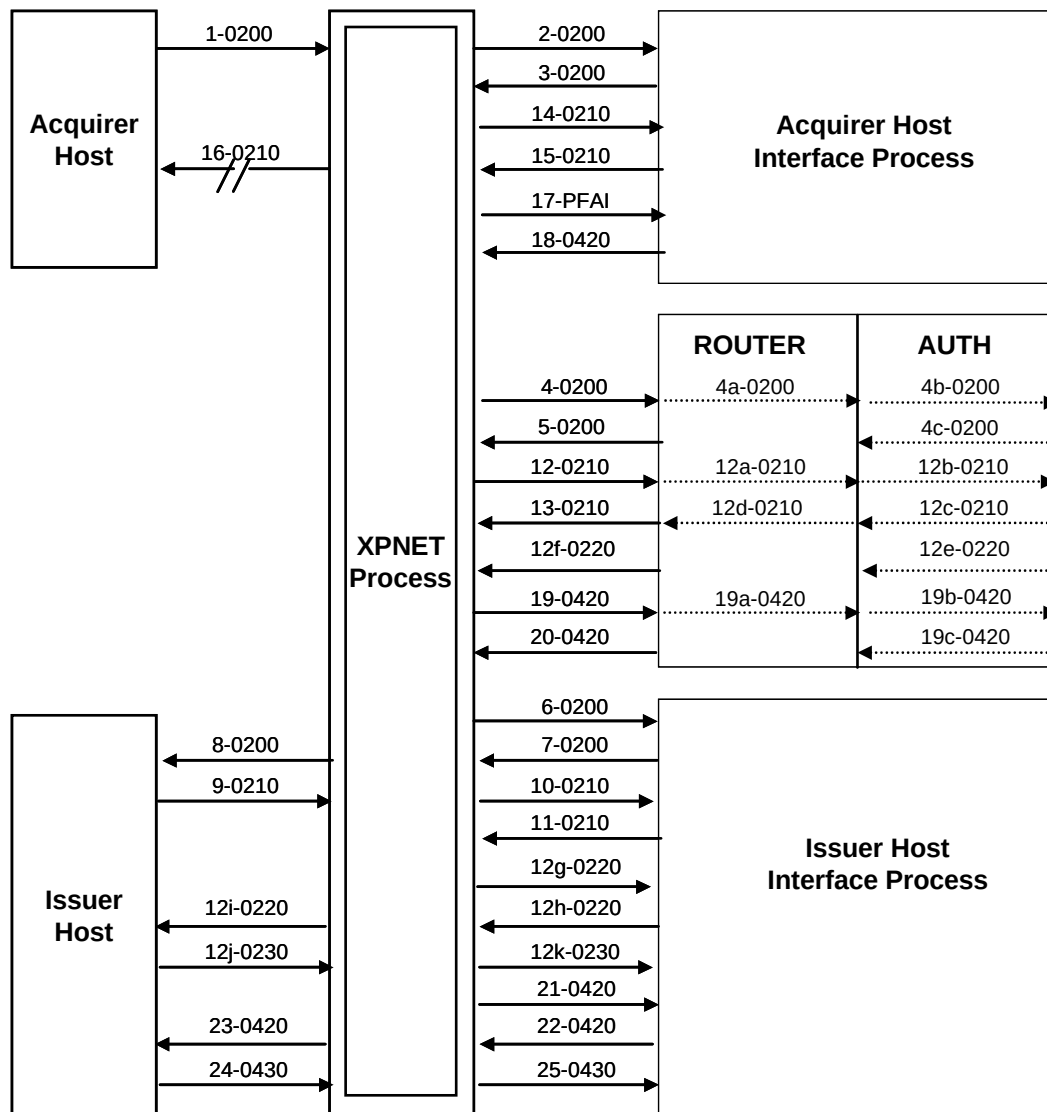
Processing

Steps	Processing
3–4	The ISO Host Interface process reformats the request message into the POS Standard Internal Message (PSTM) format, sets a response timer, and sends the 0200 request message to the Router/Authorization process. (When transactions are received from a host, the Device Handler module is not required.)
4a–4c	The Router module finds the retailer’s record in the POS Retailer Definition File (PRDF), locates a Card Prefix File (CPF) record for the card involved in the transaction, and determines AST routing. The Router module then passes the message to the Authorization module. From the CPF, the Authorization module locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 1. The request passes all prescreening checks.
5–6	The Router/Authorization process sends the 0200 message to the issuer ISO Host Interface process.
7–8	The issuer ISO Host Interface process reformats the request into an external message format, sets an outbound request timer, and sends the request as a 0200 message to the issuer host.
9–10	The issuer host approves the transaction request and returns a 0210 response message to the issuer ISO Host Interface process.
11–12	The issuer ISO Host Interface process deletes its outbound request timer, reformats the external message into a PSTM, and sends the response as a 0210 message to the Router/Authorization process.
12a–12d	The Authorization module updates its authorization files if they are being maintained by the BASE24-pos system and logs the response to the POS Transaction Log File (PTLF).
12e–12g	The Router/Authorization process formats and sends a 0220 completion message to the issuer ISO Host Interface process.
12h–12i	The issuer ISO Host Interface process reformats the 0220 completion message into an external message format and sends the completion message to the host. The completion message notifies the host that the transaction completed as authorized.

Steps	Processing
13–14	The Router/Authorization process formats a 0210 response message and sends it to the acquirer ISO Host Interface process.
15–16	The acquirer ISO Host Interface process reformats the 0210 response into external message format, deletes the response timer, and sends the response to the acquirer host.

Reversal With Host Acknowledgments

In the diagram below, the transaction is a financial transaction request that must be reversed because the communications to the acquirer host went down after the transaction was approved but before the transaction response reached the acquirer host. AUTHH is the primary authorizer and completions are required for this transaction type. In response to completion messages the host generates acknowledgment messages.



Steps	Processing
1–2	A host acquires the transaction request. This acquirer host sends the transaction request to the BASE24-pos system in external message format, where it is received by the acquirer ISO Host Interface process.
3–4	The acquirer ISO Host Interface process reformats the request message into the POS Standard Internal Message (PSTM) format, sets a response timer, and sends the 0200 request message to the Router/Authorization process. (When transactions are received from a host, the Device Handler module is not required.)
4a–4c	The Router module finds the retailer’s record in the POS Retailer Definition File (PRDF), locates a Card Prefix File (CPF) record for the card involved in the transaction, and determines AST routing. The Router module then passes the message to the Authorization module. From the CPF, the Authorization module locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 1. The request passes all prescreening checks.
5–6	The Router/Authorization process sends the 0200 message to the issuer ISO Host Interface process.
7–8	The issuer ISO Host Interface process reformats the request into an external message format, sets an outbound request timer, and sends the request as a 0200 message to the host.
9–10	The issuer host approves the transaction request and returns a 0210 response message to the issuer ISO Host Interface process.
11–12	The issuer ISO Host Interface process deletes its outbound request timer, reformats the external message into a PSTM, and sends the response as a 0210 message to the Router/Authorization process.
12a–12d	The Authorization module updates its authorization files if they are being maintained by the BASE24-pos system and logs the response to the POS Transaction Log File (PTLF).
12e–12g	The Router/Authorization process formats and sends a 0220 completion message to the issuer ISO Host Interface process.

Steps	Processing
12h–12i	The issuer ISO Host Interface process reformats the 0220 completion message into an external message format, sets a response timer, and sends the completion message to the host. The completion message notifies the host that the transaction completed as authorized.
12j–12k	The issuer host formats a 0230 advice response and sends the advice to the issuer ISO Host Interface process. This 0230 advice response notifies the issuer ISO Host Interface process that the issuer host received the transaction. The ISO Host Interface process deletes the response timer.
13–14	The Router/Authorization process formats a 0210 response message and sends it to the acquirer ISO Host Interface process.
15–16	The acquirer ISO Host Interface process reformats the 0210 response into external message format, deletes the response timer, and attempts to send the response to the acquirer host. Communications with the acquirer host, however, are down and the response message cannot reach the acquirer host to complete the transaction.
17	The XPNET process marks the 0210 response message as undeliverable and returns the message to the acquirer ISO Host Interface process.
18–19	The acquirer ISO Host Interface process formats a 0420 reversal message to notify the authorizer that the transaction did not complete as authorized and sends it to the Router/Authorization process.
19a–19c	The Authorization module updates its authorization files if they are being maintained by the BASE24-pos system. The Router/Authorization process determines where the original transaction was authorized and formats a 0420 reversal message to notify the authorizer that the transaction did not complete as authorized.
20–21	The Router/Authorization process sends the 0420 reversal message to the issuer ISO Host Interface process.
22–23	The issuer ISO Host Interface process reformats the 0420 reversal message into external message format, sets a response timer, and sends the 0420 reversal message to the issuer host. The issuer host can use this reversal message to update its database appropriately.

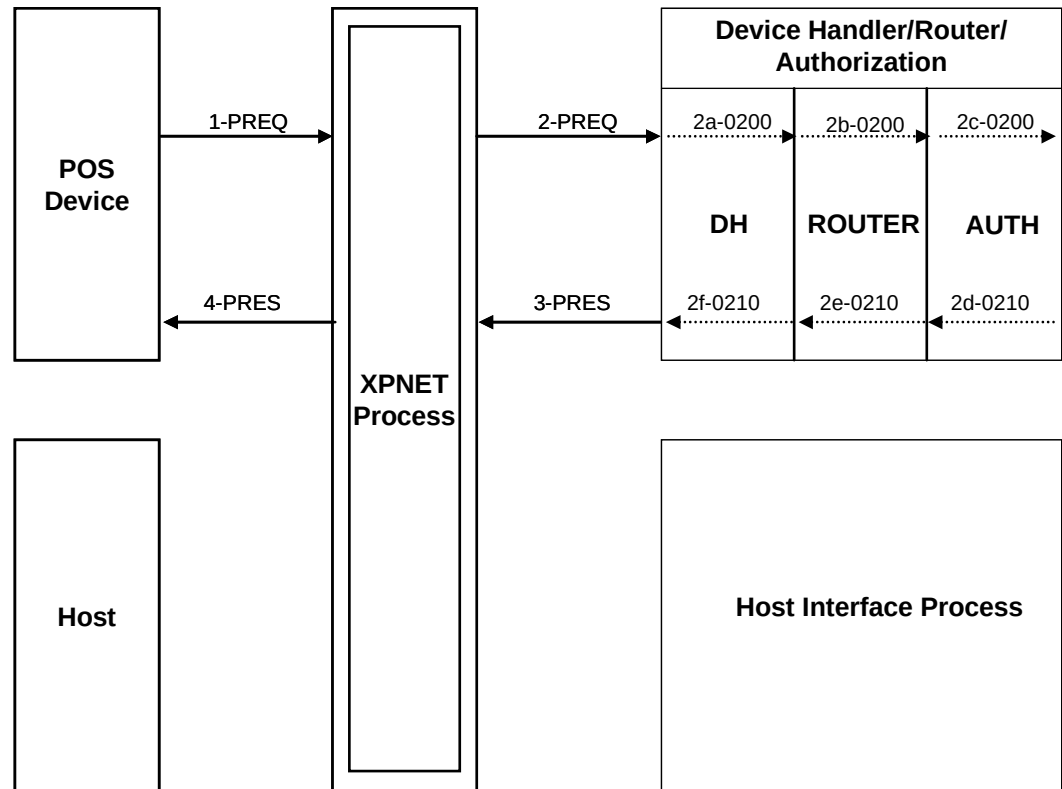
Steps	Processing
24–25	The issuer host formats a 0430 reversal response and sends the 0430 reversal response to the issuer ISO Host Interface process. This reversal response notifies the issuer ISO Host Interface process that the host received the reversal. The ISO Host Interface process deletes the response timer.

Authorization Level 2

The following pages contain message flows for transactions being processed by the BASE24-pos system using level 2 (offline) authorization processing. Authorization level 2 requires that the BASE24-pos system authorize all transactions. The host is not involved in authorizing transactions, although it might receive completions.

Approved Without Completions

In the diagram below, the transaction is a financial transaction request. AUTHH is the primary authorizer and the BASE24-pos system approves the transaction. Completions are not required for this transaction type.



Steps

Processing

- | | |
|-------|---|
| 1–2 | The POS device sends the transaction request to the BASE24-pos system as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Device Handler/Router/Authorization process. |
| 2a–2c | The Device Handler module reads the BASE24-pos Terminal Data files record for the POS device to determine how to process the request. The Device Handler module then reformats the transaction request into a POS Standard Internal Message (PSTM), sets a response timer, and passes the transaction request as a 0200 message to the Router module. |

Steps**Processing**

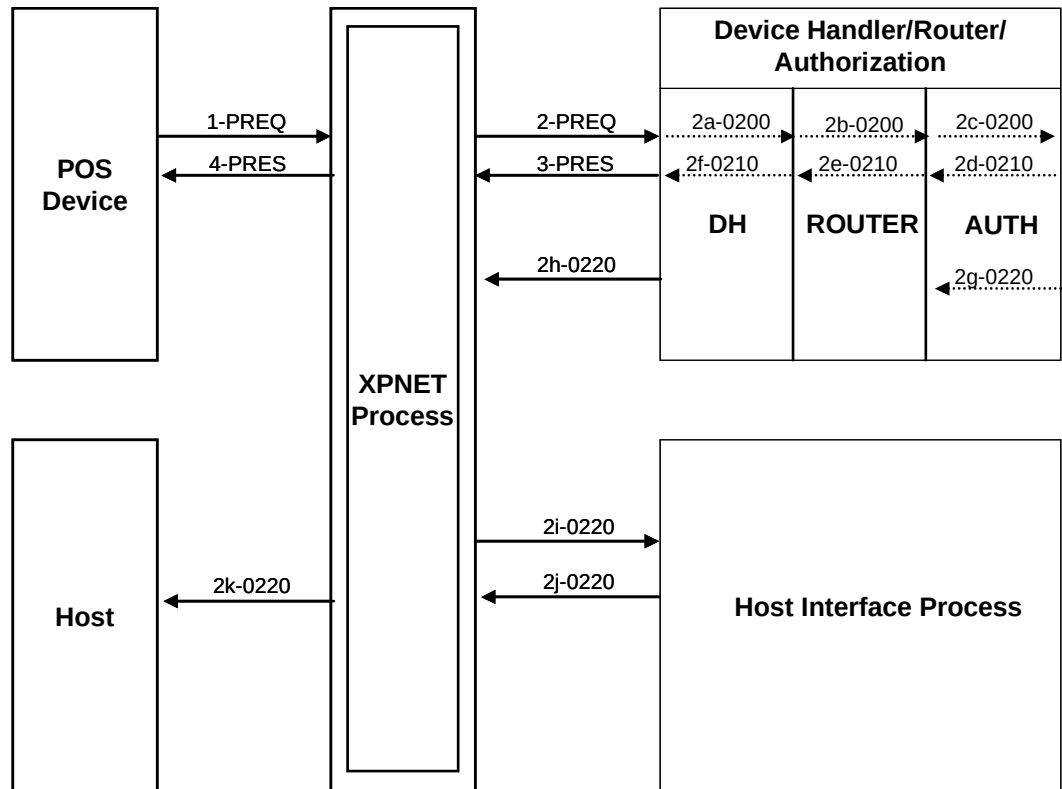
The Router module finds the retailer's record in the POS Retailer Definition File (PRDF), locates a Card Prefix File (CPF) record for the card involved in the transaction, and determines AST routing. The Router module then passes the message to the Authorization module.

From the CPF, the Authorization module locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 2.

- | | |
|-------|---|
| 2d–2f | The Authorization module approves the request, updates the database files appropriately, logs the response to the POS Transaction Log File (PTLF), and formats a 0210 response message. |
| 3–4 | The Device Handler module reformats the response message into native-mode format, deletes the response timer, and sends the response to the POS device. |

Approved With Completions

In the diagram below, the transaction is a financial transaction request. AUTHH is the primary authorizer and the BASE24-pos system approves the transaction. Completions are required for this transaction type.



Steps

Processing

- 1–2 The POS device sends the transaction request to the BASE24-pos system as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Device Handler/Router/Authorization process.
- 2a–2c The Device Handler module reads the BASE24-pos Terminal Data files record for the POS device to determine how to process the request. The Device Handler module then reformats the transaction request into a POS Standard Internal Message (PSTM), sets a response timer, and passes the transaction request as a 0200 message to the Router module.

Steps**Processing**

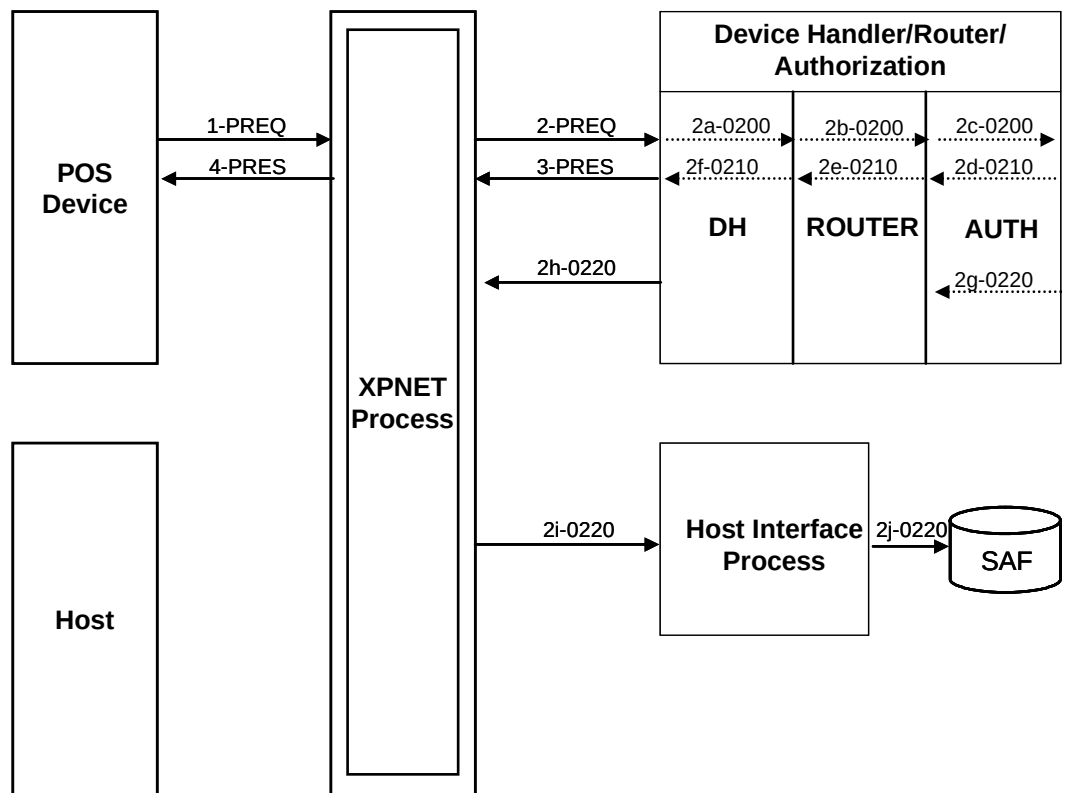
The Router module finds the retailer's record in the POS Retailer Definition File (PRDF), locates a Card Prefix File (CPF) record for the card involved in the transaction, and determines AST routing. The Router module then passes the message to the Authorization module.

From the CPF, the Authorization module locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 2.

- | | |
|-------|--|
| 2d–2f | The Authorization module approves the request and updates the cardholder files appropriately. |
| 2g–2i | The Authorization module formats a 0220 advice and sends the message to the ISO Host Interface process. |
| 2j–2k | The ISO Host Interface process reformats the 0220 advice into external message format and send the message to the host. This advice notifies the host of the transaction that the BASE24-pos system just authorized. |
| 3–4 | The Device Handler module reformats the response message into native-mode format, deletes the response timer, and sends the response to the POS device. |

Approved With Store-and-Forward Completions

In the diagram below, the transaction is a financial transaction request. AUTHH is the primary authorizer and the BASE24-pos system approves the transaction request. Completions are required for this transaction type; however, communications to the host are down and the BASE24-pos system, therefore, cannot send the completion message to the host. The BASE24-pos system instead places the completion message in the Store-and-Forward File (SAF) to be transmitted to the host at a later time.



Steps

1–2

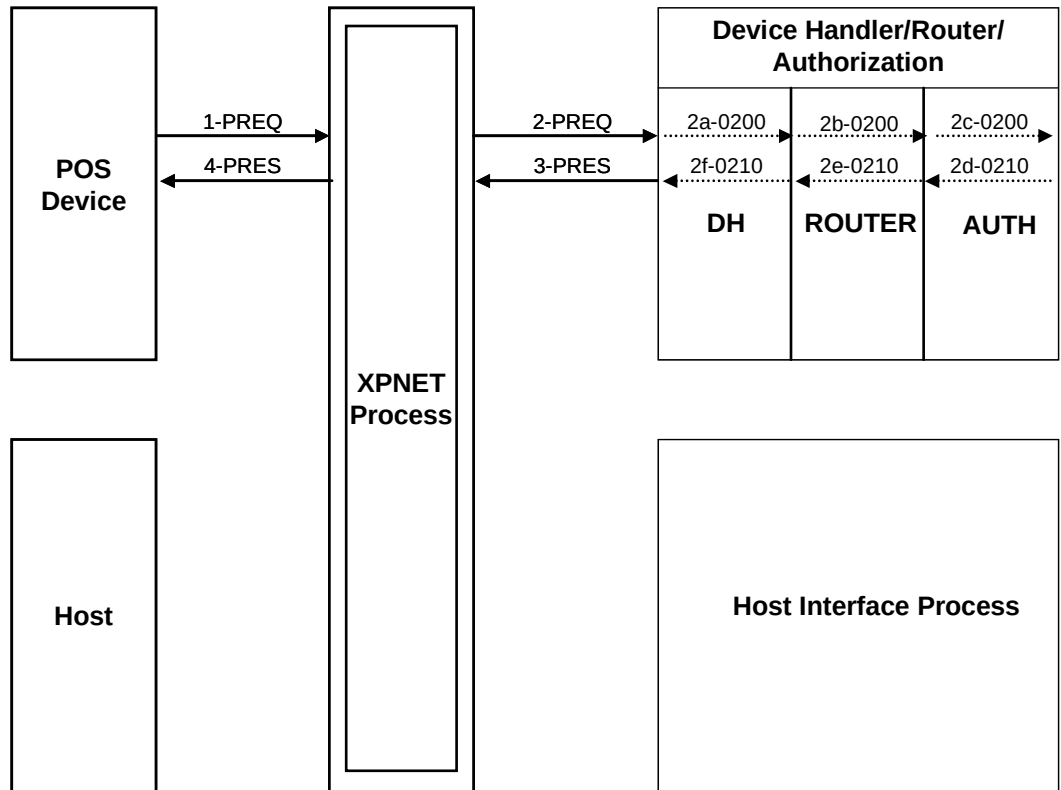
The POS device sends the transaction request to the BASE24-pos system as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Device Handler/Router/Authorization process.

Processing

Steps	Processing
2a–2c	The Device Handler module reads the BASE24-pos Terminal Data files record for the POS device to determine how to process the request. The Device Handler module then reformats the transaction request into a POS Standard Internal Message (PSTM), sets a response timer, and passes the transaction request as a 0200 message to the Router module. The Router module finds the retailer’s record in the POS Retailer Definition File (PRDF), locates a Card Prefix File (CPF) record for the card involved in the transaction, and determines AST routing. The Router module then passes the message to the Authorization module. From the CPF, the Authorization module locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 2.
2d–2f	The Authorization module authorizes the request and updates the cardholder files appropriately.
2g–2i	The Authorization module formats a 0220 advice and sends the message to the ISO Host Interface process.
2j	The ISO Host Interface process is aware that communications to the host are down. Therefore, it does not attempt to send the advice to the host. Instead, the ISO Host Interface process logs the message to the Store-and-Forward File (SAF) for storage. Later, when communications with the host are restored, the transactions in the SAF will be transmitted to the host.
3–4	The Device Handler module reformats the response message into native-mode format, deletes the response timer, and sends the response to the POS device.

Denied Transaction

In the diagram below, the transaction is a financial transaction request. AUTHH is the primary authorizer, but the BASE24-pos system denies the transaction request. Although completions to the host are required for this transaction type, the BASE24-pos system does not generate a completion because the transaction was denied.



Steps

Processing

- 1–2 The POS device sends the transaction request to the BASE24-pos system as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Device Handler/Router/Authorization process.
- 2a–2c The Device Handler module reads the BASE24-pos Terminal Data files record for the POS device to determine how to process the request. The Device Handler module then reformats the transaction request into a POS Standard Internal Message (PSTM), sets a response timer, and passes the transaction request as a 0200 message to the Router module.

Steps**Processing**

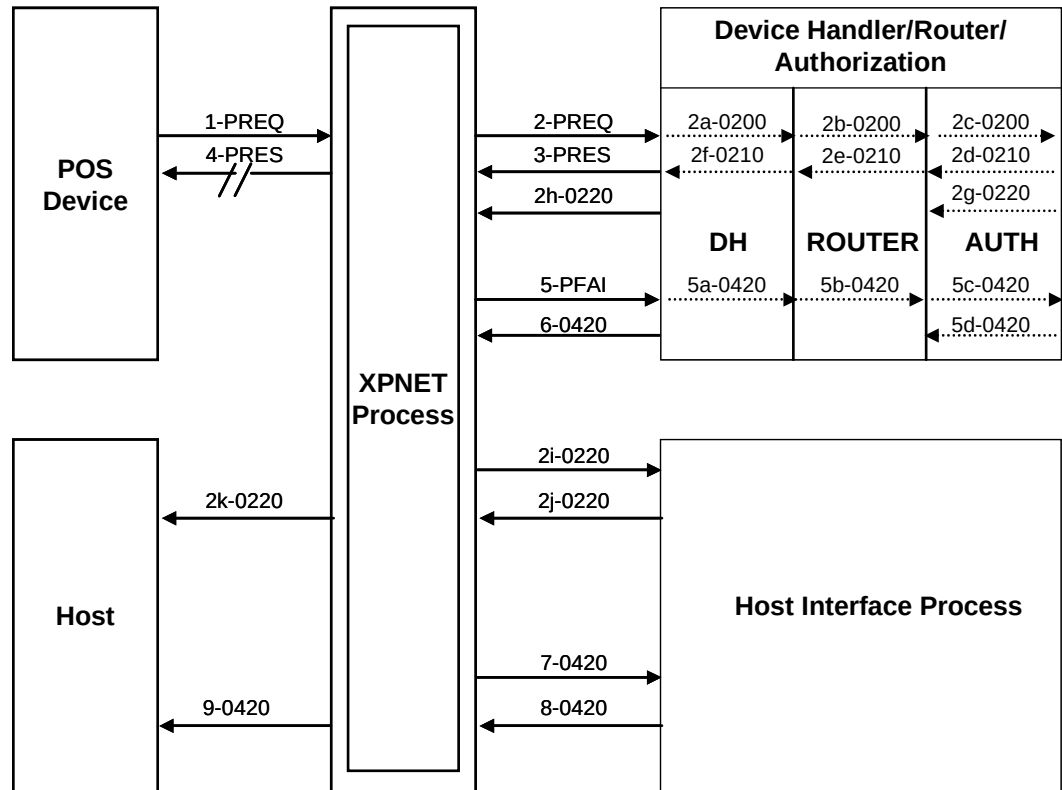
The Router module finds the retailer's record in the POS Retailer Definition File (PRDF), locates a Card Prefix File (CPF) record for the card involved in the transaction, and determines AST routing. The Router module then passes the message to the Authorization module.

From the CPF, the Authorization module locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 2.

- | | |
|-------|--|
| 2d–2f | The Authorization module denies the request and formats a 0210 response message. The cardholder authorization files are not updated for denied requests. |
| 3–4 | The Device Handler module reformats the response message into native-mode format, deletes the response timer, and sends the response to the POS device. |

Reversal After Completion is Sent

In the diagram below, the transaction is a financial transaction request. AUTHH is the primary authorizer, and the BASE24-pos system approves the transaction. Completions are required for this transaction type. However, after the completion is sent to the host, the response is unable to reach the device. Therefore, the BASE24-pos system must generate a reversal.



Steps

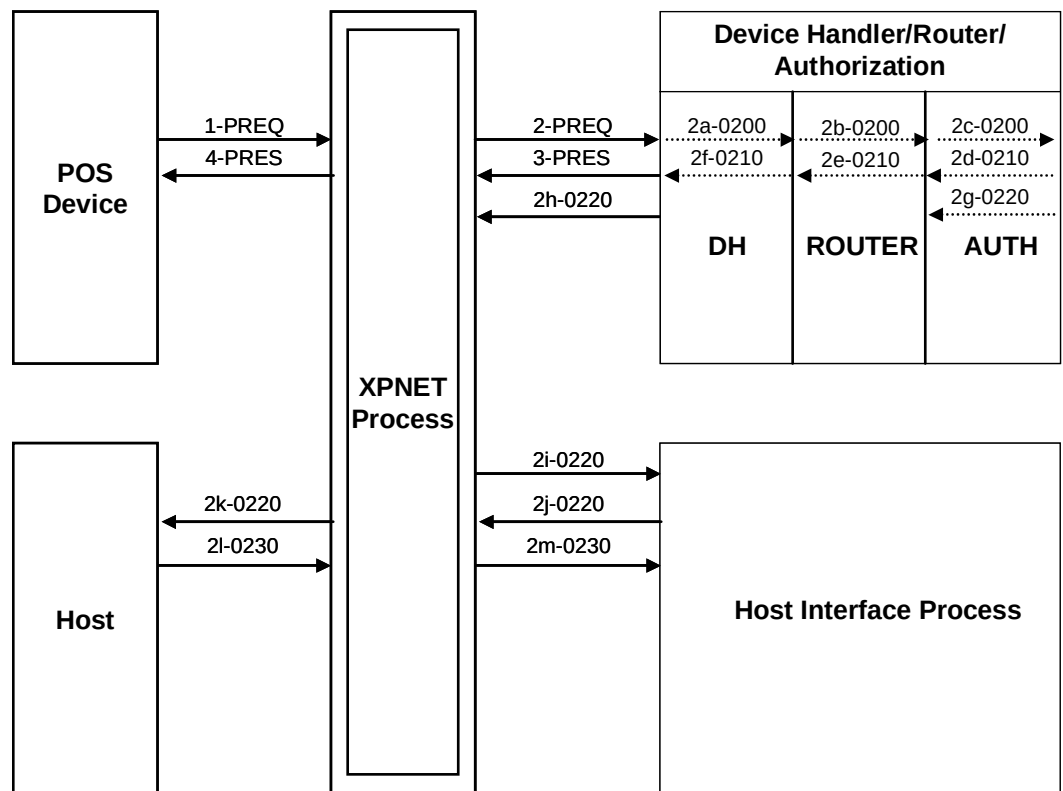
Processing

- 1–2 The POS device sends the transaction request to the BASE24-pos system as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Device Handler/Router/Authorization process.
- 2a–2c The Device Handler module reads the BASE24-pos Terminal Data files record for the POS device to determine how to process the request. The Device Handler module then reformats the transaction request into a POS Standard Internal Message (PSTM), sets a response timer, and passes the transaction request as a 0200 message to the Router module.

Steps	Processing
	The Router module finds the retailer's record in the POS Retailer Definition File (PRDF), locates a Card Prefix File (CPF) record for the card involved in the transaction, and determines AST routing. The Router module then passes the message to the Authorization module. From the CPF, the Authorization module locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 2.
2d–2f	The Authorization module authorizes the request, formats a 0210 response message, and updates the cardholder files appropriately.
2g–2i	The Authorization module formats a 0220 advice and sends the message to the ISO Host Interface process.
2j–2k	The ISO Host Interface process reformats the advice message into external message format and sends the message to the host.
3–4	The Device Handler module reformats the response message into native-mode format, deletes the response timer, and attempts to send the response to the POS device. However, communications with the POS device are down.
5	The XPNET process marks the response message as undeliverable and sends it back to the Device Handler/Router/Authorization process.
5a–5d	The Device Handler/Router/Authorization process determines where the original transaction request was authorized, generates a 0420 reversal message, and updates the cardholder files appropriately.
6–7	The Device Handler/Router/Authorization process sends the 0420 reversal message to the ISO Host Interface process for the host that authorized the original transaction request.
8–9	The ISO Host Interface process reformats the reversal message into external message format and sends the message to the host. This message notifies the host that the transaction did not complete as authorized and enables the host to update its database appropriately.

Approved With Host Acknowledgments

In the diagram below, the transaction is a financial transaction request. AUTHH is the primary authorizer, and the BASE24-pos system approves the transaction. Completions are required for this transaction type. The host sends acknowledgment messages in response to completion messages.



Steps

Processing

- 1–2 The POS device sends the transaction request to the BASE24-pos system as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Device Handler/Router/Authorization process.
- 2a–2c The Device Handler module reads the BASE24-pos Terminal Data files record for the POS device to determine how to process the request. The Device Handler module then reformats the transaction request into a POS Standard Internal Message (PSTM), sets a response timer, and passes the transaction request as a 0200 message to the Router module.

Steps**Processing**

The Router module finds the retailer's record in the POS Retailer Definition File (PRDF), locates a Card Prefix File (CPF) record for the card involved in the transaction, and determines AST routing. The Router module then passes the message to the Authorization module.

From the CPF, the Authorization module locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 2.

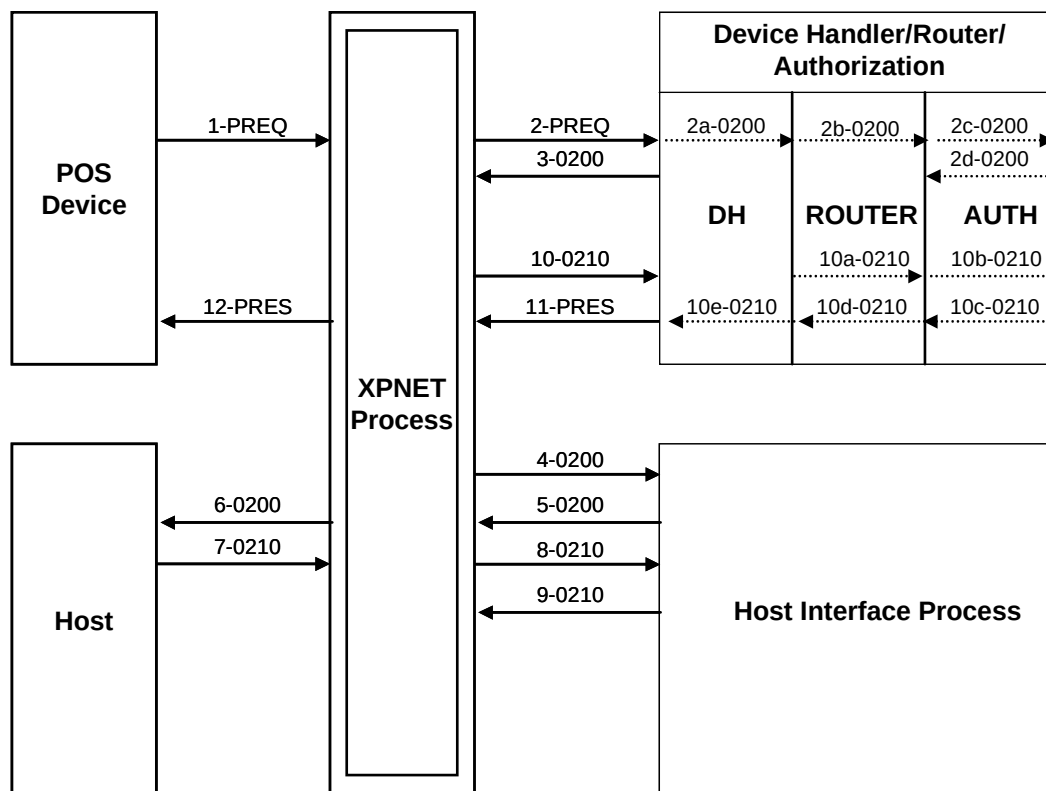
- | | |
|-------|--|
| 2d–2f | The Authorization module authorizes the request, formats a 0210 response message, and updates the cardholder files appropriately. |
| 2g–2i | The Authorization module formats a 0220 advice and sends the message to the ISO Host Interface process. |
| 2j–2k | The ISO Host Interface process reformats the advice message into external message format, sets a response timer, and sends the message to the host. |
| 2l–2m | The host generates a 0230 advice response and sends it to the ISO Host Interface process. The advice response message notifies the ISO Host Interface process that the host received the advice. The ISO Host Interface process then deletes the response timer. |
| 3–4 | The Device Handler module reformats the response message into native-mode format, deletes the response timer, and sends the response to the POS device. |

Authorization Level 3

The following pages contain message flows for transactions being processed by the BASE24-pos system using level 3 (online/offline) authorization processing. Authorization level 3 requires that transactions be sent to a host for authorization, but allows the BASE24-pos system to stand in when the host is not available.

Approved Online

In the transaction diagram below, the transaction is a financial transaction request. AUTHH is the primary authorizer, and the host approves the transaction. Completions are not required for this transaction type.



Steps

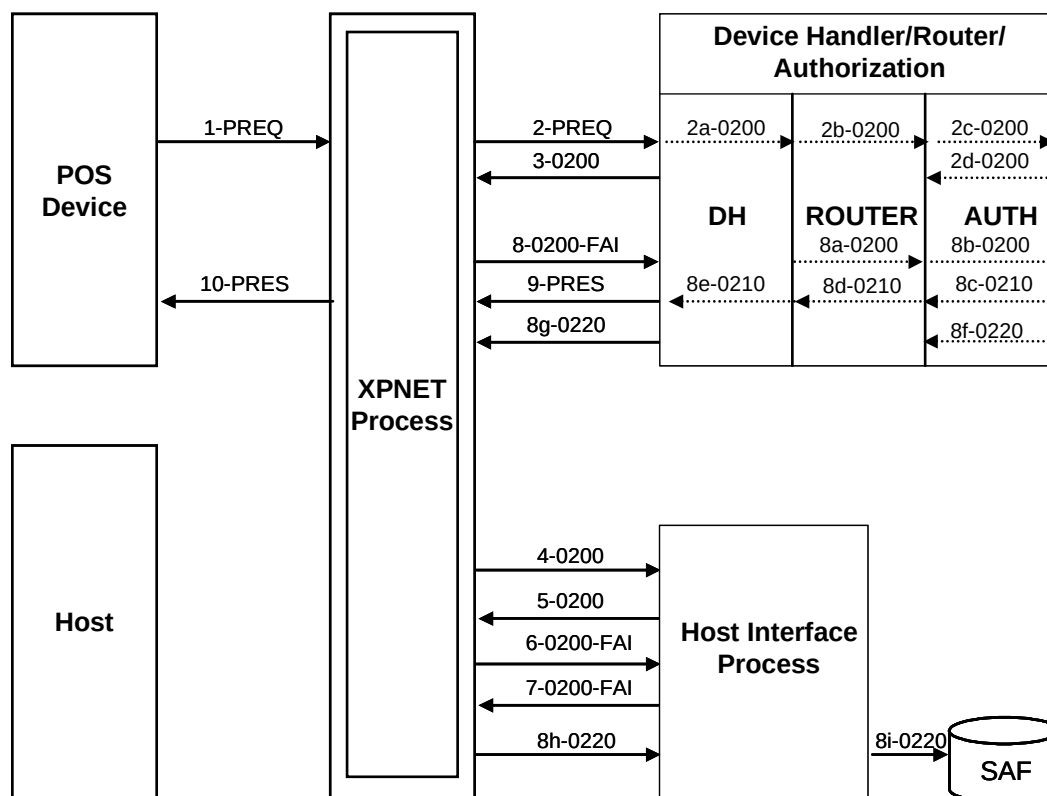
Processing

- 1–2 The POS device sends the transaction request to the BASE24-pos system as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Device Handler/Router/Authorization process.
- 2a–2d The Device Handler module reads the BASE24-pos Terminal Data files record for the POS device to determine how to process the request. The Device Handler module then reformats the transaction request into a POS Standard Internal Message (PSTM), sets a response timer, and passes the transaction request as a 0200 message to the Router module.

Steps	Processing
	The Router module finds the retailer's record in the POS Retailer Definition File (PRDF), locates a Card Prefix File (CPF) record for the card involved in the transaction, and determines AST routing. The Router module then passes the message to the Authorization module. From the CPF, the Authorization module locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 3.
3–4	The Device Handler/Router/Authorization process sends the 0200 request message to the ISO Host Interface process.
5–6	The ISO Host Interface process reformats the 0200 request message into external message format, sets a response timer, and sends the message to the host.
7–8	The host approves the transaction request, formats a 0210 response message, and sends the message to the ISO Host Interface process.
9–10	The ISO Host Interface process reformats the 0210 response message into a PSTM 0210 response message, deletes the response timer, and sends the response message to the Device Handler/Router/Authorization process.
10a–10e	The Authorization module logs the transaction to the POS Transaction Log File (PTLF) and updates the cardholder files appropriately.
11–12	The Device Handler module reformats the 0210 response message into native-mode format, deletes the response timer, and sends the response to the POS device.

Approved Offline With Completions

In the diagram below, the transaction is a financial transaction request. AUTHH is the primary authorizer. The host is not available to authorize the transaction so the BASE24-pos system approves the transaction offline. Because the BASE24-pos system approved the transaction offline, the transaction completion is logged to the Store-and-Forward File (SAF).



Steps

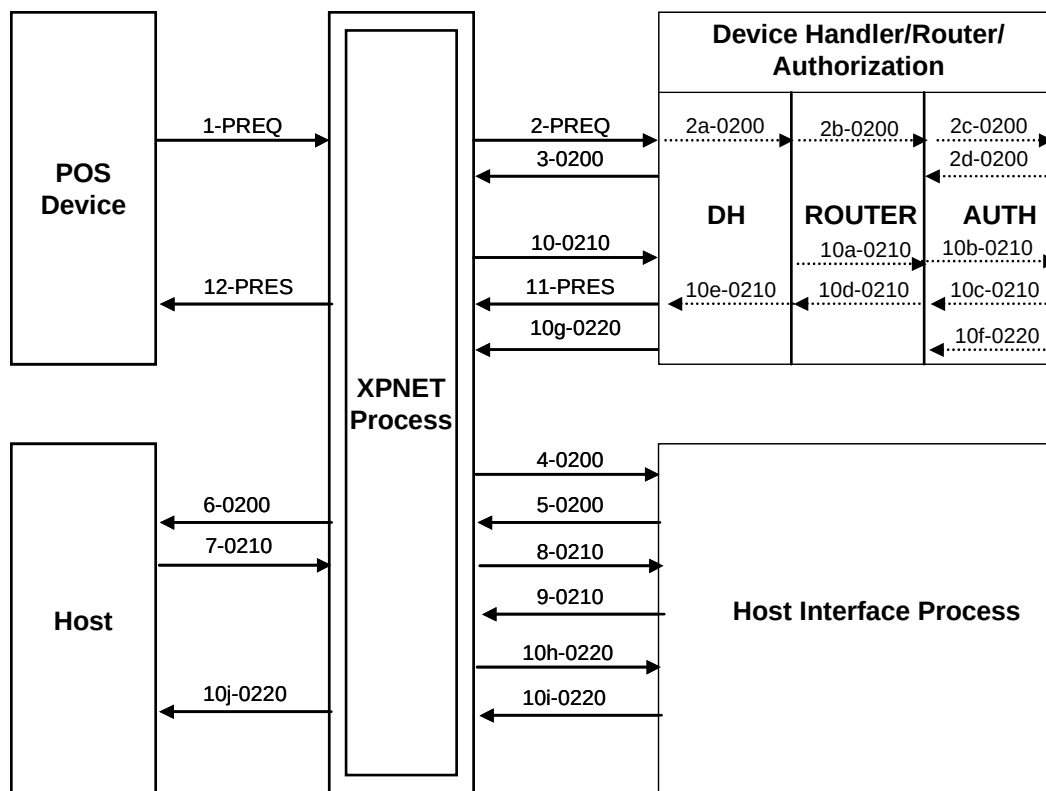
Processing

- 1–2 The POS device sends the transaction request to the BASE24-pos system as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Device Handler/Router/Authorization process.
- 2a–2d The Device Handler module reads the BASE24-pos Terminal Data files record for the POS device to determine how to process the request. The Device Handler module then reformats the transaction request into a POS Standard Internal Message (PSTM), sets a response timer, and passes the transaction request as a 0200 message to the Router module.

Steps	Processing
	The Router module finds the retailer's record in the POS Retailer Definition File (PRDF), locates a Card Prefix File (CPF) record for the card involved in the transaction, and determines AST routing. The Router module then passes the message to the Authorization module. From the CPF, the Authorization module locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 3.
3–4	The Authorization module sends the 0200 request message to the ISO Host Interface process.
5–6	The ISO Host Interface process reformats the 0200 request message into external message format and attempts to send the message to the host. However, communications with the host are down. Therefore, the XPNET process marks the 0200 request message as undeliverable and sends it back to the ISO Host Interface process.
7–8	The ISO Host Interface process reformats the 0200 failed request message into internal message format and sends it to the Device Handler/Router/Authorization process.
8a–8e	The Device Handler/Router/Authorization process approves the transaction request offline by doing stand-in authorization. The Authorization module logs the transaction to the POS Transaction Log File (PTLF) and updates the BASE24-pos cardholder files appropriately.
8f–8h	The Authorization module formats a 0220 advice message and sends it to the ISO Host Interface process.
8i	The ISO Host Interface process logs the 0220 advice message to the Store-and-Forward File (SAF) to be forwarded to the host when communications between the BASE24-pos system and the host have resumed.
9–10	The Device Handler module reformats the 0210 response message into native-mode format, deletes the response timer, and sends the response message to the POS device.

Approved Online with Completions

In the diagram below, the transaction is a financial transaction request. The host approves the transaction online. Completions are required for this transaction type.

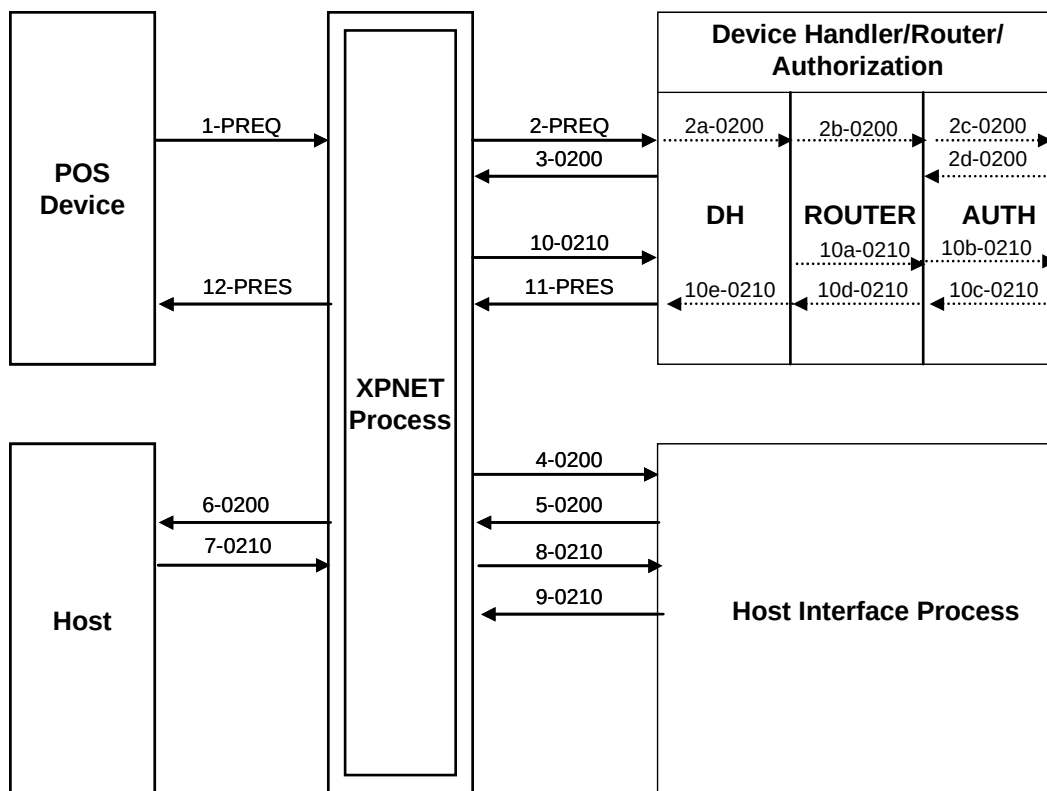


Steps	Processing
1–2	The POS device sends the transaction request to the BASE24-pos system as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Device Handler/Router/Authorization process.
2a–2d	The Device Handler module reads the BASE24-pos Terminal Data files record for the POS device to determine how to process the request. The Device Handler module then reformats the transaction request into a POS Standard Internal Message (PSTM), sets a response timer, and passes the transaction request as a 0200 message to the Router module.

Steps	Processing
	The Router module finds the retailer's record in the POS Retailer Definition File (PRDF), locates a Card Prefix File (CPF) record for the card involved in the transaction, and determines AST routing. The Router module then passes the message to the Authorization module. From the CPF, the Authorization module locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 3.
3–4	The Device Handler/Router/Authorization process sends the 0200 request message to the ISO Host Interface process.
5–6	The ISO Host Interface process reformats the 0200 request message into external message format, sets a response timer, and sends the message to the host.
7–8	The host approves the transaction, formats a 0210 response message, and sends the message to the ISO Host Interface process.
9–10	The ISO Host Interface process reformats the 0210 response message into internal message format, deletes the response timer, and sends the response message to the Device Handler/Router/Authorization process.
10a–10e	The Device Handler/Router/Authorization process logs the transaction to the POS Transaction Log File (PTLF) and updates its cardholder files appropriately.
10f–10h	The Authorization module formats a 0220 advice message and sends it to the ISO Host Interface process.
10i–10j	The ISO Host Interface process reformats the 0220 advice message into the external message format and sends it to the host. The advice message notifies the host that the transaction completed as authorized.
11–12	The Device Handler module reformats the 0210 response message into native-mode format, deletes the response timer, and sends the response to the POS device.

Denied Online

In the diagram below, the transaction is a financial transaction request. The host denies the transaction. Although completions are required for this transaction type, no completion is sent because the transaction was denied.



Steps

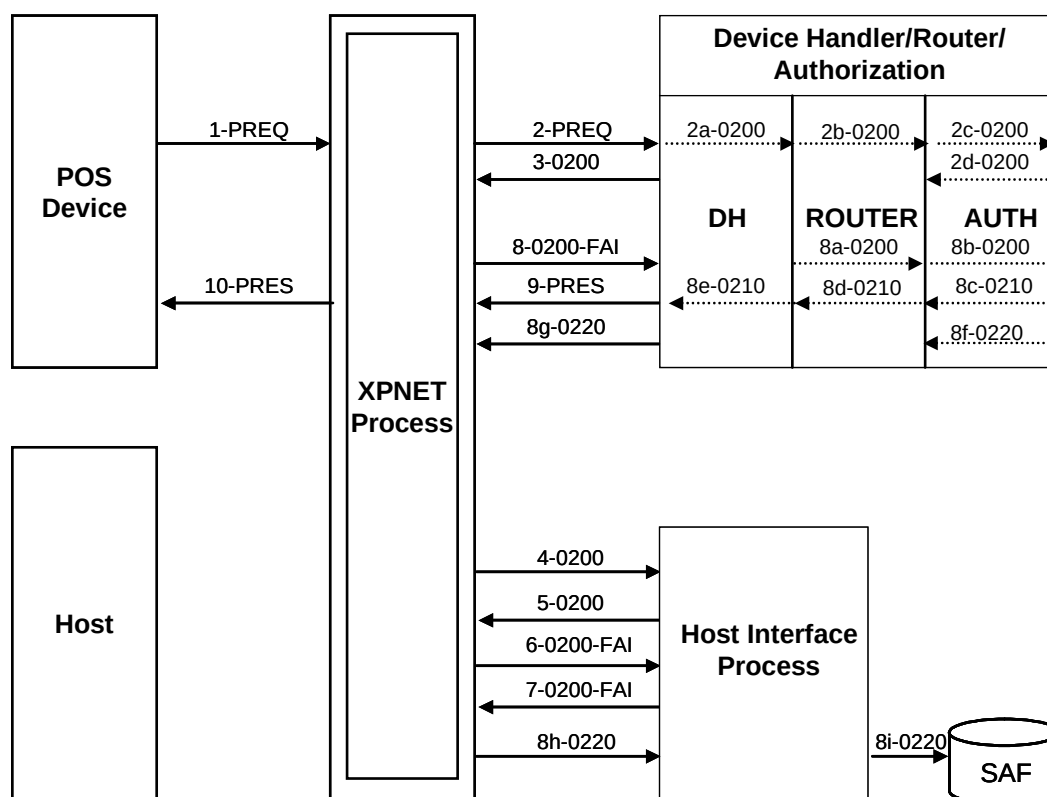
Processing

- 1–2 The POS device sends the transaction request to the BASE24-pos system as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Device Handler/Router/Authorization process.
- 2a–2d The Device Handler module reads the BASE24-pos Terminal Data files record for the POS device to determine how to process the request. The Device Handler module then reformats the transaction request into a POS Standard Internal Message (PSTM), sets a response timer, and passes the transaction request as a 0200 message to the Router module.

Steps	Processing
	The Router module finds the retailer's record in the POS Retailer Definition File (PRDF), locates a Card Prefix File (CPF) record for the card involved in the transaction, and determines AST routing. The Router module then passes the message to the Authorization module. From the CPF, the Authorization module locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 3.
3–4	The Device Handler/Router/Authorization process sends the 0200 request message to the ISO Host Interface process.
5–6	The ISO Host Interface process reformats the 0200 request message into external message format, sets a response timer, and sends the message to the host.
7–8	The host denies the transaction, formats a 0210 response message, and sends the message to the ISO Host Interface process.
9–10	The ISO Host Interface process reformats the 0210 response message into internal message format, deletes the response timer, and sends the response message to the Device Handler/Router/Authorization process.
10a–10e	The Device Handler/Router/Authorization process logs the transaction to the POS Transaction Log File (PTLF). The Device Handler/Router/Authorization process does not update its cardholder files when the transaction is denied.
11–12	The Device Handler module reformats the 0210 response message into native-mode format, deletes the response timer, and sends the response message to the POS device.

Approved Offline With SAF Completion

In the diagram below, the transaction is a financial transaction request. Communications to the host are down so the BASE24-pos system stands in to authorize the transaction. The BASE24-pos system approves the transaction and places the completion message to the host in the Store-and-Forward File.



Steps

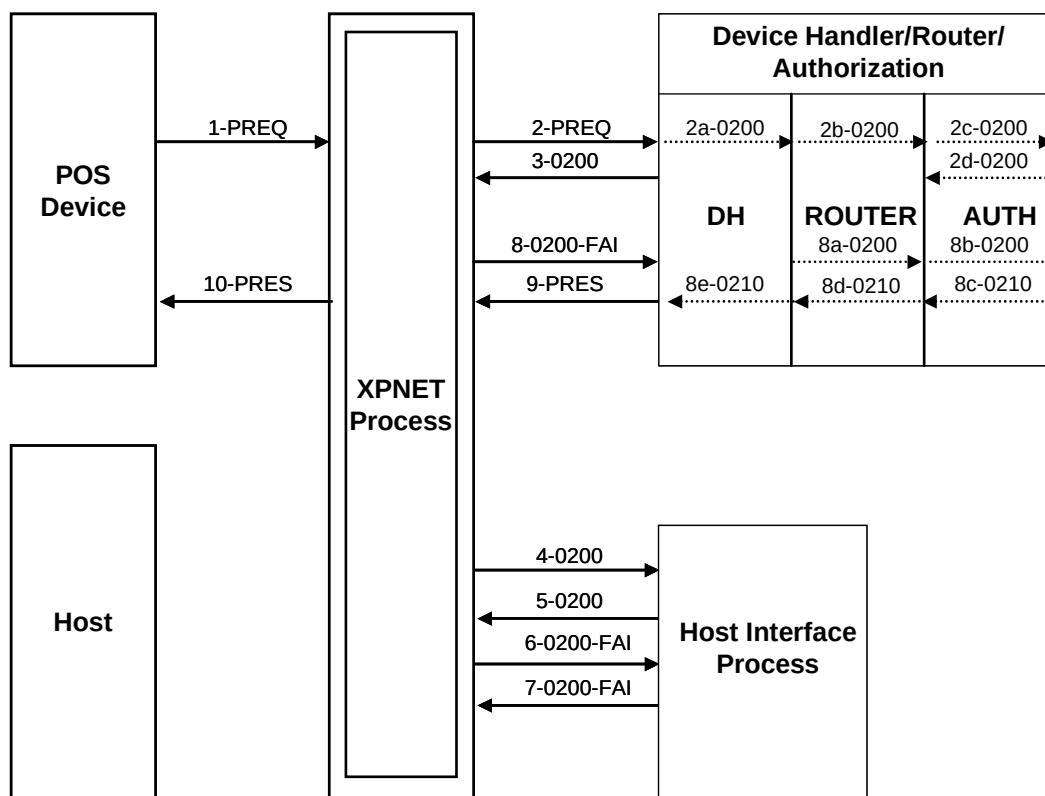
Processing

- 1–2 The POS device sends the transaction request to the BASE24-pos system as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Device Handler/Router/Authorization process.
- 2a–2d The Device Handler module reads the BASE24-pos Terminal Data files record for the POS device to determine how to process the request. The Device Handler module then reformats the transaction request into a POS Standard Internal Message (PSTM), sets a response timer, and passes the transaction request as a 0200 message to the Router module.

Steps	Processing
	The Router module finds the retailer's record in the POS Retailer Definition File (PRDF), locates a Card Prefix File (CPF) record for the card involved in the transaction, and determines AST routing. The Router module then passes the message to the Authorization module. From the CPF, the Authorization module locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 3.
3–4	The Device Handler/Router/Authorization process sends the 0200 request message to the ISO Host Interface process.
5–6	The ISO Host Interface process reformats the 0200 request message into external message format and attempts to send the message to the host. However, communications with the host are down. Therefore, the XPNET process marks the 0200 request message as failed and sends it back to the ISO Host Interface process.
7–8	The ISO Host Interface process reformats the 0200 failed request into internal message format and sends it to the Device Handler/Router/Authorization process.
8a–8e	The Device Handler/Router/Authorization process performs stand-in authorization and approves the transaction request. The Authorization module logs the transaction to the POS Transaction Log File (PTLF) and updates the BASE24-pos cardholder files appropriately.
8f–8h	The Authorization module formats a 0220 advice message and sends it to the ISO Host Interface process.
8i	The ISO Host Interface process reformats the 0220 advice message to the external message format. Because communications with the host are down, the ISO Host Interface process logs the advice message to the Store-and-Forward File (SAF). The messages in the SAF will be forwarded to the host later when communications between the BASE24-pos system and the host are restored.
9–10	The Device Handler module reformats the 0210 response message into native-mode format, deletes the response timer, and sends the response message to the POS device.

Denied Offline

In the diagram below, the transaction is a financial transaction request. Communications to the host are down, so the BASE24-pos system stands in to authorize the transaction. The BASE24-pos system denies the transaction. Although completions are required for this transaction type, the BASE24-pos system does not send a completion because the transaction was denied.



Steps

Processing

- 1–2 The POS device sends the transaction request to the BASE24-pos system as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Device Handler/Router/Authorization process.
- 2a–2d The Device Handler module reads the BASE24-pos Terminal Data files record for the POS device to determine how to process the request. The Device Handler module then reformats the transaction request into a POS Standard Internal Message (PSTM), sets a response timer, and passes the transaction request as a 0200 message to the Router module.

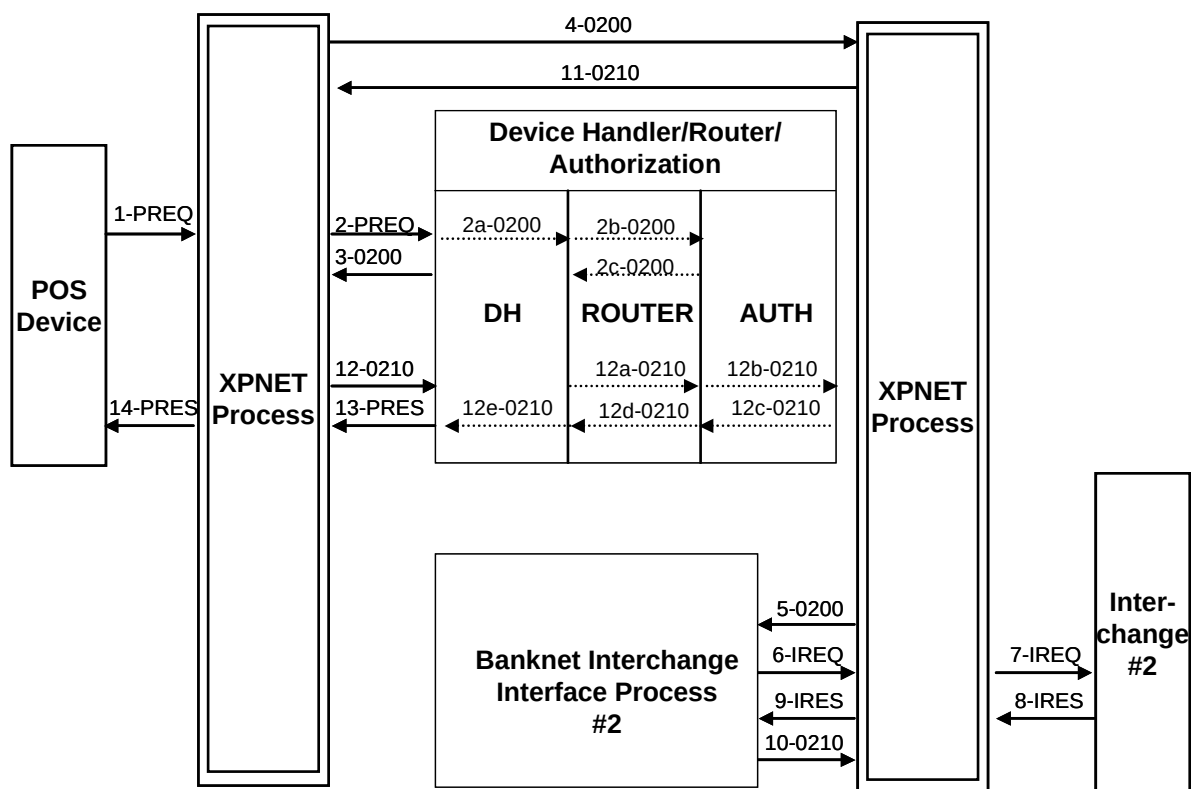
Steps	Processing
	The Router module finds the retailer's record in the POS Retailer Definition File (PRDF), locates a Card Prefix File (CPF) record for the card involved in the transaction, and determines AST routing. The Router module then passes the message to the Authorization module. From the CPF, the Authorization module locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 3.
3–4	The Device Handler/Router/Authorization process sends the 0200 request message to the ISO Host Interface process.
5–6	The ISO Host Interface process reformats the 0200 request message into external message format and attempts to send the message to the host. However, communications with the host are down. Therefore, the XPNET process marks the 0200 request message as failed and sends it back to the ISO Host Interface process.
7–8	The ISO Host Interface process reformats the 0200 failed request into internal message format and sends it to the Device Handler/Router/Authorization process.
8a–8e	The Device Handler/Router/Authorization process performs stand-in authorization and denies the transaction request. The Authorization module logs the transaction to the POS Transaction Log File (PTLF). The Device Handler/Router/Authorization module does not update the BASE24-pos cardholder files because the transaction was denied.
9–10	The Device Handler module reformats the 0210 response message into native-mode format, deletes the response timer, and sends the response message to the POS device.

Multi-Nodal Transactions

The following pages contain a transaction diagram for transactions which require traffic between two XPNET nodes.

Approved on Another XPNET Node

In the diagram below, the transaction is a not-on-us credit account transaction. The Router module cannot determine the card type from the Card Prefix File (CPF) and, therefore, determines an authorizer by performing an algorithm or by using SPROUTE routing. The Router module then sends the transaction request to a second XPNET node from which Banknet Interchange 2 authorizes the transaction and returns a response.



Steps	Processing
1–2	The POS device sends the transaction request to the BASE24-pos system as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Device Handler/Router/Authorization process.
2a–2b	The Device Handler module reads the BASE24-pos Terminal Data files record for the POS device to determine how to process the request. The Device Handler module then reformats the transaction request into a POS Standard Internal Message (PSTM), sets a response timer, and passes the transaction request as a 0200 message to the Router module. Router cannot determine the card type from the CPF. Instead, Router determines the authorizer by performing an algorithm check or by using SPROUTE and using the card type to perform AST routing.
2c–4	The Router module sends the request to the XPNET node that contains the determined authorizer.
5	The XPNET process in the second XPNET node sends the 0200 request to the Banknet Interchange Interface process.
6–7	The Banknet Interchange Interface process reformats the 0200 request message into external message format, sets an outbound message timer, and sends the message to the Banknet Interchange.
8–9	The interchange authorizes the transaction, formats a response message, and sends the message to the Banknet Interchange Interface process.
10	The Banknet Interchange Interface process deletes the outbound message timer, reformats the response message into a 0210 internal response message, and sends the message to the XPNET process.
11–12	The XPNET process sends the 0210 response message to the XPNET process in the first XPNET node. The XPNET process then routes the response to the Device Handler/Router/Authorization process.
12a–12e	The Device Handler/Router/Authorization process logs the transaction to the POS Transaction Log File (PTLF) and updates its cardholder files appropriately.

Steps

Processing

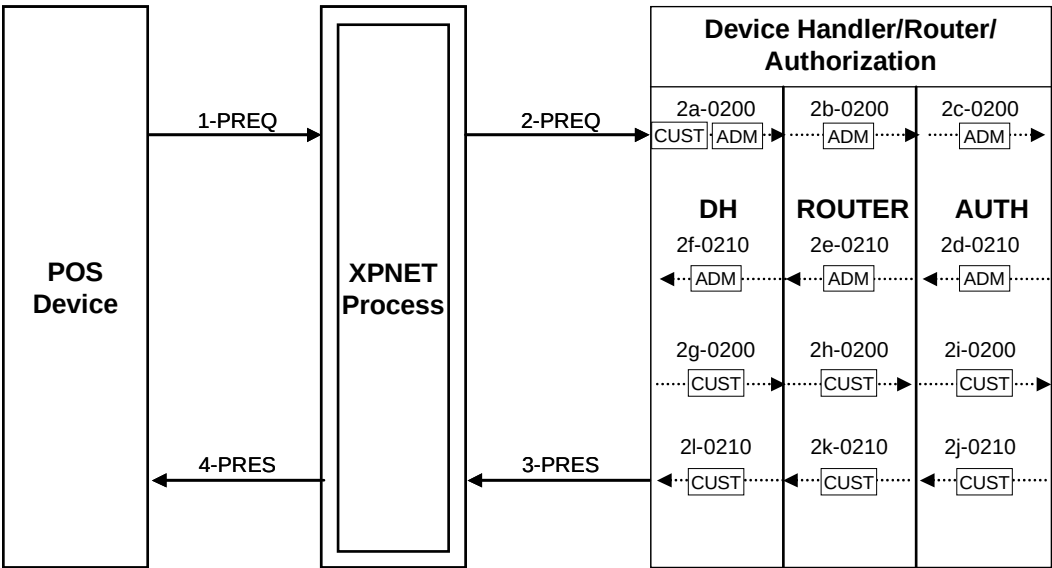
- 13–14 The Device Handler module reformats the 0210 response message into native-mode format, deletes its response timer, and sends the response message to the POS device.

Administrative Transactions

The diagrams on the following pages illustrate administrative transactions as they flow through the BASE24-pos system.

Approved Transaction

In the diagram below, the transaction is an administrative transaction request. The BASE24-pos system approves both the use of the administrative card and the actual transaction request itself.



Steps

Processing

- 1–2 The POS device sends the transaction request to the BASE24-pos system as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Device Handler/Router/Authorization process.
- 2a–2f The Device Handler module reads the BASE24-pos Terminal Data files record for the POS device to determine how to process the request. The Device Handler module then reformats the administrative portion of the transaction request into a POS Standard Internal Message (PSTM), sets a response timer, and passes the transaction request as a 0200 message to the Router module.

Steps**Processing**

The Router module finds the retailer's record in the POS Retailer Definition File (PRDF), locates a Card Prefix File (CPF) record for the card involved in the transaction, and determines AST routing. The Router module then passes the message to the Authorization module.

2g–2l

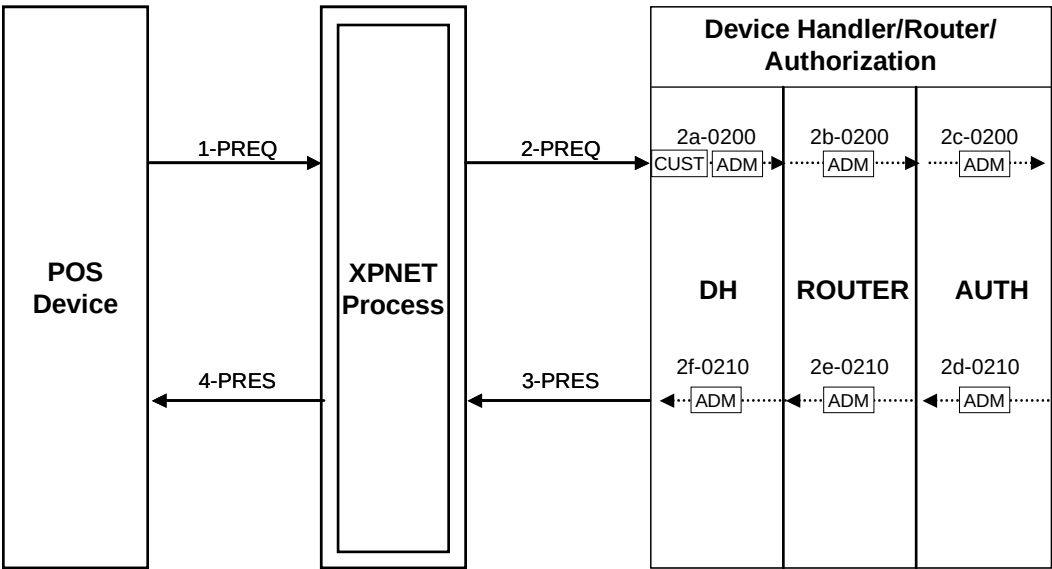
The Device Handler/Router/Authorization process then processes the customer portion of the transaction in a manner similar to that described above. The BASE24-pos system approves the transaction request. The Authorization module logs the transaction to the POS Transaction Log File (PTLF) and updates the BASE24 database appropriately.

3–4

The Device Handler module reformats the response message into native-mode format, deletes its response timer, and sends the response message to the POS device.

Denied Transaction

In the diagram below, the transaction is an administrative transaction request. The BASE24-pos system denies the use of the administrative card.



Steps

Processing

- 1–2 The POS device sends the transaction request to the BASE24-pos system as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Device Handler/Router/Authorization process.
- 2a–2f The Device Handler module reads the BASE24-pos Terminal Data files record for the POS device to determine how to process the request. The Device Handler module then reformats the administrative portion of the transaction request into a POS Standard Internal Message (PSTM), sets a response timer, and passes the transaction request as a 0200 message to the Router module.
- The Router module finds the retailer’s record in the POS Retailer Definition File (PRDF), locates a Card Prefix File (CPF) record for the card involved in the transaction, and determines AST routing. The Router module then passes the message to the Authorization module.
- The use of this administrative card is denied. Therefore, the customer portion of the transaction is not allowed to proceed.

Steps

Processing

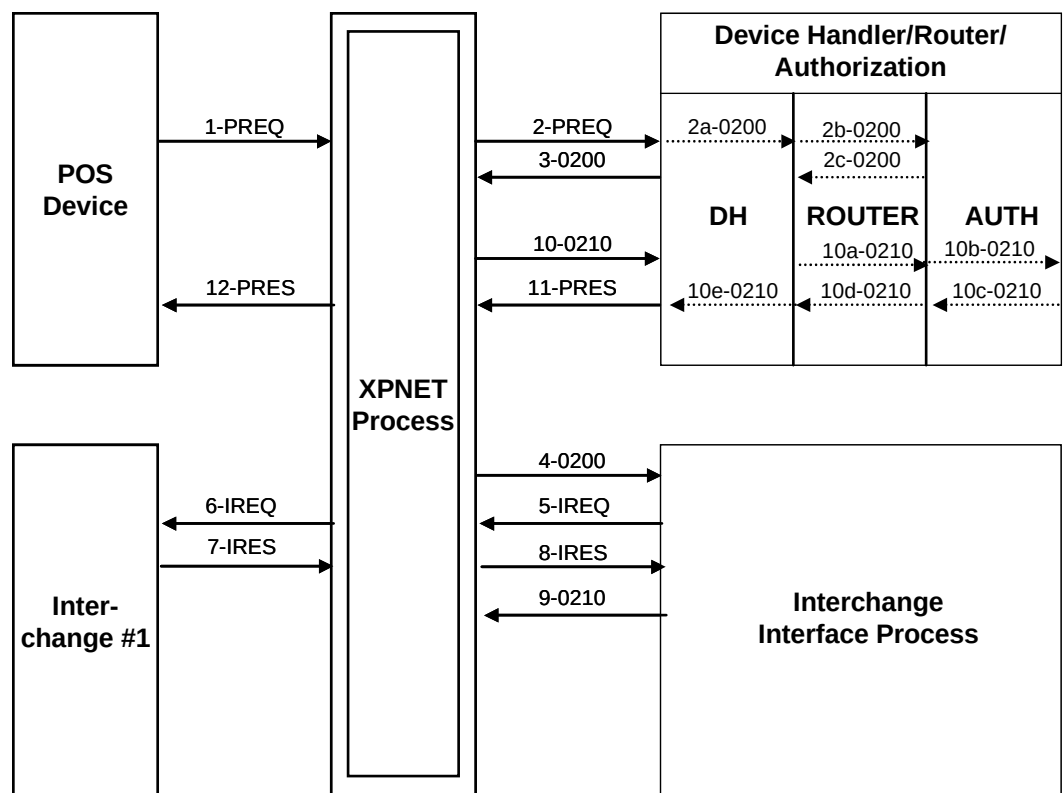
- 3–4 The Device Handler module reformats the response message into native-mode format, deletes its response timer, and sends the response message to the POS device.

Interchange Transactions

The diagrams on the following pages illustrate situations in which transactions are sent to an interchange for authorization.

Approved Transaction

In the diagram below, the transaction is a financial transaction request. The interchange approves the transaction.



Steps

Processing

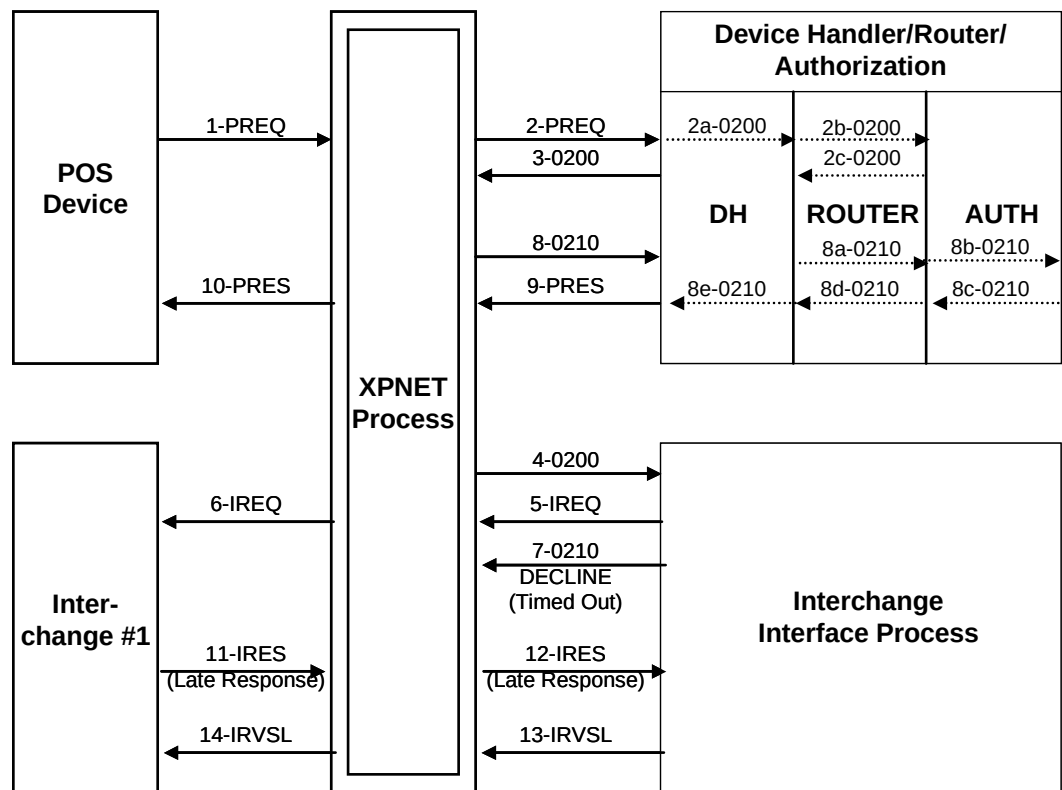
1–2

The POS device sends the transaction request to the BASE24-pos system as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Device Handler/Router/Authorization process.

Steps	Processing
2a–2c	The Device Handler module reads the BASE24-pos Terminal Data files record for the POS device to determine how to process the request. The Device Handler module then reformats the transaction request into a POS Standard Internal Message (PSTM), sets a response timer, and passes the transaction request as a 0200 message to the Router module. The Router module finds the retailer’s record in the POS Retailer Definition File (PRDF), locates a Card Prefix File (CPF) record for the card involved in the transaction, and determines AST routing.
3–4	The Device Handler/Router/Authorization process sends the 0200 request message to the Interchange Interface process.
5–6	The Interchange Interface process reformats the 0200 request into external message format, sets an outbound message timer, and sends the request to the interchange.
7–8	The interchange approves the transaction request, formats a response, and sends the response to the Interchange Interface process.
9–10	The Interchange Interface process deletes the outbound message timer, reformats the response into a 0210 internal response message, and sends the message to the Device Handler/Router/Authorization process.
10a–10e	The Device Handler/Router/Authorization process logs the transaction to the POS Transaction Log File (PTLF) and updates its cardholder files appropriately.
11–12	The Device Handler module reformats the 0210 response message into native-mode format, deletes its response timer, and sends the response message to the POS device.

Reversal on Timed-Out Transaction

In the diagram below, the transaction is a financial transaction request. Interchange #1 is the primary authorizer. Although Interchange #1 approves the transaction, the response is late and the transaction times out. When the transaction times out, the Interchange Interface process automatically denies the transaction. Therefore, when the BASE24-pos system receives the late approval from Interchange #1, the BASE24-pos system sends a reversal to the interchange because the transaction was already denied.



Steps

1–2

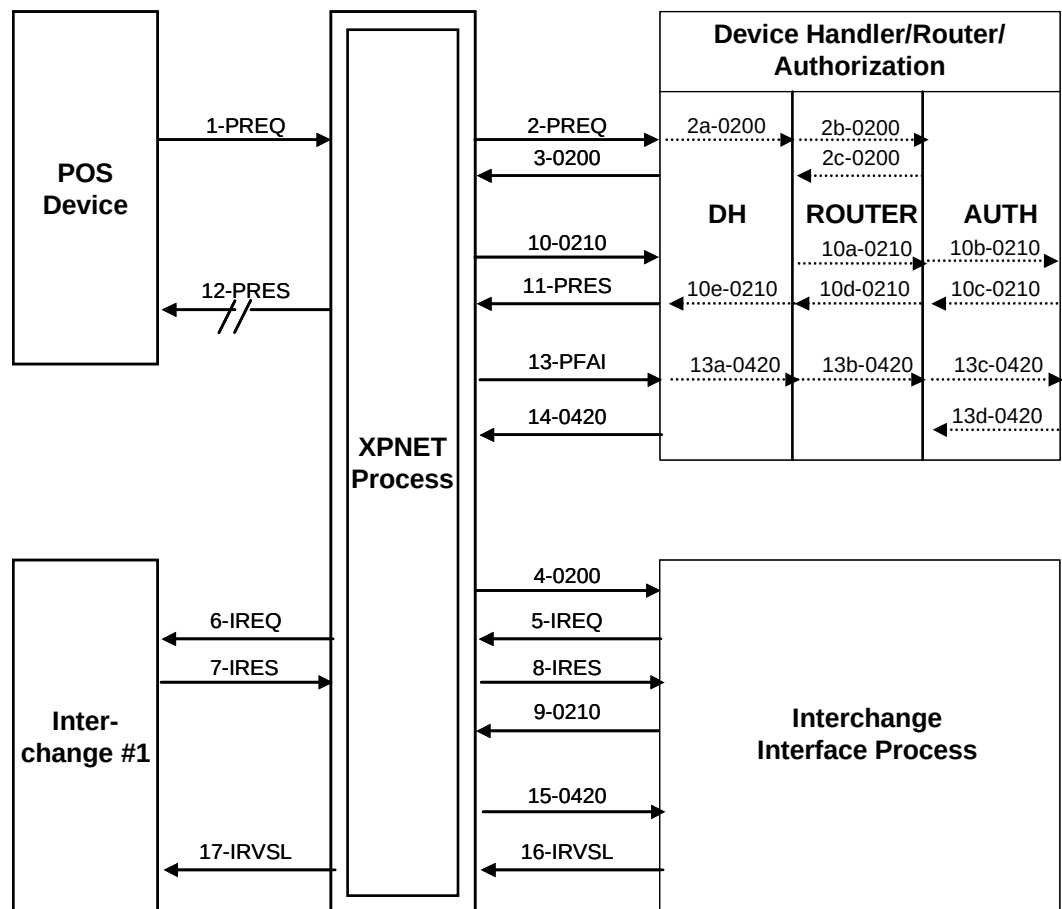
The POS device sends the transaction request to the BASE24-pos system as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Device Handler/Router/Authorization process.

Processing

Steps	Processing
2a–2c	The Device Handler module reads the BASE24-pos Terminal Data files record for the POS device to determine how to process the request. The Device Handler module then reformats the transaction request into a POS Standard Internal Message (PSTM), sets a response timer, and passes the transaction request as a 0200 message to the Router module. The Router module finds the retailer’s record in the POS Retailer Definition File (PRDF), locates a Card Prefix File (CPF) record for the card involved in the transaction, and determines AST routing.
3–4	The Device Handler/Router/Authorization process sends the 0200 request to the Interchange Interface process.
5–6	The Interchange Interface process reformats the 0200 request message into external message format, sets an outbound request timer, and sends the message to the interchange.
7–8	The outbound request timer expires before the interchange returns a response. Therefore, the Interchange Interface process takes the action specified in the TIMEOUT ACTION field on screen 10 of the Interchange Configuration File (ICF) or Enhanced Interchange Configuration File (ICFE). In this case, that action is to deny the transaction request. The Interchange Interface process generates a 0210 response message in internal message format and sends the 0210 response message to the Device Handler/Router/Authorization process.
8a–8e	The Device Handler/Router/Authorization process logs the transaction to the POS Transaction Log File (PTLF). The Device Handler/Router/Authorization process does not update the BASE24-pos cardholder files because the transaction was denied.
9–10	The Device Handler module reformats the 0210 response message into native-mode format, deletes the response timer, and sends the response message to the POS device.
11–12	The interchange sends a late response to the Interchange Interface process.
13–14	Because the transaction request was already denied, the Interchange Interface process generates a reversal and sends the reversal to the interchange. The reversal allows the authorizer to update its database appropriately.

Reversal Due to Downed Device

In the diagram below, the transaction is a financial transaction request. The interchange is the transaction authorizer. Although the interchange approves the transaction, communications between the BASE24 system and the device have gone down while the transaction was in progress. Therefore, the Device Handler/Router/Authorization process generates and sends a reversal to the interchange. This reversal notifies the interchange that the transaction did not complete as authorized.



Steps

1–2

The POS device sends the transaction request to the BASE24-pos system as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Device Handler/Router/Authorization process.

Processing

Steps	Processing
2a–2c	The Device Handler module reads the BASE24-pos Terminal Data file record for the POS device to determine how to process the request. The Device Handler module then reformats the transaction request into a POS Standard Internal Message (PSTM), sets a response timer, and passes the transaction request as a 0200 message to the Router module. The Router module finds the retailer’s record in the POS Retailer Definition File (PRDF), locates a Card Prefix File (CPF) record for the card involved in the transaction, and determines AST routing. The Router module then passes the message to the Authorization module.
3–4	The Device Handler/Router/Authorization process sends the 0200 request message to the Interchange Interface process.
5–6	The Interchange Interface process reformats the 0200 request into external message format, sets an outbound message timer, and sends the request to the interchange.
7–8	The interchange approves the transaction request, formats a response, and sends the response to the Interchange Interface process.
9–10	The Interchange Interface process deletes the outbound message timer, reformats the response into a 0210 internal response message, and sends the message to the Device Handler/Router/Authorization process.
10a–10e	The Device Handler/Router/Authorization process logs the transaction to the POS Transaction Log File (PTLF) and updates other database files appropriately.
11–12	The Device Handler module reformats the 0210 response message into native-mode format, deletes its response timer, and attempts to send the response message to the POS device. However, communications between the POS device and the BASE24-pos system are down.
13	The XPNET process marks the response message as undeliverable and sends it to the Device Handler/Router/Authorization process.
13a–13d	The Device Handler module reformats the response message into an internal 0420 reversal message. The Router and Authorization modules determine where the original transaction request was authorized.

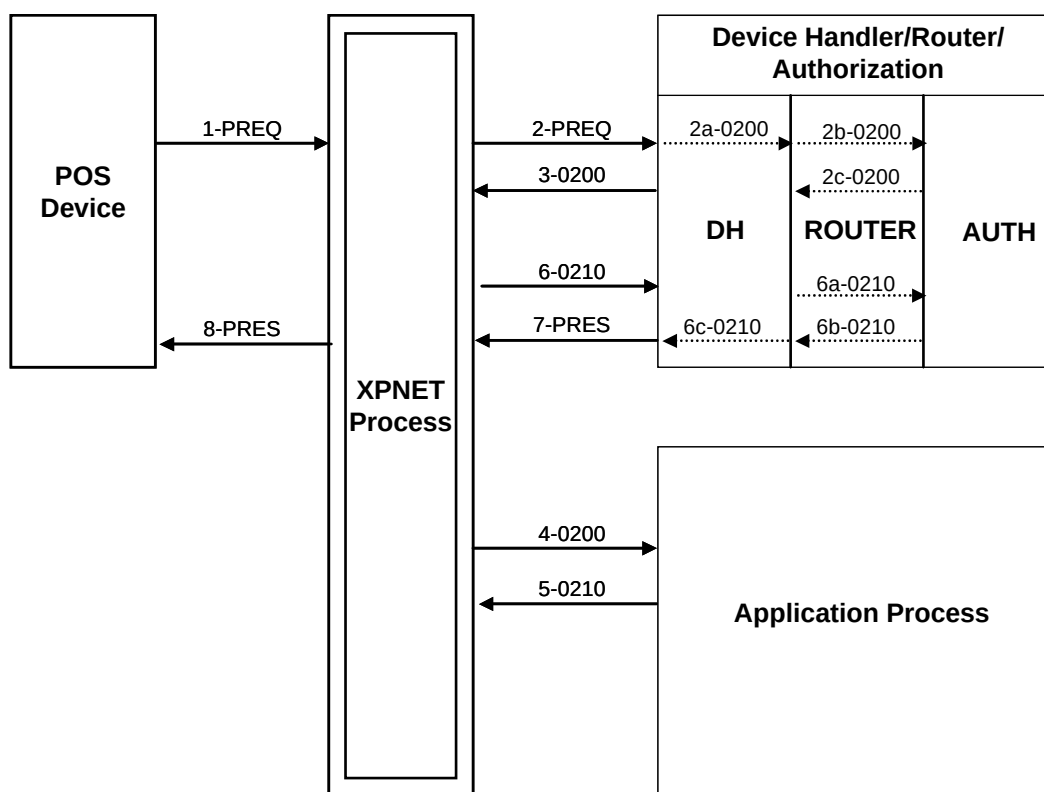
Steps	Processing
14–15	The Device Handler/Router/Authorization process sends the reversal to the appropriate Interchange Interface process.
16–17	The Interchange Interface process reformats the 0420 reversal into external message format and sends the message to the interchange. The interchange can then use this reversal message to update its database as needed.

APCF-Routed Transactions

The diagrams on the following pages illustrate situations in which transactions can be routed to an application process for authorization using the Acquirer Processing Code File (APCF) extended memory table.

Approved Transaction from a Device

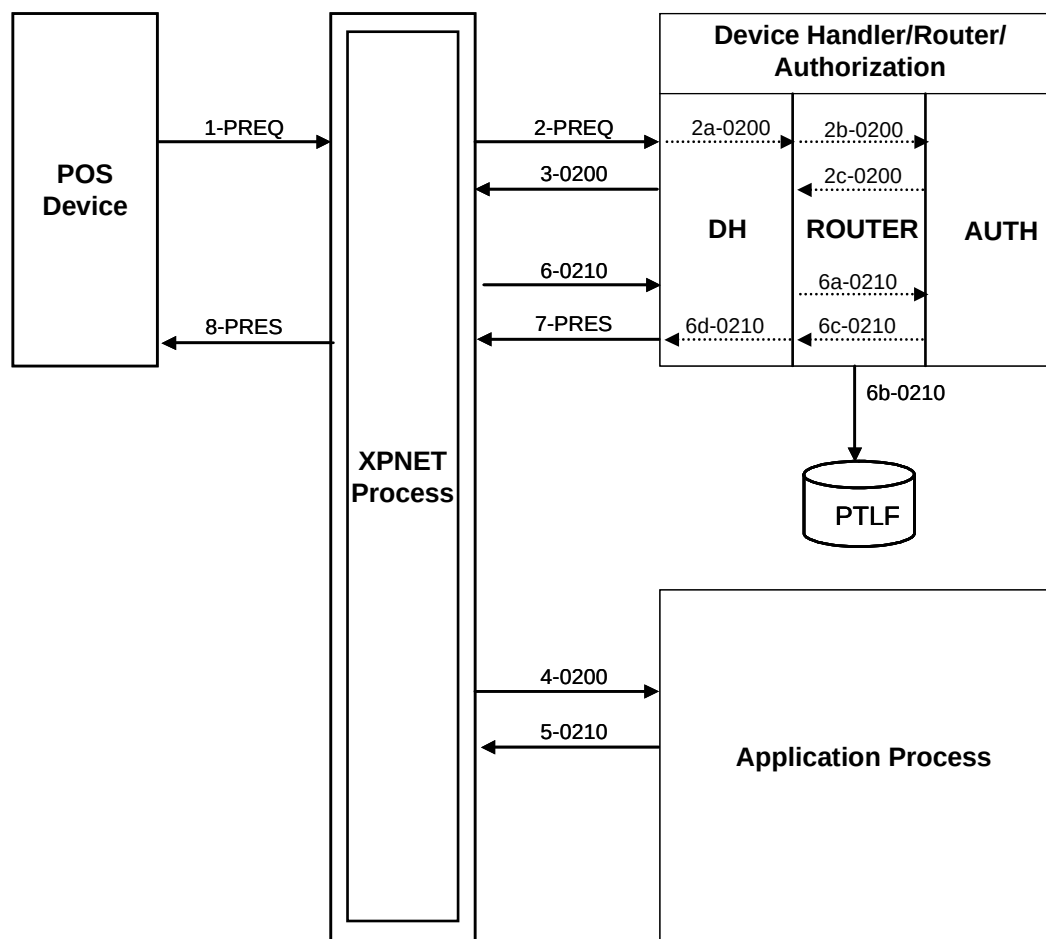
In the diagram below, the transaction is not a BASE24-pos transaction request, even though it was acquired from a POS device. An application process approves the transaction and the response is not logged to the POS Transaction Log File (PTLF).



Steps	Processing
1–2	The POS device sends the transaction request to the BASE24-pos system as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Device Handler/Router/Authorization process.
2a–2c	The Device Handler module reads the BASE24-pos Terminal Data file record for the POS device to determine how to process the request. The Device Handler module reads the APCF extended memory table for the acquirer transaction profile to determine whether the transaction is allowed and whether it is to be authorized by the BASE24-pos system. In this case, the transaction is allowed, but it is to be routed to an application process for authorization and the response is not to be logged to the PTLF. The Device Handler module then reformats the transaction request into a POS Standard Internal Message (PSTM), adds the Acquirer Routing token to the PSTM, sets a response timer, and passes the transaction request as a 0200 message to the Router module. The Router module determines that the Acquirer Routing token is present in the transaction request message and sends the message to the symbolic destination specified in the AUTH-DEST field of the Acquirer Routing token.
3–4	The Device Handler/Router/Authorization process sends the 0200 request message to the application process.
5–6	The application process authorizes the request, formats a 0210 response message, and sends the response back to the Device Handler/Router/Authorization process.
6a–6c	The Router module checks the LOG-AUTH-RESP-DEST field in the Acquirer Routing token to determine whether the response is to be logged to the PTLF. In this case, the field is set to a value of N, so the response is not logged. The Router module passes the response to the Device Handler module.
7–8	The Device Handler module reformats the 0210 response message into native-mode format, deletes the response timer, and sends the response message to the POS device.

Approved Transaction from a Device with Logging

In the diagram below, the transaction is not a BASE24-pos transaction request, even though it was acquired from a POS device. An application process approves the transaction and the response is logged to the POS Transaction Log File (PTLF).



Steps

1–2

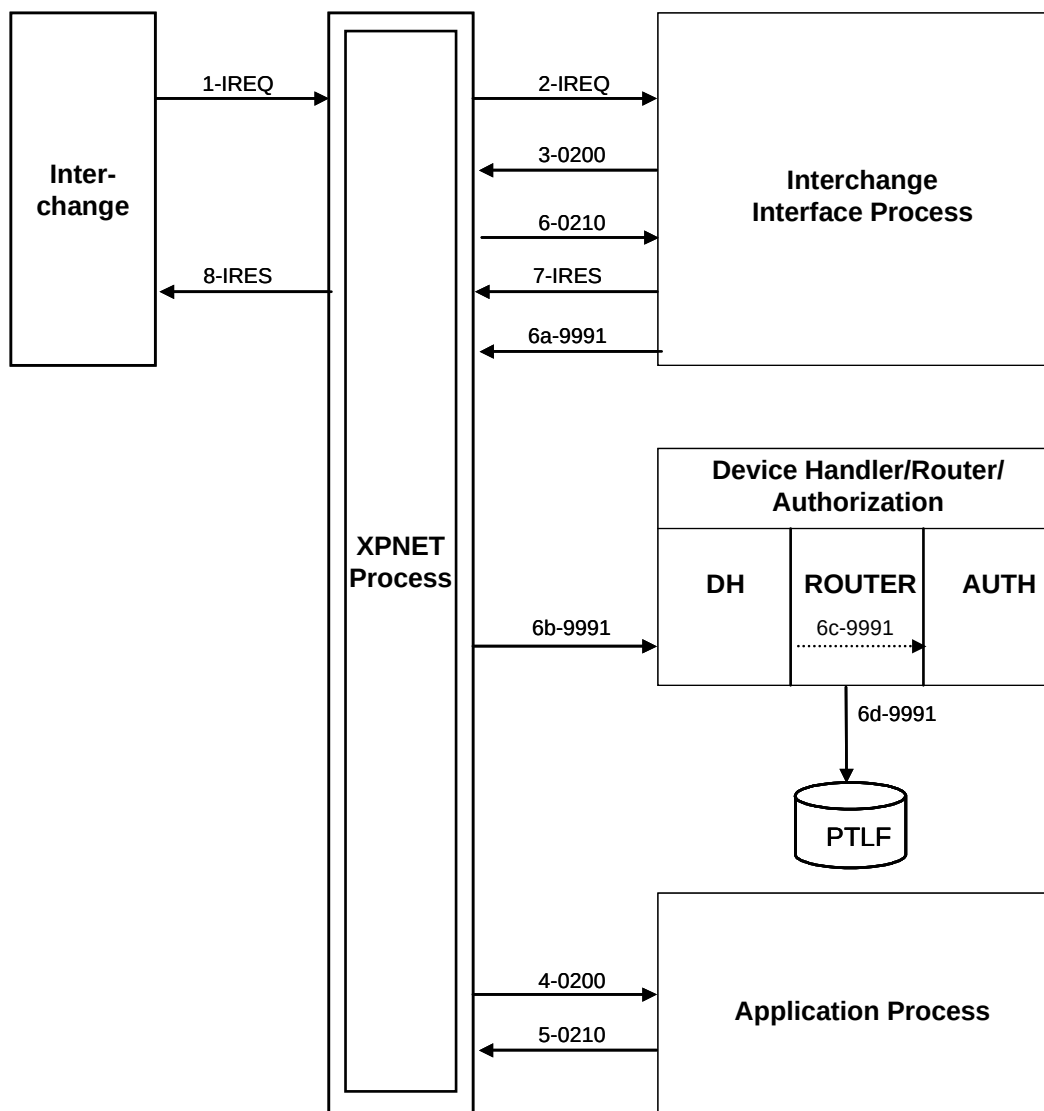
The POS device sends the transaction request to the BASE24-pos system as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Device Handler/Router/Authorization process.

Processing

Steps	Processing
2a–2c	The Device Handler module reads the BASE24-pos Terminal Data file record for the POS device to determine how to process the request. The Device Handler module reads the APCF extended memory table for the acquirer transaction profile to determine whether the transaction is allowed and whether it is to be authorized by the BASE24-pos system. In this case, the transaction is allowed, but it is to be routed to an application process for authorization and the response is not to be logged to the PTLF. The Device Handler module then reformats the transaction request into a POS Standard Internal Message (PSTM), adds the Acquirer Routing token to the PSTM, sets a response timer, and passes the transaction request as a 0200 message to the Router module. The Router module determines that the Acquirer Routing token is present in the transaction request message and sends the message to the symbolic destination specified in the AUTH-DEST field of the Acquirer Routing token.
3–4	The Device Handler/Router/Authorization process sends the 0200 request message to the application process.
5–6	The application process authorizes the request, formats a 0210 response message, and sends the response back to the Device Handler/Router/Authorization process.
6a–6d	The Router module checks the LOG-AUTH-RESP-DEST field in the Acquirer Routing token to determine whether the response is to be logged to the PTLF. In this case, the field is set to a value of Y, so the response is logged to the POS Transaction Log File (PTLF). The Router module then passes the response to the Device Handler module.
7–8	The Device Handler module reformats the 0210 response message into native-mode format, deletes the response timer, and sends the response message to the POS device.

Approved Transaction from an Interchange with Logging

In the diagram below, the transaction is not a BASE24-pos transaction request that was acquired from an interchange. An application process approves the transaction and the response is logged to the POS Transaction Log File (PTLF).



Steps

- 1–2 The interchange sends the transaction request as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Interchange Interface process.

Processing

Steps	Processing
2	The Interchange Interface process reads the Enhanced Interchange Configuration File (ICFE) record for the interchange to determine how to process the request. The Interchange Interface process reads the APCF extended memory table for the acquirer transaction profile to determine whether the transaction is allowed and whether it is to be authorized by the BASE24-pos system. In this case, the transaction is allowed, but it is to be routed to an application process for authorization and the response is to be logged to the PTLF. The Interchange Interface process then reformats the transaction request into a POS Standard Internal Message (PSTM), adds the Acquirer Routing token to the PSTM, sets a response timer, and sends the message to the symbolic destination specified in the AUTH-DEST field of the Acquirer Routing token.
3–4	The Interchange Interface process sends the 0200 request message to the application process.
5–6	The application process authorizes the request, formats a 0210 response message, and sends the response back to the Interchange Interface process.
6a–6d	The Interchange Interface process checks the LOG-AUTH-RESP-DEST field in the Acquirer Routing token to determine whether the response is to be logged to the PTLF. In this case, this field is set to a value of Y, so the Interchange Interface process sends a copy of the response message as a log-only (9991) message to the Device Handler/Router/Authorization process specified in the ICFE. The Router/Authorization module then logs the response to the POS Transaction Log File (PTLF).
7–8	The Interchange Interface process reformats the 0210 response message into native-mode format for the interchange, deletes the response timer, and sends the response message to the interchange.

Mobile Top-Up Transactions

The diagrams on the following pages illustrate the manner in which the BASE24-pos product handles various mobile top-up transactions. All of the diagrams assume authorization level 2 (offline) without completions, even though other authorization levels and the use of completions can be configured.

The BASE24-pos Standard POS Device Handler (SPDH) module is the only BASE24-pos Device Handler module for which an add-on Mobile Top-Up module is available.

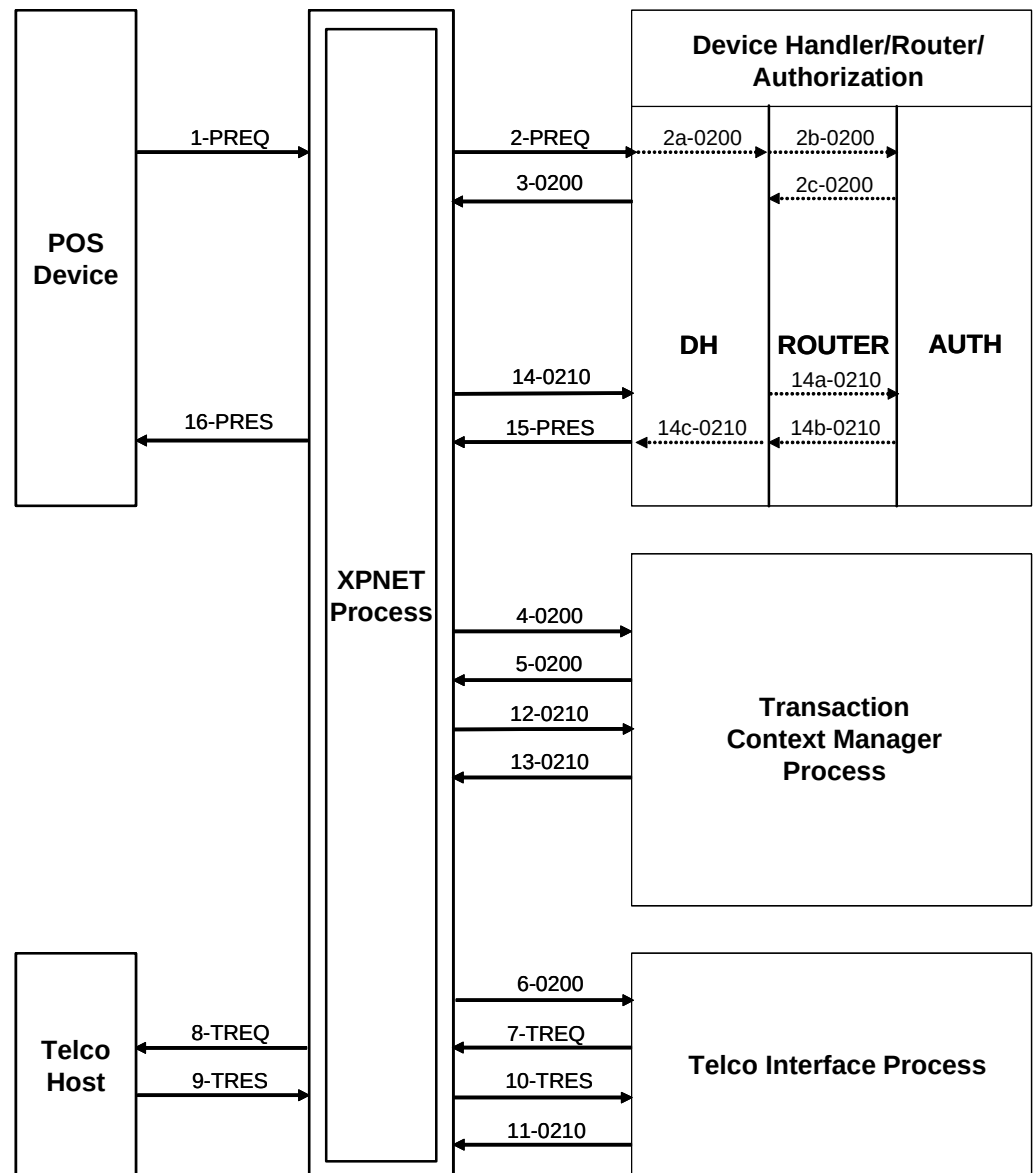
The BASE24-pos Transaction Context Manager process is responsible for receiving a mobile top-up transaction from the BASE24-pos Device Handler/Router/Authorization process and splitting the transaction into funds and service provider (e.g., telephone company) portions. The Transaction Context Manager process routes the required information to the funds authorizer for approval, then routes the required information to the service provider for approval and logs the service provider's response to the POS Transaction Log File (PTLF). Finally, the Transaction Context Manager process formats the responses into a single message that can be returned to the Device Handler/Router/Authorization process.

The SPDH module uses the POS-SPLIT-TXN-RTG-DEST param in the LCONF to identify which Transaction Context Manager process receives its mobile top-up transactions. Refer to the ***BASE24-pos Standard POS Device Configuration Manual*** for details on configuring the SPDH module for mobile top-up transactions.

The Transaction Context Manager process uses the Mobile Operator File (MOF) and Split Transaction Routing File (STRF) to determine how to split and route the transactions it receives. The STRF requester and server processes use the Transaction Code/Subtype Relationship File (TSRF). Refer to the ***BASE24 Base Files Maintenance Manual*** for details on configuring each of these files.

Mobile Top-Up With Cash, Approved by Telco Host

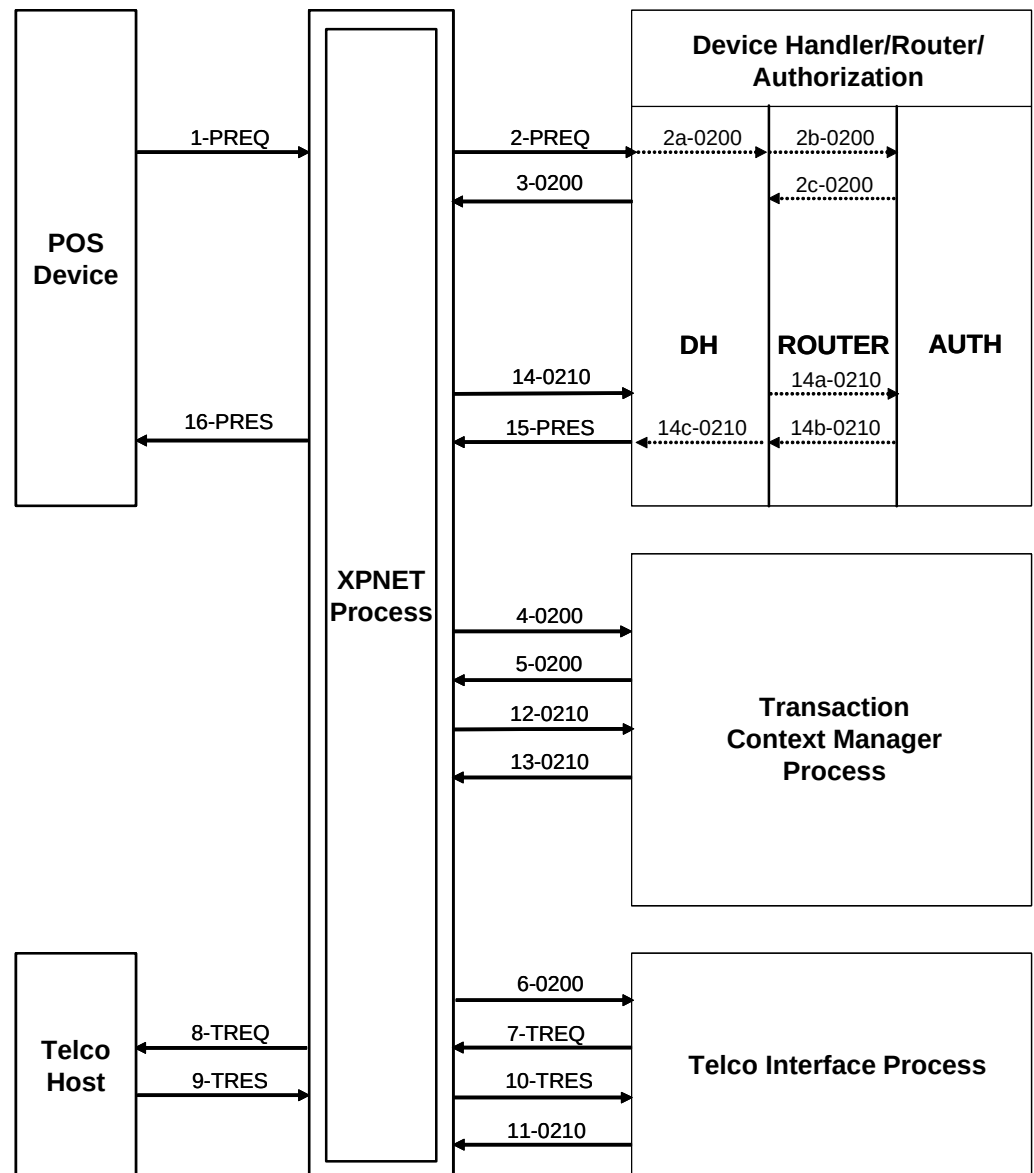
In the diagram below, the mobile top-up with cash transaction is approved by the telephone company (telco) host. Because the transaction is performed with only cash, the BASE24-pos Authorization module is not involved in authorizing the transaction.



Steps	Processing
1–2	The POS device sends the transaction request to the BASE24-pos system as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Device Handler/Router/Authorization process.
2a–2c	The Device Handler module reads the BASE24-pos Terminal Data files record for the POS device to determine how to process the request. The Device Handler module then reformats the transaction request into a POS Standard Internal Message (PSTM), sets a response timer, and passes the transaction request as a 0200 message to the Router module.
3–4	The Device Handler/Router/Authorization process sends the 0200 message to the Transaction Context Manager process.
5–6	The Transaction Context Manager process sends a 0200 message to the Telco Interface process configured in the Mobile Operator File (MOF) and Split Transaction Routing File (STRF).
7–8	The Telco Interface process reformats the 0200 message into an external message format, sets an outbound message timer, and sends the request to the telco host.
9–10	The telco host approves the transaction request, formats a response, and sends the response to the Telco Interface process.
11–12	The Telco Interface process deletes the outbound message timer, reformats the response into a 0210 internal response message, logs the 0210 message to the Interchange Log File (ILF), and sends the message to the Transaction Context Manager process.
13–14	The Transaction Context Manager process logs the 0210 message to the PTLF as a mobile top-up with cash transaction and sends the message to the Device Handler/Router/Authorization process.
14a–14c	The Router module receives the 0210 message and passes it to the Device Handler module. The Authorization module is not involved in processing so it does not log a 0210 message to the PTLF.
15–16	The Device Handler module reformats the response message into native-mode format, deletes the response timer, and sends the response to the POS device.

Mobile Top-Up With Cash, Declined by Telco Host

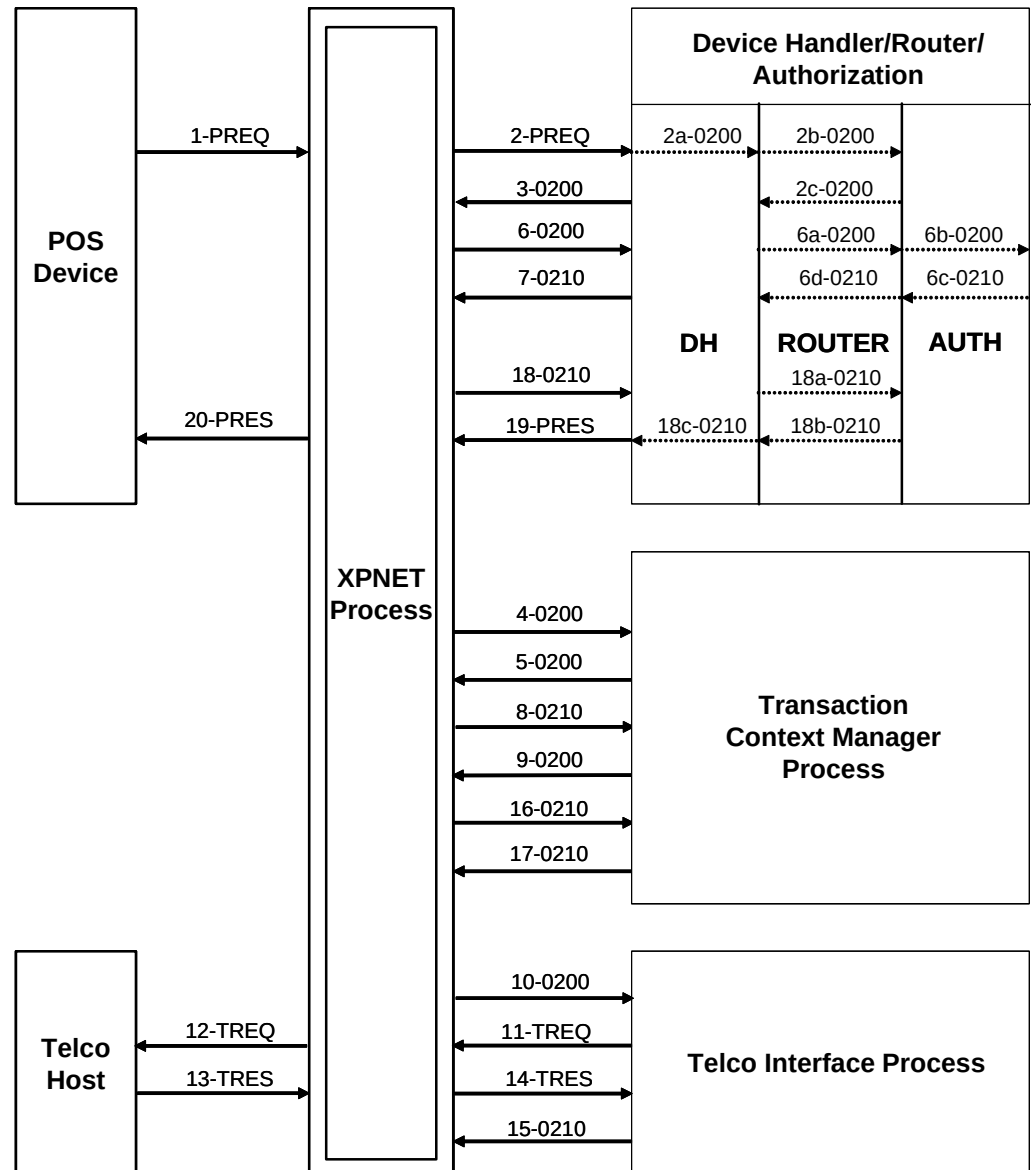
In the diagram below, the mobile top-up with cash transaction is declined by the telephone company (telco) host. Because the transaction is performed with only cash, the BASE24-pos Authorization module is not involved in authorizing the transaction.



Steps	Processing
1–2	The POS device sends the transaction request to the BASE24-pos system as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Device Handler/Router/Authorization process.
2a–2c	The Device Handler module reads the BASE24-pos Terminal Data files record for the POS device to determine how to process the request. The Device Handler module then reformats the transaction request into a POS Standard Internal Message (PSTM), sets a response timer, and passes the transaction request as a 0200 message to the Router module.
3–4	The Device Handler/Router/Authorization process sends the 0200 message to the Transaction Context Manager process.
5–6	The Transaction Context Manager process sends a 0200 message to the Telco Interface process configured in the Mobile Operator File (MOF) and Split Transaction Routing File (STRF).
7–8	The Telco Interface process reformats the 0200 message into an external message format, sets an outbound message timer, and sends the request to the telco host.
9–10	The telco host declines the transaction request, formats a response, and sends the response to the Telco Interface process.
11–12	The Telco Interface process deletes the outbound message timer, reformats the response into a 0210 internal response message, logs the 0210 message to the Interchange Log File (ILF), and sends the message to the Transaction Context Manager process.
13–14	The Transaction Context Manager process logs the 0210 message to the PTLF as a mobile top-up with cash transaction and sends the message to the Device Handler/Router/Authorization process.
14a–14c	The Router module receives the 0210 message and passes it to the Device Handler module. The Authorization module is not involved in processing so it does not log a 0210 message to the PTLF.
15–16	The Device Handler module reformats the response message into native-mode format, deletes the response timer, and sends the response to the POS device.

Mobile Top-Up With Funds, Approved by BASE24 and Telco Host

In the diagram below, the mobile top-up with funds transaction is approved by the BASE24-pos Authorization module and the telephone company (telco) host.

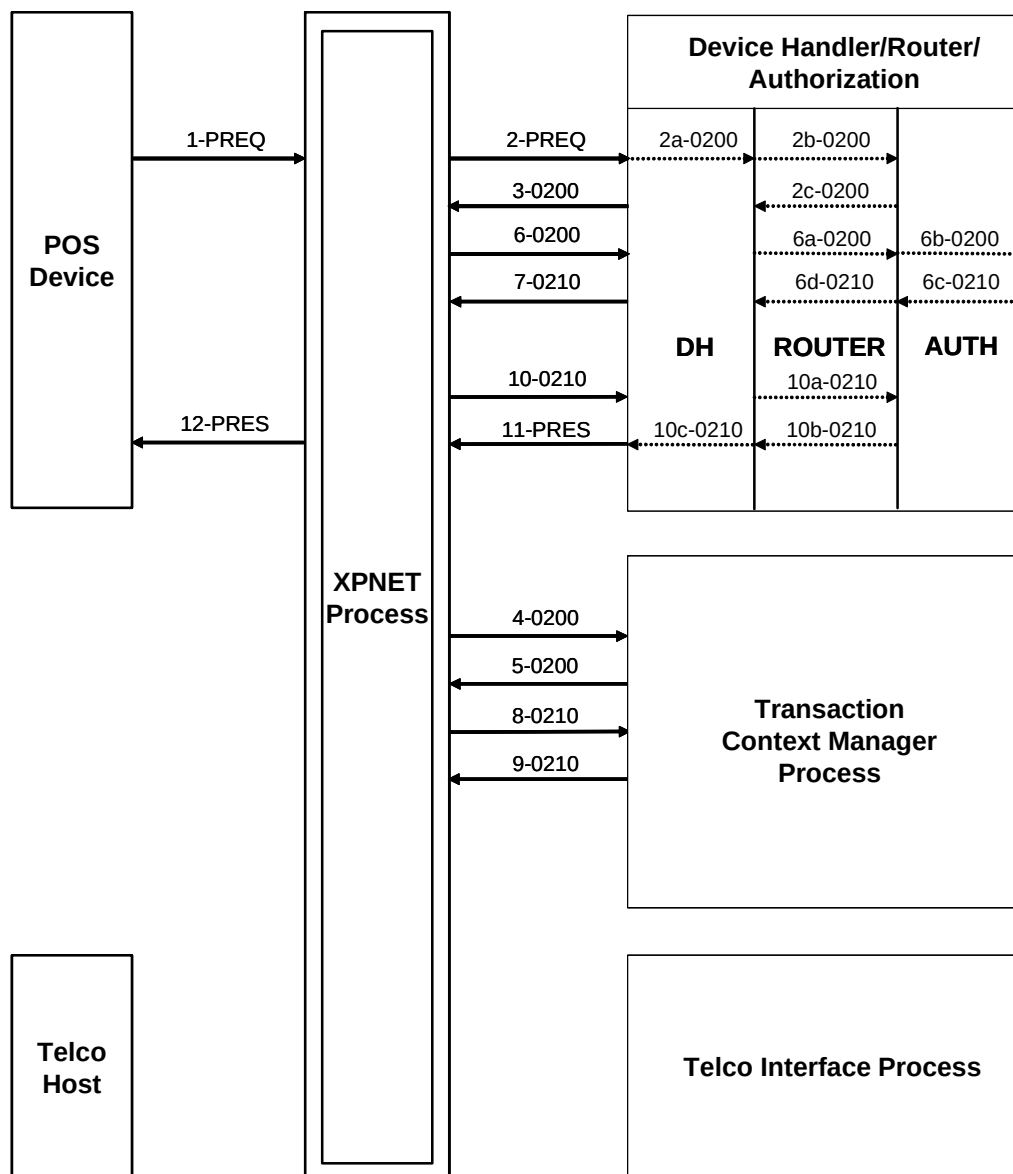


Steps	Processing
1–2	The POS device sends the transaction request to the BASE24-pos system as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Device Handler/Router/Authorization process.
2a–2c	The Device Handler module reads the BASE24-pos Terminal Data files record for the POS device to determine how to process the request. The Device Handler module then reformats the transaction request into a POS Standard Internal Message (PSTM), sets a response timer, and passes the transaction request as a 0200 message to the Router module.
3–4	The Device Handler/Router/Authorization process sends the 0200 message to the Transaction Context Manager process.
5–6	The Transaction Context Manager process formats a 0200 message for a Purchase transaction, and passes it to the original Device Handler/Router/Authorization process.
6a–6d	The Router module locates a Card Prefix File (CPF) record for the card involved in the transaction and determines AST routing. The Router module then passes the message to the Authorization module. From the CPF, the Authorization module locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 2. The Authorization module approves the request, updates the database files appropriately, logs the Purchase transaction response to the POS Transaction Log File (PTLF), and formats a 0210 response message.
7–8	The Device Handler/Router/Authorization process sends the 0210 message to the Transaction Context Manager process.
9–10	The Transaction Context Manager process formats a 0200 request message for a mobile top-up with funds transaction and passes it to the Telco Interface process configured in the Mobile Operator File (MOF) and Split Transaction Routing File (STRF).

Steps	Processing
11–12	The Telco Interface process reformats the 0200 request into an external message format, sets an outbound message timer, and sends the request to the telco host.
13–14	The telco host approves the transaction request, formats a response, and sends the response to the Telco Interface process.
15–16	The Telco Interface process deletes the outbound message timer, reformats the response into a 0210 internal response message, logs the 0210 message to the Interchange Log File (ILF), and sends the message to the Transaction Context Manager process.
17–18	The Transaction Context Manager process logs the 0210 message to the PTLF as a mobile top-up with funds transaction and sends the message to the Device Handler/Router/Authorization process.
18a–18c	The Router module receives the 0210 message and passes it to the Device Handler module. The Authorization module is not involved in processing at this point so it does not log another 0210 message to the PTLF.
19–20	The Device Handler module reformats the response message into native-mode format, deletes the response timer, and sends the response to the POS device.

Mobile Top-Up With Funds, Declined by BASE24 Authorization Module

In the diagram below, the mobile top-up with funds transaction is declined by the BASE24-pos Authorization module.

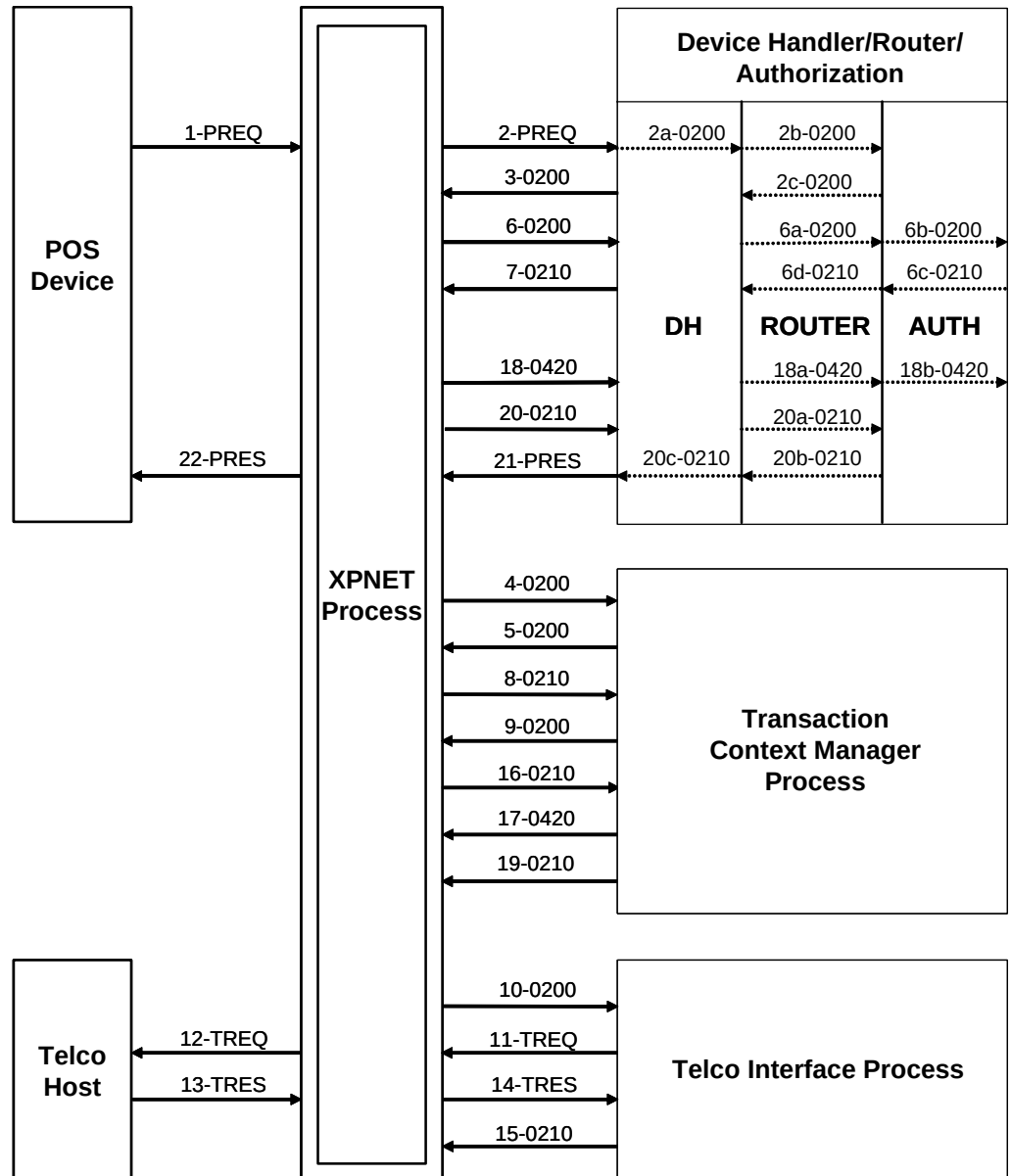


Steps	Processing
1–2	The POS device sends the transaction request to the BASE24-pos system as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Device Handler/Router/Authorization process.
2a–2c	The Device Handler module reads the BASE24-pos Terminal Data files record for the POS device to determine how to process the request. The Device Handler module then reformats the transaction request into a POS Standard Internal Message (PSTM), sets a response timer, and passes the transaction request as a 0200 message to the Router module.
3–4	The Device Handler/Router/Authorization process sends the 0200 message to the Transaction Context Manager process.
5–6	The Transaction Context Manager process formats a 0200 message for a Purchase transaction, and passes it to the original Device Handler/Router/Authorization process.
6a–6d	The Router module locates a Card Prefix File (CPF) record for the card involved in the transaction and determines AST routing. The Router module then passes the message to the Authorization module. From the CPF, the Authorization module locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 2. The Authorization module declines the request, logs the Purchase transaction response to the POS Transaction Log File (PTLF), and formats a 0210 response message. The cardholder authorization files are not updated when the Authorization module declines a request.
7–8	The Device Handler/Router/Authorization process sends the 0210 message to the Transaction Context Manager process.
9–10	The Transaction Context Manager process sends the 0210 message to the Device Handler/Router/Authorization process.

Steps	Processing
10a–10c	The Router module receives the 0210 message and passes it to the Device Handler module. The Authorization module is not involved in processing at this point so it does not log another 0210 message to the PTLF.
11–12	The Device Handler module reformats the response message into native-mode format, deletes the response timer, and sends the response to the POS device.

Mobile Top-Up With Funds, Declined by Telco Host

In the diagram below, the mobile top-up with funds transaction is declined by the telephone company (telco) host.



Steps	Processing
1–2	The POS device sends the transaction request to the BASE24-pos system as a native-mode message, where it is received by the XPNET process and forwarded to the appropriate Device Handler/Router/Authorization process.
2a–2c	The Device Handler module reads the BASE24-pos Terminal Data files record for the POS device to determine how to process the request. The Device Handler module then reformats the transaction request into a POS Standard Internal Message (PSTM), sets a response timer, and passes the transaction request as a 0200 message to the Router module.
3–4	The Device Handler/Router/Authorization process sends the 0200 message to the Transaction Context Manager process.
5–6	The Transaction Context Manager process formats a 0200 message for a Purchase transaction, and passes it to the original Device Handler/Router/Authorization process.
6a–6d	The Router module locates a Card Prefix File (CPF) record for the card involved in the transaction and determines AST routing. The Router module then passes the message to the Authorization module. From the CPF, the Authorization module locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 2. The Authorization module approves the request, updates the database files appropriately, logs the Purchase transaction response to the POS Transaction Log File (PTLF), and formats a 0210 response message.
7–8	The Device Handler/Router/Authorization process sends the 0210 message to the Transaction Context Manager process.
9–10	The Transaction Context Manager process formats a 0200 request message for a mobile top-up with funds transaction and passes it to the Telco Interface process configured in the Mobile Operator File (MOF) and Split Transaction Routing File (STRF).

Steps	Processing
11–12	The Telco Interface process reformats the 0200 message into an external message format, sets an outbound message timer, and sends the request to the telco host.
13–14	The telco host declines the transaction request, formats a response, and sends the response to the Telco Interface process.
15–16	The Telco Interface process deletes the outbound message timer, reformats the response into a 0210 internal response message, logs the 0210 message to the Interchange Log File (ILF), and sends the message to the Transaction Context Manager process.
17–18	The Transaction Context Manager process formats a 0420 reversal message for the Purchase transaction and sends it to the Device Handler/Router/Authorization process.
18a–18b	The Router module passes the 0420 message to the Authorization module. The Authorization module logs the reversal to the PTLF, updates the database files appropriately, and drops the message.
19–20	The Transaction Context Manager process logs the 0210 message to the PTLF as a mobile top-up with funds transaction and sends the message to the Device Handler/Router/Authorization process.
20a–20c	The Router module receives the 0210 message and passes it to the Device Handler module. The Authorization module is not involved in processing at this point so it does not log a 0210 message to the PTLF.
21–22	The Device Handler module reformats the response message into native-mode format, deletes the response timer, and sends the response to the POS device.

ACI Worldwide, Inc.

6: BASE24-pos Transaction Impacts on the Database

This section describes the impact of BASE24-pos transactions on the BASE24-pos database. It describes how transactions affect the balance and accumulation totals in the Positive Balance File (PBF), Cardholder Authorization File (CAF), and Usage Accumulation File (UAF). It also describes how transactions affect totals in the BASE24-pos Terminal Data files.

PBF and CAF/UAF Impacts

The table later in this subsection shows the impacts of approved transactions on PBF balances and CAF/UAF accumulation totals. The balance and accumulation totals included in the impact table are described below.

PBF Totals

The balance totals, whose transaction impacts are included in the impact table, are shown below. The captions used in the impact table match the field names of the corresponding totals on the PBF screens.

The totals on PBF screen 1 are Base totals and are impacted by all products (BASE24-atm, BASE24-pos, and BASE24-teller) using the PBF. Several PBF totals have different meanings depending on whether the PBF record represents a credit or non-credit account. The totals on PBF screen 8 are BASE24-pos totals and are impacted only by BASE24-pos transactions. Refer to the ***BASE24 Base Files Maintenance Manual*** for more information about the PBF and its fields.

PBF Totals		
Screen 1	Noncredit	Credit
	AVAILABLE BALANCE	AVAILABLE CREDIT
	LEDGER BALANCE	CREDIT LIMIT
	AMOUNT ON HOLD	CREDIT BALANCE
	CASH OUT	
Screen 8	CURRENT FLOAT	
	TOTAL FLOAT	

CAF/UAF Totals

The CAF/UAF accumulation totals, whose transaction impacts are included in the impact table, are summarized below. Not every CAF/UAF totals field is listed in the table. Rather, the impact table includes representative CAF/UAF totals, which can be tied to corresponding fields on the CAF and UAF screens.

Functionally, CAF and UAF totals are identical. They are simply in different files used with different authorization methods. Throughout this section they are treated identically.

Also, CAF and UAF accumulation fields include both total and offline accumulators. Again, for the purpose of impacts, these are identical. Total versus offline accumulation is based in the authorization level being used and whether the BASE24 system or a host approved the transaction. When the total and offline accumulators are impacted by a transaction, the method and amount by which they are impacted are the same.

Finally, CAF and UAF accumulation fields include Base fields, which total transaction accumulation information for all BASE24 products, and POS-specific fields, which total transaction accumulation information for only the BASE24-pos product. Base fields are located on CAF screen 1 and UAF screen 1. BASE24-pos fields are located on CAF screen 10 and UAF screen 6. When BASE24 products process a POS financial transaction, both Base fields and BASE24-pos fields are impacted.

Cash Withdrawal Column

The Cash Withdrawal column in the impact table describes how transactions impact the following CAF and UAF fields:

CAF screen 1 — TOTAL CASH WDL
CAF screen 1 — OFFLINE CASH WDL

CAF screen 10 — TOTAL CASH WDL
CAF screen 10 — OFFLINE CASH WDL

UAF screen 1 — TOTAL CASH WDL
UAF screen 1 — OFFLINE CASH WDL

UAF screen 6 — TOTAL CASH WDL
UAF screen 6 — OFFLINE CASH WDL

Cash Advance Column

The Cash Advance column in the impact table describes how transactions impact the following CAF and UAF fields:

CAF screen 1 — TOTAL CASH ADV
CAF screen 1 — OFFLINE CASH ADV

CAF screen 10 — TOTAL CASH ADV
CAF screen 10 — OFFLINE CASH ADV

UAF screen 1 — TOTAL CASH ADV
UAF screen 1 — OFFLINE CASH ADV

UAF screen 6 — TOTAL CASH ADV
UAF screen 6 — OFFLINE CASH ADV

Purchase Column

The Purchase column in the impact table describes how transactions impact the following CAF and UAF fields:

CAF screen 10 — TOTAL PURCHASE
CAF screen 10 — OFFLINE PURCHASE

UAF screen 6 — TOTAL PURCHASE
UAF screen 6 — OFFLINE PURCHASE

Refund Column

The Refund column in the impact table describes how transactions impact the following CAF and UAF fields:

CAF screen 10 — TOTAL REFUND
CAF screen 10 — OFFLINE REFUND

UAF screen 6 — TOTAL REFUND
UAF screen 6 — OFFLINE REFUND

Number of Refunds Column

The Number of Refunds column in the impact table describes how transactions impact the following CAF and UAF fields:

CAF screen 10 — NUMBER OF REFUNDS THIS PERIOD

UAF screen 6 — NUMBER OF REFUNDS THIS PERIOD

Times Used

The Times Used column in the impact table describes how transactions impact the following CAF and UAF fields:

CAF screen 10 — TIMES USED THIS PERIOD

UAF screen 6 — TIMES USED THIS PERIOD

Transaction Impact Table

The table beginning on the following page describes the impacts of approved transactions on the PBF and CAF/UAF balance and accumulation fields.

The impacts are shown using the amount fields from the POS Standard Internal Message (PSTM), which carry the pertinent amounts for each POS transaction. The PSTM is described in appendix A.

The values used in the impact table are described below.

Value	Description
A1	Represents amount 1 (TRAN.AMT-1) in the PSTM. A plus sign (+) or minus sign (–) indicates whether the amount is added to or subtracted from the total.
A2	Represents amount 2 (TRAN.AMT-2) in the PSTM. A plus sign (+) or minus sign (–) indicates whether the amount is added to or subtracted from the total.

Value	Description
a	Indicates for an adjustment that the impact only takes place if the new, adjusted amount is zero. For example, the original transaction was for an amount of \$100, but the adjusted amount of the transaction is zero.
b	Indicates that the impact shown is the net impact. In the case of PBF available credit balances, the value in the field is actually recomputed for each transaction; however, the net effect on a transaction-by-transaction basis is what is shown in the table.
c	Indicates that the impact occurs only if the transaction has a draft capture flag of 1 (authorize and capture).
d	Indicates that the impact occurs only if the transaction has a draft capture flag of 0 (authorize only) or 2 (authorize with electronic follow-up).
e	Indicates that the impact occurs only if the BASE24-pos eCheck add-on product is installed and configured to impact the Positive Balance File (PBF).

Note: When a value appearing in the TRAN CODE (TCTAAC) column of the following table can be any valid value for that portion of the transaction code, a hyphen is used in place of an actual value. For example, the hyphen in the T (card type) portion of the following transaction code indicates that the impacts shown apply to transactions initiated by either a debit or credit card:

Tran Code			
TC	T	AA	C
10	-	31	0

Tran Code (TCTAAC)	Transaction Description	PBF Totals						CAF/UAf Totals					
		Total Float	Current Float	Cash Out	Amount On Hold / Credit Balance	Ledger Balance / Credit Limit	Available Balance / Available Credit	Times Used	Number of Refunds/ Replenishments	Refunds/ Replenishments	Purchase	Cash Advance	Cash Withdrawal
10 - 31 0	Normal purchase; credit account; debit or credit card.	+A1 ^d	+A1 ^d		+A1 ^c		−A1 ^b	+1			+A1		
10 2 -- 0	Normal purchase; checking or savings account; debit card.					−A1	−A1	+1					+A1
10 - 31 1	Normal sales draft purchase; credit account; debit or credit card.	+A1 ^d	+A1 ^d		+A1 ^c		−A1 ^b	+1			+A1		
10 2 -- 1	Normal sales draft purchase; checking or savings account; debit card.					−A1	−A1	+1					+A1
10 - 31 2	Representation of a normal purchase; credit account; debit or credit card.	+A1 ^d	+A1 ^d		+A1 ^c		+A1 ^b	+1			+A1		
10 2 -- 2	Representation of a normal purchase; checking or savings account; debit card.					−A1 +A2	+A1 −A2	+1					+A1 −A2
10 - 31 3	Chargeback of a normal purchase; credit account; debit or credit card.	−A1 ^d +A2 ^d	−A1 ^d +A2 ^d		−A1 ^c +A2 ^c		−A ^b	−1 ^a			−A1 +A2		
10 2 -- 3	Chargeback of a normal purchase; checking or savings account; debit card.					+A1 −A2	+A1 −A2	−1 ^a					−A1 +A2
11 - 31 0	Preauthorization of purchase; credit account; debit or credit card.												

Tran Code (TCTAAC)	Transaction Description	PBF Totals						CAF/UAF Totals					
		Total Float	Current Float	Cash Out	Amount On Hold / Credit Balance	Ledger Balance / Credit Limit	Available Balance / Available Credit	Times Used	Number of Refunds/ Replenishments	Refunds/ Replenishments	Purchase	Cash Advance	Cash Withdrawal
11 2 -- 0	Preauthorization of purchase; checking or savings account; debit card.												
12 - 31 0	Completion of a preauthorized purchase; credit account; debit or credit card.	+A1 ^d	+A1 ^d		+A1 ^c		-A1 ^b	+1			+A1		
12 2 -- 0	Completion of a preauthorized purchase; checking or savings account; debit card.					-A1	-A1	+1					+A1
12 - 31 1	Completion of a preauthorized sales draft purchase; credit account; debit or credit card.	+A1 ^d	+A1 ^d		+A1 ^c		-A1 ^b	+1			+A1		
12 2 -- 1	Completion of a preauthorized sales draft purchase; checking or savings account; debit card.					-A1	-A1	+1					+A1
13 - 31 0	Mail or telephone purchase; credit account; debit or credit card.	+A1 ^d	+A1 ^d		+A1 ^c		-A1 ^b	+1			+A1		
13 2 -- 0	Mail or telephone purchase; checking or savings account; debit card.					-A1	-A1	+1					+A1
13 - 31 1	Mail or telephone sales draft purchase; credit account; debit or credit card.	+A1 ^d	+A1 ^d		+A1 ^c		-A1 ^b	+1			+A1		

Tran Code (TCTAAC)	Transaction Description	PBF Totals						CAF/UAF Totals					
		Total Float	Current Float	Cash Out	Amount On Hold / Credit Balance	Ledger Balance / Credit Limit	Available Balance / Available Credit	Times Used	Number of Refunds/ Replenishments	Refunds/ Replenishments	Purchase	Cash Advance	Cash Withdrawal
13 2 -- 1	Mail or telephone sales draft purchase; checking or savings account; debit card.					−A1	−A1	+1					+A1
14 - 31 0	Merchandise return; credit account; debit or credit card.	−A1 ^d			−A1 ^c		+A1 ^b	+1	+1	+A1	−A1		
14 2 -- 0	Merchandise return; checking or savings account; debit card.					+A1	+A1	+1	+1	+A1			−A1
14 - 31 1	Sales draft merchandise return; credit account; debit or credit card.	−A1 ^d			−A1 ^c		+A1 ^b	+1	+1	+A1	−A1		
14 2 -- 1	Sales draft merchandise return; checking or savings account; debit card.					+A1	+A1	+1	+1	+A1			−A1
15 - 31 0	Cash advance; credit account; debit or credit card.	+A1 ^d	+A1 ^d	+A1	+A1 ^c		−A1 ^b	+1				+A1	
15 2 -- 0	Cash advance; checking or savings account; debit card.			+A1		−A1	−A1	+1					+A1
15 - 31 1	Sales draft cash advance; credit account; debit or credit card.	+A1 ^d	+A1 ^d	+A1	+A1 ^c		−A1 ^b	+1				+A1	
15 2 -- 1	Sales draft cash advance; checking or savings account; debit card.			+A1		−A1	−A1	+1					+A1

Tran Code (TCTAAC)	Transaction Description	PBF Totals						CAF/UAF Totals					
		Total Float	Current Float	Cash Out	Amount On Hold / Credit Balance	Ledger Balance / Credit Limit	Available Balance / Available Credit	Times Used	Number of Refunds/ Replenishments	Refunds/ Replenishments	Purchase	Cash Advance	Cash Withdrawal
16 - 00 0	Verification of a debit or credit card.												
17 - 31 0	Credit account balance inquiry; debit or credit card.												
17 2 -- 0	Checking or savings account balance inquiry; debit card.												
18 2 -- 0	Purchase with cash back; checking or savings account; debit card.			+A2		−A1	−A1	+1					+A1
18 2 -- 1	Sales draft purchase with cash back; checking or savings account; debit card.			+A2		−A1	−A1	+1					+A1
19 0 00 -	Check verification												
20 0 00 -	Check guarantee			+A1 ^e		−A1 ^e	−A1 ^e						
21 - 31 0	Purchase adjustment; credit account; debit or credit card.	−A1 ^d +A2 ^d	−A1 ^d +A2 ^d		−A1 ^c +A2 ^c		+A1 −A2 ^b	−1 ^a			−A1 +A2		
21 2 -- 0	Purchase adjustment; checking or savings account; debit card.					+A1 −A2	+A1 −A2	−1 ^a					−A1 +A2
21 - 31 1	Sales draft purchase adjustment; credit account; debit or credit card.	−A1 ^d +A2 ^d	−A1 ^d +A2 ^d		−A1 ^c +A2 ^c		+A1 −A2 ^b	−1 ^a			−A1 +A2		

Tran Code (TCTAAC)	Transaction Description	PBF Totals						CAF/UAF Totals					
		Total Float	Current Float	Cash Out	Amount On Hold / Credit Balance	Ledger Balance / Credit Limit	Available Balance / Available Credit	Times Used	Number of Refunds/ Replenishments	Refunds/ Replenishments	Purchase	Cash Advance	Cash Withdrawal
21 2 -- 1	Sales draft purchase adjustment; checking or savings account; debit card.					+A1 -A2	+A1 -A2	-1 ^a					-A1 +A2
22 - 31 0	Merchandise return adjustment; credit account; debit or credit card.	+A1 ^d -A2 ^d			+A1 ^c -A2 ^c		-A1 +A2 ^b	-1 ^a	-1 ^a	-A1 +A2	+A1 -A2		
22 2 -- 0	Merchandise return adjustment; checking or savings account; debit card.					-A1 +A2	-A1 +A2	-1 ^a	-1 ^a	-A1 +A2			+A1 -A2
22 - 31 1	Sales draft merchandise return adjustment; credit account; debit or credit card.	+A1 ^d -A2 ^d			+A1 ^c -A2 ^c		-A1 +A2 ^b	-1 ^a	-1 ^a	-A1 +A2	+A1 -A2		
22 2 -- 1	Sales draft merchandise return adjustment; checking or savings account; debit card.					-A1 +A2	-A1 +A2	-1 ^a	-1 ^a	-A1 +A2			+A1 -A2
23 - 31 0	Cash advance adjustment; credit account; debit or credit card.	-A1 ^d +A2 ^d	-A1 ^d +A2 ^d	-A1 +A2	-A1 ^c +A2 ^c		+A1 -A2 ^b	-1 ^a				-A1 +A2	
23 2 -- 0	Cash advance adjustment; checking or savings account; debit card.			-A1 +A2		+A1 -A2	+A1 -A2	-1 ^a					-A1 +A2
23 - 31 1	Sales draft cash advance adjustment; credit account; debit or credit card.	-A1 ^d +A2 ^d	-A1 ^d +A2 ^d	-A1 +A2	-A1 ^c +A2 ^c		+A1 -A2 ^b	-1 ^a				-A1 +A2	

Draft—For ACI Worldwide Use Only 2012-03-28

BASE24-pos Transaction Impacts on the Database

Tran Code (TCTAAC)	Transaction Description	PBF Totals						CAF/UAF Totals					
		Total Float	Current Float	Cash Out	Amount On Hold / Credit Balance	Ledger Balance / Credit Limit	Available Balance / Available Credit	Times Used	Number of Refunds/ Replenishments	Refunds/ Replenishments	Purchase	Cash Advance	Cash Withdrawal
23 2 -- 1	Sales draft cash advance adjustment; checking or savings account; debit card.			-A1 +A2		+A1 -A2	+A1 -A2	-1 ^a					-A1 +A2
24 2 -- 0	Cash back amount adjustment; checking or savings account; debit card.			-A1 +A2									
24 2 -- 1	Sales draft cash back amount adjustment; checking or savings account; debit card.			-A1 +A2									
25 2 1 -	Card activation; checking account					+A1	+A1		+1	+A1			
27 2 1 -	Card replenishment; checking account					+A1	+A1	+1	+1	+A1			
28 2 1 -	Full redemption; checking account			+A1		-A1	-A1	+1					+A1

BASE24-pos Terminal Data Files Impacts

The BASE24-pos Terminal Data files contain terminal, clerk, and service transaction totals. The terminal totals are displayed on screen 4, the clerk totals are displayed on screen 5, and the service totals are displayed on screens 11, 12, 13, 14, 15, and 16. Each of these totals include debit counts and amounts, credit counts and amounts, and adjustment counts and amounts. The clerk totals also include cash out counts and amounts and check counts and amounts.

The terminal totals are kept according to whether the transaction is a draft capture transaction or a non-draft capture transaction. They are also kept according to batch, shift, day, and network totals. The appropriate batch, shift, day, and current network totals are all updated for each transaction. For example, a \$20 draft capture purchase transaction impacts the debit, draft capture fields for batch, shift, day, and current network totals.

The clerk totals are kept according to the operator initiating transactions at the terminal. The BASE24-pos product keeps clerk totals only if the terminal owner has configured the system to do so.

The service totals are kept according to the type of service, or card type, by which the transaction is initiated. For example, all transactions with a service type of American Express impact the totals for that service, while all transactions with a service type of Carte Blanche impact the totals for that service.

Each transaction that the BASE24-pos product receives impacts the debit, credit, adjustment, cash out, and check totals on each totals screen as described on the following page. Refer to the **BASE24-pos Files Maintenance Manual** for more information about the BASE24-pos Terminal Data files totals screens. Refer to the **BASE24-pos Settlement and Reporting Manual** for more information about how BASE24-pos Terminal Data files totals are accumulated.

The values used in the impact table are described below.

Value	Description
A1	Represents amount 1 (TRAN.AMT-1) in the PSTM. A plus sign (+) or minus sign (–) indicates whether the amount is added to or subtracted from the total.
A2	Represents amount 2 (TRAN.AMT-2) in the PSTM. A plus sign (+) or minus sign (–) indicates whether the amount is added to or subtracted from the total.

Value	Description
a	Applies to check verification and check guarantee transactions only when the BASE24-pos eCheck add-on product is installed and the check information has been converted to electronic format (i.e., the conversion flag in the Check Guarantee/ Verification 2 token is set to a value of 1 or 2).
b	Updates clerk totals only for both check verification and check guarantee transactions.

Transaction	Debit Amount	Debit Count	Credit Amount	Credit Count	Adjust Amount	Adjust Count	Cash Out Amount	Cash Out Count	Check Amount	Check Count
Normal Purchase	+A1	+1								
Representation										
Chargeback										
Preauthorization Purchase										
Preauthorization Purchase Completion	+A1	+1								
Mail or Telephone Order Purchase	+A1	+1								
Merchandise Return			+A1	+1						
Cash Advance	+A1	+1					+A1	+1		
Card Verification										
Balance Inquiry										
Purchase with Cash Back	+A1	+1					+A2	+1		
Check Verification/Guarantee	+A1 ^a	+1 ^a							+A1 ^b	+1 ^b
Purchase Adjustment (clerk totals)					-A1 +A2	+1				
Purchase Adjustment (non-clerk totals)	+A2	+1			+A1	+1				

Draft—For ACI Worldwide Use Only 2012-03-28

Transaction	Debit Amount	Debit Count	Credit Amount	Credit Count	Adjust Amount	Adjust Count	Cash Out Amount	Cash Out Count	Check Amount	Check Count
Merchandise Return Adjustment (clerk totals)					-A1 +A2	+1				
Merchandise Return Adjustment (non-clerk totals)			+A2	+1	-A1	+1				
Cash Advance Adjustment (clerk totals)					-A1 +A2	+1				
Cash Advance Adjustment (non-clerk totals)	+A2	+1			+A1	+1				
Purchase with Cash Back Adjustment (clerk totals)						+1	-A1 +A2	+1		
Purchase with Cash Back Adjustment (non-clerk totals)						+1				
Mobile Top-Up With Cash	+A1	+1								
Mobile Top-Up With Funds	+A1	+1								

A: BASE24-pos Internal Message

BASE24-pos internal messages consist of a series of core fields—known as the BASE24-pos Standard Internal Message (PSTM)—followed by any additional message tokens that are required for processing.

This appendix describes the fields in the PSTM. Tokens are mentioned briefly; however, in-depth descriptions of the various tokens supported by the BASE24-pos product are contained in the ***BASE24 Tokens Manual***.

Data entered on BASE24-pos Terminal Data file (PTD) screens is stored in different files, depending on the type of terminal being defined. For Hypercom and Standard POS Device Handler terminals, data is stored in the POS Terminal Data Dynamic File—general data (PTDD1) and POS Terminal Data Static File—general data (PTDS1). Dynamic data is stored in the PTDD1 file while static data is stored in the PTDS1 file. For all other types of POS terminals, data from the PTD screens is stored in the POS Terminal Data File (PTDF). Throughout this appendix, field references to data entered or displayed on PTD screens specify the actual field and file from which the data is retrieved instead of the name of the field on the PTD screen.

Note: Generally when file maintenance is performed to the static data in the PTDS1 for Standard POS Device Handler terminals, a warmboot is required for this data to be used in processing. However, a parameter in the Logical Network Configuration File (LCONF) can override this requirement. When the POS-DH-PTD-STATIC-REPL parameter is set to a value of Y and a change to static data in the PTDS1 is made, the latest modified record is used in the current transaction without a warmboot. Refer to the ***BASE24 Logical Network Configuration File Manual*** for additional information on this parameter.

Tokens and the BASE24-pos Internal Message

As transactions are processed and additional information is required that is not carried in the PSTM, BASE24-pos processes add tokens to the internal message. A token is a collection of related data required to perform a specific function. Tokens can be made up of one field or of a series of related fields.

Tokens are only added to the BASE24-pos internal message if they are required to process a specific transaction. Therefore, it is possible for an internal message to have several tokens associated with it—or none at all. When the first token is added to the internal message, the BASE24 system adds two tokens: the first is a Header Token and the second is the specific token needed for processing. Once the Header Token is in place, subsequent tokens can be added to the message individually as required. The general structure of an internal message with message tokens is illustrated below.

BASE24-pos Internal Message with Tokens

PSTM	Header Token	Token	Token	Token	...
------	--------------	-------	-------	-------	-----

All of the tokens that can be carried in BASE24-pos internal messages are described in the ***BASE24 Tokens Manual***.

POS Standard Internal Message (PSTM)

The following pages describe the POS Standard Internal Message (PSTM) used by the BASE24-pos product. The PSTM is 1170 to 1372 bytes in length, depending on the DATA-FLAG field and the length of the USER-DATA field, if appropriate.

Position	Level	Field Name and Description	Data Type
1–4	02	TYP Indicates the type of message described by this PSTM. Valid values are as follows: 0200 = Transaction Request 0210 = Transaction Response 0220 = Transaction Advice (Completion or Force Post) 0402 = Card Issuer Reversal Request (Chargeback) 0412 = Card Issuer Reversal Response 0420 = Reversal 9991 = Log only The Device Handler module, Authorization module, ISO Host Interface process, or Interchange Interface process sets this field. All processes use this field.	PIC 9(4)
5–6	02	PROD-ID Indicates the BASE24 product originating this transaction. The valid value for the BASE24-pos product is 02. The Device Handler module, ISO Host Interface process, or Interchange Interface process sets this field. This field is used by the ISO Host Interface process and the Interchange Interface process.	PIC X(2)
7–8	02	DPC-NUM The DPC number identifying the DPC to which the transaction is to be sent. Valid values are 0 through 9999. For check guarantee and check verification transactions, the Authorization module sets this field using the RTE-TBL.DPC field in the Check Authorization Routing Table File (CART). For all other transactions, the Authorization module sets this	TYPE BINARY 16

Position	Level	Field Name and Description	Data Type
----------	-------	----------------------------	-----------

field according to the RT-TBL.PRI-DPC field in the BASE24-pos segment of the IDF. Different DPCs can be designated for various prefixes and account types. This field is used by the ISO Host Interface process.

9–10	02	RTE-STAT	PIC 9(2)
------	----	----------	----------

Indicates the status of a message at the system level. Valid values are as follows:

- 00 = No error
- 01 = Processing error
- 02 = Security device failure
- 11 = Destination not available
- 12 = Line down
- 20 = Originator not available
- 21 = Unknown message type
- 22 = Unknown card number

If the value is 00, the response code field (TRAN.RESP-CDE) is used to determine the disposition of the message. Nonzero values indicate a failed message.

The ISO Host Interface process sets this field. This field is used by the Authorization module.

11–26	02	ORIG-PRO-NAME	PIC X(16)
-------	----	---------------	-----------

The symbolic process name of the process through which the transaction entered the BASE24 system.

The Device Handler module, ISO Host Interface process, or Interchange Interface process sets this field. This field is used by the Authorization module for return routing.

For transactions that originate from terminals connected to the BASE24-pos system, the symbolic name of the Device Handler module is placed in this field.

For transactions from a host, the symbolic name of the ISO Host Interface process is placed in this field.

For transactions from an interchange, the symbolic name of the Interchange Interface process is placed in this field.

Position	Level	Field Name and Description	Data Type
27–42	02	AST-RTN-PRO-NAME The symbolic name of the primary authorizer for this transaction, as defined in the Authorization Selection Table (AST). The primary authorizer can vary by card type, issuer, and transaction amount. The Router module sets this field.	PIC X(16)
43–58	02	ROUTER1-NAME The symbolic process name of the first Router module to handle this transaction. The Router module sets this field and uses it for routing purposes. For example, if a transaction passes to another logical network, the Router module in the second logical network checks this field to determine where to return the transaction.	PIC X(16)
59–74	02	ROUTER2-NAME The symbolic process name of the second Router module to handle this transaction. The Router module sets this field and uses it for routing purposes. For example, if a transaction passes to a second logical network for authorization, the Router module in the first logical network checks this field to determine where to send the completion.	PIC X(16)
75	02	MSG-ORIG-IND (NOT USED)	PIC X

Position	Level	Field Name and Description	Data Type
76	02	ORIGINATOR Indicates where this transaction originated. This field applies to 0200, 0220, and 0420 transactions. Valid values are as follows: 1 = Device controlled by the BASE24-pos system 2 = Device Handler module 3 = Authorization module 4 = ISO Host Interface process 5 = Host 6 = Interchange Interface process 7 = Interchange 8 = Resubmittal 9 = Device controlled by the BASE24-pos system using the derived unique key per transaction (DUKPT) encryption method and Thales e-Security (Racal) host security modules Values 1, 5, 7, 8, and 9 are valid for 0200 and 0220 messages. Values 1, 2, 4, 5, 6, 7, 8, and 9 are valid for 0420 messages. The Device Handler module, ISO Host Interface process, or Interchange Interface process sets this field.	PIC X
77	02	RESPONDER Indicates where the transaction was authorized. Valid values are as follows: 0 = Default; applicable only until 0210 is created 1 = Device controlled by the BASE24-pos system 2 = Device Handler module 3 = Authorization module 4 = ISO Host Interface process 5 = Host 6 = Interchange Interface process 7 = Interchange S = Secondary authorizer This field is set by the generator of the 0210 response.	PIC X

Position	Level	Field Name and Description	Data Type
78–83	02	POST-DAT The date this transaction will be settled by the BASE24 system. The Device Handler module, ISO Host Interface process, or Interchange Interface process sets this field. The Device Handler module places the value from the POST-DAT field of the POS Terminal Data File (PTDF) or POS Terminal Dynamic Data File—general data (PTDD1) in this field. The Interchange Interface process places the current BASE24 business date in this field. If a value is not sent from the acquiring host, the ISO Host Interface process determines the value to place in this field. This field is used by the Authorization module.	PIC X(6)
84–87	02	TERM-LN The logical network to which the owner of the terminal belongs. The terminal referenced is the terminal from which this transaction originated. This field is set by the Device Handler module, the Interchange Interface process, and ISO Host Interface process. The Device Handler module sets this field using the value from the TERM-OWNER.LN field in the PTDF or PTDD1. The ISO Host Interface process sets this field from element P-60.	PIC X(4)
88–91	02	TERM-FIID The FIID of the institution that owns the terminal from which this transaction originated. The Device Handler module, ISO Host Interface process, or Interchange Interface process sets this field. The Device Handler module retrieves the value for this field from the TERM-OWNER.FIID field in the PTDF or PTDD1. The ISO Host Interface process receives the value for this field	PIC X(4)

Position	Level	Field Name and Description	Data Type
----------	-------	----------------------------	-----------

from the acquiring host. The Interchange Interface process retrieves the value for this field from the PRI-KEY.FIID field in the ICF or ICFE.

92–107	02	TERM-ID	PIC X(16)
--------	----	---------	-----------

The terminal ID of the terminal where the transaction originated.

The Device Handler module, ISO Host Interface process, or Interchange Interface process sets this field. The Device Handler module retrieves the value for this field from the TERM-ID field in the PTDF or PTDD1. The ISO Host Interface process and Interchange Interface process receive the value for this field in the external message.

108–132	02	TERM-NAME-LOC	PIC X(25)
---------	----	---------------	-----------

The name and location of the terminal from which this transaction originated. The name and location are printed on receipts to comply with Regulation E.

The Device Handler module, ISO Host Interface process, or Interchange Interface process sets this field.

The Device Handler module retrieves the value for this field from the TERM-NAM-LOC field in the PTDF or POS Terminal Static Data File—general data (PTDS1). The ISO Host Interface process retrieves the value for this field from the external message. The Interchange Interface process retrieves the value for this field from the interchange message.

133–154	02	TERM-OWNER-NAME	PIC X(22)
---------	----	-----------------	-----------

The name of the financial institution that owns the terminal from which this transaction originated.

The Device Handler module, ISO Host Interface process, or Interchange Interface process sets this field.

The Device Handler module retrieves the value for this field from the TERM-OWNER-NAM field in the PTDF or PTDS1. The ISO Host Interface process retrieves the value for this

Position	Level	Field Name and Description	Data Type
		field from the external message. The Interchange Interface process retrieves the value for this field from the interchange message.	
155–167	02	TERM-CITY	PIC X(13)
		The city in which the terminal is located. The terminal referenced is the terminal from which this transaction originated.	
		The Device Handler module, ISO Host Interface process, or Interchange Interface process sets this field.	
		The Device Handler module retrieves the value for this field from the TERM-CITY-ST.CITY field in the PTDF or PTDS1. The ISO Host Interface process retrieves the value for this field from the external message. The Interchange Interface process retrieves the value for this field from the interchange message.	
168–170	02	TERM-ST	PIC X(3)
		The state or province in which the terminal is located. The terminal referenced is the terminal at which this transaction originated.	
		The Device Handler module, ISO Host Interface process, or Interchange Interface process sets this field.	
		The Device Handler module retrieves the value for this field from the TERM-CITY-ST.ST field in the PTDF or PTDS1. The ISO Host Interface process retrieves the value for this field from the external message. The Interchange Interface process retrieves the value for this field from the interchange message.	
171–172	02	TERM-CNTRY-CDE	PIC X(2)
		The country in which the terminal is located. The terminal referenced is the terminal from which the transaction originated.	

Position	Level	Field Name and Description	Data Type
		The Device Handler module, ISO Host Interface process, or Interchange Interface process sets this field. The Device Handler module retrieves the value for this field from the CNTRY-CDE field in the PTDF or PTDS1. The ISO Host Interface process retrieves the value for this field from the external message. The Interchange Interface process retrieves the value for this field from the interchange message.	
173–174	02	TERM-TYP	PIC X(2)
		The type of terminal from which this transaction originated. This field is set by the Device Handler module, which retrieves the value from the TERM-TYP field in the PTDF or PTDD1.	
175–178	02	BRCH-ID	PIC X(4)
		This field is no longer used.	
179–181	02	ORIG-CRNCY-CDE	PIC 9(3)
		The currency code applicable to this transaction. For inbound requests from a POS device, the value in this field is retrieved from the CRNCY-CDE field in the PTDF or PTDD1. For inbound requests from an interchange, the value in this field is retrieved from the CRNCY-CDE field in the Base segment of the ICF or ICFE. For inbound requests from an acquirer host, the value in this field is retrieved from the external message.	
182	02	AUTH-IND2	PIC X
		The authorization source for this transaction. Valid values are as follows: P = Primary authorizer 1 = First alternate authorizer 2 = Second alternate authorizer 4 = Default authorizer	

Position	Level	Field Name and Description	Data Type
		9 = Default, no authorizer determined yet F = SPROUTE primary authorizer f = SPROUTE first alternate authorizer 0 = None The Router module or the Interchange Interface process sets this field. This field is used by the Router module, the Authorization module, and the Interchange Interface process.	
183–184	02	TERM-TIM-OFST	TYPE BINARY 16
		The time difference between the terminal at which the transaction originated and the BASE24 processor location. It is the signed (+ or –) number of minutes to be added to the BASE24 system time in order to obtain the terminal time. The Device Handler module or the ISO Host Interface process sets this field. The Device Handler module retrieves the value for this field from the TIM-OFST field in the PTDF or PTDD1. The ISO Host Interface process retrieves the value for this field from the external message.	
185–190	02	CLERK-ID	PIC X(6)
		The clerk ID of the clerk who initiated this transaction. The Device Handler module or the ISO Host Interface process sets this field. The Device Handler module retrieves the value for this field from the CLERK-ID field in the PTDF or PTDD1. The ISO Host Interface process retrieves the value for this field from the external message.	
191–202	02	CRT-AUTH	
		The CRT Authorization operator group and user ID. The values in the fields that follow are set by the CRT Authorization Device Handler module, which retrieves the values from Pathway.	

Position	Level	Field Name and Description	Data Type
191–194	04	GRP The Pathway group number of the operator reviewing this transaction using the CRT Authorization Device Handler module.	PIC X(4)
195–202	04	USER-ID The Pathway user ID of the operator reviewing this transaction using the CRT Authorization Device Handler module.	PIC X(8)
203–214	02	SEQ-NUM The sequence number associated with this transaction. If the sequence number is generated by the Device Handler module, the sequence number is increased by one for each transaction within a batch. The value in this field is right-justified and zero filled to the left and is set by the Device Handler module. The value is from the message received from the terminal or from the TRAN-SEQ-NUM field in the PTDF or PTDD1. Note: This field can contain non-numeric data. Therefore, the BASE24-pos system does not validate this field for numeric data.	PIC X(12)
215–226	02	PRE-AUTH-SEQ-NUM The sequence number associated with this preauthorization transaction. The sequence number is increased by one for each transaction within a batch. The value in this field is right-justified and zero filled to the left and is set by the Device Handler module, ISO Host Interface process, or Interchange Interface process. Each of these processes retrieves the value from the message received from the device, host, or interchange.	PIC X(12)

Position	Level	Field Name and Description	Data Type
227–236	02	INVOICE-NUM The invoice number associated with this transaction, if the invoice number is sent from the terminal. An invoice number is commonly preprinted on the charge slip used for this transaction. The use of invoice numbers is optional and determined by retailers. When a retailer does not use invoice numbers, this field contains zeros. The value in this field is left-justified and blank-filled to the right. This field is set by the Device Handler module or ISO Host Interface process. The Device Handler module sets this field from the message received from the device or from the Device Handler module's internal data. The ISO Host Interface process sets this field using information from the external message.	PIC X(10)
237–246	02	ORIG-INVOICE-NUM The original transaction invoice number associated with this transaction, if the invoice number is sent from the terminal. This field is used for merchandise return transactions to indicate the invoice number of the original transaction in which this merchandise was purchased. The value in this field is left-justified and blank-filled to the right. This field is set by the Device Handler module or ISO Host Interface process. The Device Handler module sets this field from the message received from the device or from the Device Handler module's internal data. The ISO Host Interface process sets this field using information from the external message.	PIC X(10)
247–249	02	BATCH-SEQ-NUM The sequence number within the batch for this transaction. The value in this field is not related to the sequence number previously listed. The value in this field is set by the Device Handler module, which retrieves the value from the SEQ-NUM field in the PTDF or PTDD1.	PIC 9(3)

Position	Level	Field Name and Description	Data Type
250–252	02	BATCH-NUM The batch number for this transaction. The value in this field is set by the Device Handler module, which retrieves the value from the message received from the terminal or from the BATCH-NUM field in the PTDF or PTDD1.	PIC 9(3)
253–255	02	SHIFT-NUM The shift number for this transaction. The value in this field is set by the Device Handler module, which retrieves the value from the message sent from the terminal or from the SHIFT-NUM field in the PTDF or PTDD1.	PIC 9(3)
256–266	02	ACQ-INST-ID-NUM The terminal owner's transit and routing number or issuer identification number, if the terminal is directly connected to the BASE24-pos system. If the terminal is not directly connected to the BASE24-pos system, this field identifies the acquiring entity that introduced the transaction into the BASE24-pos system. The value in this field is right-justified and zero filled to the left and is set by the Device Handler module, ISO Host Interface process, or Interchange Interface process. The Device Handler module retrieves the value from the RTTN field in the PTDF or PTDD1. The ISO Host Interface process and the Interchange Interface process retrieve the value from the message received from the host or the interchange.	PIC 9(11)

Position	Level	Field Name and Description	Data Type
267–277	02	RCV-INST-ID-NUM The card issuer's transit and routing number or issuer identification number, if it is known to the BASE24-pos system. If the transaction is an outbound 0200 message bound to an interchange, the authorizing interchange can return information for mapping into this field for the 0210 response. The value in this field is right-justified and zero filled to the left. The Authorization module sets the value in this field for on-us transactions, not-on-us transactions, inbound acquirer 0200 transactions, and inbound interface 0200 transactions. For all other transactions, the value in this field is set by the Interchange Interface process. The Authorization module retrieves the value for this field from the INST-ID-NUM field in the Base segment of the IDF. The Interchange Interface process retrieves the value for this field from the interchange message.	PIC 9(11)
278–296	02	RETL-ID An identifier for the retailer or merchant that owns or operates the terminal at which this transaction originated. The value in this field is set by the Device Handler module, the ISO Host Interface process, or the Interchange Interface process. If the Device Handler module formats this transaction, the value is obtained from the TERM-OWNER. RETAILER-ID field in the PTDF or PTDD1. If the ISO Host Interface process formats this transaction, the value is obtained from the acquirer host. If the Interchange Interface process formats this transaction, this field contains the value obtained from the external message, a default value from the ICF or ICFE, or zeros.	PIC X(19)
297–300	02	RETL-REGN The region associated with the retailer or merchant that owns or operates the terminal at which this transaction originated.	PIC X(4)

Position	Level	Field Name and Description	Data Type
----------	-------	----------------------------	-----------

The value in this field is set by the Device Handler module or the ISO Host Interface process. The Device Handler module retrieves the value from the TERM-OWNER.RETAILER-REGN field in the PTDF or PTDD1. The ISO Host Interface process obtains the value from the acquirer host.

If the Interchange Interface process formats this transaction, this field is zero filled.

301–304	02	RETL-GRP	PIC X(4)
---------	----	----------	----------

The group associated with the retailer or merchant that owns or operates the terminal at which the transaction originated.

The value in this field is set by the Device Handler module or the ISO Host Interface process. The Device Handler module retrieves the value from the TERM-OWNER.RETAILER-GRP field in the PTDF or PTDD1. The ISO Host Interface process obtains the value from the acquirer host.

If the Interchange Interface process formats this transaction, this field is zero filled.

305–308	02	RETL-SIC-CDE	PIC X(4)
---------	----	--------------	----------

The Standard Industrial Classification (SIC) code indicating the retailer's line of business. This field is set by the Device Handler module or the Authorization module.

For mail or telephone order transactions, the Device Handler module sets this field using the MAIL-PO-SIC-CDE field in the PTDF or PTDS1. For all other transactions, the Device Handler module sets this field using the TERM-SIC-CDE field in the PTDF or PTDS1.

The Authorization module only sets this field if it contains zeros or blanks. For mail or telephone order transactions, the Authorization module sets this field using the MAIL-PO-SIC-CDE field in the POS Retailer Definition File (PRDF). For all other transactions, the Authorization module sets this field using the RETAILER-SIC-CDE field in the PRDF.

Position	Level	Field Name and Description	Data Type
309–496	02	TRAN The following fields identify this transaction. The fields include Track 2, card issuer, logical network, transaction type, amounts involved, response codes, preauthorization transaction hold data, a referral phone number, and the primary and alternate authorizers.	
309–314	04	TRAN-CDE The transaction type associated with this transaction. The transaction type is identified by <i>TCTAAC</i> , where <i>TC</i> is the transaction code, <i>T</i> is the card type, <i>AA</i> is the account type, and <i>C</i> is the category.	
309–310	06	TC The transaction code for this transaction. Not all of the transaction codes are valid for all terminals. This field is set by the Device Handler module based on the transaction entered at the terminal. Valid values are as follows: 10 = Normal purchase (includes funds portion of a mobile top-up with funds transaction) 11 = Preauthorization transaction purchase 12 = Preauthorization transaction purchase completion 13 = Mail or telephone order 14 = Merchandise return 15 = Cash advance 16 = Card verification 17 = Balance inquiry 18 = Purchase with cash back 19 = Check verification ¹ 20 = Check guarantee ¹ 21 = Adjustment — purchase 22 = Adjustment — merchandise return 23 = Adjustment — cash advance 24 = Adjustment — cash back amount on purchase with cash back 25 = Card activation 26 = Additional card activation 27 = Replenishment	PIC XX

Position	Level	Field Name and Description	Data Type
----------	-------	----------------------------	-----------

28 = Full redemption

29 = Mobile top-up with cash ²

30 = Mobile top-up with funds ²

31 = Refund mobile top-up with cash (future use) ²

32 = Refund mobile top-up with funds (future use) ²

¹ The BASE24-pos product supports check verification and check guarantee transactions in a pass-through mode only (i.e., they can be sent to a host or interchange for processing), unless the BASE24-pos eCheck add-on product is installed. When the BASE24-pos eCheck add-on product is installed, these transactions can be authorized on the BASE24-pos system.

² The BASE24-pos Standard POS Device Handler module, Router module, Transaction Context Manager process, and Telco Interface process support mobile top-up transactions. However, the BASE24-pos Authorization module processes these transactions as purchases.

311	06	T	PIC X
-----	----	---	-------

The type of card associated with this transaction. Valid values are as follows:

0 = None

1 = Credit

2 = Debit

312–313	06	AA	PIC XX
---------	----	----	--------

The type of account associated with this transaction. Valid values are as follows:

00 = None

01–09 = Checking account

11–19 = Savings account

31–39 = Credit account

96 = Cash benefit account

98 = Food stamp benefit account

Position	Level	Field Name and Description	Data Type
314	06	C	PIC X
The category associated with this transaction. This field is set by the Device Handler module based on the transaction entered at the terminal. Valid values are as follows: 0 = Normal 1 = Sales draft 2 = Representation 3 = Chargeback 4 = Personal check/cash 5 = Personal check/amount of purchase 6 = Personal check/amount of purchase with cash back 7 = Government check 8 = Payroll check 9 = Electronic check A–Z = Reserved for future use The Device Handler module or the Interchange Interface process sets the value in this field. It is used by all application processes.			
315–318	04	ORIG	PIC X(4)
The origin of this transaction. This field is used by the Router module. For transactions that originate at a device connected to the BASE24 system, the Device Handler module sets this field to B24. For transactions that originate at a device connected to a host, the ISO Host Interface process sets this field to HOST. For pass-through transactions that originate at a device connected to an acquirer BASE24 system, the Device Handler module sets this field to ML. For transactions that originate at a device connected to an interchange, the Interchange Interface process sets this field.			

Position	Level	Field Name and Description	Data Type
319–322	04	DEST The authorizer or destination for this transaction. The Device Handler module, the Authorization module, ISO Host Interface process, or Interchange Interface process sets the value in this field. It is used by the Router module.	PIC X(4)
323–326	04	CRD-LN The logical network associated with the card issuer for this transaction. The Authorization module or the Interchange Interface process sets the value in this field. The Authorization module retrieves the value from the LOGICAL-NET param in the LCONF. The Interchange Interface process retrieves the value from the LN field in the ICF or ICFE.	PIC X(4)
327–330	04	CRD-FIID The card owner as determined by the Authorization module from internal tables. The Authorization module uses this field as a key to the IDF. The Authorization module sets this field using the value in the FIID field in the Base segment of the CPF. If a CPF record cannot be found and the transaction is a check authorization or check guarantee transaction, the Authorization module sets this field using the value in the FIID field of the Check Authorization Routing Table File (CART).	PIC X(4)
331–349	04	ACCT The number of the account affected by this transaction. The Device Handler module, the Authorization module, ISO Host Interface process, or Interchange Interface process sets the value in this field. Each of these processes retrieves the appropriate value from the message received from the device, the host, or the interchange.	PIC X(19)

Position	Level	Field Name and Description	Data Type
350–352	04	RESP-CDE The response code as determined by the authorizer for this transaction. A complete list of the BASE24-pos response codes is provided in appendix B of this manual. The Router module, the Authorization module, the ISO Host Interface process, or the Interchange Interface process sets the value in this field.	PIC X(3)
353–354	04	RSN-CDE The reason code listed in the Negative Card File (NEG) or the Cardholder Authorization File (CAF) for the card that originated this transaction. The reason code listed in the RSN-CDE field in the NEG indicates the reason the card is in the NEG. The reason code listed in the RSN-CDE field in the BASE24-pos segment of the CAF indicates the account status. The Authorization module sets the value in this field.	PIC X(2)
355–394	04	TRACK2 The Track data or manually entered card data, as sent by the terminal to the Device Handler module. If the card number was entered manually instead of being read from the magnetic stripe, the first byte contains an M followed by the card number, a field separator, the expiration date (YYMM), and an end sentinel. This is true regardless of whether the magnetic stripe contains Track 1 data, Track 2 data, or both. If the information was read only from Track 2 of the magnetic stripe on the card, the data is the information encoded on Track 2 of the magnetic stripe on the back of the card originating the transaction, excluding start and end sentinel and longitudinal redundancy check (LRC) characters. The content of Track 2 data (between the start sentinel and end sentinel) is specified in the ISO 7813 standard, Identification Cards—Financial Transaction Cards .	PIC X(40)

Position	Level	Field Name and Description	Data Type
----------	-------	----------------------------	-----------

1. Primary account number (PAN), left justified (up to 19 digits)
2. Field separator (=)
3. Country code (if present; 3 digits)
4. Expiration date (YYMM)
5. Service code (if present; 3 digits)
6. Discretionary data (balance of available space)

If the information was read only from Track 1 of the magnetic stripe on the card, this field is constructed in the same way as it is for manually entered card data, and the Track 1 data is moved to the Track 1 token.

If information was read from both Track 1 and Track 2 of the magnetic stripe on the card, the Track 2 data is entered in this field in the standard Track 2 format, as described above, and the Track 1 data is moved to the Track 1 token.

If full track data is not available (e.g., in an advice or reversal message) or if discretionary data suppression is implemented, this field is constructed from either the suppressed track or other information in the incoming message. The first byte contains an M, followed by the card number, a field separator, the expiration date (YYMM), and an end sentinel. If the expiration date is not available, it is replaced with the value 0000.

The Device Handler module, ISO Host Interface process, or Interchange Interface process sets the value in this field. It is used by the Authorization module.

395–397	04	MBR-NUM	PIC 9(3)
---------	----	---------	----------

The member number for the card which initiated this transaction. When multiple cards are issued with the same card number, the member number distinguishes the cards.

The value in this field is right-justified and zero filled. It is set and used by the Authorization module.

Position	Level	Field Name and Description	Data Type
398–401	04	EXP-DAT The expiration date (YYMM) of the card which initiated this transaction. The card does not expire until the month listed has passed. The Authorization module sets the value in this field from Track 2 or Track 1 data. It is used by the Authorization module.	PIC X(4)
402	04	USER-FLD2	PIC X
403–410	04	AMT-1 The primary amount involved in this transaction. For purchase with cash back transactions, this field contains the total amount (the purchase amount plus the cash amount). For balance inquiry transactions, this field contains balance amount. For adjustment transactions, this field contains the original completed amount. For preauthorization transactions, this field contains the requested amount to be held. For preauthorization completion transactions, this field contains the completed amount. The Device Handler module, ISO Host Interface process, or Interchange Interface process sets the value in this field. These processes retrieve the value from the message received from the device, host, or interchange. This field is used by all the application processes.	TYPE BINARY 64
411–418	04	AMT-2 The secondary amount involved with this transaction. For purchase with cash back transactions, this field contains the cash back amount.	TYPE BINARY 64

Position	Level	Field Name and Description	Data Type
		For adjustment transactions, this field contains the new amount.	
		For chargeback transactions, this field contains the replacement amount.	
		The Device Handler module, ISO Host Interface process, or Interchange Interface process sets the value in this field. These processes retrieve the value from the message received from the device, host, or interchange. This field is used by all the application processes.	
419–448	04	MULT-CRNCY	
		The following group of fields is not used. It is redefined as a user-field.	
419–421	06	AUTH-CRNCY-CDE (NOT USED)	PIC 9(3)
422–424	06	SETL-CRNCY-CDE (NOT USED)	PIC 9(3)
425–432	06	AUTH-CONV-RATE (NOT USED)	PIC 9(8)
433–440	06	SETL-CONV-RATE (NOT USED)	PIC 9(8)
441–448	06	CONV-DAT-TIM (NOT USED)	TYPE BINARY 64
419–448	04	USER-FLD3	PIC X(30) REDEFINES MULT-CRNCY
449–450	04	PRE-AUTH-HLD	TYPE BINARY 16
		For preauthorization transactions, this field contains the time limit for holding the preauthorized funds.	
		This field consists of two subfields. The first subfield contains one digit and determines the hold increment. Valid values for the first subfield are listed below.	
		0 = Minutes	
		1 = Hours	
		2 = Days	

Position	Level	Field Name and Description	Data Type
----------	-------	----------------------------	-----------

The second subfield contains two digits and determines the actual preauthorization hold units (that is, the number of minutes, hours, or days, as determined by the first field). Valid values for the second subfield are 00 through 99.

The Device Handler module, ISO Host Interface process, or Interchange Interface process sets the value in this field. The Device Handler module retrieves the value from the PRE-AUTH-HLD field in the PTDF or PTDS1. The ISO Host Interface process retrieves the value from the message received from the host. The Interchange Interface process retrieves the value from the message received from the interchange or from the PRE-AUTH-HLD field in the ICF or ICFE. This field is used by the Authorization module.

451	04	APPRV-CDE-LGTH	PIC X
-----	----	----------------	-------

The maximum length of the approval code that can be handled by the device at which this transaction originated.

Approval codes must be at least 2, but not more than 6, positions in length.

The Device Handler module, ISO Host Interface process, or Interchange Interface process sets the value in this field. The Authorization module can modify the value in this field.

The Device Handler module retrieves the value for this field from the APPRV-CDE-LGTH field in the PTDF or PTDS1. The ISO Host Interface process retrieves the value for this field from the POS message. The Interchange Interface process retrieves the value for this field from the message received from the interchange or from the APPRV-CDE-LGTH field in the BASE24-pos segment of the ICF or ICFE. The Authorization module can reduce the value in this field to match the value in the APPRV-CDE-LGTH field in the BASE24-pos segment of the CPF.

This field is used by the Device Handler module, Interchange Interface process, and ISO Host Interface process.

Position	Level	Field Name and Description	Data Type
452–459	04	APPRV-CDE The approval code for this transaction. If this transaction was sent to an interchange that returns approval codes to be printed on the receipt, the returned approval code is placed in this field. If this transaction was approved by the Authorization module or a host, the value in this field is generated by the Authorization module or the host. If the BASE24-pos product generates the code and the transaction was a Visa transaction approved by checking the National Negative Files, an X is inserted into the last byte of this field. Otherwise, when the BASE24-pos product generates the code, a B is inserted into the last byte of this field. The value in this field is left-justified and blank-filled to the right. The Authorization module, the ISO Host Interface process, or the Interchange Interface process sets the value in this field. The Authorization module generates the value it places in this field. The ISO Host Interface process and the Interchange Interface process retrieve the value for this field from the message received from the host or the interchange. This field is used by the Device Handler module, the ISO Host Interface process, and the Interchange Interface process.	PIC X(8)
460–467	04	ICHG-RESP The interchange response values, if an interchange was involved in the processing of this transaction. Since different interchanges respond with different values (e.g., response code, reply code, reason code), all of these values are included in this field if returned by the interchange. The values are put into this field in the same order as they exist in the interchange message. This field is left-justified and space-filled to the right. The Interchange Interface process sets the value in this field using the message received from the interchange. This field is used by the ISO Host Interface process.	PIC X(8)

Position	Level	Field Name and Description	Data Type
468–471	04	PSEUDO-TERM-ID The pseudo ID associated with the terminal from which this transaction originated. The pseudo ID allows terminals to be identified in a way different from the value in TERM-ID field in the PSTM. The interchange sets the value in this field for messages coming from the interchange. The Interchange Interface process sets the value in this field for messages going to the interchange. This field is used by the ISO Host Interface process for informational purposes only.	PIC X(4)
472	04	DFT-CAPTURE-FLG A value indicating whether this transaction involves draft capture and how this transaction is settled. Draft capture is settling a transaction with an electronic message instead of paper. Valid values are as follows: 0 = Authorize only and use a paper receipt to settle the transaction later. 1 = Authorize and capture; transaction includes the information needed for immediate settlement. 2 = Authorize only and expect electronic follow-up to settle the transaction later. 3 = Electronic follow-up to settle a previously authorized transaction. 9 = Authorize only; settlement method could not be determined. (The card type was unavailable because the algorithm check was not performed and the SPROUTE File entry used for routing information did not contain a card type. The Router module requires a card type before it can obtain the draft capture flag value from the SRV.TRAN-PROFILE field in the PTDF, PTDD1, or PRDF.) The Router module or the ISO Host Interface process sets the value in this field. It is used by the Authorization module.	PIC 9

Position	Level	Field Name and Description	Data Type
473	04	SETL-FLG Specifies whether the Router module should modify the POST-DAT field, which causes the Device Handler module to cut over the terminal. The Device Handler module sets this field using the TERM-CUTOVER-TC field in the PTDF or PTDD1.	PIC 9
474–493	04	RFRL-PHONE The referral phone number to be used if applicable for this transaction. The Interchange Interface process sets the value in this field using the message received from the interchange. The ISO Host Interface process sets this field using the message received from the host. If no referral phone number is provided in the external message, the ISO Host Interface process sets this field using the RFRL-PHONE field in the BASE24-pos segment of the Host Configuration File (HCF).	PIC X(20)
494–495	04	SAVE-ACCT-TYP The account type value to be saved for this transaction. The ranges of valid values are as follows: 01–09 = Checking account types 11–19 = Savings account types 31–39 = Credit account types The value in this field is set by the Authorization module from the ACCT.TYP field in the CAF.	PIC X(2)
496	04	OVRRDE-FLG A flag that identifies an override performed at a CRT Authorization terminal. This flag also identifies whether the original transaction was performed at a POS device or a CRT Authorization terminal. The CRT Authorization Device Handler module sets this flag whenever a transaction originates at a CRT Authorization terminal or a transaction originates at a POS device and its	PIC X

Position	Level	Field Name and Description	Data Type
----------	-------	----------------------------	-----------

response is overridden at a CRT Authorization terminal. Both of these situations require special handling by the Router module.

When a transaction involving a debit card originates or has its response overridden at a CRT Authorization terminal, the Router module checks the DEBIT-SUPPORT-FLAG field in the PRDF record to determine whether the retailer allows a debit card transaction to be processed at a CRT Authorization terminal.

Also, when a transaction originates at a CRT Authorization terminal, the Router module uses information in the RETAILER.POSTAL-CDE, RETAILER.CITY, RETAILER.CNTRY, RETAILER.ADDR, and RETAILER.NAME fields in the PRDF record to complete the POSTAL-CDE, TERM-CITY, TERM-CNTRY-CDE, TERM-NAM-LOC, and TERM-OWNER-NAME terminal information fields in the PSTM. If the RETAILER.CITY field in the PRDF contains 14–18 characters, characters 14–18 are placed in the CITY-EXT field of the PS50 token.

The Router module performs the debit card check and completes the terminal information fields in the PSTM based on the value in this field. Valid values are as follows:

- 0 = Normal transaction; CRT Authorization not involved (default).
- 1 = CRT Authorization Override (0220) of a transaction originating at a POS device; the Router module checks debit cards.
- 3 = Normal transaction originating at a CRT Authorization terminal; the Router module checks debit cards and uses terminal information from the PRDF.

The value in this field is used by CRT Authorization and the Router/Authorization module.

Position	Level	Field Name and Description	Data Type
497–562	02	RTE The following fields contain information used for transaction routing.	
497–498	04	SRV A code identifying the type of service associated with this transaction. Service (card type) codes used in this field are either reserved by BASE24 products or are user-defined. These codes are defined in section 1.	PIC X(2)
499	04	RETL-PROG The retailer programs in which the retailer associated with this transaction participates. Valid values are as follows: 0 = No program A = MasterCard B = MasterCard and Visa N = Non-dial R = Regular T = Visa Z = All programs The Router module sets the value in this field using the RETAILER-PROG field in the PRDF. This field is used by the Router module.	PIC X
500–501	04	ISS A user-defined indicator identifying the issuer associated with this transaction. Values indicating on-us cards must fall within a certain range, as do those indicating not-on-us cards. The ranges of values are as follows: 00–29 = On-us (has an associated IDF record) 30–99 = Not-on-us (has no associated IDF record)	PIC X(2)

Position	Level	Field Name and Description	Data Type
		The Router module sets the value in this field using the ISS field in the CPF. If the Router module cannot find a CPF record for the card, the Router module sets this field to 99. This field is used by the Router module.	
502	04	OVER-LMT Indicates whether the amount associated with this transaction is over or under the floor limit for this service. Valid values are as follows: 0 = Under 1 = Over The Router module sets the value in this field. This field is used by the Router module.	PIC X
503–506	04	ONL-LMT The online limit allowed for this type of transaction, as determined by the merchant associated with this transaction. The Router module sets the value in this field using the AST. This field is used by the Router module.	TYPE BINARY 32
507–510	04	OFFL-LMT The offline limit allowed for this type of transaction, as determined by the merchant associated with this transaction. The Router module sets the value in this field using the AST. This field is used by the Router module.	TYPE BINARY 32
511–526	04	PRI The primary authorizer for this transaction. The Router module sets this field from the AST, SPROUTE, or the AUTH-DEST field in the Acquirer Routing token. This is the authorization entity to which this transaction will be sent.	PIC X(16)

Position	Level	Field Name and Description	Data Type
527–542	04	ALT1 The first alternate authorizer for this transaction. The Router module sets this field from the AST or SPROUTE. This field is used by the Router module to determine where to send the transaction if the primary authorizer is not available.	PIC X(16)
543–558	04	ALT2 The second alternate authorizer for this transaction. The Router module sets this field from the AST. This field is used by the Router module to determine where to send the transaction if the primary authorizer and the first alternate authorizer are not available.	PIC X(16)
559	04	AUTH-IND Indicates the authorizer which authorized this transaction. Valid values are as follows: P = Primary 1 = Alternate 1 2 = Alternate 2 4 = Default authorized 9 = No authorizers F = SPROUTE primary f = SPROUTE alternate 1 When authorizers are determined with the assistance of SPROUTE, a second alternate authorizer is not determined. The Router module or the Interchange Interface process sets the value in this field. It is used by the Router module, the Authorization module, or the Interchange Interface process.	PIC X

Position	Level	Field Name and Description	Data Type
560	04	DFLT The default action to take if no authorization can be performed. This action is determined from the AST. Valid values are as follows: 0 = Decline 1 = Authorize blind 2 = Issue call for authorization The Router module sets the value in this field. It is used by the Authorization module.	PIC X
561	04	HIT-NNEG Indicates whether the merchant wants to check the National Negative Files (NNFs) before sending this transaction to the interchange for authorization. The NNFs include the American Express National Negative File (ANEG), Base National Negative File (BNEG), Base National Range File (BNR), INAS National Negative File (INEG), and INAS National Range File (INR). Valid values are as follows: 0 = False; do not access NNFs before sending transaction. 1 = True; do access NNFs before sending transaction, and decline if on file. The Router module or the Authorization module sets the value in this field using the SRV.HIT-NNEG field in the PRDF.	PIC X
562	04	AST-FND A flag that indicates whether the Router module found an AST record for this transaction. Valid values are as follows: 0 = AST search was not performed 1 = AST was found 2 = AST was not found	PIC X

Position	Level	Field Name and Description	Data Type
563	02	COMPL-REQ A flag indicating whether the host associated with this transaction requires a completion message from the Authorization module. Valid values are as follows: 0 = No completion required. The Authorization module sets this value using either the IPCF or IDF. 1 = Completion required. The Authorization module sets this value using either the IPCF or IDF. 2 = Host is offline; completion required if authorization level is 3 (online/offline). The ISO Host Interface process sets this value. 4 = Completion required for all transactions. The Authorization module sets this value using the IPCF. 5 = Completion required for every approved transaction authorized by BASE24. ¹ 6 = Completion required for every approved or declined transaction authorized by BASE24. ¹ ¹ Authorizing a transaction is making the decision to approve or decline it. The Authorization module uses the value in this field.	PIC X
564	02	ADJ-FLG If this transaction is an adjustment transaction, this field indicates whether the terminal allows adjustment transactions when AMT-2 is greater than AMT-1. Valid values are as follows: 0 = Adjustment is not allowed. 1 = Adjustment is allowed. The Device Handler module sets the value in this field using either the ALLOWED-TRANS(24) field in the PTDF or the ADJ-FLG field in the PTDS1. The ISO Host Interface process sets this field using the ADJ-FLG field in the BASE24-pos segment of the HCF. This field is used by the Router module. The Authorization module checks the ADJ-FLG field in the BASE24-pos segment of the IDF to determine whether the institution allows adjustment transactions when AMT2 is greater than AMT1.	PIC X

Position	Level	Field Name and Description	Data Type
----------	-------	----------------------------	-----------

564	02	PRE-AUTH-OPT	PIC X REDEFINES ADJ-FLG
-----	----	--------------	----------------------------

Device Handlers and interchange interfaces that support authorization for a lesser amount or a greater amount set this field when transaction requests come in to BASE24-pos. This field is also set and used by the HISO Interface. To set this field to P, the Device Handler or interface must have been modified to support authorization for a lesser amount. To set this field to G, the Device Handler or interface must have been modified to support authorization for a greater amount.

This field indicates whether the transaction authorizer supports the authorization of the transaction for a lesser or a greater amount than the amount requested. If authorization for a lesser amount is allowed, a transaction request for an amount in excess of transaction limits or available credit is processed for the maximum amount that can be approved. If authorization for a greater amount is allowed, a transaction request can be approved for any amount up to a limit defined at the card prefix level, subject to transaction limits applicable to the card and available credit in the account. If authorization for a lesser amount or greater amount is not allowed, a transaction request for an amount in excess of transaction limits or available credit is declined. Valid values are as follows:

- | | | |
|---|---|--|
| P | = | Transaction can be authorized for a lesser amount than was requested. |
| G | = | Transaction can be authorized for a greater amount than was requested. Valid for Pre-auth purchase, purchase and mail-phone order transactions. |
| D | = | Host-acquired request where BASE24 will determine whether transaction can be authorized for a lesser or a greater amount than was requested. Valid for Pre-auth purchase, purchase, mail-phone order, purchase with cash back and cash advance transactions. |

Position	Level	Field Name and Description	Data Type
----------	-------	----------------------------	-----------

		Any other value = Transaction cannot be authorized for a lesser amount than was requested. Valid for Pre-auth purchase, purchase, mail-phone order, purchase with cash back and cash advance transactions.	
--	--	--	--

This field is used by BASE24-pos Router and Authorization modules, Device Handler modules, and interchange interface processes.

565–566	02	ADMIN-SEC-IND	TYPE BINARY 16
---------	----	---------------	----------------

Indicates when the administrative security should be checked. The Device Handler module must return the value in this field when a second transaction is required, such as for merchandise returns. The Authorization module does or does not validate the administrative card again depending on the value in this field. Valid values are as follows:

0 = Not validated

1 = Validated

The Device Handler module sets the value in this field. It is used by the Authorization module.

567–572	02	TRAN-DAT	PIC X(6)
---------	----	----------	----------

The date this transaction began. The date is adjusted for time offset, if applicable.

The Device Handler module, ISO Host Interface process, or Interchange Interface process sets the value in this field.

This field is used by the Authorization module.

573–580	02	TRAN-TIM	PIC X(8)
---------	----	----------	----------

The time this transaction entered the BASE24-pos system from a device, interchange or acquirer host.

Position	Level	Field Name and Description	Data Type
		This field contains the actual device time as calculated by the Device Handler module, provided by the interchange, or provided by an acquiring host. The Device Handler module determines the device time by using the system time and adding the time offset. The Device Handler module, ISO Host Interface process, or Interchange Interface process sets the value in this field. This field is used by the Authorization module.	
581–586	02	ACQ-ICHG-SETL-DAT	PIC X(6)
		The date this transaction will be settled by the interchange, if an interchange originated the transaction. If an interchange did not originate this transaction, this field is zero filled. The Interchange Interface process sets the value in this field using the POST-DAT field in the ICF or ICFE. This field is used by the Interchange Interface process.	
587–592	02	ISS-ICHG-SETL-DAT	PIC X(6)
		The date this transaction will be settled by the interchange, if an interchange authorized this transaction. If an interchange did not authorize this transaction, this field is zero filled. The Interchange Interface process sets the value in this field using the POST-DAT field in the ICF or ICFE. This field is used by the Interchange Interface process.	
593–594	02	RVSL-CDE	PIC 9(2)
		A reason code explaining why this transaction is a reversal or an adjustment. Valid values for reversals are as follows: 01 = Time-out 02 = Command reject 03 = Destination not available 08 = Customer canceled 10 = Hardware error 19 = System error	

Position	Level	Field Name and Description	Data Type
----------	-------	----------------------------	-----------

		20 = Suspect reversal	
		21 = Message Authentication Code (MAC) failure	
		22 = KMAC (Key MAC) synchronization error	
		23 = Message replay error	
		24 = Invalid MAC	
		40 = Split transaction routing enabled; secondary service not approved	

Valid values for adjustments are as follows:

	00 = Reason unknown
	12 = Original amount incorrect
	14 = Suspicious reversal override
	15 = Misdispense reversal override
	16 = Duplicate transaction
	17 = Reconciliation error
	18 = Reserved
	19 = System error

The Device Handler module, ISO Host Interface process, or Interchange Interface process sets the value in this field. This field is used by the Interchange Interface process.

595–596	02	PT-SRV-COND-CDE	PIC 9(2)
---------	----	-----------------	----------

Specifies the origin of this transaction. This code indicates a special condition which exists at the time a transaction is initiated. Valid values are as follows:

00	= Normal presentment
01	= Customer not present
02	= Unattended terminal able to retain card
03	= Merchant suspicious
04	= Electronic cash register interface
05	= Customer present but card not present
06	= Preauthorization transaction requests
07	= Telephone device request
08	= Mail or telephone order
09	= Security alert
10	= Customer identity verified
11	= Suspected fraud
12	= Security reasons
13	= Representment of item

Position	Level	Field Name and Description	Data Type
----------	-------	----------------------------	-----------

14	=	Public utility terminal
15	=	Customer terminal (home terminal)
16	=	Administration terminal
17	=	Returned item (chargeback)
18	=	No check in envelope/all returned
19	=	Deposit out-of-balance/all returned
20	=	Payment out-of-balance/all returned
21	=	Manual reversal
22	=	Terminal error/counted
23	=	Terminal error/not counted
24	=	Deposit out-of-balance/applied contents
25	=	Payment out-of-balance/applied contents
26	=	Withdrawal had error/reversed
27	=	Unattended terminal unable to retain card
28–40	=	Reserved for ISO use
41–50	=	Reserved for national use
51	=	Address verification
52–99	=	Reserved for private use

The Device Handler module, ISO Host Interface process, and Interchange Interface process set the value in this field using the message received from the device, host, or interchange. This field is used by the Device Handler module, ISO Host Interface process, and Interchange Interface process.

597–599	02	PT-SRV-ENTRY-MDE	PIC 9(3)
---------	----	------------------	----------

A code indicating the manner in which this transaction entered the BASE24-pos system.

Valid values for the first and second digits are listed below.

00	=	Unspecified
01	=	Manually
02	=	Magnetic stripe
03	=	Bar code
04	=	OCR
05	=	Integrated circuit card
06	=	Reserved for ISO use
07	=	Integrated circuit card contactless
08–60	=	Reserved for ISO use
61–80	=	Reserved for national use

Position	Level	Field Name and Description	Data Type
----------	-------	----------------------------	-----------

81–90 = Reserved for private use
91 = Magnetic stripe contactless
92–99 = Reserved for private use

Valid values for the third digit are listed below.

0 = Unspecified
1 = PIN entry capability
2 = No PIN entry capability
3–5 = Reserved for ISO use
6–7 = Reserved for national use
8–9 = Reserved for private use

The Interchange Interface process and Device Handler module set the value in this field using the message received from the interchange or device. This field is used by the Device Handler module and the Interchange Interface process.

600–615	02	PIN	PIC X(16)
---------	----	-----	-----------

The PIN (personal identification number) that was entered at the device during this transaction. If the PIN has already been verified, this field contains blanks.

The Device Handler module, ISO Host Interface process, or Interchange Interface process sets the value in this field. All of these processes retrieve the value from the message received from the device, host, or interchange. This field is used by the Authorization module, the ISO Host Interface process, and the Interchange Interface process.

616–617	02	PIN-SIZE	PIC 9(2)
---------	----	----------	----------

The length of the PIN entered during this transaction. This field is right justified and zero filled to the left.

The Device Handler module, ISO Host Interface process, or Interchange Interface process sets the value in this field. All of these processes retrieve the value from the message received from the device, host, or interchange.

This field is used by the Authorization module, the ISO Host Interface process, and the Interchange Interface process.

Position	Level	Field Name and Description	Data Type
618	02	PIN-TRIES The number of PIN attempts made at the terminal when this transaction was being entered. This field is initialized by the Device Handler module and increased by the Authorization module on each unsuccessful PIN entry. Once a PIN has been verified, the Authorization module sets this field to Z. The ISO Host Interface process sets this field to Z if the PIN field an incoming request contains all pad characters. For any other value in the PIN field, the ISO Host Interface process sets this field to zeros.	PIC 9
619–634	02	PIN-KEY An encrypted form of the key used to protect the PIN (e.g., E.MFK/1[KPE]). This field is used only when the PIN-FRMT field in the PSTM is set to 1, 2, or 3. The value in this field is used by the Device Handler module, the Authorization module, the Interchange Interface process, and the ISO Host Interface process.	PIC X(16)
635–650	02	USER-KEY The encrypted form of the previous key used by an interchange to encrypt PINs. The Interchange Interface process sets the value in this field using the message sent by the interchange. This field is used by the Authorization module.	PIC X(16)
651	02	PIN-FRMT Identifies the encrypted form used to protect the PIN. Valid values are as follows: 0 = Clear PIN 1 = Encrypted ANSI (PIN/PAN) PIN block 3 = Encrypted PIN/PAD PIN block	PIC 9

Position	Level	Field Name and Description	Data Type
		The Interchange Interface process or ISO Host Interface process sets the value in this field using the PIN-BLK field in the KEYF. The Device Handler module sets the value in this field based on the ENCR-PIN.PIN-ENCRYPT-TYP field in the PTDF or PTDD1. This field is used by the Device Handler module, the Authorization module, the ISO Host Interface process, and the Interchange Interface process.	
652	02	PINPAD-CHAR	PIC X
		The hexadecimal digit used to fill the PIN block when the PIN/PAD PIN block is used to encrypt the PIN (PIN-FRMT field in the PSTM is set to 3). The hexadecimal digit can be A through F. The Interchange Interface process or Device Handler module sets the value in this field. This field is used by the Device Handler module, the Authorization module, the ISO Host Interface process, and the Interchange Interface process.	
653–654	02	ANSI-OFST	TYPE BINARY 16
		The starting position of the twelve PAN digits when the ANSI (PIN/PAN) PIN block is used to encrypt the PIN (PIN-FRMT field in the PSTM is set to 1). The first digit in the PAN has a positional value of 0. The Interchange Interface process or Device Handler module sets the value in this field. This field is used by the Device Handler module, the Authorization module, the ISO Host Interface process, and the Interchange Interface process.	
655–660	02	HOST-TRACE-NUM	PIC 9(6)
		The transaction trace number generated by the ISO Host Interface process and the Interchange Interface process. This field is used by the ISO Host Interface process.	

Position	Level	Field Name and Description	Data Type
661–668	02	ENTRY-TIM The timestamp indicating when this transaction entered the BASE24-pos system for the first time, whether it entered the system through the Device Handler module, the ISO Host Interface process, or the Interchange Interface process. The value in this field is expressed in Greenwich mean time (GMT). The Device Handler module, ISO Host Interface process, or Interchange Interface process sets the value in this field using the JULIANTIMESTAMP system procedure.	TYPE BINARY 64
669–676	02	EXIT-TIM The timestamp indicating when this transaction left the BASE24-pos system for authorization, either through the ISO Host Interface process or the Interchange Interface process. The value in this field is expressed in Greenwich mean time (GMT). The ISO Host Interface process or the Interchange Interface process sets the value in this field using the JULIANTIMESTAMP system procedure.	TYPE BINARY 64
677–684	02	RE-ENTRY-TIM The timestamp indicating when the ISO Host Interface process or the Interchange Interface process received the response to its original request from the authorizing entity. The value in this field is expressed in Greenwich mean time (GMT). The ISO Host Interface process or the Interchange Interface process sets the value in this field using the JULIANTIMESTAMP system procedure.	TYPE BINARY 64
685–695	02	FRWD-INST-ID-NUM The value of the LCONF param POS-FORWARD-INST-ID. This value equates to a routing transit number representing the BASE24-pos system.	PIC 9(11)

Position	Level	Field Name and Description	Data Type
		This field is set by the Authorization module and is used for logging purposes only. If data is present, the value is right-justified and zero filled to the left.	
696–706	02	CRD-ACCPT-ID-NUM	PIC 9(11)
		A code identifying the card acceptor for this transaction, if it originated at an acquirer host. This field is used only in cases where the acquirer is the actual card acceptor. If the acquirer is not the actual card acceptor for this transaction, this field is left blank. The value in this field is right-justified and zero filled and is retrieved from the POS message. This field is transmitted for logging purposes only.	
707–717	02	CRD-ISS-ID-NUM	PIC 9(11)
		A code identifying the actual card issuer in the response from an authorizing host for this transaction, if the receiving institution is not the actual card acceptor. Otherwise, this field is left blank. The value in this field is right-justified and zero filled and is retrieved from the POS message. This field is transmitted for logging purposes only.	
718–749	02	ORIG-DATA	
		The following fields contain the original data elements associated with reversal transactions. Because the data is dependent on the originator of the transaction, these fields are not always filled. This information is used by the ISO Host Interface process, the Interchange Interface process, and CRT Administration Device Handler module.	
718–721	04	MSG-TYP	PIC X(4)
		The original message type.	

Position	Level	Field Name and Description	Data Type
722–729	04	TRN-TIM The original transaction time.	PIC X(8)
730–733	04	TRN-DAT The date of the original transaction.	PIC X(4)
734–745	04	TRAN-SEQ-NUM The sequence number of the original transaction.	PIC X(12)
746–749	04	B24-POST-DAT The BASE24-pos posting date of the original transaction.	PIC X(4)
750–751	02	REA-FOR-CHRGBCK The reason for the chargeback, if the transaction is a chargeback. Valid values are as follows: 03 = Invalid merchant 12 = Invalid transaction 18 = Customer dispute 59 = Suspected fraud 63 = Security violation 64 = Original transaction amount incomplete 68 = Supporting documentation received too late 93 = Transaction in violation of the law 94 = Duplicate transaction 96 = System malfunction The value in this field is set by the CRT Administration Device Handler module. This field is used for chargeback transactions by the ISO Host Interface process, the Interchange Interface process, and the CRT Administration Device Handler module.	PIC X(2)
752	02	NUM-OF-CHRGBCK The number of times this transaction has been charged back, if the transaction is a chargeback.	PIC 9

Position	Level	Field Name and Description	Data Type
		The value in this field is set by the CRT Administration Device Handler module. This field is used for chargeback transactions by the ISO Host Interface process, the Interchange Interface process, and the CRT Administration Device Handler module.	
753–754	02	CHECK-ID-TYP (NOT USED)	PIC X(2)
753–754	02	ADDR-TYP REDEFINES CHECK-ID-TYP	PIC X(2)
		A code indicating whether address verification is to be performed by the BASE24 system. If this field contains a value of 98, address verification is performed by the BASE24 system. Any other value indicates address verification is not performed by the BASE24 system.	
		The field is set by the Merchant Host Interface module, the ISO Host Interface process, the VisaNet Interchange Interface process, the Banknet Interchange Interface process, or a BASE24 Interchange Interface process. This field is used by the Authorization module.	
755–778	02	CHECK-ID-NUM (NOT USED)	PIC X(24)
755–778	02	ADDR-FLDS REDEFINES CHECK-ID-NUM	
		The following fields support the BASE24-pos address verification function.	
755–774	04	ADDR The cardholder's billing address.	PIC X(20)

Position	Level	Field Name and Description	Data Type
775	04	ADDR-VRFY-STAT A code identifying the result of comparing address verification information received in the transaction and address verification information contained in the processor's database. Valid values are as follows: A = Address—Addresses matched, but ZIP codes did not match. E ¹ = Error—The transaction was not eligible for address verification or an editing error occurred while attempting to process the message. N = No—Addresses did not match and ZIP codes did not match. R ¹ = Retry—Primary and secondary authorizers were unavailable or declined the transaction and address verification was not performed by the BASE24-pos system. S ¹ = Service Not Supported—the BASE24-pos product authorized the transaction, but did not have the add-on Address Verification module. U ¹ = Unavailable—Address information was not available to the processor performing address verification. W = Whole ZIP—Nine-digit ZIP codes matched, but addresses did not match. X = Exact—Addresses and nine-digit ZIP codes matched. Y = Yes—Addresses and five-digit ZIP codes matched. Z = ZIP—Five-digit ZIP codes matched, but addresses did not match. ␣ ¹ = Address verification information was not included in this transaction (␣ denotes a blank character).	PIC X

Position	Level	Field Name and Description	Data Type
----------	-------	----------------------------	-----------

0 ¹	=	Address verification information was included in this transaction, but was not verified. A transaction to be verified by a host or interchange carries this code. A transaction to be verified on the BASE24-pos system, but declined before address verification could be performed also carries this code.	
----------------	---	--	--

¹ Identifies a reason that a comparison was not made. Each interchange specifies the value it uses to identify this reason. Interchange Interface processes substitute the interchange-specific value before sending the message from the BASE24-pos system to the interchange. Refer to the ***BASE24-pos Address Verification Manual*** for a listing of interchange-specific values. Interchange Interface processes do not change a value received from the interchange.

The following values are also valid but are used only by Visa. They are not set by the BASE24-pos address verification function.

- | | | | |
|---|---|--|--|
| B | = | Street address match. Postal code is not verified because of incompatible formats. (Acquirer sends both street address and postal code.) | |
| C | = | Street address and postal code is not verified because of incompatible formats. (Acquirer sends both street address and postal code.) | |
| D | = | Street addresses and postal codes match. | |
| G | = | Global non-participant in the Address Verification Service. Address is not verified for international transactions. | |
| I | = | Address information is not verified for international transactions. | |

Position	Level	Field Name and Description	Data Type
		M = Street address and postal code match.	
		P = Postal code match. Street is not verified because of incompatible formats. (Acquirer sends both street address and postal code.)	
776–778	04	USER-FLD1	PIC X(3)
779–780	02	STATE-CDE (NOT USED)	PIC X(2)
781–786	02	BIRTH-DAT (NOT USED)	PIC X(6)
787–795	02	ZIP-CDE This field is used only for address verification and contains the ZIP code for the cardholder's billing address. The ZIP code must be five or nine digits long. If the ZIP code is five digits long, it is left-justified and blank-filled.	PIC X(9)
796	02	USER-FLD9	PIC X
797–806	02	POSTAL-CDE The postal ZIP code associated with the terminal at which this transaction originated. The Device Handler module or the ISO Host Interface process sets the value in this field. The Device Handler module retrieves the value from the POSTAL-CDE field in the PTDF or PTDS1. The ISO Host Interface process retrieves the value from the message received from the host.	PIC X(10)

Position	Level	Field Name and Description	Data Type
807	02	USER-FLD10	PIC X
808	02	DATA-FLAG Indicates whether the USER-DATA field is appended to this message. Valid values are as follows: 0 = No, the USER-DATA field is not appended to this message. 1 = Yes, the USER-DATA field is appended to this message. The value in this field is retrieved from the POS message. This field is used by the Router module, the Authorization module, the ISO Host Interface process, the Interchange Interface process, and the CRT Administration Device Handler module.	PIC X
809–810	02	NUM-SERVICES The number of services allowed at the terminal from which this transaction originated. The Device Handler module, ISO Host Interface process, or Interchange Interface process sets the value in this field.	TYPE BINARY 16
811–1170	02	SRVCS Allowed service information for the terminal at which this transaction originated. Each occurrence is 12 positions in length. The Device Handler module, ISO Host Interface process, or Interchange Interface process sets the value in this field. The Device Handler module retrieves the values from the SRV fields in the PTDF or PTDD1. The ISO Host Interface process retrieves the value from the ALLOWED-SRVCS field in the BASE24-pos segment of the HCF. The Interchange Interface process retrieves the value from the ALLOWED-SRVCS field in the BASE24-pos segment of the ICF or ICFE. This field is used by the Router module.	OCCURS 30 TIMES

Position	Level	Field Name and Description	Data Type
04	TYP		PIC X(2)
		The type of service allowed at the terminal at which this transaction originated. Service (card type) codes used in this field are either reserved by BASE24 products or are user-defined. These codes are defined in section 1.	
04	TRAN-PROFILE		PIC X
		The transaction profile. This profile indicates whether the transaction is a draft capture transaction or a non-draft capture transaction.	
		The Device Handler module, ISO Host Interface process, or Interchange Interface process sets the value in this field. The Device Handler module retrieves the value from the SRV. TRAN-PROFILE field in the PTDF or PTDD1, or from the message received from the device. The ISO Host Interface process sets this field to 0. The Interchange Interface process retrieves the value from the message received from the interchange. This field is used by the Router module to determine the value placed in the DFT-CAPTURE-FLG field in the PSTM.	
04	ELEC-FOLLOWUP		PIC X
		A flag that indicates whether the BASE24 system should post the authorized 0220 message.	
		The Device Handler module, ISO Host Interface process, or Interchange Interface process sets the value in this field. The Device Handler module retrieves the value from the SRV. ELEC-FOLLOWUP field in the PTDF or PTDD1. The ISO Host Interface process sets this field to 0. The Interchange Interface process retrieves the value from the message received from the interchange. This field is used by the Router module. Valid values are as follows:	
		0 = Post the transaction (providing that it is a 0220 message which has been authorized).	
		1 = Do not post the transaction; the terminal is sending the electronic follow-up.	
		2–9 = Reserved for future use.	

Position	Level	Field Name and Description	Data Type
04		FLR-LMT The floor limit associated with the type of service. The Device Handler module sets the value in this field using the SRV.NP-FLR-LMT, SRV.CA-FLR-LMT, or SRV.MO-FLR-LMT field in the PTDF or PTDD1. This field is used by the Router module.	TYPE BINARY 32
04		TRAN-LMT The transaction limit associated with the type of service. The Device Handler module sets the value in this field using the SRV.TRAN-LMT field in the PTDF or PTDD1. This field is used by the Router module.	TYPE BINARY 32

BASE24-pos USER-DATA

The following fields are appended to the PSTM if the DATA-FLAG field in the PSTM is set to 1. If used, these fields immediately follow the last occurrence of the SRVCS fields in the PSTM and immediately precede any tokens in the message.

Position	Level	Field Name and Description	Data Type
02		USER-DATA	
		User-defined information that the BASE24-pos product can carry in its internal message but does not recognize and does not use for processing.	
		This user data is available for all messages, but is appended to the message only if the DATA-FLAG field is set to 1.	
04		LEN	TYPE BINARY 16
		The length of the variable information in the USER-DATA. INFO field.	
04		INFO	PIC X(200)
		User-defined information that the BASE24 system can carry in its internal message.	

ACI Worldwide, Inc.

B: BASE24-pos Authorization Response Codes

This appendix contains descriptions of the authorization response codes supported by the BASE24-pos product. These codes are used by transaction authorizers to indicate how they want transactions to be handled. When a transaction is authorized, the authorizer returns a response code to inform the party requesting the authorization of the actions to take. The response code is carried in the RESP-CDE field of the POS Standard Internal Message (PSTM).

Response codes specific to the BASE24-pos EMV add-on product are documented in the ***BASE24-pos EMV Support Manual***.

Note: When the BASE24-pos Multiple Currency add-on product is installed, any transaction rejected due to currency conversion errors is reported using one of the standard response codes (e.g., 101 — issue a call).

The following abbreviations have been used throughout this appendix for the specified file names and authorization methods:

File Names

ADMN	=	Administrative Card File
ANEG	=	American Express National Negative Card File
AST	=	Authorization Selection Table
BNEG	=	Base1 (Visa) National Negative Card File
CAF	=	Cardholder Authorization File
CAPF	=	Card Authorization Parameters File
CPF	=	Card Prefix File
IDF	=	Institution Definition File
INEG	=	INAS (MasterCard) National Negative Card File
KEY1	=	Key Authorization File
NEG	=	Negative Card File
NNEG	=	National Negative Card File
PBF	=	Positive Balance File
PRDF	=	POS Retailer Definition File

PTD = BASE24-pos Terminal Data files
PTLF = POS Transaction Log File
UAF = Usage Accumulation File

Authorization Methods

NEG-only = Negative Authorization without Usage Accumulation method
NEG/UAF = Negative Authorization with Usage Accumulation method
CAF-only = Positive Authorization method
CAF/PBF = Positive Balance Authorization method
CAPF = Parametric Authorization method¹

¹ The BASE24-pos product performs the same checks for the Parametric Authorization method as for the CAF/PBF method, and then performs additional checks against information in the CAPF. Therefore, any response code that is generated under CAF/PBF can also be generated under Parametric Authorization.

Authorization Response Codes

Authorization response codes are three-character codes that indicate whether a transaction authorizer approved or declined a transaction and, if declined, why the transaction was declined. Authorization response codes are carried in the RESP-CDE field of the PSTM.

Each response code the BASE24-pos product supports is described below.

Code	Description
000	APPROVED WITH BALANCES The transaction was a balance inquiry transaction and was approved by the BASE24 system using the CAF/PBF method. Balances were checked in the PBF.
001	APPROVED WITHOUT BALANCES The transaction was approved without checking the PBF. • The transaction was approved by the BASE24 system using the NEG-only, NEG/UAF, or CAF-only method, or was a transaction other than a balance inquiry transaction approved under the CAF/PBF method. • The transaction was approved by an ISO interchange with or without balances.
002	APPROVED WITH COUNTRY CLUB STATUS The transaction was approved because the card used in the transaction is defined to the BASE24 system with a “country club” status, which means that transactions should be approved. • For the CAF-only and CAF/PBF authorization methods, the CARD STATUS field on CAF screen 1, CARD STATUS field on CAF screen 7, or REASON CODE field on CAF screen 10 is set to E for the card. • For the NEG-only and NEG/UAF authorization methods, the REASON ON FILE field on NEG screen 1 is set to 7 for the card.

Code	Description
003	APPROVED WITH MAYBE STATUS (ID REQUIRED) The transaction was approved because the card used in the transaction is defined to the BASE24 system with a “maybe” status. This allows the retailer to request further identification from the consumer. • For the CAF-only and CAF/PBF authorization methods, the CARD STATUS field on CAF screen 1, CARD STATUS field on CAF screen 7, or REASON CODE field on CAF screen 10 is set to B for the card. • For the NEG-only and NEG/UAF authorization methods, the REASON ON FILE field on NEG screen 1 is set to 4 for the card.
004	APPROVED WITH PENDING IDENTIFICATION (SIGNED PAPER DRAFT REQUIRED) The transaction was approved under one of the following conditions: • The card used in the transaction is defined to the BASE24 system with a “signature required” status. This allows the retailer to request a signature to support the transaction. For the CAF-only and CAF/PBF authorization methods, the CARD STATUS field on CAF screen 1, CARD STATUS field on CAF screen 7, or REASON CODE field on CAF screen 10 is set to D for the card. For the NEG-only and NEG/UAF authorization methods, the REASON ON FILE field on NEG screen 1 is set to 6 for the card. • The transaction was entered at an attended terminal (PT-SRV-COND-CDE field in the PSTM does not contain the value 02 or 27) that did not support PIN entry (PT-SRV-ENTRY-MDE field contains the value 2 in the second byte), and the PIN PROCESSING FLAG field on CPF screen 7 is set to 2. These values indicate the transaction should be approved with ID.

Code	Description
005	APPROVED BLIND The transaction was approved blind because the selected default action on AST screens 2, 4, or 6 indicates that the transaction should be authorized blind if the BASE24 system cannot authorize it. The default action is used when the primary and the alternate authorizers are not available or DFLT is specified as the authorizer. Approved blind means the transaction was approved without the use of balance files or with preestablished authorization criteria.
006	APPROVED WITH VIP STATUS The transaction was approved because the credit card used in the transaction is defined to the BASE24 system with a “VIP” status, which means that the consumer is considered to be a privileged customer who is not subject to normal credit checks. • For the CAF-only and CAF/PBF authorization methods, the CARD STATUS field on CAF screen 1 or CAF screen 7 is set to 5 for the card. • For the NEG-only and NEG/UAF authorization methods, the REASON ON FILE field on NEG screen 1 is set to 10 for the card. In addition, the card can have been defined as a “VIP” card in the INEG, BNEG, or ANEG.
007	APPROVED ADMINISTRATIVE TRANSACTION The administrative transaction was approved.
008	APPROVED AFTER CHECKING THE NATIONAL NEGATIVE FILE The transaction was approved after checking the National Negative Files available in the system. This response code is normally used by credit card processors when the host system is unavailable.

Code	Description
009	APPROVED WITH COMMERCIAL STATUS The transaction was approved because the card used in the transaction is defined to the BASE24 system as being used by a preapproved commercial customer. • For the CAF-only and CAF/PBF authorization methods, the CARD STATUS field on CAF screen 1, CARD STATUS field on CAF screen 7, or REASON CODE field on CAF screen 10 is set to G for the card. • For the NEG-only and NEG/UAF authorization methods, the REASON ON FILE field on NEG screen 1 is set to 9 for the card.
010	APPROVED WITH PARTIAL BALANCE The transaction was approved with a partial balance because it was approved by an issuer system and the authorized amount is the same as the requested amount. This is an internal code that is never set by the BASE24-pos Router/Authorization process.
050	DECLINED The transaction was denied for one of the following reasons: • For the NEG-only and NEG/UAF authorization methods, the REASON ON FILE field on NEG screen 1 is set to 5 or 11 for the card used in the transaction. The value of 5 indicates that any transaction initiated with this card is to be declined. This value does not indicate why all transactions for this card are to be declined. The value of 11 indicates that the account is closed. Therefore, all transactions initiated with this card are declined. • For the CAF-only and CAF/PBF authorization methods, the CARD STATUS field on CAF screen 1, CARD STATUS field on CAF screen 7, or REASON CODE field on CAF screen 10 is set to C for the card used in the transaction. The value C indicates that any transaction initiated with this card is to be declined. This value does not indicate why all transactions for this card are to be declined. Therefore, all transactions initiated with this card are declined. • Under any authorization method, no authorization could be performed and the RTE.DFLT field in the PSTM was set to 0, indicating the transaction is to be denied when no authorizer is available. <i>continued on the following page</i>

Code	Description
050	DECLINED <i>continued</i> Under the CAPF authorization method, the transaction was not a credit transaction (that is, the TRAN.TRAN-CDE.T field in the PSTM did not contain the value 1 or the TRAN.TRAN-CDE.AA field in the PSTM did not contain a value in the range of 31 to 39).
051	EXPIRED CARD The transaction was denied because the card that initiated the transaction has expired. The card was returned.
052	PIN TRIES EXCEEDED The transaction was denied because the cardholder has exceeded the maximum number of PIN tries allowed for the current usage accumulation period. The card was returned.
053	SHARING IS NOT ALLOWED The transaction was denied because it was an administrative transaction and the card that initiated the transaction and the terminal owner are not affiliated with the same FIID. The FIID field on PTD screen 1 for the terminal and the FIID field on CPF screen 1 for the card prefix used in the transaction do not match.
054	NO SECURITY BOX The transaction was denied because security module verification was required with the transaction, and the security module was not available.
055	INVALID TRANSACTION The transaction has been declined because the request was not a valid BASE24-pos transaction.
056	NO SUPPORT The transaction was denied because the institution does not support this transaction as defined in the IDF.

Code	Description
057	LOST OR STOLEN CARD STATUS The transaction was denied because the card used in the transaction is defined to the BASE24 system as lost or stolen. • For the CAF-only and CAF/PBF authorization methods, the CARD STATUS field on CAF screen 1 or CAF screen 7 is set to 2 for the card. • For the NEG-only and NEG/UAF authorization methods, the REASON ON FILE field on NEG screen 1 is set to 1 or 2 for the card.
058	INVALID STATUS The transaction was denied because the status of the card that initiated the transaction or the phone card was invalid. This response code is used as a default response code if none of the other response codes apply.
059	RESTRICTED STATUS The transaction was denied because the card used in the transaction is defined to the BASE24 system with a restricted status, which means that the specific account linked to the card does not allow for withdrawals. The CARD STATUS field on CAF screen 1 or CAF screen 7 is set to a value of 4 for the card or the ACCOUNT STATUS field on PBF screen 1 is set to a value of 2 or 4 for the account.
060	ACCOUNT NOT ON THE CAF The transaction was denied because the authorizer was unable to identify a CAF account to process based on card number. This is an authorizer database configuration problem to be resolved by the authorizer.
061	PBF RECORD NOT FOUND The transaction was denied because a PBF record could not be found. The specific account record may have been available in the PBF, but a read error or an edit error has occurred, which has prevented recovery of the specific account record.
062	PBF UPDATE ERROR The transaction was denied because of a write error that occurred which prevented updating of the specific account record in the PBF.

Code	Description
063	INVALID AUTHORIZATION TYPE The transaction was denied because the selected AUTH TYPE field on IDF screen 16 did not contain a value of 0 through 4 or a value of 6. The authorizer should correct the configuration error.
064	BAD TRACK DATA The transaction was denied because the Track 1 or Track 2 data could not be processed. This response code is used when the Track 1 or Track 2 data is not present or when a Track 1 or Track 2 field contains invalid data. This response code is also returned when invalid or missing card verification data is detected during the request message processing. Refer to the CRD-VRFY-FLG field in the BASE24-pos Release 5.0 token (token ID 04) in the BASE24 Tokens Manual for additional information about card verification results.
065	ADJUSTMENT IS NOT ALLOWED The adjustment transaction was denied because the institution does not allow adjustment transactions.
066	INVALID CREDIT CARD ADVANCE INCREMENT The transaction was denied because the credit card advance amount does not meet increment requirements. This response code is not currently used.
067	INVALID TRANSACTION DATE The transaction was denied because the posting date in the PSTM was not equal to the current or the next posting dates.
068	PTLF ERROR The transaction was denied because an unrecoverable error occurred while writing to the PTLF. This response code is not currently used.

Code	Description
069	BAD MESSAGE EDIT The transaction was denied because invalid data was identified in the PSTM. For check guarantee/verification transactions, the transaction was denied because the check ID type was not sent in the message from the device, and the default check ID in the BASE24-pos Terminal Data files was blank.
070	NO IDF The transaction was denied because the BASE24-pos product was unable to find an IDF record for this transaction or a fatal error has previously occurred for this IDF record. This is the result of a configuration error.
071	INVALID ROUTING TO AUTHORIZATION The transaction was denied because the AUTH-IND2 field in the PSTM was not equal to P, 1, or 2. This response code is not currently used.
072	CARD FOUND ON THE NATIONAL NEGATIVE FILE The transaction was denied because the card that initiated the transaction was found on a National Negative File.
073	INVALID ROUTING TO AUTHORIZATION SERVICE FIELD The transaction was denied because the system attempted to access the National Negative Files and the RTE.SRV field in the PSTM is not set to Mb, Vb, MM, VV, MD, VD, or AX, where b is a blank character.
074	UNABLE TO AUTHORIZE The transaction was denied. This is a general response when the other denial response codes do not apply. The Transaction Context Manager process uses this response code if it denies a transaction for any reason. For the BASE24-pos Europay, MasterCard, and Visa (EMV) add-on, this response code is generated when the CPF EMV segment does not exist or the CPF EMV Issuer Application Data Format flag contains an invalid value.

Code	Description
075	INVALID PAN LENGTH The transaction was denied because the PAN length does not match the expected length.
076	INSUFFICIENT FUNDS The transaction was denied because the PBF balances are less than the requested transaction amount.
077	PREAUTHORIZATION LIMIT REACHED The transaction was denied because the cardholder already has the maximum number of preauthorization transactions outstanding (specified in the MAXIMUM NUMBER OF PRE-AUTH HOLDS fields on CPF screen 7). Additional preauthorization transactions will continue to be denied until at least one of the preauthorization transactions expires or is removed, or until the maximum number of preauthorization holds is increased. With secure Internet validation processing, this response code is returned when a transaction is denied because the cardholder already has the maximum number of entries in the E-Commerce Authentication File (EAF).
078	DUPLICATE TRANSACTION The transaction was denied because it was a duplicate transaction. Another identical transaction was previously received and, thus, this transaction is dropped. This response code is used by the system to prevent processing of transactions that have been forwarded twice. Normally this response code only occurs during a communications failure.
079	MAXIMUM ONLINE REFUND LIMIT REACHED The transaction was denied because the cardholder has reached the refund limit for the current usage accumulation period.
080	MAXIMUM OFFLINE REFUND LIMIT REACHED The transaction was denied because the cardholder has reached the refund limit for the current usage accumulation period.

Code	Description
081	MAXIMUM CREDIT PER REFUND REACHED The transaction was denied because the transaction amount was greater than the per refund credit maximum.
082	MAXIMUM NUMBER OF TIMES THE CARD IS ALLOWED TO BE USED HAS BEEN REACHED The transaction was denied because the cardholder has reached the maximum number of times the card can be used during the current usage accumulation period.
083	MAXIMUM REFUND CREDIT LIMIT EXCEEDED The transaction was denied because the cardholder has reached the maximum refund credit limit allowed for refunds during the current usage accumulation period.
084	CUSTOMER SELECTED NEG REASON The transaction was denied because the cardholder has requested that the use of the card be disallowed.
085	INQUIRY NOT ALLOWED The transaction was denied because the transaction was a balance inquiry request and balances are not maintained by the BASE24 system.
086	OVER THE FLOOR LIMIT The transaction was denied because the transaction amount exceeds the offline limits and the default action in the AST is to be taken. The default action in the AST is used when the primary and alternate authorizers are not available. The particular default action used depends upon whether the transaction amount is over or under the limits set in the AST.
087	MAXIMUM NUMBER OF REFUND CREDITS REACHED The transaction was denied because the cardholder has reached the maximum number of refund credits allowed during the current usage accumulation period.

Code	Description
088	PLACE CALL FOR REFERRAL The transaction was denied by the BASE24-pos system, but the authorizer wants the retailer to call an authorization center. This response code is not currently used.
089	CARD STATUS INACTIVE OR CLOSED The transaction was denied because the card status was inactive or closed due to a request by the customer or the authorizer.
090	REFERRAL FILE (PRF) FULL The transaction was denied. The transaction typically would be referred, but the transaction cannot be written to the Referral File because it is full.
091	NEGATIVE CARD FILE PROBLEM The transaction was denied because an error occurred while reading the NEG.
092	CASH ADVANCE LESS THAN SPECIFIED MINIMUM The transaction was denied because the transaction amount identified for a cash advance was less than the minimum amount specified in the CPF.
093	DELINQUENT The transaction was denied because the cardholder's account was delinquent as checked in the CAPF.
094	OVER LIMIT TABLE The transaction was denied with the Parametric Authorization method because the transaction amount exceeds the amount of funds available.
095	AMOUNT OVER MAXIMUM The transaction was denied with the Parametric Authorization method because the transaction amount exceeds the limits in the CAPF.
096	PIN REQUIRED The transaction was denied because a PIN was required for this transaction.

Code	Description
097	MOD 10 CHECK The transaction was denied because of a MOD 10 check digit calculation error.
098	FORCE POST The transaction was denied because it could not be routed to the correct authorization entity. This code is also used as an Exception Reason Code for a declined advice/force post that could not be forwarded to the correct host due to a routing problem.
099	BAD PBF The transaction was denied because the account status was inactive or closed.
100	UNABLE TO PROCESS THE TRANSACTION The transaction was referred and the merchant is to use individual discretion as to whether to authorize the transaction. This is the result of exhausting all attempts to authorize.
101	ISSUE A CALL The transaction was referred and recommends that the merchant issue a call. This is a general purpose response code, covering several circumstances. For example, this response code can be generated as the result of the transaction amount not exceeding the offline limits and the default action in the AST being set to indicate a call should be issued for authorization. The default action in the AST is used when the primary and the alternate authorizers are not available or DFLT is specified as the authorizer.
102	CALL The transaction was referred because the REASON CODE field on CAF screen 10 is set to A or the REASON ON FILE field on NEG screen 1 is set to 3 for the card used in the transaction. The values of 3 and A indicate the transaction is to be referred.
103	NEGATIVE CARD FILE PROBLEM The transaction was referred due to an error that occurred when reading the NEG.

Code	Description
104	CAF FILE PROBLEM The transaction was referred because of an error that occurred when attempting to retrieve a CAF record.
105	CARD NOT SUPPORTED The transaction was referred because the card was not supported. This occurs when the value in the RTE.SRV field in the PSTM does not match defined card types.
106	AMOUNT IS OVER THE MAXIMUM The transaction was referred because the transaction amount exceeds CAPF, CAF, or other limits.
107	DAILY USAGE LIMIT REACHED The transaction was referred because the usage limit for the current usage accumulation period has been reached by the cardholder.
108	CAPF NOT FOUND The transaction was referred due to a problem when reading the CAPF.
109	CASH ADVANCE LESS THAN SPECIFIED MINIMUM The transaction was referred due to the transaction amount for a cash advance being less than the minimum amount specified in the CAPF.
110	NUMBER OF TIMES USED LIMIT EXCEEDED The transaction was referred because the cardholder has exceeded the allowable number of times used for the current usage accumulation period.
111	DELINQUENT The transaction was referred because the account failed the CAPF delinquency check.
112	OVER LIMIT TABLE The transaction was referred because the transaction amount was greater than the CAPF credit limit.

Code	Description
113	TIMEOUT The transaction was denied because it timed out before a response code could be received. The BASE24-pos Authorization module does not generate this response code.
115	PTLF FULL The transaction was referred because the PTLF is full.
120	UAF FILE PROBLEM The transaction was referred because of a UAF problem. This response code is generated only when the NEG/UAF authorization method is used.
121	ADMIN FILE PROBLEM The transaction was referred because of a problem with the ADMN.
122	UNABLE TO VALIDATE PIN The transaction was denied because the BASE24-pos system was not able to validate the PIN entered with the transaction and a PIN was required. The BASE24-pos Authorization module does not generate this response code.
130	REQUEST CRYPTOGRAM FAILURE (REFERRAL) The transaction was referred because the authorization request cryptogram included in the transaction data did not match the authorization request cryptogram generated by BASE24, and the action configured was to refer the transaction.
131	CVR REFERRAL The transaction was referred because it failed the card verification results (CVR) test based on preset rules defined for a card type, and the action configured was to refer the transaction.
132	TVR REFERRAL The transaction was referred because it failed the terminal verification results (TVR) test based on preset rules defined for a card type, and the action configured was to refer the transaction.

Code	Description
133	REASON ONLINE CODE REFERRAL The transaction was referred because the transaction failed a reason online code check based on preset rules defined for a card type, and the action configured was to refer the transaction.
134	FALLBACK REFERRAL The transaction was referred because the terminal could not read an integrated chip card, and generated a “fallback” transaction. This can occur as a result of a faulty chip reader or a faulty chip, but can also be an indication of a fraudulent card. The action configured was to refer the transaction.
150	MERCHANT IS NOT ON FILE The transaction was denied because the retailer at which the transaction was initiated did not have a record in the PRDF.
200	INVALID ACCOUNT The transaction was denied due to a transaction error. This response code is not currently used by the BASE24-pos Authorization module, however, some Interchange Interface processes may map external response codes to this response code.
201	INCORRECT PIN The transaction was denied due to a transaction error. The PIN was found to be incorrect.
202	CASH ADVANCE LESS THAN THE MINIMUM REQUIRED The transaction was denied due to a transaction error. The transaction cash advance amount is less than the minimum cash advance amount specified in the CPF.
203	TRANSACTION REQUIRES ADMINISTRATIVE CARD The transaction was denied due to a transaction error. The transaction initiated requires an administrative card.

Code	Description
204	ENTER LESSER AMOUNT The transaction was denied due to a transaction error. The transaction amount exceeds the customer's maximum current usage accumulation period limit. This indicates that another transaction entered by the cardholder may be approved, if the transaction amount is reduced. This code also is used when the transaction amount exceeds the PRDF ceiling limits.
205	INVALID ADVANCE AMOUNT The transaction was denied due to a transaction error. The transaction cash advance amount is not in an increment that complies with the cash advance increment restrictions set in the CPF or the mobile top-up amount is not valid.
206	CAF NOT FOUND The transaction was denied due to a transaction error. The authorizer could not access a CAF.
207	INVALID TRANSACTION DATE The transaction was denied due to a transaction error. The PSTM transaction date has invalid data.
208	INVALID EXPIRATION DATE The transaction was denied due to a transaction error. The authorizer identified an expiration date that is not a valid date. This typically occurs on key-entered transactions.
209	INVALID TRANSACTION CODE The transaction was denied due to an invalid transaction code. The transaction code is not one that the institution can accept. The BASE24-pos Authorization module does not generate this response code.
251	PURCHASE WITH CASH BACK The transaction was denied due to insufficient funds for the cash back amount.

Code	Description
252	DEBIT CARD PROCESSING NOT SUPPORTED THROUGH CRT AUTHORIZATION The transaction was denied because debit card processing is not supported through CRT Authorization.
400	REQUEST CRYPTOGRAM FAILURE (DECLINE) The transaction was denied because the authorization request cryptogram included in the transaction data did not match the authorization request cryptogram generated by BASE24, and the action configured was to decline the transaction.
401	SECURITY MODULE PARAMETER ERROR The transaction was denied because the security module returned an error message when used to verify a transaction.
402	SECURITY MODULE FAILURE The transaction was denied because the security module did not respond when used to verify a transaction.
403	KEYI RECORD NOT FOUND The transaction was denied because a valid KEYI record for this card prefix could not be located.
404	ATC CHECK FAILURE The transaction was denied because it failed the application transaction counter (ATC) test. One ATC is maintained by the microchip on the card and a corresponding value is held in the CAF. A comparison of these values is designed to detect fraudulent cards. This response code is used for contactless magnetic stripe transactions as well.
405	CVR DECLINE The transaction was denied because it failed the card verification results (CVR) test based on preset rules defined for a card type, and the action configured was to decline the transaction.
406	TVR DECLINE The transaction was denied because it failed the terminal verification results (TVR) test based on preset rules defined for a card type.

Code	Description
407	REASON ONLINE CODE DECLINE The transaction was denied because the transaction failed a reason online code check. The Reason Online Code Table (ROCTBL) defines the reason online code checks and actions that are to be tested during authorization of EMV transactions and the actions to be taken for a specific reason code.
408	FALLBACK DECLINE The transaction was denied because the terminal could not read an integrated chip card and generated a “fallback” transaction. This can occur as a result of a faulty chip reader or a faulty chip, but can also be an indication of a fraudulent card. The action configured was to decline the transaction.
900	PIN TRIES EXCEEDED The transaction was denied and the card is to be captured. This is the result of the cardholder entering his or her PIN incorrectly more than the maximum number of times during the current usage accumulation period.
901	EXPIRED CARD The transaction was denied and the card is to be captured. This is the result of the card no longer being valid because the card has expired.
902	NEG CAPTURE CARD The transaction was denied and the card is to be captured. This results when the transaction was denied and the CAPTURE CODE field on NEG screen 1 is set to a value other than 0.
903	CAF STATUS 3 The transaction was denied and the card is to be captured. This results when the CARD STATUS field on CAF screen 1 or CAF screen 7 is set to 3, indicating that the card is to be captured.
904	CASH ADVANCE LESS THAN SPECIFIED MINIMUM The transaction was denied and the card is to be captured. This results when the transaction cash advance amount is less than the specified minimum cash advance amount.

Code	Description
905	NUMBER OF TIMES USED LIMIT EXCEEDED The transaction was denied and the card is to be captured. This results when the card has been used more times during the current usage accumulation than is allowed. Someone other than the owner may be using this card.
906	DELINQUENT The transaction was denied and the card is to be captured. The transaction account failed CAPF delinquency checks because the cardholder has failed to make payments.
907	OVER LIMIT TABLE The transaction was denied and the card is to be captured. The authorizer, using the Parametric Authorization method, has determined that the cardholder has overextended his or her credit line.
908	AMOUNT IS OVER THE MAXIMUM The transaction was denied and the card is to be captured. This occurs when the transaction amount exceeds the cardholder's allowed limit and fraud is suspected.
909	CAPTURE The transaction was denied and the card is to be captured. This response code is used when an error occurs during card verification and the appropriate BAD CV ACTION field on CPF screen 2 is set to 2 (retain card).
910	REQUEST CRYPTOGRAM FAILURE (CAPTURE) The transaction was denied and the card is to be captured. This response code is used when the authorization request cryptogram included in the transaction data did not match the authorization request cryptogram generated by BASE24 and the action configured was to capture the transaction.
911	CVR CAPTURE The transaction was denied and the card is to be captured. This response code is used when the transaction failed the card verification results (CVR) test based on preset rules defined for a card type and the action configured was to decline the transaction.

Code	Description
912	TVR CAPTURE The transaction was denied and the card is to be captured. This response code is used when the transaction failed the terminal verification results (TVR) test based on preset rules defined for a card type and the action configured was to decline the transaction.
950	ADMINISTRATIVE CARD NOT FOUND The transaction was denied because the administrative card that initiated this transaction has no corresponding record in the ADMN.
951	ADMINISTRATIVE CARD NOT ALLOWED The transaction was denied because the administrative card that initiated this transaction was not allowed to perform this transaction.
952	APPROVED ADMINISTRATIVE REQUEST IN THE WINDOW The transaction was approved and the transaction was received within the cutover window.
953	APPROVED ADMINISTRATIVE REQUEST OUTSIDE THE WINDOW The transaction was approved and the transaction was received outside the cutover window.
954	APPROVED ADMINISTRATIVE REQUEST ANYTIME The transaction was approved. This response code is not currently used.
955	CHARGEBACK APPROVED The chargeback transaction was approved and the file has been updated.
956	CHARGEBACK APPROVED The chargeback transaction was approved, the file has been updated, but the acquirer could not be found.
957	CHARGEBACK ACCEPTED—BAD PREFIX The chargeback transaction was accepted, but the transaction contained an incorrect prefix number.

Code	Description
958	CHARGEBACK BAD RESPONSE OR PREFIX The chargeback transaction has been denied, since the transaction contained an incorrect response code or the CPF is missing a prefix or has a bad prefix.
959	CHARGEBACK NOT SUPPORTED The transaction was denied because chargeback transactions are not supported. This response code is not currently used.
960	CHARGEBACK APPROVED The chargeback transaction was approved, but the files have not been updated.
961	CHARGEBACK APPROVED The chargeback transaction was approved, but the files have not been updated and the acquirer has not been found.
962	CHARGEBACK ACCEPTED The chargeback transaction was accepted, but an incorrect authorization destination was included in the transaction.

ACI Worldwide, Inc.

C: BASE24-pos Parametric Authorization Method

The Parametric Authorization method is one of the authorization methods by which the BASE24-pos product can process transactions. This method uses the Cardholder Authorization File (CAF), the Positive Balance File (PBF), and the Card Authorization Parameters File (CAPF).

The Parametric Authorization method is similar to the Positive Balance Authorization method and differs only in that the PBF is used in relation to the CAPF. This appendix explains the processing that the Authorization module performs using the PBF and the CAPF. It also provides a working example of this processing.

PBF/CAPF Processing Overview

The Authorization module first verifies transactions by using the card status and account activity during the current usage accumulation period as maintained in the CAF. If the card status and account activity are valid according to the CAF, the Authorization module then compares the delinquency history as maintained in the PBF with the delinquency table in the CAPF. Next, the Authorization module computes the amount of available credit and uses the CAPF overlimit table in an attempt to authorize the transaction if sufficient credit is not available. Finally, the Authorization module checks the single transaction limits in the CAPF before approving the transaction.

The Parametric Authorization method is used to authorize transactions if all of the following conditions are true:

- The Parametric Authorization method is specified as the method under which the transaction must be authorized.
- The account accessed in this transaction is a credit account.
- The authorization level being used is offline, online/offline and the transaction must be screened, or online/offline and the authorizer is AUTHF.

The card status and the account activity are first verified just as they would be using the Positive Balance Authorization method. Once these steps are complete and the transaction has met all checks, the Authorization module proceeds with authorization steps specific to the Parametric Authorization method as described below.

Comparing Delinquency History Information

The Authorization module compares delinquency information in the CAPF to delinquency information in the PBF. The information compared includes the number of days this account has been delinquent, the number of months this account has been active, the number of times this account has been delinquent one cycle, the number of times this account has been delinquent two cycles, and the number of times this account has been delinquent three cycles.

The corresponding fields in which this information is contained are listed below for both the PBF and the CAPF.

PBF (Screen 8)	CAPF (Delinquent Table)
DAYS DELINQUENT	DAYS
MONTHS ACTIVE	ACTIVE
# OF TIMES 1 CYCLE DELINQUENT	CYCLE 1
# OF TIMES 2 CYCLES DELINQUENT	CYCLE 2
# OF TIMES 3 CYCLES DELINQUENT	CYCLE 3

The fields in the CAPF occur in up to three sets, or tables: TABLE A, TABLE B, and TABLE C. The fields in TABLE B are used only if the transaction does not meet the authorization requirements of TABLE A. The fields in TABLE C, subsequently, are used only if the transaction does not meet the authorization requirements of TABLE B. If the Parametric Authorization method is being used, only the fields in TABLE A must be configured. The fields in TABLE B and TABLE C are optional.

The transaction meets the authorization requirements of TABLE A, TABLE B, or TABLE C if the values in the PBF and CAPF fields meet the following conditions:

- The value in the MONTHS ACTIVE field on PBF screen 8 is greater than or equal to the value in the ACTIVE field of the CAPF.
- The value in the DAYS DELINQUENT field on PBF screen 8 is less than or equal to the value in the DAYS field of the CAPF.
- The value in the # OF TIMES 1 CYCLE DELINQUENT field on PBF screen 8 is less than or equal to the value in the CYCLE 1 field of the CAPF.
- The value in the # OF TIMES 2 CYCLES DELINQUENT field on PBF screen 8 is less than or equal to the value in the CYCLE 2 field of the CAPF.
- The value in the # OF TIMES 3 CYCLES DELINQUENT field on PBF screen 8 is less than or equal to the value in the CYCLE 3 field of the CAPF.

If the account for this transaction fails to meet the DAYS DELINQUENT or the MONTHS ACTIVE requirements in a CAPF table (TABLE A, TABLE B, or TABLE C), the transaction is denied or referred according to the value in the RESPONSE field for that table. The remaining tables are not checked.

If the account for this transaction fails to meet any of the other requirements in a CAPF table, the account is checked against the next available table. If no more tables are available, the transaction is denied or referred according to the value in the RESPONSE field for the current table.

The DAYS DELINQUENT and MONTHS ACTIVE requirements are most lenient in TABLE A. As a result, if the account fails either of these tests in one table, it will fail the test in any remaining tables. The CYCLE 1, CYCLE 2, and CYCLE 3 requirements are most restrictive in TABLE A, so the account may fail these tests in one table and pass them in the next table.

Comparing Available Credit with Overlimit Amounts

The Authorization module then determines if the requested transaction would cause an overlimit situation if approved. A transaction that causes an overlimit situation can still be approved using the Parametric Authorization method as long as the transaction does not cause the credit limit to be exceeded by more than a certain percentage. This percentage is defined in the CAPF.

The credit limit in the CREDIT LIMIT field on PBF screen 1 is used as a basis for determining whether the requested transaction would cause an overlimit situation. An overlimit situation occurs when the sum of the following amounts exceeds the credit limit in the CREDIT LIMIT field.

- The balance of the account. This amount is in the CREDIT BALANCE field on PBF screen 1.
- The total float amount for this account. This amount is in the TOTAL FLOAT field on PBF screen 8.
- The total of all preauthorized holds currently standing on this account. These amounts can be found on screen 5 of the Cardholder Authorization File (CAF), screen 3 of the PBF, or screen 2 of the Usage Accumulation File (UAF).
- The requested transaction amount.

Calculating the Percentage of Overlimit

If the Authorization module determines that this transaction would cause an overlimit situation, it calculates the percentage that this transaction would cause the account to be overlimit as follows:

$$\frac{100 \times (\text{credit balance} + \text{total float} + \text{total of preauthorized holds} + \text{requested transaction amount} - \text{credit limit})}{\text{credit limit}}$$

Using the Calculated Overlimit Percentage

The Authorization module then compares this calculated percentage to the allowed overlimit percentages in the OVERLIMIT TABLE in the CAPF. These overlimit percentages are in the PERCENT fields of the CAPF.

The Authorization module also compares overlimit requirements in the CAPF with those defined in the PBF. The overlimit requirements in the CAPF and the PBF include the number of months this account has been active, the number of times this account has been one cycle delinquent, the number of times this account has been two cycles delinquent, and the number of times this account has been three cycles delinquent.

The corresponding fields in which this information is contained are listed below for both the PBF and the CAPF.

PBF (Screen 8)	CAPF (Overlimit Table)
MONTHS ACTIVE	ACTIVE
# OF TIMES 1 CYCLE DELINQUENT	CYCLE 1
# OF TIMES 2 CYCLES DELINQUENT	CYCLE 2
# OF TIMES 3 CYCLES DELINQUENT	CYCLE 3

The fields in the CAPF occur in up to three sets, or tables: TABLE A, TABLE B, and TABLE C. The fields in TABLE B are used only if the transaction does not meet the authorization requirements of TABLE A. The fields in TABLE C, subsequently, are used only if the transaction does not meet the authorization requirements of TABLE B. If the Parametric Authorization method is being used, only the fields in TABLE A must be configured. The fields in TABLE B and TABLE C are optional.

The transaction meets the authorization requirements of TABLE A, TABLE B, or TABLE C if the values in the PBF and CAPF fields meet the following conditions:

- The calculated overlimit percentage is less than or equal to the percentage in the PERCENT field of the CAPF.
- The value in the MONTHS ACTIVE field on PBF screen 8 is greater than or equal to the value in the ACTIVE field of the CAPF.
- The value in the # OF TIMES 1 CYCLE DELINQUENT field on PBF screen 8 is less than or equal to the value in the CYCLE 1 field of the CAPF.
- The value in the # OF TIMES 2 CYCLES DELINQUENT field on PBF screen 8 is less than or equal to the value in the CYCLE 2 field of the CAPF.
- The value in the # OF TIMES 3 CYCLES DELINQUENT field on PBF screen 8 is less than or equal to the value in the CYCLE 3 field of the CAPF.

If the account for this transaction fails to meet the MONTHS ACTIVE requirement in a CAPF table (TABLE A, TABLE B, or TABLE C), the transaction is denied or referred according to the value in the RESPONSE field for that table. The remaining tables are not checked.

If the account for this transaction fails to meet any of the other requirements in a CAPF table, the account is checked against the next available table. If no more tables are available, the transaction is denied or referred according to the value in the RESPONSE field for the current table.

The MONTHS ACTIVE requirement is most lenient in TABLE A. As a result, if the account fails this test in one table, it will fail the test in any remaining tables. The CYCLE 1, CYCLE 2, and CYCLE 3 requirements are most restrictive in TABLE A, so the account may fail these tests in one table and pass them in the next table.

Using Preauthorization for a Lesser Amount

If the requested transaction is a preauthorization purchase request and the account fails the percentage overlimit check, the BASE24-pos product can adjust the amount of the transaction and authorize it for a lesser amount than was requested. To do this, the Authorization module calculates the maximum amount that will meet the percentage overlimit check and still allow the transaction to be approved.

Note: A preauthorization transaction can be authorized for a lesser amount only if it was initiated at an Interchange Interface process or Device Handler module that has been modified to support preauthorization for a lesser amount. The preauthorization transaction is also supported by the HISO Interface. Refer to the discussion about preauthorization for a lesser amount in section 3 of this manual for more information.

Verifying Single Transaction Limits

Finally, the Authorization module verifies that the requested transaction amount is less than or equal to the purchase limit amount or advance limit amount defined in the CAPF and PBF. To determine the purchase or advance limit amount, the Authorization module uses the values in the following fields:

SINGLE PURCHASE LIMIT in the CAPF
SINGLE PURCHASE PERCENT in the CAPF
SINGLE ADVANCE LIMIT in the CAPF
SINGLE ADVANCE PERCENT in the CAPF
CREDIT LIMIT on screen 1 of the PBF

Determining Applicable Transaction Limits

The Authorization module uses the transaction code to determine which single transaction limit applies to the transaction.

Single advance limits are used for transactions having transaction code 15 (cash advance). Single purchase limits are used for transactions with the following transaction codes:

10 (purchase)
11 (preauthorization purchase)
13 (mail or telephone purchase)

Determining the Limit for Purchase Transactions

The purchase limit amount for purchase transactions is either the amount contained in the SINGLE PURCHASE LIMIT field of the CAPF or the result of the following calculation, whichever is greater.

$$(\text{CREDIT LIMIT}) \times (\text{SINGLE PURCHASE PERCENT})$$

If the requested transaction amount is less than the purchase limit amount, the transaction is approved. If the requested transaction amount is greater than the purchase limit amount, the transaction is a preauthorization purchase request, and preauthorization for a lesser amount is an option, the Authorization module calculates the maximum amount for which the transaction can be approved and uses that calculated lesser amount. If the requested transaction amount is greater than the purchase limit amount and preauthorization for lesser amount is not an option, the transaction is denied or referred according to the value in the SINGLE PURCHASE RESPONSE field in the CAPF.

Determining the Limit for Cash Advance Transactions

The cash advance limit amount for cash advance transactions is either the amount contained in the SINGLE ADVANCE LIMIT field of the CAPF or the result of the following calculation, whichever is greater.

$$(\text{CREDIT LIMIT}) \times (\text{SINGLE ADVANCE PERCENT})$$

If the requested transaction amount is less than the cash advance limit amount, the transaction is approved. If the requested transaction amount is greater than the cash advance limit amount, the transaction is denied or referred according to the value in the SINGLE ADVANCE RESPONSE field in the CAPF.

PBF/CAPF Processing Example

The following example illustrates the PBF/CAPF processing portion of the Parametric Authorization method. Actual processing steps depend on the type of transaction, transaction amount, and the limits established by the institution. This is an example only and is not intended to show recommended field settings or recommended transaction and credit limits.

The transaction in this example is a preauthorization purchase request for \$50 from a credit account. The institution is using the Parametric Authorization method, and all transactions are authorized at the offline authorization level. The entity that originated this transaction is a device or interchange that supports preauthorization for a lesser amount.

Checking the Account's Delinquency History

The credit account for this transaction has the delinquency history shown in the table below. This information is from the Positive Balance File (PBF).

PBF Screen 8	
Field	Field Value
DAYS DELINQUENT	15
MONTHS ACTIVE	25
# OF TIMES 1 CYCLE DELINQUENT	15
# OF TIMES 2 CYCLES DELINQUENT	0
# OF TIMES 3 CYCLES DELINQUENT	0

For the Parametric Authorization method, the institution has established delinquency limits in the Card Authorization Parameters File (CAPF). These delinquency limits are summarized in the following table.

CAPF Delinquent Table			
Field	TABLE A	TABLE B	TABLE C
DAYS	30	20	0
ACTIVE	36	24	0
CYCLE 1	24	36	—
CYCLE 2	12	24	—
CYCLE 3	0	12	—

The institution has configured only TABLE A and TABLE B of the CAPF Delinquent Table. TABLE C is not used.

The Authorization module compares the delinquency information in the PBF with the delinquency limitations established in the CAPF. The comparisons it makes are summarized earlier in this appendix. TABLE A in the CAPF cannot be used because the account has not been active at least 36 months (the value in the MONTHS ACTIVE field of the PBF is less than the value in the ACTIVE field of TABLE A of the CAPF). The account for this transaction does, however, meet the requirements established in TABLE B.

Checking for an Overlimit Situation

The Authorization module next determines if this transaction will cause an overlimit situation if it is approved. To determine this, the Authorization module uses information from the PBF. This information is summarized in the table below.

PBF	
Field	Amount
CREDIT LIMIT	\$500
CREDIT BALANCE	\$480

PBF	
Field	Amount
TOTAL FLOAT	\$5
sum of preauthorized holds	\$10

An overlimit situation exists if the credit limit is less than the sum of the following equation:

$$\text{CREDIT BALANCE} + \text{TOTAL FLOAT} + \text{sum of preauthorized holds} + \text{transaction amount}$$

Entering the account information for this transaction in the equation illustrates that the transaction will cause an overlimit situation if it is approved.

$$(480 + 5 + 10 + 50) > \text{credit limit of } \$500$$

Calculating the Percent Overlimit

An account can be a certain percentage over its credit limit and the Authorization module can still approve transactions. The institution sets this percentage in the PERCENT fields of the CAPF. Therefore, the Authorization module now calculates the percentage overlimit that the account will be if this transaction is approved.

To calculate the percent overlimit, the Authorization module uses the following equation:

$$\frac{100 \times (\text{credit balance} + \text{total float} + \text{total of preauthorized holds} + \text{requested transaction amount} - \text{credit limit})}{\text{credit limit}}$$

Entering the account information for this transaction in the equation illustrates that the transaction will cause the account to be 9 percent overlimit if it is approved.

$$\frac{100 \times (480 + 5 + 10 + 50 - 500)}{500}$$

Looking back at the Delinquent Tables of the CAPF, the institution allows accounts to be overlimit up to the percentages shown in the following table. This transaction is using TABLE B because the account did not meet the requirements established in TABLE A.

CAPF PERCENT		
TABLE A	TABLE B	TABLE C
10%	5%	—

According to TABLE B, the account can be up to 5 percent overlimit. Without preauthorization for a lesser amount, this transaction would be denied because the 9 percent overlimit exceeds the 5 percent overlimit established in TABLE B. However, because this transaction is a preauthorization purchase transaction and the transaction originator supports preauthorization for a lesser amount, the Authorization module can attempt to approve this transaction for a lesser amount than requested.

Calculating a Lesser Amount

To perform the preauthorization for a lesser amount function, the Authorization module calculates the maximum transaction amount that can be approved and still meet the percentage overlimit requirement set in the CAPF.

To calculate this amount, the Authorization module uses the following equation:

$$\begin{aligned} &(\text{PERCENT} \times \text{CREDIT LIMIT}) + ((\text{CREDIT LIMIT}) \\ &\quad - (\text{CREDIT BALANCE} + \text{TOTAL FLOAT} \\ &\quad \quad + \text{sum of preauthorized holds})) \end{aligned}$$

Entering the account information in this equation produces a maximum transaction amount of \$30. Therefore, the Authorization module uses a transaction amount of \$30 in place of the requested \$50 amount.

$$(.05 \times 500) + ((500) - (480 + 5 + 10))$$

Verifying the Single Transaction Limit

The final step in PBF/CAPF processing ensures that the transaction amount does not exceed the limits set for single transactions in the CAPF. For this example, the Authorization module uses the fields and corresponding field values from the CAPF and PBF as shown in the following table.

CAPF		PBF	
Field	Value	Field	Value
SINGLE PURCHASE LIMIT	\$60	CREDIT LIMIT	\$500
SINGLE PURCHASE PERCENT	15%		

The Authorization module compares the transaction amount with either the SINGLE PURCHASE LIMIT or the result of the following equation, whichever is greater.

$$\text{CREDIT LIMIT} \times \text{SINGLE PURCHASE PERCENT}$$

In this example, the SINGLE PURCHASE LIMIT is \$60 and the calculated limit is \$75. Therefore, the Authorization module uses \$75 as the comparison amount. Because the preauthorization purchase request is for \$30, the Authorization module can approve the transaction based on PBF/CAPF processing.

ACI Worldwide, Inc.

D: BASE24-pos Draft Capture Processing

This appendix explains the processing involved in determining whether a transaction is draft capture or non-draft capture. The BASE24-pos product provides the draft capture options listed below.

- Authorization only with paper follow-up
- Authorization and draft capture
- Authorization with electronic follow-up
- Draft capture option determined by the terminal

Institutions set the above options for each card type using the TP (transaction profile) field on screens 8, 9, and 10 of the BASE24-pos Terminal Data files (PTD). Depending on the option set in the TP field, the institution may also need to set the EF (electronic followup) field on screens 8, 9, and 10 of the PTD.

Processing According to the Draft Capture Flag

The Authorization module of the Device Handler/Router/Authorization process processes each transaction according to the value of a draft capture flag in the POS Standard Internal Message (PSTM) for that transaction. Based on this draft capture flag, a transaction is settled or not settled immediately.

The draft capture flag is located in the DFT-CAPTURE-FLG field of the PSTM. (Refer to appendix A for the format and field descriptions of the PSTM.) All Device Handler modules, except the CRT Authorization Device Handler module, set the draft capture flag using the draft capture option set in the TP (transaction profile) field on screens 8, 9, and 10 of the PTD. The CRT Authorization Device Handler module sets the draft capture flag using the value entered by an operator or the value in the DRAFT CAPTURE field on screens 2 and 3 of the POS Retailer Definition File (PRDF).

The DFT-CAPTURE-FLG field in the PSTM indicates that the Authorization module should process the transaction in one of the following ways: authorize the transaction, authorize and capture the transaction for settlement (draft capture), authorize the transaction and expect an electronic follow-up, or treat this transaction as an electronic follow-up.

When the DFT-CAPTURE-FLG field in the PSTM indicates to authorize the transaction, the Authorization module of the Device Handler/Router/Authorization process authorizes the transaction but the transaction is not settled. A paper receipt arrives later to settle the transaction. The Authorization module adjusts the balances maintained by the BASE24-pos product for all transactions except preauthorization purchase request transactions.

When the DFT-CAPTURE-FLG field in the PSTM indicates to authorize and capture the transaction, the Authorization module treats the transaction as a draft capture transaction. No further paper or electronic activity is necessary to settle the transaction. The transaction is considered settled at the time transaction processing completes. The Authorization module adjusts the balances maintained by the BASE24-pos product for all transactions except preauthorization purchase request transactions.

When the DFT-CAPTURE-FLG field in the PSTM indicates to authorize the transaction and expect an electronic follow-up, the Authorization module authorizes the transaction but the host does not consider the transaction to be settled until the Authorization module receives an electronic follow-up. The Authorization module adjusts the balances maintained by the BASE24-pos product for all transactions except preauthorization purchase request transactions.

When the DFT-CAPTURE-FLG field in the PSTM indicates that the transaction is an electronic follow-up of a previously authorized transaction, the host uses the electronic follow-up transaction for settlement.

With a preauthorization purchase request transaction, no balances are affected until the Authorization module receives a preauthorization purchase completion.

Note: Although the BASE24-pos balances have been impacted, institutions might not consider the transactions settled. The BASE24-pos product does not support settlement of individual POS transactions, but it does supply information that enables hosts to determine whether transactions are settled.

Setting the Draft Capture Flag in the PSTM

The draft capture flag indicates whether BASE24-pos balances should immediately reflect transaction activity. All Device Handler modules, except the CRT Authorization Device Handler module, set the draft capture flag using the value in the TP field on screens 8, 9, and 10 of the PTD or the transaction profile value sent by the terminal. The CRT Authorization Device Handler module sets the draft capture flag using a value entered by an operator or the value in the DRAFT CAPTURE field on screens 2 and 3 of the POS Retailer Definition File (PRDF).

CRT Authorization Device Handler Module

The CRT Authorization Device Handler module determines the value for the DFT-CAPTURE-FLG field in the PSTM using one of the following two methods:

- A CRT operator enters a value using the Online Transaction screen (OLT) to select the draft capture method used for a specific transaction.

The possible options for the DRAFT CAPTURE field on the OLT screen are authorize the transaction, authorize and capture the transaction, and use the draft capture value from the PRDF.

- If a value has not been set using the OLT, the BASE24-pos product uses the draft capture option that the financial institution chose in the DRAFT CAPTURE field on screens 2 and 3 of the PRDF. The possible options for the DRAFT CAPTURE field in the PRDF are to authorize the transaction or authorize and capture the transaction.

Note: Draft capture values in the PRDF are set by the financial institution for each card type at the retailer level. The value in the DRAFT CAPTURE field of the PRDF is a default value for the CRT Authorization process only. Refer to the ***BASE24-pos CRT Authorization and CRT Administration Manual*** for more information about CRT Authorization's support of draft capture processing.

Other Device Handler Modules

Device Handler modules such as Visa I and Visa II Device Handler modules, BASE24 Standard POS Device Handler modules, and Hypercom Device Handler modules, use the option specified in the TP field on screens 8, 9, and 10 in the PTD or the transaction profile value sent by the terminal to set the TRAN-PROFILE field in the PSTM. The Router module then moves the value in the

TRAN-PROFILE field to the DFT-CAPTURE-FLG field in the PSTM. The values for the TP field on PTD screens and the DFT-CAPTURE-FLG field in the PSTM are shown in the table below:

Possible Field Values and Options	
Transaction Profile (TP) from the PTD	DFT-CAPTURE-FLG
0 = Authorize only	0 = Authorize only
1 = Authorize and capture	1 = Authorize and capture
2 = Authorize and expect an electronic follow-up	2 = Authorize and expect an electronic follow-up
3 = Use draft capture option sent from terminal	3 = Treat as an electronic follow-up of a previous transaction

The transaction profile values and corresponding draft capture flag settings used by each type of Device Handler module are described below.

Visa I Device Handler

The following table shows the transaction profile values and the corresponding draft capture flag settings used by the Visa I Device Handler module.

TP	DFT-CAPTURE-FLG	Transaction Type
0, 2, or 3	0	All transactions
1	0	Card verification transactions
	1	All transactions, except card verification transactions

Visa II Device Handler

The following table shows the transaction profile values and the corresponding draft capture flag settings used by the Visa II Device Handler module.

TP	DFT-CAPTURE-FLG	Transaction Type
0 or 3	0	All transactions
1	0	Card verification transactions
	1	All transactions except card verification transactions
2	0	Card verification transactions
	2	All transactions except card verification transactions

Note: Visa II draft capture is a two-transaction process and 0220 messages are sent in batch mode. If the request does not complete as expected, the 0220 message acts as an adjustment transaction rather than a completion. In that case, the DFT-CAPTURE-FLG field in the PSTM is set to 1 to adjust balances.

Visa II devices can also perform stand-in authorization. When they do, a force-post transaction is generated. Force-post transactions are indicated by a 1 in the DFT-CAPTURE-FLG field in the PSTM. This setting informs the BASE24-pos product that the transaction request does not have a follow-up.

BASE24 Standard POS Device Handler (SPDH)

The following table shows the transaction profile values and the corresponding draft capture flag settings used by the BASE24 Standard POS Device Handler (SPDH) module.

TP	DFT-CAPTURE-FLG	Transaction Type
0	0	All transactions.

TP	DFT-CAPTURE-FLG	Transaction Type
1	0	Preauthorization purchase, card verification, check verification, check guarantee, and balance inquiry transactions.
	1	Normal purchase, mail or telephone order, refund, cash advance, purchase with cash back, adjustment, and preauthorization purchase completion transactions.
2	1	A value of 2 is invalid for the SPDH. The draft capture flag defaults to 1.
3	Value sent from the terminal	All transactions.

Hypercom Device Handler

The following table shows the transaction profile values and the corresponding draft capture flag settings used by the Hypercom Device Handler module.

TP	DFT-CAPTURE-FLG	TRANSACTION TYPE
0	0	All transactions.
1	0	Preauthorization purchase, card verification, balance inquiry, and 0100 transactions.
	1	Normal purchase, mail or telephone order, refund, cash advance, purchase with cash back, adjustment, and preauthorization purchase completion transactions.

TP	DFT-CAPTURE- FLG	TRANSACTION TYPE
2	2	All transaction types from a Visa II terminal.
	1	A value of 2 is invalid for Hypercom terminals. The draft capture flag defaults to 1.
3	0	All transactions with a message type of 0100.
	1	All transactions with a message type of 0200 or 0220.

E: Pregon Utility

This appendix describes the Pregon utility and the procedures used to build a new or revised SPROUTE File (SPROUTE). It describes the commands used by the utility, the procedures required to run the utility, and the possible error messages that the utility can generate. It also describes the procedures for warmbooting SPROUTE after it is built.

The Pregen Utility

The PREMAP File (PREMAP) and SPROUTE File (SPROUTE) contain the transaction routing logic used by the BASE24-pos Router/Authorization module to route transactions. The Pregen utility builds the routing files from information contained in the Pregen Command File (PRECMD). The Pregen utility creates and purges a temporary SPREFIX File (SPREFIXN) when building a new or revised SPROUTE.

The Pregen utility should be run anytime after you add or delete card or interchange prefixes or you modify the routing logic. No system down-time is required to run the utility. It can be executed while the XPNET system is running.

The Pregen utility is a command-driven utility. The utility reads the commands from PRECMD. The commands provide the Pregen utility with the information it needs to create the routing files. There are six Pregen commands:

BUILDFILESIZE	Specifies the size of the primary and secondary extents for PREMAP and SPREFIXN.
BUILDSPROUTE	Specifies the location for PREMAP and SPROUTE.
PRADD	Specifies the transaction routing logic.
PREND	Indicates the end of the PRADD commands.
REMOTESPROUTE	Specifies the location for PREMAP and SPROUTE on remote HP NonStop systems.
SPROUTEND	Indicates the end of the Pregen command file.

These commands are described below in the order that they appear in PRECMD.

BUILDFILESIZE Command

The BUILDFILESIZE command alters the number of pages used for the primary and secondary extents of PREMAP or SPREFIXN. Each page can contain 2048 bytes of data. This command is optional. If this command is used, it must be the first command in PRECMD. If this command is not included in the command file, the Pregen utility uses a default of 5 pages for both the primary and secondary extents for PREMAP and 1 page for both the primary and secondary extents for SPREFIXN. You may need to change the default settings for large configurations.

Syntax

BUILDFILESIZE [PREMAP | SPREFIXN] , *primary-extents* , *secondary-extents* ;

primary-extents — The number of pages for the first extent of the file. Valid values are from 0 to 65535 pages.

secondary-extents — The number of pages for the secondary extents of the file. Valid values are from 0 to 65535 pages.

Example

BUILDFILESIZE PREMAP, 10, 10;

BUILDFILESIZE SPREFIXN, 4, 4;

BUILDSPROUTE Command

The BUILDSPROUTE command directs the Pregon utility to build SPROUTE. This command is required.

Syntax

BUILDSPROUTE *location*;

location — Specifies the HP NonStop system name (optional), volume, and subvolume where the Pregon utility should create SPROUTE. Typically, SPROUTE is placed on the XPNET system data subvolume. The LCONF param SPANCTL must be set to the same SPROUTE location.

Example

BUILDSPROUTE \$DATA1.PRODDATA;

PRADD Command

The PRADD command provides the physical description of how transactions are to be routed from source processes within the named logical network to interchanges and destination processes throughout the system. It identifies a source logical network or interchange, source process, and routing group (that is, the key fields) to the system as a transaction source. For each transaction source, the command also identifies valid symbolic destinations to which the source process can route transactions, along with the method of routing to each

destination. Also, for transactions to be routed using the LGNT method, the prefix file and file type (that is, Card Prefix File or Interchange Prefix File) is indicated for each destination. There must be one PRADD command for each logical network, source process, and routing group combination in the XPNET system.

For more information on PREMAP routing entries, see the topic “SPROUTE Routing” in section 4 of this manual.

Syntax

*PRADD source-ln , source-process , routing-group ,
method , [card-type] , dest , [file-name] , [type] , [alt-dest] ,
[method , [card-type] , dest , [file-name] , [type] , [alt-dest]] ;*

source-ln — The name of a logical network (or interchange) from which a transaction can originate. Logical network (or interchange) names can be up to 4 characters in length. Valid source logical network names include all logical networks configured in the XPNET system and interchange logical networks. Together with the source process, this element uniquely identifies the source of a transaction.

source-process — The name of a process that invokes its own routing utilities to determine the destination for a transaction, and then routes the transaction accordingly. Typically, source process names are one of the following:

- The symbolic name of a BASE24-pos Device Handler/Router/Authorization process. These processes invoke their routing utilities (SKEL) when a CPF record for the card cannot be found in the local logical network.
- An asterisk (*). An asterisk indicates that routing for a specific process is not required.

Symbolic process names are up to 16 characters long.

routing-group — A routing group associated with a POS terminal or an interchange.

A point-of-sale device is assigned to a routing group in its BASE24-pos Terminal Data files record. Each routing group defined in the BASE24-pos Terminal Data files can have a corresponding entry in the PREMAP table, where the routing methodology for all devices in the routing group is defined.

An asterisk can be entered in this field if routing based on the routing group is not required. The asterisk is used with the last two entries in the selection hierarchy when the BASE24-pos Router module is searching SPROUTE for a PREMAP entry. For more information on the selection hierarchy, see the “Selection Hierarchy” topic in section 4 of this manual.

Valid routing groups for POS terminals are 1 to 11 alphanumeric characters in length, while valid routing groups for interchanges are 1 to 11 numeric characters in length.

method — A routing method, or algorithm, to use in determining whether to route the transaction using the current set of action fields. Valid routing methods are as follows:

AMEX	= American Express algorithm
BNET or INAS	= Mastercard algorithm
DSCV	= Discover algorithm
LGNT	= Logical network routing method
VISA or BASE	= Visa algorithm
DFLT	= Default routing method

For more information on routing methods and algorithms, see the “Routing Methods” topic in section 4 of this manual.

card-type — The card type to be used for the transaction. Valid card types are 1 to 2 characters in length and always begin with a letter. For more information on reserved card types and for a description of valid values, see section 1.

If a card type is not included in the entry, the comma is still required as a placeholder.

dest — The symbolic name of the process to which the transaction is to be forwarded by the BASE24-pos Router module if the routing method tests are passed. The primary destination must be a symbolic process name defined in the XPNET system. For example, when an interchange is the primary destination, the symbolic name of the Interchange Interface process is specified in this action field.

file-name — The fully-qualified name of the prefix file to be checked for the LGNT routing method. The routing utilities check this file for the card prefix associated with the transaction to determine whether to use this set of action fields.

Fully qualified prefix file names can be from 1 to 36 characters in length. It is recommended that prefix file name entries be compatible with the HP NonStop Expand product to accommodate communication between HP NonStop systems.

If the prefix file name is included in the entry, the file type must also be included in the entry. If a prefix file name is not included in the entry, the comma is still required as a placeholder.

type — The type of file named in the *file-name* variable. Valid values are as follows:

CP = Card Prefix File (CPF)

IP = Interchange Prefix File (IPF)

If the prefix file name is included in the entry, a value must be included for this variable. If this variable is not included in the entry, the comma is still required as a placeholder.

alt-dest — The symbolic name of the process to which a transaction is to be forwarded by the BASE24-pos Router module if the primary destination is not available to authorize the transaction.

Valid combinations for the action variables (method, card type, destination, file name, type, and alternate destination) are shown in the table below:

<i>method</i>	<i>card-type</i>	<i>dest</i>	<i>file-name</i>	<i>type</i>	<i>alt-dest</i>
AMEX	blanks	any symbolic destination	blanks	blanks	any symbolic destination or blanks
BNET or INAS	blanks	any symbolic destination	blanks	blanks	any symbolic destination or blanks
DSCV	blanks	any symbolic destination	blanks	blanks	any symbolic destination or blanks
VISA or BASE	blanks	any symbolic destination	blanks	blanks	any symbolic destination or blanks
LGNT	blanks	any symbolic destination	fully qualified CPF name	CP	any symbolic destination or blanks

<i>method</i>	<i>card-type</i>	<i>dest</i>	<i>file-name</i>	<i>type</i>	<i>alt-dest</i>
LGNT	blanks	any symbolic destination	fully qualified IPF name	IP	any symbolic destination or blanks
DFLT	any	any symbolic destination	blanks	blanks	any symbolic destination or blanks

Examples

```

PRADD  PRO1,      *      , 000000000000,
      LGNT, , P2A^SPDH1, \SYS.$DATA.PRO2DATA.CPF, CP, P2A^SPDH2,
      DFLT, , P1A^CIRR ,      , ,      ;

PRADD  PRO2,      *      , 000000000000,
      LGNT, , P1A^SPDH1, \SYS.$DATA.PRO1DATA.CPF, CP, P1A^SPDH2,
      DFLT, , P1A^CIRR ,      , ,      ;

```

PREND Command

The PREND command marks the end of the PREMAP routing commands. This command is required.

Syntax

```
PREND;
```

REMOTESPROUTE Command

The REMOTESPROUTE command specifies the locations on remote HP NonStop systems where the Pregon utility should place a copy of PREMAP and SPROUTE. There must be one REMOTESPROUTE command for each remote HP NonStop system in the XPNET system. The command can be repeated for each remote HP NonStop system. These commands are optional and must follow the PREND command.

After the Pregon utility successfully creates SPROUTE on the local system, the Pregon utility creates and obeys a FUP obey file called PREFUPIN to copy SPROUTE to each of the specified remote HP NonStop systems.

Syntax

REMOTESPROUTE *location* ;

location — Specifies the HP NonStop system name, volume, and subvolume on which PREMAP and SPROUTE should be placed. Typically, this command specifies the XPNET system data subvolume on the remote HP NonStop system.

Example

REMOTESPROUTE \SYS2.\$DATA1.PRODDATA;

SPROUTEND Command

The SPROUTEND command marks the end of the SPROUTE commands. This command is required and must be the last command in the Pregen Command File.

Syntax

SPROUTEND;

Pregen Command File

The Pregen Command File (PRECMD) is composed of a series of Pregen commands used to create SPROUTE. PRECMD is used as the input file when the Pregen utility is executed. It resides on the XPNET system control subvolume (e.g., \$DATA.PRODCNTL).

The following is the PRECMD command order required to create SPROUTE:

```
[BUILDFILESIZE]
BUILDSPROUTE
PRADD
PREND
[REMOTESPROUTE]
SPROUTEND
```

An example of a PRECMD is shown below. System managers are strongly urged to use a comment box like the one at the top of this example. The comment box helps you identify the files and processes impacted by the Pregon utility.

```

!*****
!*                               PREGEN COMMAND FILE (PRECMD)                               *
!*****
!*                               S O U R C E                                             *
!* LN          PROCESS          ROUTING GROUP                                           *
!*                               *                                                       *
!* ORDERED    CARD    PRIMARY          ALT                                             *
!* METHOD      TYPE    DEST              DEST              PREFIX FILE NAME             *
!*-----*-----*-----*-----*-----*-----*-----*-----*-----*-----*
!* PRO1      P1A^SPDH1  000000000001                                                    *
!*                               *                                                       *
!* LGNT      (none)   P1A^CIRR   (none)  \SYS1.$DATA.PRO1DATA.IPFCIRR                 *
!* LGNT      (none)   P1A^PLUS   (none)  \SYS1.$DATA.PRO1DATA.IPFPLUS                 *
!* LGNT      (none)   P1A^BICI   (none)  \SYS1.$DATA.PRO1DATA.IPFBICI                 *
!*                               *                                                       *
!* PRO1      *          000000000002                                                    *
!*                               *                                                       *
!* LGNT      (none)   P1A^VNET   (none)  \SYS1.$DATA.PRO1DATA.IPFVISA                 *
!* DFLT      (none)   P1A^PLUS   (none)  \SYS1.$DATA.PRO1DATA.IPFPLUS                 *
!*****
BUILDSPROUTE $DATA.PRODDATA;
PRADD        PRO1,  P1A^SPDH1  , 000000000001,
  LGNT, ,  P1A^CIRR   , \SYS1.$DATA.PRO1DATA.IPFCIRR,IP, ,
  LGNT, ,  P1A^PLUS   , \SYS1.$DATA.PRO1DATA.IPFPLUS,IP, ,
  LGNT, ,  P1A^BICI   , \SYS1.$DATA.PRO1DATA.IPFBICI,IP, ,;
PRADD        PRO1,  *          , 000000000002,
  LGNT, ,  P1A^VNET   , \SYS1.$DATA.PRO1DATA.IPFVISA,IP, ,
  DFLT, ,  P1A^PLUS   , \SYS1.$DATA.PRO1DATA.IPFPLUS,IP, ,;
PREND;
REMOTESPROUTE \SYS2.$DATA.PRODDATA;
REMOTESPROUTE \SYS3.$DATA.PRODDATA;
SPROUTEND;

```

Pregen FUP Obey File

Each time you run the Pregen utility, it creates and executes a FUP obey file to perform the following:

- Rename the old SPROUTE to SPROUTEO
- Rename the new SPROUTE created during the run from SPROUTEN to SPROUTE
- Purge the temporary SPREFIX File called SPREFIXN
- Copy the new SPROUTE to any remote HP NonStop system specified in a REMOTESPROUTE command

An example of the PREFUPIN FUP obey file created and executed by the Pregen utility is shown below. This example assumes that one REMOTESPROUTE command was found in PRECMD. The VOLUME, PURGE, RENAME, and DUP commands are repeated for each remote HP NonStop system to distribute the new SPROUTE. The number of errors and warnings allowed is increased by four for each remote HP NonStop system. The Pregen utility creates and executes this file after it successfully creates a new SPROUTE. The file is placed on the XPNET system control subvolume.

```
ALLOW 012 ERRORS
ALLOW 012 WARNINGS
CONFIGURE ECHO OBEY
VOLUME \SYS1.$DATA.PRODDATA
PURGE SPROUTEO!
RENAME SPROUTE, SPROUTEO
RENAME SPROUTEN, SPROUTE
PURGE SPREFIXN!
VOLUME \SYS2.$DATA.PRODDATA
PURGE SPROUTEO!
RENAME SPROUTE, SPROUTEO
DUP \SYS1.$DATA.PRODDATA.SPROUTE, *, SOURCEDATE
```


Running the Pregon Utility

You can execute the Pregon utility directly from a TACL prompt or by obeying the Pregon Go File (GOPRE) located on the XPNET system control subvolume. GOPRE identifies PRECMD as the input file.

GOPRE should contain the following TACL commands:

```
RUN $SYSTEM.XPNET.PREGEN/OUT $S.#PREG, IN PRECMD,NAME/  
PERUSE
```

When execution of GOPRE completes, a HP NonStop PERUSE utility prompt is displayed. You can view or print the Pregon report from the PERUSE prompt. An example of a Pregon report is shown below.

Pregon Report

Each time the Pregon utility is run, a report is generated. The report basically reflects what the Pregon utility did during its run. It does this by producing a copy of PRECMD and then embedding messages after appropriate commands. These messages indicate how files or processes were affected and where the files are located. Each message generated by the Pregon utility on the report is preceded by two asterisks (**).

The Pregon report also indicates any errors that the Pregon utility encountered. The Pregon error messages are listed and described later in this appendix.

The structure of the report reflects the steps executed by the Pregon utility during its run. The steps are as follows:

1. Open PRECMD.
2. If an error occurs, print the error to the home terminal andabend.
3. Open the report location.
4. If an error occurs, print the error on the home terminal andabend.
5. Print the VERSION line on the report.
6. Process the Pregon build commands. Echo the comments and commands on the report.
7. If an error occurs, print the error on the report and stop.

8. Process the remainder of PRECMD, echoing commands on the report. Print the action taken where relevant.
9. Create the PREFUPIN command file and start a FUP process to obey the PREFUPIN commands.
10. Print an informational message on the report for each FUP command executed and operation performed, as well as any messages for warnings or errors encountered by FUP.
11. Print the END line on the report.
12. Stop.

An example of the report produced by the Pregen utility is shown on the following pages.

```

** PREGON VERSION 3.00   June 24, 1997

*****
!* PREGON COMMAND FILE (PRECMD)
*****
!* S O U R C E
!* LN PROCESS ROUTING GROUP
!*
!* ORDERED CARD PRIMARY ALT
!* METHOD TYPE DEST PREFIX FILE NAME
!*=====
!* PRO1 P1A^SPDH1 000000000001
!*
!* LGNT (none) P1A^CIRR (none) \SYS1.$DATA.PRO1DATA.IPFCIRR
!* LGNT (none) P1A^PLUS (none) \SYS1.$DATA.PRO1DATA.IPFPLUS
!* LGNT (none) P1A^BICI (none) \SYS1.$DATA.PRO1DATA.IPFBICI
!*
!* PRO1 * 000000000002
!*
!* LGNT (none) P1A^VNET (none) \SYS1.$DATA.PRO1DATA.IPFVISA
!* DFLT (none) P1A^PLUS (none) \SYS1.$DATA.PRO1DATA.IPFPLUS
*****
BUILDSPROUTE \SYS1.$DATA.PRODATA;

** PREGON NEXT VERSION REPORT: Tuesday, October 3, 2000 - 10:16:05 AM USED COMMAND FILE: $DATA.PRODCNTL.PRECMD

** PREMAP FILE: \SYS1.$DATA.PRODDATA.PREMAP CREATED
PRADD PRO1, P1A^SPDH1 , 000000000001,
LGNT, , P1A^CIRR , \SYS1.$DATA.PRO1DATA.IPFCIRR,IP, ,
LGNT, , P1A^PLUS , \SYS1.$DATA.PRO1DATA.IPFPLUS,IP, ,
LGNT, , P1A^BICI , \SYS1.$DATA.PRO1DATA.IPFBICI,IP, ,;
PRADD PRO1, * , 000000000002,
LGNT, , P1A^VNET , \SYS1.$DATA.PRO1DATA.IPFVISA,IP, ,
DFLT, , P1A^PLUS , \SYS1.$DATA.PRO1DATA.IPFPLUS,IP, ,;
PREND;

** PREMAP CONTENTS:

PRO1 P1A^SPDH1 000000000001
0 LGNT P1A^CIRR IP IP000000 \SYS1.$DATA.PRO1DATA.IPFCIRR
1 LGNT P1A^PLUS IP IP000001 \SYS1.$DATA.PRO1DATA.IPFPLUS
2 LGNT P1A^BICI IP IP000002 \SYS1.$DATA.PRO1DATA.IPFBICI
PRO1 * 000000000002
0 LGNT P1A^VNET IP IP000000 \SYS1.$DATA.PRO1DATA.IPFVISA
1 DFLT P1A^PLUS IP IP000001 \SYS1.$DATA.PRO1DATA.IPFPLUS

```

```
** BUILDING SPREFIX USING PREFIX FILE(S) :
```

```
\SYS1.$DATA.PRO1DATA.IPFICIR  
\SYS1.$DATA.PRO1DATA.IPFPLUS  
\SYS1.$DATA.PRO1DATA.IPFBICI  
\SYS1.$DATA.PRO1DATA.IPFVISA  
\SYS1.$DATA.PRO1DATA.IPFPLUS
```

```
** SPROUTE FILE: \SYS1.PRODDATA.SPROUTEN CREATED
```

```
SPROUTEND;  
09:30:33 (I) - 3801 FUP command/response : OBEY \SYS1.$PRODCNTL.PREFUPIN  
09:30:33 (I) - 3801 FUP command/response : VOLUME \SYS1.$DATA.PRODDATA  
09:30:33 (I) - 3801 FUP command/response : PURGE SPROUTE!  
09:30:33 (E) - 3680 Fup warning message is : \SYS1.$PRODDATA.SPROUTE: ERR 11  
09:30:33 (E) - 3690 PREGEN will continue to run  
09:30:33 (E) - 3700 Please evaluate fup message  
09:30:33 (E) - 3710 PREGEN may have to be run again  
09:30:33 (I) - 3801 FUP command/response : 0 FILES PURGED  
09:30:33 (I) - 3801 FUP command/response : RENAME SPROUTE, SPROUTE  
09:30:33 (I) - 3801 FUP command/response : RENAME SPROUTEN, SPROUTE  
09:30:33 (I) - 3801 FUP command/response : PURGE SPREFIXN!  
09:30:33 (I) - 3801 FUP command/response : 1 FILE PURGED
```

```
** END OF PREGEN BUILD NEXT VERSION RUN Tuesday, October 3, 2000 - 10:16:18 AM
```

Using a New SPROUTE File

Whenever you add or delete card or interchange prefixes, or change routing logic, a new SPROUTE must be created and warmbooted as follows:

1. Add or delete prefixes from the CPF or IPF as required.
2. Modify the routing logic in PRECMD as required using the commands explained earlier in this appendix.
3. Execute the GOPRE obey file to run the Pregon utility. Once run, verify the output.
4. Warmboot all Device Handler/Router/Authorization processes to start using the new SPROUTE. Enter the following commands from a network control facility:

```
CONTROL NODE xpobj, WARMBOOTSPROUTE  
DELIVER PROCESS object-name, "9501****"
```

xpobj — Identifies the specific XPNET node or nodes affected by the command.

object-name — The symbolic name assigned to the Device Handler/Router/Authorization process in the XPNET node. You can enter the fully qualified symbolic name of a specific process, a symbolic name that identifies one or more processes with the same name, or an object name that uses wildcard characters to identify one or more processes.

For more information on the CONTROL NODE and DELIVER PROCESS commands, see the ***NET24-XPNET Network Control Command Reference Manual***.

Error and Status Messages

All error and status conditions that affect normal program processing result in messages being interspersed in the report file by the Pregen utility. The system manager can then take corrective action as necessary.

The following pages provide a description of each error or status message that can be generated by the Pregen utility. The operator action required in response to each message is also provided.

Message Identification

Each error and status message from the Pregen utility is grouped into one of four categories:

Critical (C)	Indicates that a processing or hardware error has occurred. The error must be corrected and the utility rerun.
Error (E)	Indicates some type of processing or hardware error has occurred that is noncritical to normal processing. The error must be corrected and the utility rerun.
Warning (W)	Indicates a potential problem that may or may not require intervention.
Information (I)	Conveys general information and usually does not require any action.

All messages are identified by a unique numeric code.

The following error messages can be generated by the Pregen utility.

(C) - 1360 Error occurred during obey of PREFUPIN - error *number*

Where: *number* = The operating system error number

Cause: An operating system error occurred while executing the PREFUPIN obey file.

Remedy: See the appropriate HP NonStop documentation for information about the operating system error. Correct the error condition and manually rerun the utility.

(C) - 1370 Correct error and manually re-run PREFUPIN

Cause: An operating system error occurred while executing the PREFUPIN obey file.

Remedy: See the appropriate HP NonStop documentation for information about the operating system error. Correct the error condition and manually rerun the utility.

(C) - 1540 ** Write of FUP OBEY FILE (SPTFUPIN) failed - error *number*

Where: *number* = The operating system error number

Cause: An operating system error occurred while attempting to create the SPTFUPIN obey file.

Remedy: See the appropriate HP NonStop documentation for information about the operating system error. Correct the error condition and rerun the utility.

Note: This message only applies when running a previous release of the BASE24-pos product under NET24-XPNET release 3.0. This message does not apply for BASE24-pos product release 5.3 or later.

(C) - 1550 ** Close of FUP OBEY FILE (SPTFUPIN) failed - error *number*

Where: *number* = The operating system error number

Cause: An operating system error occurred while attempting to close the SPTFUPIN obey file.

Remedy: See the appropriate HP NonStop documentation for information about the operating system error. Correct the error condition and rerun the utility.

Note: This message only applies when running a previous release of the BASE24-pos product under NET24-XPNET release 3.0. This message does not apply for BASE24-pos product release 5.3 or later.

(C) - 3010 BUILD command - no Buildsproute found *string*

Where: *string* = Data from the command file

Cause: An unexpected command was found in the command file where the required BUILDSPROUTE command was expected.

Remedy: Correct the command file and rerun the Pregen utility.

(E) - 3175 An error occurred while updating *filename*'s filecode

Where: *filename* = The name of a file

Cause: The Pregen utility was unable to mark a file as valid.

Remedy: Correct the error condition and rerun the Pregen utility.

(C) - 3220 PREMAPCREATE - Premap *filename* create error *number*

Where: *filename* = The name of a file
number = The operating system error number

Cause: An error occurred while attempting to create a file due to an operating system error.

Remedy: See the appropriate HP NonStop documentation for information about the operating system error. Correct the error condition and rerun the Pregen utility.

(C) - 3370 BUILDSPREFIX - Sprefix *filename* create error *number*

Where: *filename* = The name of the new SPREFIX File (SPREFIXN)
number = The operating system error number

Cause: The Pregen utility was unable to create SPREFIXN due to an operating system error.

Remedy: See the appropriate HP NonStop documentation for information about the operating system error. Correct the error condition and rerun the Pregen utility.

(C) - 3390 BUILDSPROUTE - no End, found *string*

Where: *string* = Data from the command file

Cause: A required semicolon (;) was missing from the end of the BUILDSPROUTE command or a REMOTESPROUTE command.

Remedy: Correct the command file and rerun the Pregen utility.

(C) - 3400 PREMAP routing - invalid command, found *string*

Where: *string* = Data from the command file

Cause: An unexpected command was found in the command file.

Remedy: Correct the command file and rerun the Pregen utility.

(C) - 3410 PREMAP routing - no empty records in PREMAPTABLE

Cause: The number of routing methods specified in the command file has exceeded the maximum (8800) allowed by the Pregen utility.

Remedy: Correct the command file and rerun the Pregen utility.

(C) - 3421 PREMAP routing - invalid METHOD *method*

Where: *method* = The routing method specified in the PRADD command

Cause: An invalid routing method was specified in the PRADD statement. Valid routing methods are LGNT, DFLT, AMEX, BASE, BNET, VISA, DSCV, or INAS.

Remedy: Correct the PRADD command and rerun the Pregen utility.

(C) - 3422 PREMAP routing - required FILENAME missing

Cause: The LGNT routing method requires an associated prefix file name.

Remedy: Correct the PRADD command and rerun the Pregen utility.

(C) - 3423 PREMAP routing - FILENAME not allowed

Cause: A file name is only allowed for the LGNT routing method.

Remedy: Correct the PRADD command and rerun the Pregen utility.

(C) - 3424 PREMAP routing - required FILE TYPE missing

Cause: The LGNT routing method requires a prefix file type of CP or IP.

Remedy: Correct the PRADD command and rerun the Pregen utility.

(C) - 3425 PREMAP routing - invalid FILE TYPE *type*

Where: *type* = The invalid file type

Cause: An invalid prefix file type was specified in a PRADD command.
Valid prefix file types are CP or IP.

Remedy: Correct the PRADD command and rerun the Pregen utility.

(C) - 3426 PREMAP routing - FILE TYPE not allowed

Cause: A prefix file type was specified with a routing method other than LGNT. Prefix file types can only be specified with the LGNT routing method.

Remedy: Correct the PRADD command and rerun the Pregen utility.

(C) - 3427 PREMAP routing - logical network name too long *name*

Where: *name* = An invalid logical network name

Cause: The source logical network name is longer than four characters.

Remedy: Correct the PRADD command and rerun the Pregen utility.

(C) - 3428 PREMAP routing - symbolic source too long *name*

Where: *name* = An invalid symbolic name

Cause: The symbolic source process name is longer than 16 characters.

Remedy: Correct the command and rerun the Pregen utility.

(C) - 3429 PREMAP routing - routing group too long *group*

Where: *group* = A routing group found in a PRADD command

Cause: The routing group entered in the PRADD command is invalid because it is too long.

Remedy: Correct the command file and rerun the Pregen utility.

(C) - 3431 PREMAP routing - alternate destination too long
destination

Where: *destination* = An alternate destination found in a PRADD command

Cause: An alternate destination specified in a PRADD command is invalid because it is too long.

Remedy: Correct the command file and rerun the Pregen utility.

(C) - 3432 PREMAP routing - card type too long *card-type*

Where: *card-type* = A card type found in a PRADD command

Cause: A card type specified in a PRADD command is invalid because it is too long.

Remedy: Correct the command file and rerun the Pregen utility.

(C) - 3480 GENERATE keys - invalid file format, found *type*

Where: *type* = The file type found in the command file

Cause: An invalid file format specification was found in the command file.

Remedy: Correct the command file and rerun the Pregen utility.

(C) - 3490 OPEN and BUILD - file open error, using *filename*

Where: *filename* = The name of a file

Cause: An error occurred while opening the file specified.

Remedy: Correct the error condition and rerun the Pregen utility.

(C) - 3500 OPEN and BUILD - fname expand error, using *filename*

Where: *filename* = The name of a file

Cause: The Pregen utility was unable to expand a file name in the command file.

Remedy: If the file name used is not the one desired, correct the command file and rerun the Pregen utility.

(C) - 3540 OPEN and BUILD - invalid format, found *format*

Where: *format* = The file format

Cause: An internal program error has occurred.

Remedy: Contact HELP24.

(C) - 3560 FILESIZE - primary extents convert error, used *number*

Where: *number* = The number used in place of the number in the command file

Cause: The primary extent size is specified incorrectly in the command file.

Remedy: Correct the command file and rerun the Pregen utility.

(C) - 3570 FILESIZE - secondary extents convert error, used *number*

Where: *number* = The number used in place of the number in the command file

Cause: The secondary extent size is specified incorrectly in the BUILDFILESIZE command.

Remedy: Correct the command file and rerun the Pregen utility.

(C) - 3580 FILESIZE - invalid file name, used *filename*

Where: *filename* = The name of a file

Cause: An invalid file name was specified in the BUILDFILESIZE command.

Remedy: Correct the command and rerun the Pregen utility.

(C) - 3590 FILE CODE FILLER - invalid filetype filecode

Cause: An internal program error has occurred.

Remedy: Contact HELP24.

(C) - 3600 FILE CODE FILLER - filetype filecode conversion error

Cause: An internal program error has occurred.

Remedy: Contact HELP24.

(E) - 3610 Error occurred during creation of FUP, error was *number*

Where: *number* = The error returned by the call to the NEWPROCESS procedure

Cause: The Pregen utility was unable to start FUP due to an error.

Remedy: Rerun the Pregen utility. If error persists, contact HELP24.

(E) - 3620 Error occurred during open of FUP

Cause: The Pregen utility was unable to open FUP.

Remedy: Contact HELP24.

(E) - 3630 Error occurred during write of startup message to FUP

Cause: The Pregen utility was unable to send a startup message to FUP.

Remedy: Contact HELP24.

(E) - 3640 Error *number* occurred during open of \$RECEIVE

Where: *number* = The operating system error number

Cause: The Pregen utility was unable to open \$RECEIVE due to an operating system error.

Remedy: See the appropriate HP NonStop documentation for information about the operating system error. Contact HELP24.

(E) - 3650 This is not a named process. Be sure to use the NAME field in the run command. Example: RUN PREGEN /IN <command file>, OUT <out file>, NAME/

Cause: The Pregen utility was not run as a named process.

Remedy: Rerun the Pregen utility as a named process.

(E) - 3660 Error *number* occurred during readupdate of \$RECEIVE

Where: *number* = The operating system error number

Cause: A read on \$RECEIVE failed due to an operating system error.

Remedy: See the appropriate HP NonStop documentation for information about the operating system error. Contact HELP24.

(E) - 3670 Fup error message is : *text*

Where: *text* = The error message from FUP

Cause: An error message was received from the FUP process started by the Pregen utility.

Remedy: Evaluate the error message, take appropriate action, and rerun the Pregen utility.

(E) - 3680 Fup warning message is : *text*

(E) - 3690 PREGEN will continue to run

(E) - 3700 Please evaluate fup message

(E) - 3710 PREGEN may have to be run again

Where: *text* = The error message from FUP

Cause: A warning message was received from the FUP process started by the Pregen utility.

Remedy: Evaluate the warning message. If necessary, take appropriate action, and rerun the Pregen utility.

(E) - 3720 Unknown FUP message : *text*

Where: *text* = The message from FUP

Cause: A message was received from the FUP process started by the Pregen utility.

Remedy: Evaluate the message. If necessary, take appropriate action, and rerun the Pregen utility.

(E) - 3800 Fup response with no text

Cause: An internal program error has occurred.

Remedy: Contact HELP24.

(I) - 3801 FUP command/response : *text*

Where: *text* = The command or response message from FUP

Cause: An informational message indicating the current command FUP is executing or an operation FUP has performed.

Remedy: No action is required.

(E) - 3803 System STOP message received

Cause: The FUP process being used by the Pregen utility has stopped.

Remedy: Rerun the Pregen utility.

(E) - 3806 System ABEND message received

Cause: The FUP process being used by the Pregen utility has stopped.

Remedy: Rerun the Pregen utility.

(E) - 3809 Unknown system message received

Cause: An unknown system message was received. It is ignored by the Pregen utility.

Remedy: No action is required.

(E) - 3810 Problem has occurred while initializing FUP

Cause: The Pregen utility encountered a problem while initializing FUP.

Remedy: Rerun the Pregen utility. If the error persists, contact HELP24.

(E) - 3820 An error occurred during EXIT from FUP

Cause: The Pregen utility encountered an error while exiting FUP.

Remedy: Rerun the Pregen utility. If the error persists, contact HELP24.

(C) - 3909 Looking for SPROUTEND - found *string*, ignoring

Where: *string* = Data from the command file

Cause: An unexpected command was found in the command file where the required SPROUTEND command was expected.

Remedy: Correct the command file and rerun the Pregen utility.

(C) - 3920 ALLOCATESEGMENT error *number*, (size *size*, file *filename*)

Where: *number* = The error returned from ALLOCATESEGMENT
size = The size (in bytes) of the extended memory segment
filename = The name of a file

Cause: The Pregen utility was unable to allocate an extended memory segment.

Remedy: Analyze the ALLOCATESEGMENT error and take appropriate action.

(C) - 3921 Open failed for *type* file *filename*, error *number*

Where: *type* = Specifies the type of file, and indicates PREMAP or SPREFIX

filename = The name of a file

number = The OPEN error number

Cause: The Pregen utility was unable to open the specified file.

Remedy: Analyze the OPEN error and take appropriate action.

(C) - 3960 Fatal error *number* while reading PREMAP file *filename*

Where: *number* = The READ error number
filename = The name of a file

Cause: The Pregen utility was unable to read PREMAP.

Remedy: Analyze the error and take appropriate action.

(C) - 3980 Fatal error *number* while reading SPREFIX file *filename*

Where: *number* = The READ error number
 filename = The name of a file

Cause: The Pregen utility was unable to read the temporary SPREFIX.

Remedy: Analyze the error and take appropriate action.

(C) - 3981 Fatal error, Sprefix record does not match a premap
sprefix key

Cause: The Pregen utility encountered a record in the temporary SPREFIX
 with an invalid key.

Remedy: Verify that the information in the Pregen Command File
 (PRECMD), Card Prefix File (CPF), and Interchange Prefix File
 (IPF) is correct and rerun the Pregen utility.

(C) - 3982 Fatal error, more sprefix keys found than table space
allocated

Cause: The Pregen utility read more records from the temporary SPREFIX
 than it allocated space for in memory.

Remedy: Contact HELP24.

(C) - 3983 Invalid PREFIX length *length* for SPREFIX key *key*

Where: *length* = The length of a record in the SPREFIX
 key = The key to the SPREFIX record

Cause: The Pregen utility encountered a SPREFIX record with an invalid
 prefix length.

Remedy: Correct the prefix length and rerun the Pregen utility.

(C) - 3984 Invalid PAN length *length* for SPREFIX key *key*

Where: *length* = The length of the primary account number (PAN)
 key = The key to the record in the SPREFIX

Cause: The Pregon utility encountered a SPREFIX record with an invalid PAN length.

Remedy: Correct the PAN length and rerun the Pregon utility.

(W) - 3990 NEXT^PARAM - end of statement, possible missing field

Cause: The Pregon utility expected to find the next parameter for a command in the command file, but instead found the end of the command.

Remedy: Correct the Pregon Command File (PRECMD) and rerun the Pregon utility.

(C) - 9999 BUILDSPROUTE - fname expand error, using *filename*

Where: *filename* = The name of a file

Cause: The Pregon utility was unable to expand the file name specified for the BUILDSPROUTE command or a REMOTESPROUTE command.

Remedy: If the file name used is not the one desired, correct the command file and rerun the Pregon utility.

ACI Worldwide, Inc.

F: Base Extended Memory Table Build Utility

This section provides detailed instructions for using the Base Extended Memory Table Build utility. This utility is a stand-alone program used to build or rebuild shared extended memory tables from disk files. The Base Extended Memory Table Build utility is executed from a TACL prompt. This section discusses the following topics:

- An overview of Base Extended Memory Table Build utility processing, including procedures to make processes reallocate (warmboot) a newly built or rebuilt extended memory table
- A description of the source files from which extended memory tables are created for the BASE24-pos product
- The configuration requirements for the Base Extended Memory Table Build utility, including a list of all Logical Network Configuration File (LCONF) assigns and params
- Procedures for starting the Base Extended Memory Table Build utility from a TACL prompt
- A description of the Extended Memory Table Build Report produced when the Base Extended Memory Table Build utility is run

Note: This appendix addresses extended memory tables used by procedural BASE24-pos processes. For information on extended memory tables used by object-oriented applications, refer to the ***BASE24 ITS Kernel Configuration Manual***.

Overview

The Base Extended Memory Table Build utility builds shared extended memory tables for use by BASE24 processes. The program can build hash tables, radix tables, or custom-designed tables.

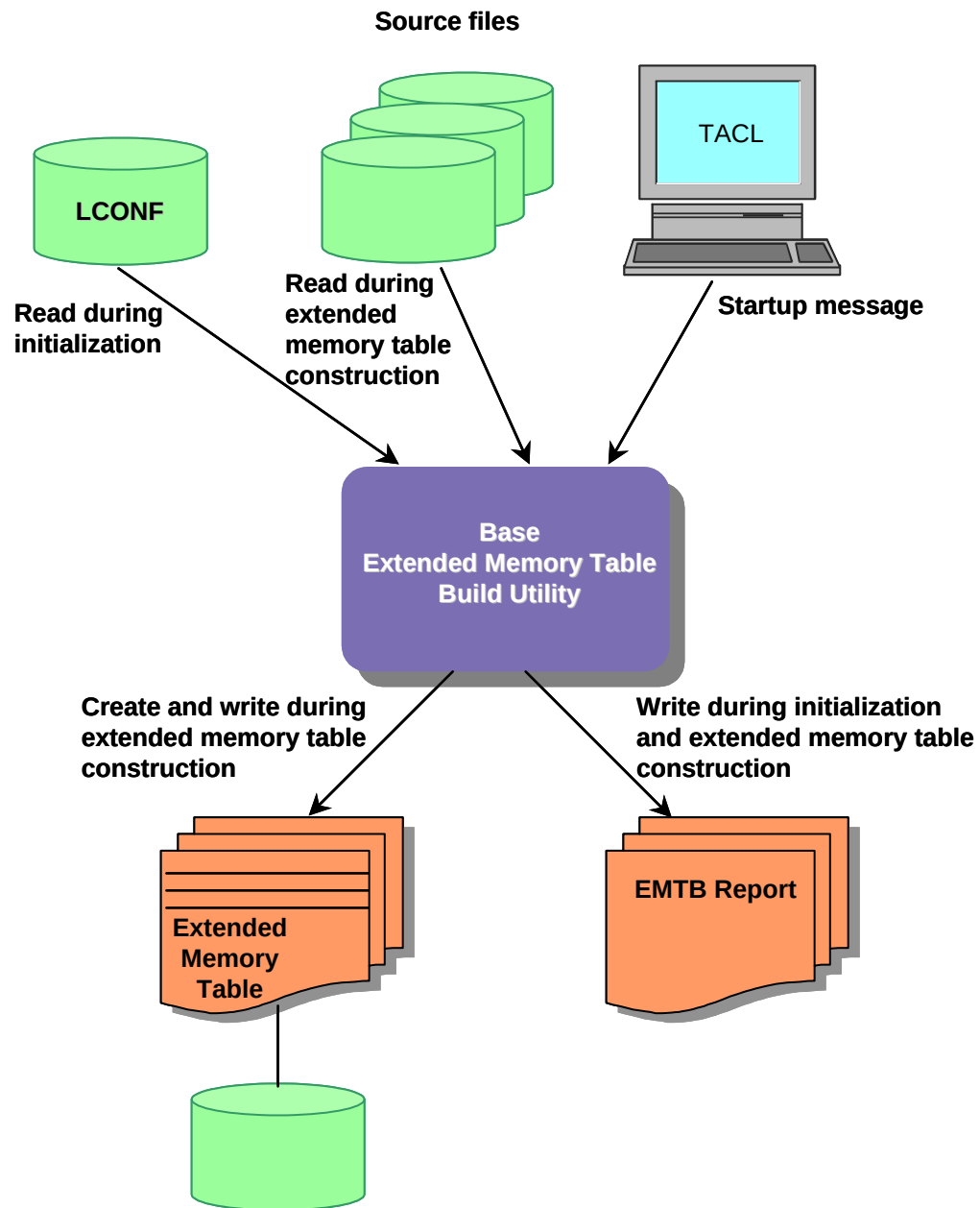
Shared extended memory tables are built in the background while the BASE24 system is processing.

The Base Extended Memory Table Build utility currently supports the following shared extended memory tables used by BASE24-pos processes.

- Acquirer Processing Code File (APCF) extended memory table
- Exchange Rate File (ERF) extended memory table (if the Multiple Currency add-on product is installed)
- Issuer Processing Code File (IPCF) extended memory table
- Response Code Description File (RCDF) extended memory table
- Split Transaction Routing File (STRF) extended memory table (used by the Transaction Context Manager process)

The Base Extended Memory Table Build utility creates an Extended Memory Table Build Report summarizing the processing performed each time it is executed. The Extended Memory Table Build Report is described later in this section.

The processing performed by the Base Extended Memory Table Build utility is depicted in the following illustration. A key to the abbreviations follows the illustration.



Key

EMTB Extended Memory Table Build utility
LCONF = Logical Network Configuration File

Allocating a Newly Built or Rebuilt Extended Memory Table

Once an extended memory table is built or rebuilt, it is not used until the application that uses the extended memory table has been informed of the change and reallocates (warmboots) the table. You can warmboot extended memory tables using the EMT Warmboot Control screen available from the Device Control Terminal (DCT) or by issuing ALTER ATTRIBUTE text commands from a network control facility. Both methods of warmbooting extended memory tables are described below.

Note: You must have network control security access to enter DELIVER PROCESS commands before you can use either of these methods.

EMT Warmboot Control Screen

Extended memory tables are reread by BASE24 processes when you issue an ALTER ATTRIBUTE text command on an extended memory table from the EMT Warmboot Control screen. For example, when you enter the APCF acronym in the ALTER ATTRIBUTE EMT field on this screen, all processes configured to use the APCF extended memory table in an APCFEMT-WARMIdentifier LCONF param reallocate the APCF extended memory table. An EMT-WARMIdentifier LCONF param exists for each extended memory table that can be warmbooted from the EMT Warmboot Control screen as shown in the following table.

Extended Memory Table	LCONF Warmboot Param
APCFEMT	APCFEMT-WARMIdentifier
ERFEMT	ERFEMT-WARMIdentifier
IPCFEMT	IPCFEMT-WARMIdentifier
RCDFEMT	RCDFEMT-WARMIdentifier
STRFEMT	STRFEMT-WARMIdentifier

After each process successfully reallocates the specified extended memory table, a *warmboot completed* message is displayed on the network log for the process. For detailed information on the EMT Warmboot Control screen, refer to the **BASE24 Device Control Manual**.

Network Control Facility

You can also allocate extended memory tables by entering DELIVER PROCESS commands directly from a network control facility. The text sent in these commands is “ALTER ATTRIBUTE *emt-name*,” where *emt-name* is either APCFEMT, ERFEMT, IPCFEMT, RCDFEMT, or STRFEMT. The command text is sent only to those processes identified as the object of the DELIVER PROCESS command. After each process to which the command is sent has successfully reallocated the specified extended memory table, a warmboot completed message is displayed on the network log for the process. For detailed information on the ALTER ATTRIBUTE text command, refer to the ***BASE24 Text Command Reference Manual***.

Source Files

Extended memory tables must be rebuilt anytime a change is made to the source files for the extended memory table. The source files for extended memory tables used by the BASE24-pos product are identified below.

Acquirer Processing Code File Extended Memory Table

The Acquirer Processing Code File (APCF) extended memory table must be rebuilt anytime a change is made to the Acquirer Processing Code File (APCF). The APCF contains one record for each combination of acquirer transaction profile, message type, and transaction code.

The following BASE24-pos modules and processes use the APCF extended memory table:

BASE24-pos Hypercom Device Handler module
BASE24-pos Standard POS Device Handler module
BASE24-pos Router/Authorization module

Banknet Interchange Interface process
BIC ISO Interchange Interface process
MDS/MDSM Interchange Interface process
PLUS ISO Interchange Interface process
VisaNet Interchange Interface process

Exchange Rate File Extended Memory Table

The Exchange Rate File (ERF) extended memory table must be rebuilt anytime a change is made to the Exchange Rate File (ERF). The ERF contains one record for each currency (except the Base currency) that is used by BASE24.

The following BASE24-pos modules and processes use the ERF extended memory table:

BASE24-pos Standard POS Device Handler module
BASE24-pos APACS 30/40 Device Handler module
BASE24-pos Router/Authorization module

Banknet Interchange Interface process
BIC ISO Interchange Interface process
MDS/MDSM Interchange Interface process
PLUS ISO Interchange Interface process
VisaNet Interchange Interface process

Issuer Processing Code File Extended Memory Table

The Issuer Processing Code File (IPCF) extended memory table must be rebuilt anytime a change is made to the Issuer Processing Code File (IPCF). The IPCF contains one record for each combination of issuer transaction profile, message type, and transaction code.

The following BASE24-pos modules and processes use the APCF extended memory table:

BASE24-pos Router/Authorization module

Banknet Interchange Interface process
BIC ISO Interchange Interface process
MDS/MDSM Interchange Interface process
PLUS ISO Interchange Interface process
VisaNet Interchange Interface process

Response Code Description File Extended Memory Table

The Response Code Description File (RCDF) extended memory table must be rebuilt anytime a change is made to the Response Code Description File (RCDF). The RCDF contains one record for each combination of terminal group, FIID, record type, account type, language indicator, and response code.

The BASE24-pos Standard POS Device Handler module uses the RCDF extended memory table.

Split Transaction Routing File Extended Memory Table

The Split Transaction Routing File (STRF) extended memory table must be rebuilt anytime a change is made to the Split Transaction Routing File (STRF). The STRF contains one record for each combination of transaction subtype and FIID.

The Transaction Context Manager process uses the STRF extended memory table.

LCONF Assigns and Params

The following LCONF assigns and params are used by the Base Extended Memory Table Build utility on behalf of BASE24 and BASE24-pos modules or processes. Other assigns and params used by the utility on behalf of BASE24-atm processes are listed in the ***BASE24-atm Transaction Processing Manual***.

Each of these assigns and params is described in detail in the ***BASE24 Logical Network Configuration File Manual***.

Assigns	Params
ALT-COLL	ALT-TIM-LMT
APCF	APCFEMT-MAX-TBL-SIZE
APCFEMT	CONSOLE-PRT
EMT-SWAPVOL	DATE-DISPLAY-FORMAT
ERF	ERFEMT-MAX-TBL-SIZE
ERFEMT	IPCFEMT-MAX-TBL-SIZE
RCDF	PRI-TIM-LMT
RCDFEMT	RCDFEMT-MAX-TBL-SIZE
IPCF	STRFEMT-MAX-TBL-SIZE
IPCFEMT	
PRI-COLL	
SPANCTL	
STRF	
STRFEMT	

Startup

This subsection describes the procedures for starting the Extended Memory Table Build utility. The utility is started from a TACL prompt and can be run using a high or low process identification number (PIN). For convenience, ACI provides an interactive TACL macro named EMTBRUN for starting the utility.

Note: If NET24-XPNET security is bound into the Extended Memory Table Build utility executable, you must have security access to the STATUS NODE network control command before using the Extended Memory Table Build utility.

EMTBRUN TACL Macro

ACI provides a TACL macro called EMTBRUN to operate the Extended Memory Table Build utility. This macro is installed on the BAXxEMTB subvolume, where *xx* is the number of the current release. If desired, you can add the EMTBRUN macro to the PMSEARCHLIST for the system or your TACL logon. When started from a TACL prompt, the macro prompts you for the individual file acronym of the extended memory table to be built (e.g., APCF, IPCF, ERF, etc.) or ALL (if you want to build all available extended memory tables). The macro also prompts you to enter the fully qualified name of the Logical Network Configuration File (LCONF) containing the Extended Memory Table Build utility assigns and params.

The Extended Memory Table Build Reports generated by the utility are sent to a spooler location of *\$\$.#emt.input-file*, where *input-file* is the acronym of the file from which the extended memory table is built.

Executing the Macro

When you run the Extended Memory Table Build utility using the EMTBRUN macro, the utility prompts you for all needed information. You provide the information required at each prompt by typing it and pressing the **Enter** key. You can exit the utility at any time by typing EXIT and pressing the **Enter** key or by pressing the **Break** key.

1. At the TACL prompt, enter the following command to start the Extended Memory Table Build utility, where *\$vol.subvol* identifies the volume and subvolume location of the TACL macro if it is not on the current volume and subvolume or defined in the PMSEARCHLIST.

```
[$vol.subvol.] EMTBRUN
```

After entering this command, the following text is displayed on the screen:

```
$vol.subvol.EMTBRUN
```

```
-----  
=  
= Welcome to the EMT Build Run TACL macro.                               =  
=  
= This TACL macro is used to build Extended Memory Table                 =  
= files (EMTs).                                                           =  
=  
= An EMT will be rebuilt using the EMT Build utility.  It                =  
= reads data from the appropriate database files on the                  =  
= logical network specified in the LCONF.  This EMT Build                =  
= utility creates extended memory table files created                   =  
= specifically for use by BASE24 applications.                           =  
=  
-----
```

Available EMT choices:

APCF ERF IDF IPCF STRF RCPT TRF RCDF

ALL or EXIT

Enter EMT choice ->

2. Enter one of the extended memory table input file (source file) acronyms or ALL for the extended memory table or tables to be built.

Prompt: Enter EMT choice ->

Entry: Source file acronym, ALL, or EXIT

After entering a valid choice, the following text is displayed on the screen:

```
Enter LCONF filename  
or EXIT
```

LCONF filename ->

3. Enter the fully qualified name of the Logical Network Configuration File (LCONF).

Prompt: LCONF filename ->

Entry: The fully qualified name of the LCONF or EXIT. If the LCONF resides on the same volume or subvolume as your current volume and subvolume, the volume or subvolume name is not required.

After entering the LCONF location, the utility starts running. The following is an example of the text displayed on the screen:

Running EMT Build Program

The LCONF file: \$volume.PRODCNTL.L1CONF
EMTS : APCFEMT
EMT Build Obj : \$volume.BA60OBJ.BEMTBLO

\$volume.BA60OBJ.BEMTBLO/NOWAIT,OUT \$S.#EMT.APCF/ LCONF
\$volume.PRODCNTL.L1CONF,APCFEMT

Check Spooler to verify completion status

Spooler Name is: \$s.#emt.apcf

4. Check the spooler reports to make sure the requested extended memory tables were successfully built.
5. Warmboot the extended memory tables built in steps 1–4 using the EMT Warmboot Control screen or the ALTER ATTRIBUTE text commands.

Extended Memory Table Build Report

The Extended Memory Table Build Report provides a summary of all processing performed by the Base Extended Memory Table Build utility. A sample of the Extended Memory Table Build Report is shown below and on the following pages. Each sample page is followed by an explanation of each field appearing on the report page.

The title of the report and the current page number are printed at the top of each report page.

In addition to the fields documented here, various self-explanatory exception messages may be written to the report at any time indicating exceptions that occur during processing. If an exception causes the build utility to abend, the phrase “*emt-name* BUILD HALTED” will be written to the report, where *emt-name* is the name of the extended memory table being built when the exception occurs.

Cover Page

The cover page is printed at the beginning of the Extended Memory Table Build Report. It summarizes the start date and time and lists the extended memory tables that were requested to be built.

BASE24	EXTENDED MEMORY TABLE BUILD REPORT	PAGE: 1
Start Date:	YY/MM/DD	
Start Time:	HH:MM:SS	
EMT Swap Files Requested:		
	APCFEMT	
	IPCFEMT	

Start Date — The date, in the format specified by the DATE-DISPLAY-FORMAT LCONF param, that the Base Extended Memory Table Build utility began processing.

Start Time — The time of day (HH:MM:SS) the Base Extended Memory Table Build utility began processing.

EMT Swap Files Requested — The names of the extended memory tables requested to be built.

Hash Table Summary Page

A Hash Table Summary report page is printed for each extended memory table built with the exception of some core extended memory tables that have their own report page format. For descriptions of any core-specific report pages, refer to the product-specific transaction processing manual.

The Hash Table Summary page details the hashing algorithms and bucket factors used to build the optimum hash table for a file or table.

BASE24	EXTENDED MEMORY TABLE BUILD REPORT	PAGE: <i>n</i>
Table Type:	APCFEMT	
Source File:	\SYS.\$DATA.PRO1DATA.APCF	
EMT Swap File:	\SYS.\$DATA.PRO1DATA.APCFEMT	
Table Size:	135470 bytes	
Total Data Records:	2356	
Maximum Record Length:	112 bytes	
Key Type:	ALL STRING	
Key Length:	24 bytes	
Key Algorithm:	DOUBLE BYTE SUMMATION	
Hash Algorithm:	MODULO DIVIDE	
Bucket Factor:	2	
Total Buckets:	29	
Total Buckets Used:	13	
Percent Buckets Used:	44%	
Total Barrels:	1	
Total Collisions:	0	
Percent Collisions:	0%	
Total Collision Chains:	0	
Longest Collision Chain:	0	
Average Collision Chain:	0	
Build Attempts:	24	

Table Type — The acronym for the extended memory table being built.

Source File — The fully qualified location of the source file or table on disk from which the extended memory table is to be built.

EMT Swap File — The fully qualified location of the swap file used to build this extended memory table. This is the name of the actual extended memory table file on disk. This file is opened by the object when it allocates the extended memory segment. The swap file is used to swap out records from disk as they are added to the table in memory.

Table Size — The total size, in bytes, required for the extended memory table.

Total Data Records — The total number of data records or rows included in the extended memory table.

Maximum Record Length — The maximum length, in bytes, of each record included in the extended memory table.

Key Type — The type of record or row key used in this extended memory table. Valid types are as follows: ALL ASCII (alphanumeric) and ALL BINARY (the binary data cannot exceed four bytes).

Key Length — The total length, in bytes, of the primary key to this extended memory table.

Key Algorithm — The algorithm used to hash the key from this file or table in building the optimum hash table. The Base Extended Memory Table Build utility tries all combinations of allowed hash algorithms, key algorithms, and bucket factors until a table with the fewest collisions is built. Valid key algorithms include the following: SINGLE BYTE XOR, DOUBLE BYTE XOR, TRIPLE BYTE XOR, QAUDRUPLE BYTE XOR, SINGLE BYTE SUMMATION, DOUBLE BYTE SUMMATION, and SINGLE BYTE POLYNOMIAL.

Hash Algorithm — The algorithm used to hash key values from this file or table in building the optimum hash table. The Base Extended Memory Table Build utility tries all combinations of allowed hash algorithms, key algorithms, and bucket factors until a table with the fewest collisions is built. Valid hash algorithms include the following: MULTIPLICATIVE, MID-SQUARE, and MODULO DIVIDE.

Bucket Factor — The bucket factor used in building the hash table. The bucket factor is the ratio of the number of buckets that will exist as opposed to the actual number of data records in the hash table. The Base Extended Memory Table Build utility tries all combinations of allowed hash algorithms, key algorithms, and bucket factors until a table with the fewest collisions is built.

Total Buckets — The total number of buckets used for all combinations of allowed hash algorithms, key algorithms, and bucket factors attempted.

Total Buckets Used — The total number of buckets used in the optimum hash table. The Base Extended Memory Table Build utility tries all combinations of allowed hash algorithms, key algorithms, and bucket factors until a table with the fewest collisions is built.

Percent Buckets Used — The percentage of buckets containing data items in the optimum hash table.

Total Barrels — The total number of barrels used in the optimum hash table. A barrel is equal to a group of buckets.

Total Collisions — The total number of collisions that occurred on the optimum hash table. A collision occurs when two or more data records or rows hash to the same bucket.

Percent Collisions — The percentage of collisions that occurred on the optimum hash table. A collision occurs when two or more data records or rows hash to the same bucket.

Total Collision Chains — The total number of collision chains that occurred on the optimum hash table. A collision chain occurs when two or more data records or rows hash to the same bucket.

Longest Collision Chain — The longest collision chain that occurred on the optimum hash table. A collision chain occurs when two or more data records or rows hash to the same bucket.

Average Collision Chain — The average length of all collision chains that occurred on the optimum hash table. A collision chain occurs when two or more data records or rows hash to the same bucket.

Build Attempts — The total number of combinations attempted to obtain the optimum hash table.

Trailer Page

The trailer page is printed at the end of the Extended Memory Table Build Report. It summarizes the total number of swap files successfully built, the total number of swap files that failed, as well as the date and time processing completed.

BASE24	EXTENDED MEMORY TABLE BUILD REPORT	PAGE: <i>n</i>
EMT Swap Files Built:		
	APCFEMT	
	IPCFEMT	
EMT Swap Files Failed:		
End Date:	YY/MM/DD	
End Time:	HH:MM:SS	

EMT Swap Files Built — The acronyms identifying extended memory tables successfully built in this run.

EMT Swap Files Failed — The acronyms identifying extended memory tables that failed to be built in this run.

End Date — The date, in the format specified by the DATE-DISPLAY-FORMAT LCONF param, that this run of the Base Extended Memory Table Build utility ended.

End Time — The time of day (HH:MM:SS) this run of the Base Extended Memory Table Build utility ended.

Index

A

Account Type Table File (ATT), [1-13](#)

Accounts

- assigning account names, [1-13](#)
- assigning account types, [1-12](#)
- credit accounts, [1-11](#)
- limits and accumulators, [1-12](#)
- noncredit accounts, [1-11](#)
- relationship between accounts and cards, [1-8](#)

Accumulation totals

- in the CAF, [6-3](#)
- in the UAF, [6-3](#)

ACI Payments Manager product, [1-41](#)

ACI Settlement Manager product, [1-41](#)

Acquirer

- definition of, [1-6](#)
- host, [1-2](#)

Acquirer Processing Code File (APCF)

- account type names, [1-13](#)
- source file for APCF extended memory table, [F-5](#)

Acquirer Processing Code File (APCF) extended memory table

- building, [F-5](#)
- processing for interchange acquirer transaction profiles, [3-17](#)
- processing for retailer and administrative card transaction profiles, [3-11](#)
- processing for terminal owner acquirer transaction profiles, [3-10](#)

Acquirer Routing token

- APCF-routed, approved, [5-75](#)
- APCF-routed, interchange, [5-79](#)
- APCF-routed, with logging, [5-77](#)

Acquirer transaction profile

- transactions allowed at the terminal, [3-10](#)
- transactions allowed from an interchange, [3-17](#)

Address verification, [3-66](#)

Adjustments

- definition of, [1-33](#)

Administrative card transaction profile, [3-11](#)

Administrative transactions

- message flow diagrams, [5-63](#)
- supported, [1-32](#)

Advice, definition of, [1-33](#)

Algorithms

- AMEX, [4-31](#)
- BASE (Visa), [4-31](#)
- BNET (MasterCard), [4-31](#)
- DFLT (Default Routing), [4-32](#)
- DSCV (Discover), [4-31](#)
- INAS (MasterCard), [4-31](#)
- LGNT (Logical Network), [4-31](#)
- VISA, [4-31](#)

ALTER ATTRIBUTE text command, [F-4](#)

APCF

- see* Acquirer Processing Code File (APCF)

APCF-routed transactions, message flow diagrams, [5-74](#)

Approval code, generating, [4-53](#)

AST

- see* Authorization Selection Table (AST)

AST routing

- example of, [4-18](#)
- for check authorization transactions, [4-48](#)
- record entry selection hierarchy, [4-11](#)
- record selection hierarchy, [4-10](#)
- selecting an AST record, [4-12](#)
- setting up the Authorization Selection Table (AST), [4-15](#)

ATT

- see* Account Type Table File (ATT)

Authorization levels

- offline authorization (level 2), [3-4](#), [5-34](#)
- online authorization (level 1), [3-3](#), [5-5](#)
- online/offline authorization (level 3), [3-4](#), [5-47](#)

Authorization message types, [1-34](#)

Authorization methods

- Negative Authorization with Usage Accumulation, [3-5](#), [3-38](#)
- Negative Authorization without Usage Accumulation, [3-6](#)
- online host processing, [3-5](#)
- Parametric Authorization, [3-6](#), [C-1](#)
- Positive Authorization, [3-6](#), [3-43](#)
- Positive Balance Authorization, [3-6](#), [3-38](#), [3-43](#)

Authorization module, functions of, [2-6](#), [4-5](#)

Authorization Selection Table (AST)

- how used, [4-18](#)
- information in the, [4-6](#)
- record entry selection hierarchy, [4-11](#)
- record selection hierarchy, [4-10](#)
- sample settings, [4-15](#)
- selecting an AST record, [4-12](#)

Authorizers
 see Transaction authorizers
Authorizing for a different amount than requested, [3-50](#)
Authorizing for a greater amount than requested, [3-51](#)

B

Balance Inquiry message types, [1-36](#)
Balance totals in PBF, [6-2](#)
BASE24 (BIC) ISO Interface process, functions of, [2-9](#)
BASE24 Standard POS Device Handler (SPDH) module, [2-5](#)
BASE24 system, as a transaction authorizer, [1-3](#)
BASE24-pos Terminal Data files
 allowed service types, [3-18](#)
 allowed transactions, [3-8](#)
 draft capture processing, [3-30](#), [3-31](#)
 transaction limits, [3-33](#), [3-37](#)
BASE24-pos Terminal Data files
 authorizing for a lesser amount than requested, [3-50](#)
 preauthorized holds processing, [3-46](#)
Batch upload message types, [1-37](#)
BIC ISO Interface
 see BASE24 (BIC) ISO Interface process
BUILDFILESIZE command, Pregen utility, [E-2](#)
BUILDSPROUTE command, Pregen utility, [E-3](#)

C

CAF
 see Cardholder Authorization File (CAF)
CAPF
 see Card Authorization Parameters File (CAPF)
Card Authorization Parameters File (CAPF)
 cardholder and transaction limits, [3-36](#)
 Parametric Authorization method processing, [3-6](#), [C-2](#)
Card Prefix File (CPF)
 cardholder and transaction limits, [3-33](#)
 chargeback processing, [3-60](#)
 preauthorized holds processing, [3-46](#)
 representment processing, [3-60](#)
 transaction routing, [4-47](#)
 transaction screening, [3-28](#)
Card status, [3-42](#)
Card types
 card type codes, [1-9](#)
 reserved card type codes, [1-9](#)
 user-defined card type codes, [1-11](#)
Card verification, [3-22](#)
Card verification digits (CVD), [3-27](#)
Cardholder Authorization File (CAF)
 accumulation totals, [6-3](#)
 address verification processing, [3-66](#)
 cardholder and transaction limits, [3-33](#), [3-35](#)
 Parametric Authorization method, [3-6](#)

 Positive Authorization method, [3-6](#)
 Positive Balance Authorization method, [3-6](#)
 preauthorized holds processing, [3-45](#)
 transaction screening, [3-26](#)
 usage accumulation, [3-38](#)

Cards

 combination cards, [1-8](#)
 credit cards, [1-8](#)
 debit cards, [1-8](#)
 dual cards, [1-8](#)
 relationship between cards and accounts, [1-8](#)

CART

see Check Authorization Routing Table File (CART)

Chargeback

 definition of, [3-54](#)
 originating from a BIC co-network, [3-57](#)
 originating from a host, [3-56](#)
 originating from an interchange, [3-58](#)
 originating from CRT Administration, [3-56](#)
 originating sources, [3-55](#)
 preauthorized hold completion, [3-44](#)
 processing example, [3-56](#), [3-57](#), [3-58](#)
 response codes for, [3-55](#)
 support by the BASE24-pos product, [3-54](#)

Check Authorization Routing Table File (CART), routing
 check authorization transactions, [4-46](#), [4-49](#)

Check authorization transactions, routing, [4-11](#), [4-46](#)

Check Guarantee/Verification 2 token, [4-51](#)

Check Guarantee/Verification token, [4-12](#)

Combination cards, [1-8](#)

Completion acknowledgment timer, [2-20](#), [2-22](#)

Completion timer, [2-22](#)

Completions

 definition, [3-61](#)
 generating to a host, [3-61](#)

Co-network

 as a transaction authorizer, [1-4](#)
 as a transaction originator, [1-3](#)

CPF

see Card Prefix File (CPF)

Credit accounts, [1-11](#)

Credit cards, [1-8](#)

CRT Administration Device Handler module, as a
 transaction originator, [1-3](#)

CRT Authorization Device Handler module, as a
 transaction originator, [1-3](#)

CRT Authorization Device Handler module, token
 handling, [3-64](#)

Cutover, terminal cutover, [2-4](#)

D

Data processing center (DPC)
 as defined to the BASE24 system, [1-7](#)
 definition, [1-7](#)

- Database, transaction impacts
 - BASE24-pos Terminal Data files, [6-13](#)
 - Cardholder Authorization File (CAF), [6-2](#)
 - Positive Balance File (PBF), [6-2](#)
 - Usage Accumulation File (UAF), [6-2](#)
 - Debit cards, [1-8](#)
 - Delinquency history, [C-2](#)
 - DELIVER PROCESS command, [F-4](#)
 - Derivation Key File (KEYD), [3-24](#)
 - Derived unique key per transaction (DUKPT) processing, [3-24](#)
 - Device Handler module
 - BASE24 Standard POS Device Handler (SPDH) module, [2-5](#)
 - CRT Administration Device Handler module, [2-6](#)
 - CRT Authorization Device Handler module, [2-5](#)
 - functions of, [2-3](#)
 - Hypercom Device Handler (HPDH) module, [2-5](#)
 - Merchant Host Interface (MHI) Device Handler module, [2-5](#)
 - timers, [2-18](#)
 - Visa I Device Handler module, [2-5](#)
 - Visa II Device Handler module, [2-5](#)
 - Device Handler timer, [2-18](#)
 - Device Handler/Router/Authorization process
 - as a processing link, [2-13](#), [2-14](#)
 - inbound messages from POS terminals, [2-11](#)
 - Downline load, [2-4](#)
 - DPC
 - see Data processing center (DPC)
 - Draft capture flag
 - as set by CRT Authorization Device Handler module, [D-4](#)
 - as set by other Device Handler modules, [D-4](#)
 - possible settings, [D-2](#)
 - processing according to, [D-2](#)
 - settling according to, [D-2](#)
 - Draft capture options
 - authorization and draft capture, [3-31](#)
 - authorization only with electronic follow-up, [3-30](#)
 - authorization with electronic follow-up, [3-31](#)
 - terminal-defined draft capture, [3-32](#)
 - Draft capture transaction, definition of, [3-30](#)
 - Dual cards, [1-8](#)
 - Dynamic key management, [3-25](#)
- ## E
- EMT Warmboot Control screen, [F-4](#)
 - EMTBRUN macro, [F-9](#)
 - Enhanced Interchange Configuration File (ICFE)
 - identifying Device Handler/Router/Authorization processes for inbound transactions, [2-14](#)
 - store-and-forward processing, [3-40](#)
 - Enhanced preauthorized holds
 - see Preauthorized holds
 - Enhanced reversal routing, [4-52](#)
 - Exchange Rate File (ERF) extended memory table, building, [F-6](#)
 - Exchange Rate File (ERF), source file for ERF extended memory table, [F-6](#)
 - Expiration date processing, [3-43](#)
 - Extended Memory Table Build Report
 - cover page, [F-12](#)
 - hash table summary page, [F-13](#)
 - introduction, [F-12](#)
 - trailer page, [F-16](#)
 - Extended Memory Table Build utility
 - allocating an extended memory table, [F-4](#)
 - EMTBRUN TACL macro, [F-9](#)
 - illustration of, [F-2](#)
 - LCONF assigns and params, [F-8](#)
 - overview, [F-2](#)
 - PIN support, [F-9](#)
 - processing summary report, [F-12](#)
 - source files, [F-5](#)
 - Extended network timer, [2-19](#), [2-21](#)
- ## F
- Financial transaction message types, [1-35](#)
 - Financial transactions, supported, [1-16](#)
 - First-generation terminal
 - capabilities of, [2-4](#)
 - draft capture capabilities, [3-31](#)
 - Force post, definition of, [1-33](#)
- ## G
- GOPRE
 - see Prgen utility Go file
- ## H
- Hard-coded authorizers
 - AUTHF, [4-8](#)
 - AUTHH, [4-8](#)
 - DFLT, [4-9](#)
 - NEGF, [4-9](#)
 - NEGN, [4-9](#)
 - HCF
 - see Host Configuration File (HCF)
 - High PIN support, Extended Memory Table Build utility, [F-9](#)
 - Host
 - acquirer, [1-2](#)
 - as a processing link, [2-13](#), [2-14](#)
 - as a transaction authorizer, [1-4](#)
 - as a transaction originator, [1-2](#)
 - definition, [1-7](#)
 - inbound messages from, [2-11](#)

Host Configuration File (HCF)
controlling store-and-forward processing, [3-40](#)
defining DPCs and hosts, [1-7](#)
identifying Device Handler/Router/Authorization processes for inbound transactions, [2-14](#)

Host Interface process
see ISO Host Interface process

Hypercom Device Handler (HPDH) module, [2-5](#)

Hypercom Device Handler timer, [2-19](#)

I

ICF
see Interchange Configuration File (ICF)

ICFE
see Enhanced Interchange Configuration File (ICFE)

IDF
see Institution Definition File (IDF)

IDF routing
example of, [4-44](#)
using the IDF routing table, [4-44](#)

IDF routing table
examples of, [4-40](#)
how used, [4-44](#)
information in the, [4-38](#)
preauthorized holds, [4-39](#)

Inbound message timer
Interchange Interface process, [2-22](#)
ISO Host Interface process, [2-20](#)

Institution Definition File (IDF)
authorization levels, [3-3](#)
authorization methods, [3-5](#)
generating completions to a host, [3-61](#)
preauthorized holds processing, [3-45](#)
routing and authorization selection, [4-37](#)
transaction screening, [3-26](#), [3-27](#), [3-28](#)
usage accumulation controls, [3-38](#)

Interchange Configuration File (ICF)
identifying Device Handler/Router/Authorization processes for inbound transactions, [2-14](#)
store-and-forward processing, [3-40](#)

Interchange Interface process
functions, [2-8](#)
inbound messages from interchanges, [2-11](#), [2-12](#)
receiving messages from a Device Handler/Router/Authorization process, [2-14](#)
sending messages to a Device Handler/Router/Authorization process, [2-14](#)
timers, [2-21](#)

Interchange transactions, message flow diagrams, [5-67](#)

Interchanges
as processing links, [2-14](#)
as transaction authorizers, [1-4](#)
as transaction originators, [1-3](#)
inbound messages from, [2-11](#)

Intermediary networks, [1-6](#)

IPCF
see Issuer Processing Code File (IPCF)

ISO Host Interface process
functions of, [2-7](#)
inbound messages from hosts, [2-11](#), [2-12](#)
receiving messages from a Device Handler/Router/Authorization process, [2-13](#)
sending messages to a Device Handler/Router/Authorization process, [2-14](#)
timers, [2-19](#)

Issuer Processing Code File (IPCF)
account type names, [1-13](#)
generating completions to a host, [3-61](#)
source file for IPCF extended memory table, [F-6](#)

Issuer Processing Code File (IPCF) extended memory table
building, [F-6](#)
processing for interchange issuer transaction profiles, [3-15](#)
processing for issuer transaction profiles, [3-13](#)

Issuer transaction profile
transactions allowed by the card issuer, [3-13](#)
transactions allowed to an interchange, [3-15](#)

Issuer, definition of, [1-6](#)

K

Key 6 File (KEY6), [3-23](#)

Key Authorization File (KEYA), [3-22](#)

Key File (KEYF), [3-23](#)

KEY6
see Key 6 File (KEY6)

KEYA
see Key Authorization File (KEYA)

KEYD
see Derivation Key File (KEYD)

KEYF
see Key File (KEYF)

L

Level 1 authorization
definition of, [3-3](#)
message flow diagrams, [5-5](#)

Level 2 authorization
definition of, [3-4](#)
message flow diagrams, [5-34](#)

Level 3 authorization
definition of, [3-4](#)
message flow diagrams, [5-47](#)

Linkages, process, [2-11](#)

Logging, transaction
message types logged, [1-40](#)
multiple logical networks, [1-41](#)
tokens logged, [1-41](#)

Logical Network Configuration File (LCONF) assigns

- ALT-COLL, [F-8](#)
- APCF, [F-8](#)
- APCFEMT, [F-8](#)
- CRT-AUTH-TQFT, [3-64](#)
- EMT-SWAPVOL, [F-8](#)
- ERF, [F-8](#)
- ERFEMT, [F-8](#)
- IPCF, [F-8](#)
- IPCFEMT, [F-8](#)
- PRI-COLL, [F-8](#)
- RCDF, [F-8](#)
- RCDFEMT, [F-8](#)
- SPANCTL, [F-8](#)
- STRF, [F-8](#)
- STRFEMT, [F-8](#)

Logical Network Configuration File (LCONF) params

- ALT-TIM-LMT, [F-8](#)
- APCFEMT-MAX-TBL-SIZE, [F-8](#)
- CONSOLE-PRT, [F-8](#)
- CRT-AUTH-MAX-TQFTS, [3-64](#)
- CRT-AUTH-TKN-RETRV-OPT, [3-64](#)
- DATE-DISPLAY-FORMAT, [F-8](#)
- DES-TRIPLE-SINGLE, [3-23](#)
- ENHNC-PRE-AUTH-HOLD-STORE-TIME, [3-48](#)
- ERFEMT-MAX-TBL-SIZE, [F-8](#)
- IPCFEMT-MAX-TBL-SIZE, [F-8](#)
- POS-CHRGBCCK-NO-RESP-ORIG, [3-55](#)
- POS-RTAU-AST-ROUTE, [4-13](#)
- POS-RVSL-ROUTE, [4-52](#)
- POS-SPLIT-TXN-RTG-DEST, [2-13](#)
- PRI-TIM-LMT, [F-8](#)
- RCDFEMT-MAX-TBL-SIZE, [F-8](#)
- STRFEMT-MAX-TBL-SIZE, [F-8](#)

M

Merchant Host Interface (MHI) Device Handler module, [2-5](#)

Message authentication, [3-23](#)

Message flow diagrams

- general information, [5-1](#)
- host multithreading processing assumption, [5-4](#)
- host text-level acknowledgment processing assumption, [5-4](#)

Message flow diagrams for administrative transactions

- approved transaction, [5-63](#)
- denied transaction, [5-65](#)

Message flow diagrams for APCF-routed transactions

- device acquired—approved transaction—logged to PTLF, [5-76](#)
- device acquired—approved transaction—not logged to PTLF, [5-74](#)
- interchange acquired—approved transaction—logged to PTLF, [5-78](#)

Message flow diagrams for authorization level 1

- alternate authorization not possible, [5-18](#)
- approved financial transaction, [5-8](#)
- approved financial transaction with completions, [5-10](#)

- approved preauthorization, [5-6](#)
- approved with completions, [5-27](#)
- denied financial transaction, [5-13](#)
- interchange as alternate authorizer (denied), [5-15](#)
- reversal with host acknowledgments, [5-24](#), [5-30](#)
- reversal without host acknowledgments, [5-21](#)

Message flow diagrams for authorization level 2

- approved with completions, [5-37](#)
- approved with host acknowledgments, [5-45](#)
- approved with store-and-forward completions, [5-39](#)
- approved without completions, [5-35](#)
- denied transaction, [5-41](#)
- reversal after completion is sent, [5-43](#)

Message flow diagrams for authorization level 3

- approved offline with completions, [5-50](#)
- approved offline with SAF completion, [5-56](#)
- approved online, [5-48](#)
- approved online with completions, [5-52](#)
- denied offline, [5-58](#)
- denied online, [5-54](#)

Message flow diagrams for interchange transactions

- approved transaction, [5-67](#)
- reversal due to downed device, [5-71](#)
- reversal on timed-out transaction, [5-69](#)

Message flow diagrams for mobile top-up transactions

- with cash, approved by telephone company host, [5-81](#)
- with cash, declined by telephone company host, [5-83](#)
- with funds, approved by BASE24 and telephone company host, [5-85](#)
- with funds, declined by Authorization module, [5-88](#)
- with funds, declined by telephone company host, [5-91](#)

Message flow diagrams for multi-nodal transactions

- approved on another XPNET node, [5-60](#)

Messages

- authorization message types, [1-34](#)
- balance inquiry message types, [1-36](#)
- batch upload message types, [1-37](#)
- external messages types, [1-33](#)
- financial transaction message types, [1-35](#)
- functions, [1-33](#)
- internal message types, [1-33](#)
- message flow diagrams, [5-1](#)
- message types logged, [1-40](#)
- network management message types, [1-39](#)
- reconciliation control message types, [1-38](#)
- reversal message types, [1-37](#)
- types, [1-33](#)

Multi-nodal transactions message flow diagrams, [5-60](#)

Multiple Logical Network token, [1-41](#)

N

National Negative Card File (NNF), [4-4](#)

NEG

- see* Negative Card File (NEG)

Negative Authorization with Usage Accumulation method
 card status, [3-27](#)
 description of, [3-5](#)
 usage accumulation totals, [3-38](#)

Negative Authorization without Usage Accumulation method
 card status, [3-27](#)
 description of, [3-6](#)

Negative Card File (NEG)
 Negative Authorization with Usage Accumulation method, [3-5](#)
 Negative Authorization without Usage Accumulation method, [3-6](#)
 transaction screening, [3-27](#)

Network management message types, [1-39](#)

Network management timer
 Interchange Interface process, [2-21](#)
 ISO Host Interface process, [2-19](#)

NNF
 see National Negative Card File (NNF)

Noncredit accounts, [1-11](#)

Non-draft capture transaction, definition of, [3-30](#)

O

Offline authorization (level 2)
 definition of, [3-4](#)
 message flow diagrams, [5-34](#)

Online authorization (level 1)
 definition of, [3-3](#)
 message flow diagrams, [5-5](#)
 transaction screening, [3-4](#)

Online host processing, [3-5](#)

Online/offline authorization (level 3)
 definition of, [3-4](#)
 message flow diagrams, [5-47](#)

Outbound message timer
 Interchange Interface process, [2-21](#)
 ISO Host Interface process, [2-19](#)

P

Parametric Authorization method
 available credit, [C-4](#)
 delinquency history information, [C-2](#)
 example of, [C-9](#)
 overlimit amounts, [C-4](#)
 processing explanation, [C-1](#)
 single transaction limits, [C-7](#)

PBF
 see Positive Balance File (PBF)

Performance period timer, [2-21](#)

PIN encryption, [3-22](#)

PIN verification methods, [3-21](#)

Point-of-sale (POS) devices
 as transaction originators, [1-2](#)
 first-generation, [2-4](#)
 inbound messages from, [2-11](#)
 second-generation, [2-4](#)
 third-generation, [2-4](#)

POS Retailer Definition File (PRDF)
 allowed service types, [3-20](#)
 routing, [4-3](#)
 transaction limits, [3-36](#)

POS Standard Internal Message (PSTM) fields
 ACQ-ICHG-SETL-DAT, [A-37](#)
 ACQ-INST-ID-NUM, [A-14](#)
 ADDR-FLDS, [A-46](#)
 ADDR-FLDS.ADDR, [A-46](#)
 ADDR-FLDS.ADDR-VRIFY-STAT, [A-47](#)
 ADDR-FLDS.USER-FLD1, [A-49](#)
 ADDR-TYP, [A-46](#)
 ADJ-FLG, [A-34](#)
 ADMIN-SEC-IND, [A-36](#)
 ANSI-OFST, [A-42](#)
 AST-RTN-PRO-NAME, [A-5](#)
 AUTH-IND2, [A-10](#)
 BATCH-NUM, [A-14](#)
 BATCH-SEQ-NUM, [A-13](#)
 BIRTH-DAT, [A-49](#)
 BRCH-ID, [A-10](#)
 CHECK-ID-NUM, [A-46](#)
 CHECK-ID-TYP, [A-46](#)
 CLERK-ID, [A-11](#)
 COMPL-REQ, [A-34](#)
 CRD-ACCPT-ID-NUM, [A-44](#)
 CRD-ISS-ID-NUM, [A-44](#)
 CRT-AUTH, [A-11](#)
 CRT-AUTH.GRP, [A-12](#)
 CRT-AUTH.USER-ID, [A-12](#)
 DATA-FLAG, [A-50](#)
 DPC-NUM, [A-3](#)
 ENTRY-TIM, [A-43](#)
 EXIT-TIM, [A-43](#)
 FRWD-INST-ID-NUM, [A-43](#)
 HOST-TRACE-NUM, [A-42](#)
 INVOICE-NUM, [A-13](#)
 ISS-ICHG-SETL-DAT, [A-37](#)
 MSG-ORIG-IND, [A-5](#)
 NUM-OF-CHRGBCK, [A-45](#)
 NUM-SERVICES, [A-50](#)
 ORIG-CRNCY-CDE, [A-10](#)
 ORIG-DATA, [A-44](#)
 ORIG-DATA.B24-POST-DAT, [A-45](#)
 ORIG-DATA.MSG-TYP, [A-44](#)
 ORIG-DATA.TRAN-SEQ-NUM, [A-45](#)
 ORIG-DATA.TRN-DAT, [A-45](#)
 ORIG-DATA.TRN-TIM, [A-45](#)
 ORIGINATOR, [A-6](#)
 ORIG-INVOICE-NUM, [A-13](#)
 ORIG-PRO-NAME, [A-4](#)
 PIN, [A-40](#)
 PIN-FRMT, [A-41](#)
 PIN-KEY, [A-41](#)
 PINPAD-CHAR, [A-42](#)

POS Standard Internal Message (PSTM) fields *continued*

PIN-SIZE, [A-40](#)
 PIN-TRIES, [A-41](#)
 POSTAL-CDE, [A-49](#)
 POST-DAT, [A-7](#)
 PRE-AUTH-OPT, [A-35](#)
 PRE-AUTH-SEQ-NUM, [A-12](#)
 PROD-ID, [A-3](#)
 PT-SRV-COND-CDE, [A-38](#)
 PT-SRV-ENTRY-MDE, [A-39](#)
 RCV-INST-ID-NUM, [A-15](#)
 REA-FOR-CHRGBCK, [A-45](#)
 RE-ENTRY-TIM, [A-43](#)
 RESPONDER, [A-6](#)
 RETL-GRP, [A-16](#)
 RETL-ID, [A-15](#)
 RETL-REGN, [A-15](#)
 RETL-SIC-CDE, [A-16](#)
 ROUTER1-NAME, [A-5](#)
 ROUTER2-NAME, [A-5](#)
 RTE, [A-30](#)
 RTE.ALT1, [A-32](#)
 RTE.ALT2, [A-32](#)
 RTE.AST-FND, [A-33](#)
 RTE.AUTH-IND, [A-32](#)
 RTE.DFLT, [A-33](#)
 RTE.HIT-NNEG, [A-33](#)
 RTE.ISS, [A-30](#)
 RTE.OFFL-LMT, [A-31](#)
 RTE.ONL-LMT, [A-31](#)
 RTE.OVER-LMT, [A-31](#)
 RTE.PRI, [A-31](#)
 RTE.RETL-PROG, [A-30](#)
 RTE.SRV, [A-30](#)
 RTE-STAT, [A-4](#)
 RVSL-CDE, [A-37](#)
 SEQ-NUM, [A-12](#)
 SHIFT-NUM, [A-14](#)
 SRVCS, [A-50](#)
 SRVCS.ELEC-FOLLOWUP, [A-51](#)
 SRVCS.FLR-LMT, [A-52](#)
 SRVCS.TRAN-LMT, [A-52](#)
 SRVCS.TRAN-PROFILE, [A-51](#)
 SRVCS.TYP, [A-51](#)
 STATE-CDE, [A-49](#)
 TERM-CITY, [A-9](#)
 TERM-CNTRY-CDE, [A-9](#)
 TERM-FIID, [A-7](#)
 TERM-ID, [A-8](#)
 TERM-LN, [A-7](#)
 TERM-NAME-LOC, [A-8](#)
 TERM-OWNER-NAME, [A-8](#)
 TERM-ST, [A-9](#)
 TERM-TIM-OFST, [A-11](#)
 TERM-TYP, [A-10](#)
 TRAN, [A-17](#)
 TRAN.ACCT, [A-20](#)
 TRAN.AMT-1, [A-23](#)
 TRAN.AMT-2, [A-23](#)
 TRAN.APPRV-CDE, [A-26](#)
 TRAN.APPRV-CDE-LGTH, [A-25](#)
 TRAN.CRD-FIID, [A-20](#)

TRAN.CRD-LN, [A-20](#)
 TRAN.DEST, [A-20](#)
 TRAN.DFT-CAPTURE-FLG, [A-27](#)
 TRAN.EXP-DAT, [A-23](#)
 TRAN.ICHG-RESP, [A-26](#)
 TRAN.MBR-NUM, [A-22](#)
 TRAN.MULT-CRNCY, [A-24](#)
 TRAN.MULT-CRNCY.AUTH-CONV-RATE, [A-24](#)
 TRAN.MULT-CRNCY.AUTH-CRNCY-CDE, [A-24](#)
 TRAN.MULT-CRNCY.CONV-DAT-TIM, [A-24](#)
 TRAN.MULT-CRNCY.SETL-CONV-RATE, [A-24](#)
 TRAN.MULT-CRNCY.SETL-CRNCY-CDE, [A-24](#)
 TRAN.ORIG, [A-19](#)
 TRAN.OVRRDE-FLG, [A-28](#)
 TRAN.PRE-AUTH-HLD, [A-24](#)
 TRAN.PSEUDO-TERM-ID, [A-27](#)
 TRAN.RESP-CDE, [A-21](#)
 TRAN.RFRL-PHONE, [A-28](#)
 TRAN.RSN-CDE, [A-21](#)
 TRAN.SAVE-ACCT-TYP, [A-28](#)
 TRAN.SETL-FLG, [A-28](#)
 TRAN.TRACK2, [A-21](#)
 TRAN.TRAN-CDE, [1-15](#), [A-17](#)
 TRAN.TRAN-CDE.AA, [1-15](#), [A-18](#)
 TRAN.TRAN-CDE.C, [1-15](#), [A-19](#)
 TRAN.TRAN-CDE.T, [1-15](#), [A-18](#)
 TRAN.TRAN-CDE.TC, [1-15](#), [A-17](#)
 TRAN.USER-FLD2, [A-23](#)
 TRAN.USER-FLD3, [A-24](#)
 TRAN-DAT, [A-36](#)
 TRAN-TIM, [A-36](#)
 TYP, [A-3](#)
 USER-FLD10, [A-50](#)
 USER-FLD9, [A-49](#)
 USER-KEY, [A-41](#)
 ZIP-CDE, [A-49](#)

Positive Authorization method

card status, [3-27](#)
 description of, [3-6](#)
 expiration date, [3-43](#)

Positive Balance Authorization method

card status, [3-27](#)
 description of, [3-6](#)
 expiration date, [3-43](#)
 usage accumulation totals, [3-38](#)

Positive Balance File (PBF), balance totals, [6-2](#)PRADD command, Pregon utility, [E-3](#)

PRDF

see POS Retailer Definition File (PRDF)

Preauthorization

establishing preauthorized holds processing, [3-45](#)
 file segments for, [3-45](#)
 files containing preauthorized holds
 information, [3-45](#)
 limiting number of preauthorized holds, [3-46](#)
 maintaining authorization files for, [3-47](#)

Preauthorization for a lesser amount, with the Parametric Authorization method, [C-7](#)Preauthorized hold transactions, [4-39](#)

Preauthorized holds
 chargebacks of, [3-44](#)
 enhanced processing, [3-44](#)

PRECMD
 see Pregen command file

PREFUPIN
 see Pregen FUP obey file

Pregen command file
 example of, [E-9](#)
 PRECMD, [E-8](#)

Pregen FUP obey file
 PREFUPIN, [E-10](#)

Pregen utility
 commands, [E-2](#)
 error and status messages, [E-16](#)
 function of, [E-2](#)
 message identification, [E-16](#)
 procedures for using, [E-15](#)
 report, [E-11](#)
 running, [E-11](#)

Pregen utility Go file (GOPRE), [E-11](#)

PREMAP table
 action field descriptions, [4-28](#)
 basics of, [4-27](#)
 example of, [4-32](#)
 key field descriptions, [4-27](#)
 selection hierarchy, [4-29](#)

PREND command, Pregen utility, [E-7](#)

Prescreening
 see Transaction screening

Processes, communication between, [2-2](#)

Processing control options
 address verification processing, [3-66](#)
 authorization levels, [3-3](#)
 authorization methods, [3-5](#), [C-1](#)
 card floor limits, [3-18](#)
 card limits, [3-33](#)
 cardholder limits, [3-33](#)
 chargeback processing, [3-60](#)
 draft capture options, [3-30](#)
 generating completions to a host, [3-61](#)
 preauthorized holds, [3-44](#)
 prescreening, [3-26](#)
 representment processing, [3-60](#)
 service types allowed at the terminal level, [3-18](#)
 service types allowed by retailer, [3-20](#)
 store-and-forward functions, [3-40](#)
 token handling, [3-62](#)
 transaction limits, [3-35](#)
 transaction screening, [3-26](#)
 transaction security, [3-21](#)
 transactions allowed at the terminal level, [3-8](#)
 transactions allowed between the BASE24 system and
 an interchange, [3-14](#)
 transactions allowed by account type, [3-12](#)
 usage accumulation totals, [3-38](#)

PTD
 see BASE24-pos Terminal Data files

PTDF
 see POS Terminal Data File

Q

Queue subtract timer, [2-20](#)

R

Reconciliation control message types, [1-38](#)

Referral code, generating, [4-55](#)

Referral processing, [3-31](#)

REMOTESPROUTE command, Pregen utility, [E-7](#)

Representment
 definition of, [3-54](#)
 support by the BASE24-pos product, [3-59](#)

Request, definition of, [1-33](#)

Response Code Description File (RCDF)
 source file for RCDF extended memory table, [F-7](#)

Response Code Description File (RCDF) extended
 memory table
 building, [F-7](#)

Response codes, list of, [B-1](#)

Response, definition of, [1-33](#)

Retailer transaction profile, [3-11](#)

Reversal message types, [1-37](#)

Reversal routing, enhanced, [4-52](#)

Reversal, definition of, [1-33](#)

Router module
 functions of, [2-6](#), [4-2](#)
 message sources, [4-2](#)
 processing overview, [4-2](#)

Routing Table File (RTBL)
 character substitution, [4-22](#)
 information in the, [4-7](#)

Routing transactions
 APCF routing, [3-11](#)
 AST routing, [4-3](#), [4-6](#), [4-12](#), [4-48](#)
 check guarantee, [4-46](#)
 check verification, [4-46](#)
 determining authorizers, [4-3](#), [4-6](#)
 determining the card prefixes, [4-3](#)
 IDF routing, [4-3](#), [4-37](#)
 PREMAP table, [4-27](#)
 SPROUTE routing, [4-4](#), [4-23](#), [4-47](#)

RTBL
 see Routing Table File (RTBL)

S

SAF
 see Store-and-Forward File (SAF)

Second card processing, [3-42](#)

Second-generation terminal, [2-4](#)

- Security
 - see Transaction security
- Security timer, 2-18
- Service type codes
 - reserved, 1-9
 - user-defined, 1-11
- Service-based routing, 2-16
- Split Transaction Routing File (STRF)
 - source file for STRF extended memory table, F-7
- Split Transaction Routing File (STRF) extended memory table
 - building, F-7
- SPROUTE retrieval, for check authorization transactions, 4-47
- SPROUTE routing
 - example, 4-32
 - for check authorization transactions, 4-47
 - PREMAP selection hierarchy, 4-29
 - PREMAP table, 4-27
 - routing methods, 4-30
 - setting up routing tables, 4-32
 - using internal algorithms, 4-24
- SPROUTEND command, Prgen utility, E-8
- Star routing
 - see Routing Table File (RTBL), character substitution
- Station, definition of, 2-11
- STATUS NODE command, F-9
- Store and forward timer
 - Interchange Interface process, 2-21
 - ISO Host Interface process, 2-19
- Store-and-Forward File (SAF), 3-40
- Store-and-forward function
 - hosts, 3-40
 - interchanges, 3-40
 - timers, 2-19, 2-21
 - with online/offline authorization, 3-4
- Supported transactions, 1-15
- T**
- Terminal cutover, 2-4
- Third-generation terminal, 2-4
- Timers
 - Device Handler timers, 2-18
 - functions of, 2-18
 - general timers, 2-18
 - Interchange Interface timers, 2-21
 - ISO Host Interface timers, 2-19
- TKN
 - see Token File (TKN)
- TLF token, 3-63
- Token File (TKN)
 - as used by BIC ISO Interface process, 3-65
 - as used by ISO Host Interface process, 3-65
 - as used by Super Extract process, 3-65
 - as used by the Authorization process, 3-62
 - screen 2, 3-62
 - screen 3, 3-65
 - screen 4, 3-65
- Token handling
 - carrying tokens in the BASE24-pos internal message, 3-65
 - Check Authorization token, 4-11
 - extracting tokens, 3-65
 - general, 3-62
 - logging tokens, 3-62
 - retrieving tokens, 3-62
 - sending tokens to a host or co-network, 3-65
- Totals
 - accumulation, 6-3
 - balance, 6-2
 - BASE24-pos Terminal Data files, 6-13
 - CAF, 6-3
 - clerk, 6-13
 - PBF, 6-2
 - service, 6-13
 - terminal, 6-13
 - UAF, 6-3
- Transaction acquirer, definition of, 1-6
- Transaction authorizers
 - application processes, 1-4
 - AUTHF, 4-8
 - AUTHH, 4-8
 - BASE24 system, 1-3
 - BIC ISO Interface process, 1-4
 - co-network, 1-4
 - determining, 4-3, 4-6
 - DFLT, 4-9
 - hard-coded authorizers, 4-7
 - host, 1-4
 - interchange, 1-4
 - NEGF, 4-9
 - NEGN, 4-9
- Transaction Context Manager process
 - functions of, 2-7
- Transaction identifiers, 1-15
- Transaction issuer, definition of, 1-6
- Transaction logging, 1-40
- Transaction screening
 - card status check, 3-26
 - card verification check, 3-27
 - definition of, 3-26
 - expiration date check, 3-27
 - PIN check, 3-28
 - with Negative Authorization with Usage
 - Accumulation method, 3-27
 - with Negative Authorization without Usage
 - Accumulation method, 3-27
 - with online authorization, 3-4
 - with online host processing, 3-5

- Transaction screening *continued*
 - with online/offline authorization, [3-4](#)
 - with Positive Authorization method, [3-26](#), [3-42](#)
 - with Positive Balance Authorization method, [3-26](#), [3-42](#)
 - withdrawal limits check, [3-28](#)
- Transaction security
 - card verification, [3-22](#)
 - data encryption, [3-24](#)
 - dynamic key management, [3-25](#)
 - message authentication, [3-23](#)
 - PIN encryption, [3-22](#)
 - PIN verification, [3-21](#)
- Transactions
 - administrative, [1-32](#)
 - database impacts of, [6-5](#)
 - defining those allowed, [3-8](#)
 - draft capture, [3-30](#)
 - financial, [1-16](#)
 - identifiers, [1-15](#)
 - non-draft capture, [3-30](#)
 - supported, [1-15](#)
- Triple Data Encryption algorithm (3DEA), [3-23](#)

U

- UAF
 - see* Usage Accumulation File (UAF)
- Usage accumulation
 - definition of usage accumulation periods, [3-38](#)
 - establishing usage accumulation periods, [3-38](#)
 - usage accumulation totals, [3-38](#)
- Usage Accumulation File (UAF)
 - accumulation totals, [6-3](#)
 - Negative Authorization with Usage Accumulation method, [3-5](#)
 - preauthorized holds processing, [3-45](#)
 - usage accumulation, [3-33](#), [3-38](#)

V

- Visa I Device Handler module, [2-5](#)
- Visa II Device Handler module, [2-5](#)

W

- Wait for traffic timer, [2-19](#), [2-21](#)

X

- XPNET process
 - communication between processes, [2-2](#)
 - communication with stations (POS terminals, hosts, interchanges), [2-11](#)
 - queue subtract timer, [2-20](#)